Chapter 1

Basic Ideas of Probability Theory

1.1 Preliminaries

1.1.1 Motivation

One wants to assign chances to outcomes or groups of outcomes of an experiment.

If the number of possible outcomes are finite, say $\omega_1, \ldots, \omega_n$, then a natural way to assign chances or probabilities is to assign for each outcome a number $\mathbf{P}(\omega_i) = p_i$, where $0 \le p_i \le 1$ and $\sum_{i=1}^n p_i = 1$. One can extend this line of thought to experiments with a countable number of outcomes.

However, things become complicated when the number of outcomes becomes uncountable. No reasonable way exists to assign positive probabilities to an uncountable number of outcomes and still make their sum 1.

Example 1 (Cannot assign probability to uncountably many items). Consider the following collection of events $\mathcal{E}_k = \{w : \mathbf{P}(w) \ge \frac{1}{k}\}$. Clearly $|\mathcal{E}_k| \le k$. (From $\sum_w P(w) = 1$, we can see that normalization is critical in this counter-example.) Let $\mathcal{E} = \cup_k \mathcal{E}_k = \{w : \mathbf{P}(w) > 0\}$. *$\mathcal{E}$ contains every outcome w with a positive probability.* This countable union construction gives a direct characterization of events with positive probabilities. (Rational number is dense in $\mathbb{R}$. Between every positive real number ϵ and 0, there exists an integer $k = \lfloor \frac{1}{\epsilon} \rfloor + 1$ such that $0 < \frac{1}{k} < \epsilon$.) However, the number of elements in $\mathcal{E}$ is countable. (Countable union over countable (here even finite) sets is countable [1] Recall a set is countable if we can set up a 1-1 correspondence between the set and the natural numbers.)

A Remedy: Instead of defining probabilities for individual outcomes, one defines probabilities for collections of outcomes (or *events*).

Let $\mathcal{A}$ denote the events for which probabilities are assigned. Then one would like $\mathcal{A}$ to have the following properties:

Definition 1.1.1. *An algebra or a field $\mathcal{A}$:*

(i) if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$ ("or" event), (with (iii), we also have $A \cap B \in \mathcal{A}$ ("and" event) by De Morgan's law $(\cup_i A_i^c)^c = \cap_i A_i$)

(ii) $\emptyset \in \mathcal{A}$, ($\Omega \in \mathcal{A}$ also by the following)

(iii) if $A \in \mathcal{A}$ then $A^c \in \mathcal{A}$

Such a collection $\mathcal{A}$ is called an algebra or a field.[2] We may also require the set of events for which we wish to assign probabilities to have the following additional property:

(i+) If $A_i \in \mathcal{F}$ then $\cup_i^\infty A_i \in \mathcal{F}$.

An algebra or a field with (closed under countable unions) is called a sigma-algebra or a sigma-field.

Example 2 (algebra but not σ-algebra). Let $\mathcal{A}$ be a collection of sets such that:

(1) $\phi \in \mathcal{A}$, $\mathbb{R} \in \mathcal{A}$

(2) $\forall a \in \mathbb{R}$, $(-\infty, a] \in \mathcal{A}$.

(3) $\forall b \in \mathbb{R}$, $(b, \infty) \in \mathcal{A}$.

(4) $\forall k \in \mathbb{N}$, $a_1 < b_1 < a_2 < b_2 < \ldots < a_k < b_k \in \mathbb{R} \cup \{-\infty, \infty\}$, $(a_1, b_1] \cup (a_2, b_2] \cup (a_k, b_k] \in \mathcal{A}$.

Following this construction, we have $\mathcal{A}$ is the collection of subsets of $\mathbb{R}$ of the form

$$A = L \ \cup \ \bigcup_{j=1}^{k} (a_j, b_j] \ \cup \ R$$

for some integer $k \ge 0$ and real numbers satisfying $-\infty < a_1 < b_1 < a_2 < b_2 < \cdots < a_k < b_k < \infty$, and where $a, b \in \mathbb{R} \cup \{-\infty, \infty\}$ are arbitrary, L is either $(-\infty, a]$ or ϕ, R is either (b, ∞) or ϕ.

It is easy to see $\phi \in \mathcal{A}$, $\mathcal{A}$ is closed under complements and closed under finite unions. Thus, $\mathcal{A}$ is an algebra.

To see $\mathcal{A}$ is not a σ-algebra, take $A_n = (0, 1 - \frac{1}{n}]$ then $\cup_{n=1}^\infty A_n = (0, 1) \notin \mathcal{A}$ [3].

Aside: Why can't we then assign probabilities to *all subsets of outcomes*, i.e., to the power set 2^Ω? Let us take $\Omega = \mathbb{R}^3$. Assume that we want the measure to be constructed to yield the "volume" of the set. There is a rather deep result that says that we must make one of the following concessions:

1. The volume of a set might change when it is rotated.
2. The volume of the union of two disjoint sets might differ from the sum of their volumes.
3. Some sets might be tagged "non-measurable", and one would need to check whether a set is "measurable" before talking about its volume.
4. The axioms of ZFC (Zermelo–Fraenkel set theory with the axiom of Choice) might have to be altered.

Most probabilists choose to accept (3), i.e., to tag certain sets as "unmeasurable." See `https://en.wikipedia.org/wiki/Non-measurable_set`

[1] https://math.stackexchange.com/questions/603456/prove-that-the-union-of-countably-many-countable-sets-is-countable

[2] German mathematicians used Körper ("body/field") for many kinds of algebraic systems closed under certain operations. Hence, Carathéodory (1907) spoke of a "Menge(n)körper" (field of sets) when he introduced measure theory. When English translations appeared, Körper became "field". Later algebraists reserved the word "field" for the additive–multiplicative structure we use in algebra today, but probabilists and analysts kept the older phrase "field of sets". When countable closure was added, one just said "σ-field" ("sigma" for "countable").

[3] One can also consider construct $[0, 1] \notin \mathcal{A}$ by taking countable intersections of $B_n = (-\frac{1}{n}, 1]$

In any case, in this class, we will talk about the standard axiomatic treatment under the assumption (3). (This deserves writing a blog to discuss.)

1.1.2 Definitions

σ-algebra and measure space

Definition 1.1.2 (σ-algebra)**.** *A σ-algebra $\mathcal{F}$ is a collection of events $A \subset \Omega$ such that*

(i) $\emptyset, \Omega \in \mathcal{F}$

(ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

(iii) $A_i \in \mathcal{F} \Rightarrow \cup_{i=1}^{\infty} A_i \in \mathcal{F}$. *(By De Morgan's Law, $A_i \in \mathcal{F} \Rightarrow \cap_{i=1}^{\infty} A_i \in \mathcal{F}$)*

The pair $(\Omega, \mathcal{F})$ is defined to be a measurable space.

Remark 1. (ii) and (iii) in the above definition give that σ-algebra (as well as algebra) is closed under set minus: $A_1, A_2 \in \mathcal{A}$, then $A_1 \setminus A_2 = A_1 \cap A_2^c \in \mathcal{A}$.

Example 3 (a few σ-algebra as examples). A few common σ-algebra:

(1) The smallest σ-algebra is $\{\phi, \Omega\}$.

(2) The largest σ-algebra: the power set For a countable space Ω (like $\mathbb{N}$ or $\mathbb{Z}$),
$$2^{\Omega} := \{A : A \subseteq \Omega\}$$
is the *power set* and is a σ-algebra. It contains *all* events. By Cantor's Theorem, even for a countable set its power set is uncountable.

(3) A coarse σ-algebra on $\mathbb{Z}$: even vs. odd Let
$$E := \{z \in \mathbb{Z} : z \text{ is even}\}, \qquad O := \{z \in \mathbb{Z} : z \text{ is odd}\}.$$
Then $\mathbb{Z} = E \sqcup O$ is a partition and
$$\mathcal{F}_{\text{parity}} := \{\emptyset,\ \mathbb{Z},\ E,\ O\}$$
is a σ-algebra. The interpretation is that you can talk about "even" vs. "odd" but not about singletons like $\{2\}$.

Definition 1.1.3 (Measure)**.** *A measure μ on the measurable space $(\Omega, \mathcal{F})$ is a mapping $\mu : \mathcal{F} \mapsto [0, \infty]$ satisfying:*

1. $\mu(\emptyset) = 0$. *(Non-negativity is implied in its definition.)*
2. *if $A_i \in \mathcal{F}$ is a countable collection of pairwise disjoint sets, then $\mu(\sqcup_i A_i) = \sum_i \mu(A_i)$, where $\sqcup$ denotes disjoint union. (Countable additivity on disjoint sets.)*

Remark 2 (Monotonicity of measures). Algebra is closed under set minus, and measure is non-negative and countable additive. Therefore if $A_1, A_2 \in \mathcal{A}$ and $A_1 \subseteq A_2$, then μ on $(\Omega, \mathcal{A})$ satisfies $\mu(A_2) = \mu(A_1) + \mu(A_2 \setminus A_1) \geq \mu(A_1)$. This is also known as monotonicity of measures.

It is also a direct consequence of sub-additivity.

Definition 1.1.4 (Probability measure)**.** *A probability measure is a measure that satisfies $\mathbf{P}(\Omega) = 1$.*

Note that we can still have a countably additive probability measure on an algebra. An algebra can contain some countable unions of certain subsets, like the $A_i \in \mathcal{F}$ in the definition 1.1.3 above.

Exercise 1.1.1. *Show that the following holds for a probability measure:*

1. $\mathbf{P}(\cup_i A_i) \leq \sum_i \mathbf{P}(A_i)$ *(sub-additivity)*
2. *if $A_i \uparrow A$, ($A_1 \subseteq A_2 \subseteq \ldots \subseteq A$, dilates up to A), then $\mathbf{P}(A_i) \uparrow \mathbf{P}(A)$ (monotone up-limits, continuity from below)*
3. *if $A_i \downarrow A$, ($A_1 \supseteq A_2 \supseteq \cdots \supseteq A$, shrinks down to A), then $\mathbf{P}(A_i) \downarrow \mathbf{P}(A)$ (monotone down-limits, continuity from above, not true for unbounded measure.)*

Proof. (1) Sub-additivity. Define the disjointified sets $B_1 := A_1$, $B_n := A_n \setminus \bigcup_{j=1}^{n-1} A_j$, $n \geq 2$. Then $\{B_n\}_{n \geq 1}$ are pairwise disjoint, $B_n \subseteq A_n$ for all n, and $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$. By countable additivity on disjoint sets,
$$\mathbf{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbf{P}\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mathbf{P}(B_n) \leq \sum_{n=1}^{\infty} \mathbf{P}(A_n).$$
while the last inequality is from index-wise monotonicity of measures.

(2) Continuity from below (monotone up-limits). Assume $A_1 \subseteq A_2 \subseteq \cdots$ and let $A = \bigcup_{n=1}^{\infty} A_n$. Define $C_1 := A_1$, $C_n := A_n \setminus A_{n-1}$, $n \geq 2$. Then $\{C_n\}$ are pairwise disjoint and for each n, $A_n = \bigcup_{k=1}^{n} C_k$, and $A = \bigcup_{k=1}^{\infty} C_k$. Hence by countable additivity,
$$\mathbf{P}(A_n) = \sum_{k=1}^{n} \mathbf{P}(C_k) \uparrow \sum_{k=1}^{\infty} \mathbf{P}(C_k) = \mathbf{P}(A).$$
(3) Continuity from above (monotone down-limits). Assume $A_1 \supseteq A_2 \supseteq \cdots$ and let $A = \bigcap_{n=1}^{\infty} A_n$. Then the complements satisfy $A_1^c \subseteq A_2^c \subseteq \cdots$ and
$$\bigcup_{n=1}^{\infty} A_n^c = \left(\bigcap_{n=1}^{\infty} A_n\right)^c = A^c.$$
Applying (2) to $\{A_n^c\}$ gives
$$\mathbf{P}(A_n^c) \uparrow \mathbf{P}(A^c).$$
Since $\mathbf{P}$ is a probability measure, $\mathbf{P}(B^c) = 1 - \mathbf{P}(B)$ for any B, so
$$\mathbf{P}(A_n) = 1 - \mathbf{P}(A_n^c) \downarrow 1 - \mathbf{P}(A^c) = \mathbf{P}(A).$$
This proves continuity from above. □

Remark 3. If A_i is a collection such that $\mathbf{P}(A_i) = 0, \forall i$ then $\mathbf{P}(\cup_i A_i) = 0$ (from sub-additivity).

Remark 4. It is shown in Homework 1 that a finite additive probability measure is countably additive if and only if it is continuous from below (or equivalently, continuous from above, as it is a probability (finite) measure).

Exercise 1.1.2. *Show that (1), (2) holds for general measures, while (3) may not.*
(Hint: To see a counter example to (3), define $B_i = p_i\mathbb{N}$ (here p_i is the ith prime) and set $A_i = \mathbb{N} \setminus \cup_{j=1}^{i} B_i$. Use counting measure as the measure and show that $\mu(A_i) = \infty, \forall i$, while $A = \{1\}$, and hence $\mu(A) = 1$.)

Proof. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space (no finiteness assumption on $\mu(\Omega)$).

(1) Sub-additivity holds for any measure. The things needed are countable additivity on disjoint sets and non-negativity of measure.

(2) Continuity from below holds for any measure. The things needed are countable additivity on disjoint sets and non-negativity of measure.

(3) Continuity from above can fail for general measures. We cannot know the behavior of $\mathbf{P}(A_n)$ from $\mathbf{P}(A_n^c)$. Consider $\Omega = \mathbb{N}$ with $\mathcal{F} = 2^{\mathbb{N}}$ and let μ be *counting measure*:

$$\mu(E) = |E| \in \{0, 1, 2, \dots, \infty\}.$$

Let p_i be the ith prime and define

$$B_i := p_i\mathbb{N} = \{p_i, 2p_i, 3p_i, \dots\}, \qquad A_i := \mathbb{N} \setminus \bigcup_{j=1}^{i} B_j.$$

(The idea is to use primes as coset "leaders".) Then $A_{i+1} \subseteq A_i$, so $A_i \downarrow A := \bigcap_{i=1}^{\infty} A_i$.

Claim 1: $\mu(A_i) = \infty$ for all i. Indeed, $\bigcup_{j=1}^{i} B_j$ is the set of integers divisible by at least one of $p_1, \dots, p_i$. Hence A_i is the set of integers *not divisible* by any of these primes. In particular, the infinite set (the "+1" trick)

$$S_i := \left\{ (\prod_{j=1}^{i} p_j \cdot k) + 1 : \ k \in \mathbb{N} \right\}$$

is contained in A_i, because every element of S_i is congruent to 1 (mod p_j) for each $j \le i$, and thus not divisible by any p_j. Therefore A_i is infinite, so $\mu(A_i) = \infty$.

Claim 2: $A = \{1\}$. If $n \ge 2$, then n has a prime divisor p_r for some r; hence $n \in B_r \subseteq \bigcup_{j=1}^{\infty} B_j$, so $n \notin A$. On the other hand, 1 is not divisible by any prime, so $1 \in A$. Thus $A = \{1\}$ and $\mu(A) = 1$.

Now we have $A_i \downarrow A$ but

$$\mu(A_i) = \infty \text{ for all } i, \qquad \text{while} \qquad \mu(A) = 1.$$

Hence $\mu(A_i) \not\downarrow \mu(A)$, showing that continuity from above fails in general.

Continuity from above *does* hold for general measures if one assumes $\mu(A_1) < \infty$ (this finiteness prevents the above pathology). □

Smallest σ-algebra generated

Consider a collection of events $\mathcal{A}$. We wish to *extend* the collection to a sigma field. Or, in other words, is there a smallest *sigma* field that contains $\mathcal{A}$? The answer is Yes. To see this, the following exercise is useful.

Exercise 1.1.3. *Let $\mathcal{F}_\alpha$ be an arbitrary collection of σ-fields. Then $\cap_\alpha \mathcal{F}_\alpha$ is also a σ-field.*

Proof. Check through the definition of σ-field,

(i) $\phi \in \cap_\alpha \mathcal{F}_\alpha$ as $\phi \in \mathcal{F}_\alpha$.

(ii) if $A \in \cap_\alpha \mathcal{F}_\alpha$, then $\forall \alpha, A \in \mathcal{F}_\alpha$, thus, $\forall \alpha, A^c \in \mathcal{F}_\alpha$, $A^c \in \cap_\alpha \mathcal{F}_\alpha$.

(iii) if $A_i \in \cap_\alpha \mathcal{F}_\alpha$, then $\forall \alpha$, $\cup_i^\infty A_i \in \mathcal{F}_\alpha$, thus $\cup_i^\infty A_i \in \cap_\alpha \mathcal{F}_\alpha$.

□

Example 4 (union of σ-field is not a σ-field). The example is trivial. Take $\mathcal{F}_A \cup \mathcal{F}_B = \{\phi, A, B, A^c, B^c, \Omega\}$ and $A \cup B$ is not in it.

From the above exercise, the intersection of all σ-algebras that contain $\mathcal{A}$ is a σ-algebra and is the smallest σ-algebra that contains $\mathcal{A}$. This is denoted as $\sigma(\mathcal{A})$, and called the σ-algebra generated by $\mathcal{A}$, where $\mathcal{A}$ is an arbitrary collection of subsets.

Supposed Ω is a topological space (i.e., equipped with open sets)[4]. Then the σ-algebra generated by the open sets is called the *Borel σ-algebra*.

1.1.3 Other classes of sets

Besides σ-algebra's and algebras, other collections of sets with certain properties also turn out to be helpful in probability theory.

Monotone class

Definition 1.1.5 (Monotone class)**.** *A collection $\mathcal{M}$, of subsets of Ω, is called a monotone class, if $\mathcal{M}$ is (1) non-empty, and satisfies the following two conditions:*

(2) If a countable increasing collection of sets, $M_1 \subseteq M_2 \subseteq \cdots$, belong to $\mathcal{M}$, then $\cup_i M_i \in \mathcal{M}$. In other words, $\mathcal{M}$ is closed under monotone up limits.

(3) If a countable decreasing collection of sets, $M_1 \supseteq M_2 \supseteq \cdots$, belong to $\mathcal{M}$, then $\cap_i M_i \in \mathcal{M}$. In other words, $\mathcal{M}$ is closed under monotone down limits.

Remark 5. A monotone class does not necessarily contain the empty set ϕ. But it needs to be non-empty itself.

Example 5 (a monotone class). An example of a monotone class on the reals is the collection $\{(0, 1], (2, 3]\}$. This is not closed under unions or intersections. (Monotone class only requires to include the closure in monotonic dilation and shrinking.) Note that increasing unions can only be formed by taking identical elements; hence, it is a monotone class.

Example 6 (not monotone classes). A few examples that are not a monotone class, mostly because countable infinity creates a "fundamental" change, alike $0.\dot{9} = 1$:

(1) Finite subsets of $\mathbb{R}$, union goes to infinity. Let

$$\mathcal{F} := \{A \subseteq \mathbb{R} : \ A \text{ is finite}\}.$$

Define an increasing sequence $A_n := \{1, 2, \dots, n\}$. Then $A_1 \subseteq A_2 \subseteq \cdots$ and $A_n \in \mathcal{F}$ for all n, but $\bigcup_{n=1}^{\infty} A_n = \mathbb{N} \notin \mathcal{F}$.

(2) Dedekind cut, union of rational intervals goes to an irrational number. Let

$$\mathcal{H}_{\mathbb{Q}} := \{\emptyset, \mathbb{R}\} \cup \{(-\infty, q] \ : \ q \in \mathbb{Q}\}.$$

Choose a sequence $(q_n)_{n\ge1} \subset \mathbb{Q}$ such that $q_n \uparrow \sqrt{2}$ and set $A_n := (-\infty, q_n]$. Then $A_1 \subseteq A_2 \subseteq \cdots$ and $A_n \in \mathcal{H}_{\mathbb{Q}}$ for all n, but

$$\bigcup_{n=1}^{\infty} A_n = (-\infty, \sqrt{2})$$

which is not in $\mathcal{H}_{\mathbb{Q}}$. Hence $\mathcal{H}_{\mathbb{Q}}$ is not a monotone class.

(3) Open sets (standard metric) in $\mathbb{R}$ shrinks to a singleton. Let

$$\mathcal{O} := \{U \subseteq \mathbb{R} : \ U \text{ is open}\}.$$

Define a decreasing sequence $U_n := \left(-\frac{1}{n}, \frac{1}{n}\right)$. Then $U_1 \supseteq U_2 \supseteq \cdots$ and $U_n \in \mathcal{O}$ for all n, but

$$\bigcap_{n=1}^{\infty} U_n = \{0\} \notin \mathcal{O}.$$

Hence $\mathcal{O}$ is not a monotone class.

[4](Wikipedia) In mathematics, a topological space is, roughly speaking, a geometrical space in which closeness is defined but cannot necessarily be measured by a numeric distance. More specifically, a topological space is a set whose elements are called points, along with an additional structure called a topology, which can be defined as a set of neighbourhoods for each point that satisfy some axioms formalizing the concept of closeness.

Remark 6. It is clear that arbitrary intersections of monotone classes in a monotone class. Therefore, given a collection of sets $\mathcal{S}$, we can talk about the monotone class generated by $\mathcal{S}$ denoted by $\mathcal{M}(\mathcal{S})$, to be the smallest monotone class that contains $\mathcal{S}$.

The beauty of the following theorem is that for any algebra $\mathcal{A}$, the monotone class and sigma-algebra it generated is the same. Note that these two notions are defined from different aspects:

m. : closure under countable monotonic dilation and shrinkage,

σ-a. : contains the empty set ϕ, closure under countable unions and taking complement.

One additional thing is that a monotone class does not necessarily contain ϕ, while an algebra does.

Theorem 1.1.4 (Monotone Class Theorem)**.** *Let $\mathcal{A}$ be an algebra. Let $\mathcal{M}(\mathcal{A})$ be the monotone class generated by $\mathcal{A}$. Then $\mathcal{M}(\mathcal{A}) = \sigma(\mathcal{A})$.*

Proof. (Assume there exists a unique smallest set $\mathcal{M}(\mathcal{A})$. Uniqueness is from that intersection of monotone classes is a monotone class.)

(i) $\sigma(\mathcal{A}) \supseteq \mathcal{M}(\mathcal{A})$ is by definition of $\sigma(\mathcal{A})$, and $\mathcal{M}(\mathcal{A})$ is the smallest monotone class contains $\mathcal{A}$.

(ii) $\sigma(\mathcal{A}) \subseteq \mathcal{M}(\mathcal{A})$, to show that $\mathcal{M}(\mathcal{A})$ is a σ-algebra:

(1) Since $\emptyset \in \mathcal{A}$ we have $\emptyset \in \mathcal{M}(\mathcal{A})$.

(2) Closed under complement: Let $\mathcal{M}_0(\mathcal{A}) = \{B : B \in \mathcal{M}(\mathcal{A})$ s.t. $B^c \in \mathcal{M}(\mathcal{A})\}$. $\mathcal{A}$ contains its subsets and all corresponding complements, and $\mathcal{M}(\mathcal{A})$ contains all sets in $\mathcal{A}$, thus $\mathcal{A} \subseteq \mathcal{M}_0(\mathcal{A})$. (non-empty)
Since $\mathcal{M}(\mathcal{A})$ is closed under monotone limits, it is clear that $\mathcal{M}_0(\mathcal{A})$ is also a monotone class. (Taking sets that have complements preserves monotone class property.) As $\mathcal{M}(\mathcal{A})$ is the smallest monotone class contains $\mathcal{A}$, we have $\mathcal{M}_0(\mathcal{A}) = \mathcal{M}(\mathcal{A})$.
(If there is an "odd" one in $\mathcal{M}(\mathcal{A})$ without a complement, then the "smallest" requirement says this "odd" one shouldn't be in $\mathcal{M}(\mathcal{A})$ in the first place.)

(3) Closed under union: If true, then $\mathcal{M}(\mathcal{A})$ automatically closed under countable union by definition of monotone class. So we only need to show closure under pairwise union.
Towards this, fix arbitrary $A \in \mathcal{A}$, and define
$$\mathcal{M}_A = \{B \in \mathcal{M}(\mathcal{A}) : A \cap B \in \mathcal{M}(\mathcal{A})\}.$$
$\forall A' \in \mathcal{A},\ A \cap A' \in \mathcal{A}, \in \mathcal{M}(\mathcal{A})$, hence $\mathcal{A} \subseteq \mathcal{M}_A$ (1. non-empty) and $\mathcal{M}_A$ is a monotone class. (2. Taking intersection with a fixed set preserves monotone class property.) Hence $\mathcal{M}_A = \mathcal{M}(\mathcal{A})$. ($\mathcal{M}(\mathcal{A})$ is the smallest.)
In words, $\forall B \in \mathcal{M}(\mathcal{A})$, for arbitrary $A \in \mathcal{A}$, we have $A \cap B \in \mathcal{M}(\mathcal{A})$.
Now fix arbitrary $C \in \mathcal{M}(\mathcal{A})$ and define
$$\mathcal{M}_C = \{B \in \mathcal{M}(\mathcal{A}) : C \cap B \in \mathcal{M}(\mathcal{A})\}.$$
Using the previous argument, for arbitrary $A \in \mathcal{A}$, for arbitrary $C \in \mathcal{M}(\mathcal{A})$, we have $C \cap A \in \mathcal{M}(\mathcal{A})$. We have $\mathcal{A} \subseteq \mathcal{M}_C$ and $\mathcal{M}_C$ is a monotone class. Hence $\mathcal{M}_C = \mathcal{M}(\mathcal{A})$.
Let C be arbitrary set in $\mathcal{M}(\mathcal{A})$, thus $\mathcal{M}(\mathcal{A})$ is closed under intersection. Combine this with closed under taking complement and De Morgan's Law, we get closed under union. (Or use union when constructing $\mathcal{M}_A$ and $\mathcal{M}_C$.)

□

Definition 1.1.6 (π-system)**.** *A collection $\mathcal{H}$, of subsets of Ω, is called a π-system, if $\mathcal{H}$ is (1) non-empty, and (2) $H_1, H_2 \in \mathcal{H}$ implies that $H_1 \cap H_2$ belongs to $\mathcal{H}$.*
("π" looks like "$\cap$".)

Definition 1.1.7 (λ-system)**.** *A collection $\mathcal{D}$, of subsets of Ω, is called a λ-system (or a d-system), if:*

(1) $\Omega \in \mathcal{D}$,

(2) $D_1, D_2 \in \mathcal{D}$ and $D_1 \subseteq D_2$ implies $(D_2 \setminus D_1) \in \mathcal{D}$,

(3) If a countable increasing collection of sets, $D_1 \subseteq D_2 \subseteq \cdots$, belong to $\mathcal{D}$, then $\cup_i D_i \in \mathcal{D}$. In other words, $\mathcal{D}$ is closed under monotone up limits (dilation).

Remark 7. Arbitrary intersections of λ-systems is a λ-system. Therefore, given a collection of sets $\mathcal{S}$, we can talk about the λ-system generated by $\mathcal{S}$ denoted by $\mathcal{D}(\mathcal{S})$, to be the smallest λ-system that contains $\mathcal{S}$.

Remark 8. Since $\Omega \in \mathcal{D}$, if $D \in \mathcal{D}$ we have $(\Omega \setminus D) = D^c \in \mathcal{D}$. Therefore, it is closed under taking complements, and consequently, $\mathcal{D}$ is also closed under monotone down limits.

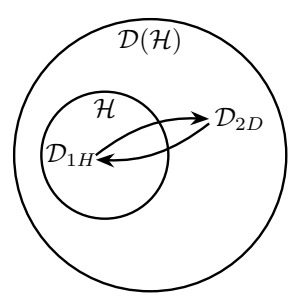
Theorem 1.1.5 (Dynkin's π-λ Theorem)**.** *Let $\mathcal{H}$ be a π-system. Let $\mathcal{D}(\mathcal{H})$ be the λ-system generated by $\mathcal{H}$. Then $\mathcal{D}(\mathcal{H}) = \sigma(\mathcal{H})$.*

Proof.

(i) Any σ-algebra is automatically a λ-system. If $D_1 \subseteq D_2$, then $D_2 \setminus D_1 = D_2 \cap D_1^c \in \sigma(\mathcal{H})$. Note that $\mathcal{D}(\mathcal{H})$ is the smallest λ-system containing $\mathcal{H}$, thus $\mathcal{D}(\mathcal{H}) \subseteq \sigma(\mathcal{H})$.

(ii) To show $\mathcal{D}(\mathcal{H}) \supseteq \sigma(\mathcal{H})$, that is, $\mathcal{D}(\mathcal{H})$ is also a σ-algebra: If $\mathcal{D}(\mathcal{H})$ was additionally a π-system, then for any $\{D_i\} \in \mathcal{D}(\mathcal{H})$, we have $\cup_{i=1}^k D_i = \left(\cap_{i=1}^k D_i^c\right)^c \in \mathcal{D}(\mathcal{H})$. Further as $\mathcal{D}(\mathcal{H})$ is closed under monotone up-limits, $\cup_i D_i = \cup_k \left(\cup_{i=1}^k D_i\right) \in \mathcal{D}(\mathcal{H})$. (Closure under complement is automatic by definition of λ-system.) Thus, it suffices to show that $\mathcal{D}(\mathcal{H})$ is also a π-system.

(iii) (Alike the proof of Monotone class theorem.) To show $\mathcal{D}(\mathcal{H})$ is also a π-system:



(a) Fix $H \in \mathcal{H}$. Define
$$\mathcal{D}_{1H} = \{D \in \mathcal{D}(\mathcal{H}) : D \cap H \in \mathcal{D}(\mathcal{H})\}.$$
Observe that $\Omega \in \mathcal{D}_{1H}$, and since $\mathcal{H}$ is a π-system, $\mathcal{H} \subseteq \mathcal{D}_{1H}$. Further, if $D_2, D_1 \in \mathcal{D}_{1H}$ and $D_2 \supseteq D_1$, as $(D_2 \setminus D_1) \cap H = (D_2 \cap H) \setminus (D_1 \cap H)$, we have that $(D_2 \setminus D_1) \in \mathcal{D}_{1H}$. In the above we used that $\mathcal{D}(\mathcal{H})$ is a λ-system, and that $(D_2 \setminus H) \supseteq (D_1 \setminus H)$,

as $D_2 \supseteq D_1$. Finally if a countable increasing collection of sets, $D_1 \subseteq D_2 \subseteq \cdots$, belong to $\mathcal{D}_{1H}$, then as $(\cup_i D_i) \cap H = \cup_i (D_i \cap H) \in \mathcal{D}$, we have that $\cup_i D_i \in \mathcal{D}_{1H}$. This implies that $\mathcal{D}_{1H} \subseteq \mathcal{D}(\mathcal{H})$ is a λ-system containing $\mathcal{H}$, implying $\mathcal{D}_{1H} = \mathcal{D}(\mathcal{H})$. This implies that for any $H \in \mathcal{H}$ and $D \in \mathcal{D}(\mathcal{H})$, $H \cap D \in \mathcal{D}(\mathcal{H})$.

(b) Fix $D \in \mathcal{D}(\mathcal{H})$. Define
$$\mathcal{D}_{2D} = \{\hat{D} \in \mathcal{D}(\mathcal{H}) : \hat{D} \cap D \in \mathcal{D}(\mathcal{H})\}.$$
Observe that $\Omega \in \mathcal{D}_{2D}$, and $\mathcal{H} \subseteq \mathcal{D}_{2D}$ from the previous part. Now, if $D_2, D_1 \in \mathcal{D}_{2D}$ and $D_2 \supseteq D_1$, as $(D_2 \setminus D_1) \cap D = (D_2 \cap D) \setminus (D_1 \cap D)$, we have that $(D_2 \setminus D_1) \in \mathcal{D}_{2D}$. As in the earlier part, we again used that $\mathcal{D}(\mathcal{H})$ is a λ-system, and that $(D_2 \setminus D) \supseteq (D_1 \setminus D)$, as $D_2 \supseteq D_1$. Finally if a countable increasing collection of sets, $D_1 \subseteq D_2 \subseteq \cdots$, belong to $\mathcal{D}_{2D}$, then as $(\cup_i D_i) \cap D = \cup_i (D_i \cap D) \in \mathcal{D}$, we have that $\cup_i D_i \in \mathcal{D}_{2D}$. This implies that $\mathcal{D}_{2D} \subseteq \mathcal{D}(\mathcal{H})$ is a λ-system containing $\mathcal{H}$, implying $\mathcal{D}_{2D} = \mathcal{D}(\mathcal{H})$.

□

1.1.4 Existence and construction of measures

Theorem 1.1.6 (Caratheodory Extension Theorem)**.** *Any countably additive probability measure $\mathbf{P}$ on an algebra $\mathcal{A}$ extends uniquely as a countably additive probability measure on $\mathcal{F} = \sigma(\mathcal{A})$.*

Proof. The proof consists of various steps. Here, we outline the main steps.

(1) Define a quantity $\mathbf{P}^*(B), B \in \mathcal{F}$ (which will turn out an outer measure) as follows:
$$\mathbf{P}^*(B) = \inf_{A_i \in \mathcal{A},\ \cup_i A_i \supseteq B} \sum_i \mathbf{P}(A_i).$$
(Note that $\cup_i^\infty A_i$ may not in $\mathcal{A}$. Though this union can still be a finite union if only allow finite number of subsets to be non-empty.)

Remark 9. Without loss of generality, we can assume the cover of B used in $\mathbf{P}^*(B)$, that is, $\cup_i A_i$, to be disjoint:

– For any cover $\cup_i A_i$ of B, we can obtain $\cup_i \tilde{A}_i \supseteq B$, which is pairwise disjoint:
$$\tilde{A}_1 \triangleq A_1, \quad \tilde{A}_i = A_i \setminus \bigcup_{j<i} A_j, \ (i \geq 2).$$
$\mathcal{A}$ is an algebra, thus $A_i \subseteq \mathcal{A} \implies \tilde{A}_i \subseteq \mathcal{A}$. This disjoint collection of sets can still cover $B \subseteq \cup_i \tilde{A}_i = \cup_i A_i$.
For each i, $\tilde{A}_i \subseteq A_i$, therefore by the monotonicity of measure, $\mathbf{P}(\tilde{A}_i) \leq \mathbf{P}(A_i)$. As $\mathbf{P}$ is a probability (finite) measure on $\mathcal{A}$,
$\lim_{n\to\infty} \sum_{i=1}^\infty \mathbf{P}(\tilde{A}_i) \leq 1$,
$$\sum_{i=1}^\infty \mathbf{P}(\tilde{A}_i) \leq \sum_{i=1}^\infty \mathbf{P}(A_i).$$

Therefore taking the infimum over covers does not change if we restrict to disjoint covers.

(i) Subadditivity: $\mathbf{P}^*(\cup_i B_i) \leq \sum_i \mathbf{P}^*(B_i)$. (Global infimum is less than the sum of local infimum. We only need to write everything down, the proof is automatically emerging.)

Fix $\epsilon > 0$. For each i, choose a disjoint cover $B_i \subseteq \cup_j A_{i,j}$, $(A_{i,j})_{j\geq 1} \subseteq \mathcal{A}$ such that
$$\mathbf{P}^*(B_i) \leq \sum_j \mathbf{P}(A_{i,j}) \leq \mathbf{P}^*(B_i) + \epsilon 2^{-i}.$$
Such a cover exists by the definition of infimum.

Note that countable union over countable sets is countable.[5] Then $\{A_{i,j} : i, j \geq 1\}$ is countable and covers $\bigcup_i B_i$, hence plugging in the definition of infimum again we have:
$$\mathbf{P}^*\Big(\bigcup_i B_i\Big) \leq \sum_{i,j} \mathbf{P}(A_{i,j}).$$
thus
$$\mathbf{P}^*\Big(\bigcup_i B_i\Big) \leq \sum_i \big(\mathbf{P}^*(B_i) + \varepsilon 2^{-i}\big) = \sum_i \mathbf{P}^*(B_i) + \varepsilon$$
Letting $\epsilon \to \infty$ gives the desired inequality.

(ii) If $A \in \mathcal{A}$ then $\mathbf{P}^*(A) = \mathbf{P}(A)$.
If $A \in \mathcal{A}$ then $\mathbf{P}^*(A) \leq \mathbf{P}(A)$. ($A \subseteq A$ is a valid cover.)
If $A \in \mathcal{A}$ then $\mathbf{P}^*(A) \geq \mathbf{P}(A)$. Let $(A_i)_{i\geq 1} \subseteq \mathcal{A}$ be any cover of A, and w.l.o.g assume (A_i) are pairwise disjoint. Then $A \cap A_i \in \mathcal{A}$, the sets $(A \cap A_i)$ are disjoint, and $A = \bigcup_i (A \cap A_i)$. By countable additivity of $\mathbf{P}$ on $\mathcal{A}$,
$$\mathbf{P}(A) = \sum_i \mathbf{P}(A \cap A_i) \leq \sum_i \mathbf{P}(A_i).$$
Taking the infimum over all covers (A_i) yields $\mathbf{P}(A) \leq \mathbf{P}^*(A)$.

(2) Define $E \in \mathcal{F}$ to be measurable (Caratheodory's criterion) if for all $B \in \mathcal{F}$:
$$\mathcal{E} = \{E : \forall B \in \mathcal{F}, \mathbf{P}^*(B) \geq \mathbf{P}^*(B \cap E) + \mathbf{P}^*(B \cap E^c)\}.$$
$\mathcal{E}$ is the set of all measurable sets.

The goal is to show $\mathcal{E}$ is a σ-field and that $\mathbf{P}^*$ is a countably additive measure on $\mathcal{E}$. Finally show that $\mathcal{A} \subseteq \mathcal{E}$ and hence $\sigma(\mathcal{A}) \subseteq \mathcal{E}$.

Remark 10.

– Note that subadditivity gives $\mathbf{P}^*(B) \leq \mathbf{P}^*(B \cap E) + \mathbf{P}^*(B \cap E^c)$ as $B = (B \cap E) \cup (B \cap E^c)$. This measurable condition is equivalent to exact additivity under splitting by E, $\forall B \in \mathcal{F}, \mathbf{P}^*(B) = \mathbf{P}^*(B \cap E) + \mathbf{P}^*(B \cap E^c)$. In the proof, when measurability is true, additivity is true.

– $(B \cap E)$ and $(B \cap E^c)$ are both in $\mathcal{F}$ and disjoint (a partition, thus additivity for measure).

(i) $\mathcal{E}$ is an algebra:
Clearly $\mathbf{P}^*(\emptyset) = \emptyset$, hence $\emptyset \in \mathcal{E}$. It is also clear that if $E \in \mathcal{E}$, then $E^c \in \mathcal{E}$. Now suppose $E_1, E_2 \in \mathcal{E}$. Then observe that $E_1 \cup E_2$ is measurable:
$$\begin{aligned}\mathbf{P}^*(B) &= \mathbf{P}^*(B \cap E_1) + \mathbf{P}^*(B \cap E_1^c) \\ &= \mathbf{P}^*(B \cap E_1) + \mathbf{P}^*(B \cap E_1^c \cap E_2) \\ &\quad + \mathbf{P}^*(B \cap E_1^c \cap E_2^c) \\ &\geq \mathbf{P}^*(B \cap (E_1 \cup E_2)) \\ &\quad + \mathbf{P}^*(B \cap E_1^c \cap E_2^c).\end{aligned}$$
Here the equalities follow from the measurability of E_1, E_2, and the inequality follows from sub-additivity of $\mathbf{P}^*(\cdot)$. This implies finite unions are measurable. We can conclude that $\mathcal{E}$ is an algebra.

[5] Zigzaging, the same idea to show rationals are countable.

(ii) $\mathcal{E}$ is a σ-algebra:

Let E_i be a pairwise disjoint collection. Clearly $G_n = \cup_{i=1}^n E_i$ (finite union) is measurable. Let $G = \cup_i E_i$. Therefore

$$\begin{aligned}\mathbf{P}^*(B) &= \mathbf{P}^*(B \cap G_n) + \mathbf{P}^*(B \cap G_n^c) \\ &\geq \mathbf{P}^*(B \cap G_n) + \mathbf{P}^*(B \cap G^c) \\ &= \mathbf{P}^*(B \cap G_n \cap E_n) + \mathbf{P}^*(B \cap G_n \cap E_n^c) \\ &\quad + \mathbf{P}^*(B \cap G^c) \\ &= \mathbf{P}^*(B \cap E_n) + \mathbf{P}^*(B \cap G_{n-1}) \\ &\quad + \mathbf{P}^*(B \cap G^c) \\ &= \sum_{i=1}^n \mathbf{P}^*(B \cap E_i) + \mathbf{P}^*(B \cap G^c).\end{aligned}$$

Let $n \to \infty$ we have $\mathbf{P}^*(B) \geq \sum_{i=1}^\infty \mathbf{P}^*(B \cap E_i) + \mathbf{P}^*(B \cap G^c) \geq \mathbf{P}^*(B \cap G) + \mathbf{P}^*(B \cap G^c)$, where the last inequality follows from sub-additivity. Hence $G = \cup_i E_i$ is measurable, $\mathcal{E}$ is an σ-algebra.

Remark 11.

* Again, countable additivity is implied: $\mathbf{P}^*(G) = \sum_{i=1}^\infty \mathbf{P}^*(E_i)$.

(iii) $\mathcal{A} \subseteq \mathcal{E}$,

For any B, take a "good" cover of B and show that this induces a cover on $B \cap A$ and $B \cap A^c$. Hence show that

$$\mathbf{P}^*(B) \geq \mathbf{P}^*(B \cap A) + \mathbf{P}^*(B \cap A^c).$$

Thus $\mathcal{A} \subseteq \mathcal{E}$ and since $\mathcal{E}$ is a sigma-algebra, we also have $\sigma(\mathcal{A}) \subseteq \mathcal{E}$. Therefore,

$$\mathcal{F} = \sigma(\mathcal{A}) \subseteq \mathcal{E} \subseteq \mathcal{F}.$$

(3) To show uniqueness, show the following: Let $\mathcal{M} = \{B \in \mathcal{F} : p_1(B) = p_2(B)\}$ be the collection of all sets in which the two countably additive probability extensions agree.

- $\mathcal{M}$ is a monotone class. (Monotone up-limits and down limits for probability measures.)
- Clearly $\mathcal{A} \subseteq \mathcal{M}$. Since $\mathcal{A}$ is an algebra and $\mathcal{M}$ is a monotone class, by monotone class theorem, $\sigma(\mathcal{A}) \subseteq \mathcal{M}$. Since $\sigma(\mathcal{A}) = \mathcal{F}$, it follows that $p_1 = p_2$ on $\mathcal{F}$. This proves uniqueness over $\mathcal{F}$.

□

Remark 12. We could also have extended the definition of $\mathbf{P}^*(B)$ to all $B \in 2^\Omega$, but it does not buy us anything in the context of this proof.

Thus, Caratheodory's extension theorem reduces the burden of constructing measures on σ-algebras to that on algebras.

Constructing countably additive probability measures on algebras

We will now see that there is a canonical way of constructing a countably additive probability measure on $\mathcal{F}_B$, the *Borel σ-algebra* on the real line.

Consider the following collection of (left open, right closed) intervals: $\mathcal{I} = \{I_{a,b} : -\infty \leq a < b \leq \infty\}$, where $I_{a,b} = (a, b]$ when $b < \infty$ and $-\infty < a$, and $I_{a,\infty} = (a, \infty)$, $I_{-\infty,b} = (-\infty, b]$.

$\mathcal{A}_B$, of finite disjoint union of members of $\mathcal{I}$ is an algebra.

Remark 13. We have shown that $\mathcal{A}_B$ is an algebra but not a sigma-algebra in example 2. In fact, the Borel σ-algebra is the smallest sigma-algebra contains $\mathcal{A}_B$: $\mathcal{F}_B = \sigma(\mathcal{A}_B)$.

Definition 1.1.8. *A function $F(x)$ is called a distribution function if:*

(i) non-decreasing,

(ii) right-continuous,

(iii) $\lim_{x \to -\infty} F(x) = 0$,

(iv) $\lim_{x \to \infty} F(x) = 1$.

Then we can define a finitely additive $\mathbf{P}$ by first defining $\mathbf{P}(I_{a,b}) = F(b) - F(a)$ for intervals, and then extending it to $\mathcal{A}_B$ by defining it as the sum for disjoint unions from $\mathcal{I}$.

We then show the Lebesgue's theorem, which shows when one can extend the finitely additive $\mathbf{P}$ to a countably additive $\mathbf{P}$.

Theorem 1.1.7 (Lebesgue)**.** *Let $\mathbf{P}$ be a finitely additive probability measure on $\mathcal{A}_B$. $\mathbf{P}$ is countably additive on $\mathcal{A}_B$ if and only if $F(x) = \mathbf{P}((-\infty, x])$ is right continuous function of x.*

Remark 14. Using Caratheodory's theorem: for every right continuous function non-decreasing $F(x)$ that satisfies $F(-\infty) = 0$ and $F(\infty) = 1$, there is a unique countably additive probability measure on $\mathcal{F}_B$; and conversely every countably additive probability measure on $\mathcal{F}_B$ induces a right continuous function.

Proof. In this proof, we use a shorthand notation $\mathcal{A} := \mathcal{A}_B$.

Suppose $\mathbf{P}$ is countably additive on $\mathcal{A}$. Then for any x and $\{\epsilon_n\} \downarrow 0$, the collection of intervals $J_n = (x, x + \epsilon_n]$ decreases to $\emptyset$. (Monotone down limits.) This means that $F(x + \epsilon_n) - F(x) \downarrow 0$ and this suffices. ($F(x)$ is right continuous and non-decreasing.) And it is trivial to show $\lim_{x \to -\infty} F(x) = \mathbf{P}((-\infty, -\infty)) = 0; \lim_{x \to \infty} F(x) = \mathbf{P}((-\infty, \infty)) = \mathbf{P}(\Omega) = 1$.

The tricky part is the reverse direction, i.e., getting countable additivity from right continuous and non-decreasing $F(x)$ with $F(-\infty) = 0$ and $F(\infty) = 1$. From $F(x)$ we construct $\mathbf{P}$ to as:

The following steps construct a contradiction: Suppose that $\mathbf{P}$ is not countably additive on $\mathcal{A}$, then exists $A_j \in \mathcal{A}, A_j \downarrow \emptyset$ such that $\mathbf{P}(A_j) \geq \delta > 0$.

(Clearly A_j are not in the form of $(-\infty, a_j] \to (-\infty, -\infty)$ as $F((-\infty, a_j]) \to F((-\infty, -\infty)) = 0$. Sets like this can be hard to deal with. Right-continuity, along with $F(-\infty) = 0$ and $F(\infty) = 1$, allow us to find a large enough interval $(-l, l)$ in the middle that has large enough probability.)

Pick l large enough that $1 - F(l) + F(-l) < \frac{\delta}{2}$ (two tails smaller than $\delta/2$), and define $B_j = A_j \cap (-l, l]$. Clearly $\mathbf{P}(B_j) \geq \frac{\delta}{2}, \forall j$. (For the fixed δ, A_j is too big to fit inside the tails $(-\infty, l]$ and (l, ∞). The part that A_j is not inside the tails has probability at least $\delta/2$.)

Since B_j is a finite disjoint union of left open right closed intervals, create $C_j \subset B_j$ by moving the left (open) endpoint of the intervals to the right (i.e., shortening each interval).

$$\begin{aligned}&B_j = (a_{j,1}, b_{j,1}] \cup \ldots \cup (a_{j,k}, b_{j,k}] \\ &\quad \text{for some } -\infty < a_{j,1} < b_{j,1} < \ldots < a_{j,k} < b_{j,k} < \infty, \\ &C_j = (\hat{a}_{j,1}, b_{j,1}] \cup \ldots \cup (\hat{a}_{j,k}, b_{j,k}] \\ &\quad \text{for some } -\infty < \hat{a}_{j,1} < b_{j,1} < \ldots < \hat{a}_{j,k} < b_{j,k} < \infty, \\ &a_{j,1} < \hat{a}_{j,1}, \ldots, a_{j,k} < \hat{a}_{j,k}.\end{aligned}$$

This can be done to guarantee that

$$\mathbf{P}(B_j \setminus C_j) \leq \frac{\delta}{3 \cdot 2^j}, \ \forall j.$$

(Countable sum of 2^{-j} goes to 1. Any countable summable coefficient here is okay.)

Define $\overline{C}_j = closure(C_j)$. Clearly $\overline{C}_j \subset B_j \subseteq A_j$. (Note that $\overline{C}_j$ is not in the algebra.)

$$\overline{C}_j = [\hat{a}_{j,1}, b_{j,1}] \cup \ldots \cup [\hat{a}_{j,k}, b_{j,k}]$$
$$\text{for some } -\infty < \hat{a}_{j,1} < b_{j,1} < \ldots < \hat{a}_{j,k} < b_{j,k} < \infty,$$

We know that B_j's are decreasing, but C_j's may not be. ($C_j \subseteq B_j, C_{j+1} \subseteq B_{j+1}$, but may not have $C_{j+1} \subseteq C_j$. Taking intersections over C_j guarantees decreasing.) Therefore define $E_j = \cap_{i=1}^{j} C_i$, and $F_j = \cap_{i=1}^{j} \overline{C}_i$. $E_j \subseteq F_j \subseteq \overline{C}_j \subset B_j \subseteq A_j$ (by construction), and observe that

$$\begin{aligned}
&\mathbf{P}(E_j) \geq \mathbf{P}(B_j) - \mathbf{P}(B_j \cap E_j^c) \\
&= \mathbf{P}(B_j) - \mathbf{P}(B_j \cap (\cup_{i=1}^{j} C_i^c)) \\
&\geq \mathbf{P}(B_j) - \sum_{i=1}^{j} \mathbf{P}(B_j \cap C_i^c) \geq \frac{\delta}{2} - \frac{\delta}{3} > 0.
\end{aligned}$$

Therefore F_j is non-empty. By definition of measure, $\mathbf{P}(\phi) = 0$.

Thus, F_j is non-empty, compact (closed and bounded over $\mathbb{R}$), and decreasing. Thus, $\cap_j^{\infty} F_j$ cannot be the $\emptyset$. However $F_j \subset A_j$ and $A_j \downarrow \emptyset$. This leads to a contradiction. (For any element a, we can find a large enough N_a such that $a \notin \cap_{i=1}^{N_a} A_i$.)

□

Remark 15. An interesting fact to prove here is that if F_j is non-empty, compact, and decreasing. Then $\cap_j^{\infty} F_j$ cannot be the $\emptyset$. This is a special case of the Cantor intersection theorem: in any complete metric space (every Cauchy sequence converges to a limit), a decreasing sequence of non-empty compact sets has non-empty intersections. A toy proof here is to explicitly write down the shrinking interval boundaries and give the sup and inf over them.

Remark 16. Aside from this proof, note that for intervals over the real line, The closure of a finite intersection of intervals does not equal the finite intersection of interval closures:

$I_1 = (0,1)$, $I_2 = (1,2)$. Then $I_1 \cap I_2 = \varnothing \quad \Rightarrow \quad \overline{I_1 \cap I_2} = \varnothing$, but $\overline{I_1} \cap \overline{I_2} = [0,1] \cap [1,2] = \{1\} \neq \varnothing$. Hence $\overline{I_1 \cap I_2} \neq \overline{I_1} \cap \overline{I_2}$.

1.1.5 Some additional stuffs

Cantor's Theorem

The largest sigma-algebra is too large. Even for a countable set, its power set is uncountable. This is known as *Cantor's Theorem.*

Theorem 1.1.8 (Cantor's Theorem.). *(Power set is strictly larger)*

For any set S, there is no surjection from S onto its power set 2^S. Equivalently,

$$|2^S| > |S|.$$

Proof. Suppose, for contradiction, that there exists a surjection $f : S \to 2^S$. Define the subset

$$D := \{\, x \in S : \; x \notin f(x) \,\} \; \subseteq S,$$

so $D \in 2^S$. Since f is surjective, there exists $y \in S$ such that $f(y) = D$. Now check whether $y \in D$:

$$y \in D \iff y \notin f(y) \iff y \notin D,$$

a contradiction. Hence no such surjection exists, and therefore $|2^S| > |S|$. □

Corollary. Taking $S = \mathbb{N}$ gives $|2^{\mathbb{N}}| > |\mathbb{N}|$, so the power set $2^{\mathbb{N}}$ is uncountable.

Dedekind cut

Definition 1.1.9. *A Dedekind cut*[6] *is a partition of the rationals $\mathbb{Q}$ into two subsets A and B such that*

1. *A is nonempty.*
2. *$A \neq \mathbb{Q}$ (equivalently, B is nonempty).*
3. *If $x, y \in \mathbb{Q}, x < y$, and $y \in A$, then $x \in A$. (A is "closed downwards".)*
4. *If $x \in A$, then there exists a $y \in A$ such that $y > x$. (A does not contain a greatest element.)*

Remark 17. By omitting the first two requirements, we formally obtain the extended real number line.

Example 7 (Dedekind construction of $\sqrt{2}$). A typical Dedekind cut of the rational numbers $\mathbb{Q}$ is given by the partition (A, B) with

$$A = \{a \in \mathbb{Q} : a^2 < 2 \text{ or } a < 0\},$$
$$B = \{b \in \mathbb{Q} : b^2 \geq 2 \text{ and } b \geq 0\}.$$

This cut represents the irrational number $\sqrt{2}$ in Dedekind's construction. The essential idea is that we use a set A, which is the set of all rational numbers whose squares are less than 2, to "represent" the number $\sqrt{2}$, and further, by defining properly arithmetic operators over these sets (addition, subtraction, multiplication, and division), these sets (together with these arithmetic operations) form the familiar real numbers.

To show A does not contain a maximum, for any positive rational x with $x^2 < 2$, we show there is a rational y with $x < y$ and $y^2 < 2$. The choice $y = \frac{2x+2}{x+2}$ works.

Note that the equality $b^2 = 2$ cannot hold since $\sqrt{2}$ is not rational.

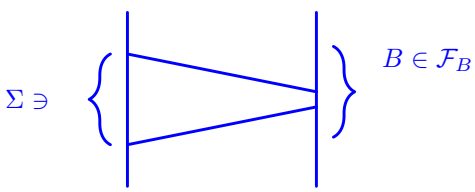
[6] credit: Wikipedia

1.2 Random variables and integration

Let $\mathcal{F}_B$ denotes the Borel σ-field(sigma-algebra) generated by the open intervals on the real line.

Definition 1.2.1 (Borel measurable)**.** *A random variable or a measurable function is a map $f : (\Omega, \Sigma) \mapsto (\mathbb{R}, \mathcal{F}_B)$ (really a mapping $\Omega \to \mathbb{R}$ but with the sigma-algebras specified) such that $\forall B \in \mathcal{F}_B$, $f^{-1}(B) \triangleq \{w : f(w) \in B\} \in \Sigma$.*

The first $\in$ in "$f(\omega) \in B$" means a real number are inside some countable union of intervals. Therefore it can be "loose" in some sense since real numbers are dense. For example, we may have f is an indicator function about $A \in \Sigma$ but B could be $(0.5, 2)$, $f(A) = 1 \in (0.5, 2)$. The second $\in$ in "$\in \Sigma$" is "exact" as Σ is "discrete"(such intuitive analogy is only true if Σ is countable, note that we can also define measurable mapping from $\mathbb{R}$ to $\mathbb{R}$). If the pre-image is not in Σ, then the function is non-measurable. Trivially, $\forall \omega \in \Omega, f(w) \in \mathbb{R}$, hence, $f^{-1}(\mathbb{R}) = \Omega$. f is really a mapping from Ω to $\mathbb{R}$, so people also write $f : \Omega \to R$.



Intuitively, Σ provides what kind of events we can describe in the sample space Ω. Note that the cardinality of $\mathcal{F}_B$ is $|\mathbb{R}|$, which is large enough and allows us to do certain operations over real field $\mathbb{R}$ such as integration.

Remark 18. Measurable by Σ is saying that the states can be described by Σ is enough to describe the mapping $X(\omega)$. In this sense, we don't really care about the real value $X(\omega)$ mapped onto. The random variable is translating the outcome to real numbers for us to do meaningful calculations.

One can see that if Σ has an even larger cardinality than the Borel sigma-algebra, then trivially any random variable defined there is measurable by Σ. It is even more complicated than the Borel sigma-algebra!

Remark 19. Trivially, in any measure space (Ω, Σ) with a non-measurable set $A \subset \Omega, A \notin \Sigma$, one can construct a non-measurable indicator function

$$1_A : \Omega \to R, \quad 1_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ 0, & \text{otherwise.} \end{cases}$$

For example, the Vitali set is not in the Borel sigma-algebra.

Remark 20. This definition (pre-image of measureable set in $(\mathbb{R}, \mathcal{F}_B)$ is a measureable set in (Ω, Σ)) resembles the definition of continuous function (over some topological space) that the pre-image of every open sets are open sets.

Exercise 1.2.1. *For a class of sets $\mathcal{A} \subset \mathcal{F}_B$, and a mapping $f : (\Omega, \Sigma) \mapsto (\mathbb{R}, \mathcal{F}_B)$, suppose it holds that $f^{-1}(A) \in \Sigma$ for all $A \in \mathcal{A}$, then show that $f^{-1}(C) \in \Sigma$ for all $C \in \sigma(\mathcal{A})$, where $\sigma(\mathcal{A})$ is the smallest σ-algebra containing $\mathcal{A}$.*

Proof. (This exercise is beneficial to understanding $f^{-1}(\cdot)$)
Define

$$\mathcal{G} := \{C \in \mathcal{F}_B : \ f^{-1}(C) \in \Sigma\}.$$

Then $\mathcal{G}$ is a σ-algebra since

- $f^{-1}(\mathbb{R}) = \Omega \in \Sigma \Rightarrow \mathbb{R} \in \mathcal{G}$, (first "$\in$" is "loose")
- $f^{-1}(C^c) = \Omega \setminus f^{-1}(C) \in \Sigma \Rightarrow C^c \in \mathcal{G}$, (if not mapped to C, then must be mapped to C^c)
- $f^{-1}(\cup_n C_n) = \cup_n f^{-1}(C_n) \in \Sigma \Rightarrow \cup_n C_n \in \mathcal{G}$. (mapped to some C_i, iff be mapped to the union)

Assumption gives $\mathcal{A} \subset \mathcal{G}$. Hence, by minimality of $\sigma(\mathcal{A})$,

$$\sigma(\mathcal{A}) \subseteq \mathcal{G},$$

so for all $C \in \sigma(\mathcal{A})$, $f^{-1}(C) \in \Sigma$. □

The above exercise show that to verify that a function is measurable, it suffices to consider the inverse images of any collection of sets that generate $\mathcal{F}_B$.

Remark 21. To verify that f is a random variable, it is enough to check the generating family $\{\omega : X(w) \le t\} \in \Sigma$, $\forall t \in \mathbb{R}$. Because the sets $(-\infty, t]$ generate the Borel sigma-algebra $\mathcal{F}_B$.

Some facts about random variables:

1. If $A \in \Sigma$ then
$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$
is bounded and measurable.

   We can classify $B \in \mathcal{F}_B$ into four classes and such classification is complete: only $1 \in B$, only $0 \in B$, both $0, 1 \in B$, and none $0, 1 \notin B$. The pre-image of each class is A, A^c, Ω, and $\emptyset$.

2. If f_1, f_2 are measurable, then $f_1 \cdot f_2$ (their pointwise product) is measurable, as well as $f_1 + f_2$, $\max(f_1, f_2)$, and $\lim_{n\to\infty} f_n$ whenever the pointwise limit exists.

   (1) Sum. For any $a \in \mathbb{R}$,
$$\begin{aligned} &\{\omega : f(\omega) + g(\omega) < a\} \\ &= \bigcup_{q \in \mathbb{Q}} \Big(\{\omega : f(\omega) < q\} \cap \{\omega : g(\omega) < a - q\} \Big) \in \Sigma, \end{aligned}$$
   so $f + g$ is measurable. Similarly $f - g$ is measurable.

   Or:

   If X_1 and X_2 are $\mathcal{G}$-measurable, then $X_1 + X_2$ is also $\mathcal{G}$-measurable.

   The reason is that the map
$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x, y) = x + y$$
   is continuous, hence Borel measurable. Since X_1 and X_2 are $\mathcal{G}$-measurable, the vector-valued map
$$\omega \mapsto (X_1(\omega), X_2(\omega))$$
   is $\mathcal{G}$-measurable, and composing with (f) gives
$$X_1 + X_2 = f(X_1, X_2),$$
   which is therefore $\mathcal{G}$-measurable.

   (2) Product. Since $x \mapsto x^2$ is continuous, f^2 preserves measurability. $(f \pm g)^2$ are measurable, and using
$$4(f \circ g) = (f + g)^2 - (f - g)^2,$$
   we get fg measurable.

   (3) Maximum. For any $a \in \mathbb{R}$,
$$\begin{aligned} &\{\omega : \max(f(\omega), g(\omega)) < a\} \\ &= \{\omega : f(\omega) < a\} \cap \{\omega : g(\omega) < a\} \in \Sigma, \end{aligned}$$
   so $\max(f, g)$ is measurable.

   (4) Pointwise limit. If $f_n \to f$ pointwise, then for any $a \in \mathbb{R}$,
$$\{f < a\} = \bigcup_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \ge N} \left\{ f_n < a - \frac{1}{k} \right\} \in \Sigma,$$
   so f is measurable.

3. If $\{A_j : 1 \le j \le n\}$ is a finite measurable partition of Ω (i.e. $A_j \in \Sigma$, $A_j \cap A_k = \emptyset$ for $j \neq k$, and $\bigcup_{j=1}^n A_j = \Omega$), then a function of the form
$$f(\omega) = \sum_{j=1}^n c_j \mathbf{1}_{A_j}(\omega),$$
with constants $c_j \in \mathbb{R}$, is called a simple function. Simple functions are bounded and measurable.

1.2.1 Approximation by simple functions

Lemma 1.2.2. *Any bounded measurable function is a uniform limit of simple functions.*

Proof. Let $f : \Omega \to \mathbb{R}$ be measurable and bounded: there exists $M > 0$ such that $|f(\omega)| < M$ for all $\omega \in \Omega$. Partition $(-M, M]$ into n disjoint intervals of equal length,
$$I_i = \Big(-M + \frac{2M}{n}(i-1),\, -M + \frac{2M}{n} i\Big], \qquad i = 1, \dots, n.$$
Let c_i be the midpoint of I_i, and define
$$A_i = \{\omega \in \Omega : f(\omega) \in I_i\}.$$
Since f is measurable and each I_i is Borel, $A_i = f^{-1}(I_i) \in \Sigma$ for all i, and the sets $\{A_i\}_{i=1}^n$ are pairwise disjoint with $\bigcup_i A_i = \bigcup_i f^{-1}(I_i) = f^{-1}(\bigcup_i I_i) = \Omega$. (By property of preimage mapping.)

Define the simple function
$$f_n(\omega) = \sum_{i=1}^n c_i \mathbf{1}_{A_i}(\omega).$$
Within each interval, for any $\omega \in \Omega$, if $f(\omega) \in I_i$ then $|f(\omega) - c_i| \le \frac{1}{2}|I_i| = \frac{M}{n}$. Hence over all intervals we have
$$|f(\omega) - f_n(\omega)| \le \frac{M}{n} \quad \text{for all } \omega \in \Omega,$$
showing that $\|f - f_n\|_\infty \le \frac{M}{n} \to 0$ as $n \to \infty$. Thus $f_n \to f$ uniformly and each f_n is a simple function. □

1.2.2 Definition of the integral (simple and bounded functions)

Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space.

1. If f is a simple function of the form
$$f(\omega) = \sum_{i=1}^n c_i \mathbf{1}_{A_i}(\omega),$$
where $A_i \in \Sigma$ are pairwise disjoint and $\bigcup_i A_i = \Omega$, define the *integral* (or expectation) of f with respect to $\mathbb{P}$ of simple functions by
$$\int f \, d\mathbb{P} = \int \Big(\sum_{i=1}^n c_i \mathbf{1}_{A_i}\Big) d\mathbb{P} = \sum_{i=1}^n c_i \mathbb{P}(A_i).$$
Note that measure is only imposed onto subsets of the sample space Ω, where the σ-algebra kicks in. So $d\mathbb{P}$ is a shorthand notation and means that when we focus on a singleton ω, we are integrating with respect to the probability measure $\mathbb{P}$.

2. If f is bounded and measurable, let $\{f_n\}$ be any sequence of simple functions converging uniformly to f. Then define
$$\int f \, d\mathbb{P} = \lim_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Well-definedness. One must show that:

- The limit $\lim_n \int f_n \, d\mathbb{P}$ exists.
- The value does not depend on the particular approximating sequence $\{f_n\}$ of simple functions.

The limit exists because $\{f_n\}$ converges uniformly to f, so $\|f_n - f_m\|_\infty \to 0$ as $n, m \to \infty$. Using the bound
$$\Big|\int f_n \, d\mathbb{P} - \int f_m \, d\mathbb{P}\Big| = \Big|\int (f_n - f_m) \, d\mathbb{P}\Big| \le \|f_n - f_m\|_\infty,$$
the sequence $\int f_n d\mathbb{P}$ is Cauchy in $\mathbb{R}$ and hence convergent.

If $\{g_n\}$ is another uniformly convergent sequence of simple functions with $g_n \to f$, then $h_n = f_n - g_n$ is a sequence of bounded simple functions with $\|h_n\|_\infty \to 0$. Thus
$$\Big|\int f_n \, d\mathbb{P} - \int g_n \, d\mathbb{P}\Big| = \Big|\int h_n \, d\mathbb{P}\Big| \le \|h_n\|_\infty \to 0,$$
so both sequences of integrals have the same limit. Therefore $\int f \, d\mathbb{P}$ is well defined for any bounded measurable f.

Exercise 1.2.1. Fill in the details of the above argument: (a) Prove rigorously that $\{\int f_n d\mathbb{P}\}$ is Cauchy whenever $f_n \to f$ uniformly and each f_n is simple. (b) Show that if $f_n \to f$ and $g_n \to f$ uniformly, with f_n, g_n simple, then $\lim_n \int f_n d\mathbb{P} = \lim_n \int g_n d\mathbb{P}$.

Proof. Use the bound: for bounded measurable u,
$$\Big|\int u \, d\mathbb{P}\Big| \le \int |u| \, d\mathbb{P} \le \|u\|_\infty \, \mathbb{P}(\Omega) = \|u\|_\infty.$$
(a) Fix $\varepsilon > 0$. Uniform convergence gives N s.t. $n \ge N \Rightarrow \|f_n - f\|_\infty < \varepsilon/2$. Then for $n, m \ge N$,
$$\|f_n - f_m\|_\infty \le \|f_n - f\|_\infty + \|f - f_m\|_\infty < \varepsilon,$$
so
$$\Big|\int f_n \, d\mathbb{P} - \int f_m \, d\mathbb{P}\Big| = \Big|\int (f_n - f_m) \, d\mathbb{P}\Big| \le \|f_n - f_m\|_\infty < \varepsilon.$$
Hence $\{\int f_n \, d\mathbb{P}\}$ is Cauchy.

(b) Fix $\varepsilon > 0$. Choose N s.t. $n \ge N$ implies $\|f_n - f\|_\infty < \varepsilon/2$ and $\|g_n - f\|_\infty < \varepsilon/2$. Then $n \ge N$ gives $\|f_n - g_n\|_\infty < \varepsilon$, hence
$$\Big|\int f_n \, d\mathbb{P} - \int g_n \, d\mathbb{P}\Big| = \Big|\int (f_n - g_n) \, d\mathbb{P}\Big| \le \|f_n - g_n\|_\infty < \varepsilon.$$
So $\int f_n \, d\mathbb{P} - \int g_n \, d\mathbb{P} \to 0$, and the limits coincide. □

Exercise 1.2.2. Let f be a bounded measurable function on $(\Omega, \mathcal{F}, \mathbb{P})$. Prove the following:

1. (Linearity)The map $f \mapsto \int f \, d\mathbb{P}$ is linear on the space of bounded measurable functions; that is, for all bounded measurable f, g and all $a, b \in \mathbb{R}$,
$$\int (af + bg) \, d\mathbb{P} = a \int f \, d\mathbb{P} + b \int g \, d\mathbb{P}.$$

2. (Abs dominance)The integral is dominated by the integral of the absolute value:
$$\Big|\int f \, d\mathbb{P}\Big| \le \int |f| \, d\mathbb{P}.$$

Proof. Standard-machine style argument: (1) For simple $s = \sum_i c_i \, 1_{A_i}$ and $t = \sum_j d_j \, 1_{B_j}$, refine by $A_i \cap B_j$ to get $\int (as + bt) \, d\mathbf{P} = a \int s \, d\mathbf{P} + b \int t \, d\mathbf{P}$. For bounded measurable f, g, take simple $s_n \to f$, $t_n \to g$ uniformly. Then $as_n + bt_n \to af + bg$ uniformly, hence
$$\int (af + bg) \, d\mathbf{P} = \lim_n \int (as_n + bt_n) \, d\mathbf{P}$$
$$= \lim_n \Big(a \int s_n \, d\mathbf{P} + b \int t_n \, d\mathbf{P}\Big)$$
$$= a \int f \, d\mathbf{P} + b \int g \, d\mathbf{P}.$$

(2) For simple s, pointwise $-|s| \le s \le |s|$, so $0 \le \int(|s| \pm s)\,d\mathbf{P} = \int |s|\,d\mathbf{P} \pm \int s\,d\mathbf{P}$, hence $\left|\int s\,d\mathbf{P}\right| \le \int |s|\,d\mathbf{P}$. For bounded measurable f, take simple $s_n \to f$ uniformly; then $|s_n| \to |f|$ uniformly and

$$\left|\int f\,d\mathbf{P}\right| = \left|\lim_n \int s_n\,d\mathbf{P}\right|$$
$$\le \lim_n \left|\int s_n\,d\mathbf{P}\right| \le \lim_n \int |s_n|\,d\mathbf{P} = \int |f|\,d\mathbf{P}.$$

□

1.2.3 Modes of convergence

Definition 1.2.2 (Pointwise convergence)**.** *A sequence of functions $\{f_n\}$ converges to f* pointwise *(or* everywhere*) if*

$$\lim_{n\to\infty} f_n(\omega) = f(\omega) \quad \textit{for all } \omega \in \Omega.$$

Definition 1.2.3 (Almost everywhere convergence)**.** *Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space. A sequence of measurable functions $\{f_n\}$ is said to converge to a measurable function f* almost everywhere *(or* almost surely*) if there exists a set $N \subset \Omega$ with $\mathbb{P}(N) = 1$ such that*

$$\lim_{n\to\infty} f_n(\omega) = f(\omega) \quad \textit{for all } \omega \in N.$$

Equivalently, $f_n(\omega) \to f(\omega)$ for all ω outside a $\mathbb{P}$-null set.

Definition 1.2.4 (Convergence in measure / in probability)**.** *A sequence of measurable functions $\{f_n\}$ is said to converge to a measurable function f* in measure *(or* in probability*, in the context of probability spaces) if for every $\varepsilon > 0$,*

$$\lim_{n\to\infty} \mathbb{P}\big(\{\omega : |f_n(\omega) - f(\omega)| > \varepsilon\}\big) = 0.$$

Convergence in measure (imposed on the measure of sequence of sets) is weaker than almost sure convergence(imposed on a single set with a measure equals to 1); almost sure convergence implies convergence in measure, but not conversely.

A quick way to identify these two convergence is for almost surely convergence, the limit is inside the measure $\mathbf{P}$ while for converge in measure/probability, the limit is outside the measure $\mathbf{P}$.

Lemma 1.2.3. *(a.e. $\implies$ m.)If a sequence of measurable functions f_n converges to f almost everywhere, then f_n converges to f in measure.*

Proof. Let $f_n \to f$ almost everywhere. Then there exists $N \subset \Omega$ with $\mathbb{P}(N) = 1$ such that $f_n(\omega) \to f(\omega)$ for all $\omega \in N$. Fix $\varepsilon > 0$ and define

$$A_n^\varepsilon = \{\omega : |f_n(\omega) - f(\omega)| > \varepsilon\}, \qquad B_n^\varepsilon = \bigcup_{m=n}^{\infty} A_m^\varepsilon.$$

(Select the bad ω, then take union to get decreasing sets.) Then B_n^ε is a decreasing sequence of sets: $B_1^\varepsilon \supseteq B_2^\varepsilon \supseteq \cdots$. Let

$$B^\varepsilon = \bigcap_{n=1}^{\infty} B_n^\varepsilon.$$

If $\omega \in N$, then $f_n(\omega) \to f(\omega)$, so there exists $n_0(\omega)$ such that $|f_n(\omega) - f(\omega)| \le \varepsilon$ for all $n \ge n_0(\omega)$. Hence $\omega \notin A_n^\varepsilon$ for all $n \ge n_0(\omega)$, so $\omega \notin B^\varepsilon$. Thus

$$B^\varepsilon \subseteq N^c,$$

and therefore $\mathbb{P}(B^\varepsilon) \le \mathbb{P}(N^c) = 0$. Since $B_n^\varepsilon \downarrow B^\varepsilon$, continuity from above implies

$$\mathbb{P}(B_n^\varepsilon) \downarrow \mathbb{P}(B^\varepsilon) = 0.$$

Finally, $A_n^\varepsilon \subseteq B_n^\varepsilon$ for each n, so $\mathbb{P}(A_n^\varepsilon) \le \mathbb{P}(B_n^\varepsilon) \to 0$ as $n \to \infty$. This is exactly convergence in measure. □

1.2.4 Example: convergence in measure without almost sure convergence

The examples of convergence in measure but not almost sure convergence in this subsection is given by a "running cloud" over the event space Ω. So that picking any $\omega \in \Omega$, over the long time span (indexed by n), it is on the ground infinitly often and on the cloud infinitely often.

These two examples also relate to the two versions of Borel-Cantelli lemma.

Example 8 (fixed cloud, shrinking size will not work.). Consider first the sequence

$$X_n(\omega) = \begin{cases} n, & \omega \in [0, 1/n], \\ 0, & \text{otherwise}, \end{cases} \qquad \omega \in [0,1],$$

on $([0,1], \mathcal{B}, \lambda)$, where λ is Lebesgue measure.

This sequence converges to 0 in probability, since for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - 0| > \varepsilon) = \mathbb{P}(X_n = n) = \lambda([0, 1/n]) = \frac{1}{n} \to 0.$$

However, this is *not* an example of convergence in probability without almost sure convergence, because in fact $X_n \to 0$ almost everywhere. Indeed, if $\omega > 0$, then for all sufficiently large n we have $1/n < \omega$, so $\omega \notin [0, 1/n]$ and hence $X_n(\omega) = 0$ eventually. The only exceptional point is $\omega = 0$, which has Lebesgue measure 0. Therefore

$$X_n(\omega) \to 0 \quad \text{for almost every } \omega.$$

Example 9 (Typewriter sequence running around). A standard counterexample is the *typewriter sequence*. For each n, write

$$n = 2^k + j, \qquad k \ge 0, \quad 0 \le j \le 2^k - 1,$$

and define

$$Y_n(\omega) = \mathbf{1}_{[j/2^k, (j+1)/2^k)}(\omega), \qquad \omega \in [0,1].$$

Then $Y_n \to 0$ in probability, because for any $0 < \varepsilon < 1$,

$$\mathbb{P}(|Y_n| > \varepsilon) = \mathbb{P}(Y_n = 1) = 2^{-k} \to 0.$$

But Y_n does not converge almost everywhere. Indeed, for each fixed $\omega \in [0,1)$ and for each k, there is exactly one dyadic interval of length 2^{-k} containing ω.

The dyadic intervals are obtained by repeatedly dividing $[0,1)$ into equal halves.

For each $k \ge 0$, the dyadic intervals of level k are

$$\left[\frac{j}{2^k}, \frac{j+1}{2^k}\right), \qquad j = 0, 1, \ldots, 2^k - 1.$$

Thus:

$$k = 0: \quad [0,1),$$

$$k = 1: \quad \left[0, \frac{1}{2}\right), \left[\frac{1}{2}, 1\right),$$

$$k = 2: \quad \left[0, \frac{1}{4}\right), \left[\frac{1}{4}, \frac{1}{2}\right), \left[\frac{1}{2}, \frac{3}{4}\right), \left[\frac{3}{4}, 1\right),$$

and so on.

In the typewriter sequence, for each n we write

$$n = 2^k + j, \qquad 0 \le j \le 2^k - 1,$$

and define

$$Y_n(\omega) = \mathbf{1}_{[j/2^k, (j+1)/2^k)}(\omega).$$

So the support of Y_n runs through all dyadic intervals of level 0, then all dyadic intervals of level 1, then all dyadic intervals of level 2, and so on, moving from left to right in each level.

Hence $Y_n(\omega) = 1$ infinitely often, and also $Y_n(\omega) = 0$ infinitely often. Therefore $Y_n(\omega)$ does not converge for any $\omega \in [0,1)$.

So $Y_n \to 0$ in probability, but not almost everywhere.

Exercise 1.2.3 (A vanishing cloud, also running). Construct a $f_n(\omega) \to 0$ in measure on $[0,1]$, but that $\lim_n f_n(\omega)$ does not exist for almost every $\omega \in [0,1]$. In particular, there is no almost sure convergence, illustrating that convergence in measure does not imply almost sure convergence.

Proof. Let

$$I_n := (H_n, H_{n+1}], \qquad \delta_n := H_{n+1} - H_n = \frac{1}{n+1}.$$

For $\omega \in [0,1]$, define

$$f_n(\omega) = \sum_{i=1}^{\infty} \mathbf{1}_{I_n}(\omega + i),$$

so $f_n(\omega)$ counts the integers $i \geq 1$ such that $\omega + i \in I_n$. Since $|I_n| = \delta_n < 1$, the interval $(H_n - \omega,\ H_{n+1} - \omega]$ has length < 1 and thus contains at most one integer.

Hence

$$f_n(\omega) \in \{0,1\} \qquad \text{for all } n, \omega.$$

$f_n \to 0$ in measure.

Define

$$A_n := \{\omega \in [0,1] : f_n(\omega) = 1\}.$$

We have $f_n(\omega) = 1$ iff there exists an integer $i \geq 1$ such that

$$H_n < \omega + i \leq H_{n+1} \quad \iff \quad \omega \in (H_n - i,\ H_{n+1} - i].$$

Reducing modulo 1, this means ω lies in an interval of length δ_n on the unit circle. Writing $\{x\}$ for the fractional part,

$$A_n \equiv \big(\{H_n\},\ \{H_n\} + \delta_n\big] \pmod 1,$$

(Think about that $(H_n, H_{n+1}]$ is extending as n grows large but we wrap them around $[0,1]$, as if there is a portal connecting 0 and 1. And A_n are the $\omega \in [0,1]$ that are within this wrapped around $(H_n, H_{n+1}]$.)

So (with wrap-around if needed) its Lebesgue measure (denotes as $\lambda(\cdot)$) is

$$\lambda(A_n) = \delta_n = \frac{1}{n+1}.$$

Since $f_n \in \{0,1\}$, for any $\varepsilon \in (0,1]$,

$$\lambda\big(\{|f_n| > \varepsilon\}\big) = \lambda(\{f_n = 1\}) = \lambda(A_n) = \frac{1}{n+1} \xrightarrow[n\to\infty]{} 0.$$

Thus $f_n \to 0$ in measure on $[0,1]$. I_n is "shrinking" to the empty set eventually.

The pointwise limit $\lim_{n\to\infty} f_n(\omega)$ does not exist (fix arbitrary ω).

Since $f_n(\omega) \in \{0,1\}$, the limit $\lim_{n\to\infty} f_n(\omega)$ exists iff the sequence is eventually constant.

Let

$$S_N(\omega) := \sum_{n=1}^{N} f_n(\omega) = \underbrace{\sum_{n=1}^{N}}_{\text{finite}} \underbrace{\sum_{i=1}^{\infty} \mathbf{1}_{I_n}(\omega + i)}_{\leq 1} = \sum_{i=1}^{\infty} \sum_{n=1}^{N} \mathbf{1}_{I_n}(\omega + i).$$

Forget about the portal stuff and consider the flat case since we are fixing ω. The intervals $I_1, \ldots, I_N$ are disjoint and

$$\bigcup_{n=1}^{N} I_n = (H_1, H_{N+1}] = (1, H_{N+1}],$$

so for each fixed i,

$$\sum_{n=1}^{N} \mathbf{1}_{I_n}(\omega + i) = \mathbf{1}_{(1, H_{N+1}]}(\omega + i).$$

Hence

$$S_N(\omega) = \#\big\{i \in \mathbb{N} :\ 1 < \omega + i \leq H_{N+1}\big\}.$$

In particular,

$$S_N(\omega) \xrightarrow[N\to\infty]{} \infty,$$

$$\frac{S_N(\omega)}{N} \leq \frac{H_{N+1}}{N} \xrightarrow[N\to\infty]{} 0 \quad (\text{since } H_{N+1} \sim \log N).$$

- Because $S_N(\omega) \to \infty$, the number of indices $n \leq N$ with $f_n(\omega) = 1$ tends to ∞, so $f_n(\omega) = 1$ for infinitely many n.
- If $f_n(\omega) = 1$ for all sufficiently large n, then $S_N(\omega) \geq N - C$ for large N, so $S_N(\omega)/N \to 1$, contradicting $S_N(\omega)/N \to 0$. Thus $f_n(\omega) = 0$ for infinitely many n.
- The idea of this counter example is a cloud shrinking size $(\frac{1}{n+1})$ to zero but sliding over $[0,1]$ over and over again. The tail behavior is jumping between the cloud and the ground.

□

Remark 22 (When converge to a constant).

$$X_n \xrightarrow{\text{a.s.}} c \implies X_n \xrightarrow{P} c,$$

but the converse is false in general. That is,

$$X_n \xrightarrow{P} c \not\Longrightarrow X_n \xrightarrow{\text{a.s.}} c.$$

A standard counterexample is the typewriter sequence on $[0,1]$, which converges to 0 in probability but not almost surely.

On the other hand, if

$$X_n \Rightarrow c,$$

that is, X_n converges in distribution (weakly) to a constant c, then

$$X_n \xrightarrow{P} c.$$

Indeed, for any $\varepsilon > 0$, by the Portmanteau theorem,

$$\limsup_{n\to\infty} \mathbb{P}(|X_n - c| \geq \varepsilon) \leq 0,$$

hence

$$\mathbb{P}(|X_n - c| > \varepsilon) \to 0.$$

Therefore weak convergence to a constant implies convergence in probability, hence also convergence in measure on a probability space.

In summary,

$$X_n \xrightarrow{\text{a.s.}} c \implies X_n \xrightarrow{P} c \impliedby X_n \Rightarrow c,$$

while

$$X_n \xrightarrow{P} c \not\Longrightarrow X_n \xrightarrow{\text{a.s.}} c.$$

On the other hand, convergence in measure does imply almost sure convergence on a sub-sequence as demonstrated by the following lemma.

Lemma 1.2.4. *(subsequence convergence)If a sequence of measurable functions $\{f_n\}$ converges to f in measure on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then there exists a subsequence $\{f_{n_i}\}$ that converges to f almost everywhere.*

Proof. For each $k \in \mathbb{N}$, define the "bad set"

$$A_n^k = \left\{\omega : |f_n(\omega) - f(\omega)| > \frac{1}{k}\right\}.$$

Convergence in measure implies $\mathbb{P}(A_n^k) \to 0$ as $n \to \infty$. Thus, we can inductively choose a subsequence $n_k \uparrow \infty$ such that

$$\mathbb{P}(A_{n_k}^k) \leq \frac{1}{2^k} \quad \text{for each } k \geq 1.$$

Now define the tail bad sets

$$B_k = \bigcup_{m=k}^{\infty} A_{n_m}^m.$$

By subadditivity,

$$\mathbb{P}(B_k) \leq \sum_{m=k}^{\infty} \mathbb{P}(A_{n_m}^m) \leq \sum_{m=k}^{\infty} \frac{1}{2^m} = \frac{1}{2^{k-1}}.$$

The sets $\{B_k\}$ are decreasing ($B_k \supset B_{k+1}$) with $\mathbb{P}(B_k) \to 0$ as $k \to \infty$, so by continuity of measure, $\mathbb{P}\left(\bigcap_k B_k\right) = 0$. Let $B = \bigcap_k B_k$.

For $\omega \notin B$, there exists $K := K(\omega)$ such that $\omega \notin B_K$, i.e., $\omega \notin A^m_{n_m}$ for all $m \geq K$. Thus, for these $\omega \notin B$,

$$|f_{n_m}(\omega) - f(\omega)| \leq \frac{1}{m} \quad \text{for all } m \geq K,$$

so $f_{n_m}(\omega) \to f(\omega)$ as $m \to \infty$. Equivalently,

$$B^c \subseteq \left\{\omega : \limsup_{k\to\infty} |f_{n_k}(\omega) - f(\omega)| > 0\right\}^c,$$

implying that the set where subsequential convergence fails has probability zero. This completes the proof. □

Remark 23. Unless explicitly stated otherwise, all functions considered below are assumed to be measurable.

1.2.5 Bounded convergence

Theorem 1.2.5 (Bounded Convergence Theorem)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Suppose $\{f_n\}$ is a sequence of measurable functions and f is a measurable function such that:*

- *$f_n \to f$ in measure, and*
- *(Bounded)there exists $M < \infty$ such that $|f_n(\omega)| \leq M$ and $|f(\omega)| \leq M$ for all $\omega \in \Omega$ and all n.*

Then

$$\lim_{n\to\infty} \int f_n \, d\mathbb{P} = \int f \, d\mathbb{P}.$$

Proof. First note that linearity of the integral gives

$$\int f_n \, d\mathbb{P} - \int f \, d\mathbb{P} = \int (f_n - f) \, d\mathbb{P}.$$

Hence

$$\left|\int f_n \, d\mathbb{P} - \int f \, d\mathbb{P}\right| = \left|\int (f_n - f) \, d\mathbb{P}\right| \leq \int |f_n - f| \, d\mathbb{P}.$$

Fix $\varepsilon > 0$ and define

$$A^\varepsilon_n = \{\omega : |f_n(\omega) - f(\omega)| > \varepsilon\}.$$

Since $f_n \to f$ in measure, $\mathbb{P}(A^\varepsilon_n) \to 0$ as $n \to \infty$. On $A^{\varepsilon\,c}_n$ we have $|f_n - f| \leq \varepsilon$, while on A^ε_n we have the uniform bound

$$|f_n(\omega) - f(\omega)| \leq |f_n(\omega)| + |f(\omega)| \leq 2M.$$

Therefore

$$\int |f_n - f| \, d\mathbb{P} \leq \varepsilon\big(1 - \mathbb{P}(A^\varepsilon_n)\big) + 2M\,\mathbb{P}(A^\varepsilon_n).$$

Taking $\limsup_{n\to\infty}$ and using $\mathbb{P}(A^\varepsilon_n) \to 0$ gives

$$\limsup_{n\to\infty} \left|\int f_n \, d\mathbb{P} - \int f \, d\mathbb{P}\right| \leq \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, the limit must be 0, and the claim follows. □

1.2.6 Integration of nonnegative functions

Definition 1.2.5 (Integral of a nonnegative function)**.** *Let $f : \Omega \to [0, \infty]$ be a nonnegative measurable function. Define*

$$\int f \, d\mathbb{P}$$
$$\triangleq \sup\left\{\int g \, d\mathbb{P} : g \text{ is bounded, measurable, and } 0 \leq g \leq f\right\},$$

where the supremum is taken over all bounded measurable functions g dominated by f.

Exercise 1.2.6 (Linearity)**.** *$f_1, f_2 : \Omega \to [0, \infty]$ are non-negative measurable functions, then for all $a, b \in \mathbb{R}, a, b \geq 0$: $\int af_1 + bf_2 \, d\mathbb{P} = a\int f_1 \, d\mathbb{P} + b\int f_2 \, d\mathbb{P}$.*

Proof.

$$\begin{aligned}
&\int af_1 + bf_2 \, d\mathbb{P} \\
&= \sup\left\{\int g \, d\mathbb{P} : g \text{ is b.m., } 0 \leq g \leq af_1 + bf_2\right\} \\
&= \sup\Big\{\int ag_1 + bg_2 \, d\mathbb{P} : g_1, g_2 \text{ are b.m.,} \\
&\qquad 0 \leq ag_1 + bg_2 \leq af_1 + bf_2\Big\} \\
&= \sup\Big\{\int ag_1 + bg_2 \, d\mathbb{P} : g_1, g_2 \text{ are b.m.,} \\
&\qquad 0 \leq ag_1 \leq af_1, 0 \leq bg_2 \leq bf_2\Big\}
\end{aligned}$$

Then apply Exercise 1.2.2. □

Remark 24. We already know how to integrate simple/bounded functions. For general $f \geq 0$, we define $\int f \, d\mathbb{P}$ by comparing it to bounded measurable $g \leq f$ whose integrals are already defined. The supremum gives the "best" lower approximation. by taking

$$\int f \, d\mathbb{P} \triangleq \sup\left\{\int g \, d\mathbb{P} : 0 \leq g \leq f,\ g \text{ bounded measurable}\right\}.$$

A few more good reasons:

- Existence and ∞ are automatic. The set is nonempty (contains $g = 0$), so the sup is well-defined in $[0, \infty]$ (could be $= \infty$), with no $\infty - \infty$ issues.
- Built-in monotonicity. If $0 \leq f \leq h$, then every $g \leq f$ is also $\leq h$, hence $\int f \, d\mathbb{P} \leq \int h \, d\mathbb{P}$.
- Designed for monotone limits. There exist bounded $g_n \uparrow f$, and then the definition yields
$$\int f \, d\mathbb{P} = \sup_n \int g_n \, d\mathbb{P},$$
which is the right setup for the Monotone Convergence Theorem.

1.2.7 Fatou's lemma and monotone convergence

Lemma 1.2.7 (Fatou)**.** *Let $\{f_n\}$ be a sequence of nonnegative measurable functions with $f_n \to f$ in measure, where $f \geq 0$ is measurable. Then*

$$\int f \, d\mathbb{P} \leq \liminf_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Proof. Let g be a bounded measurable function such that $0 \leq g \leq f$. Define

$$h_n = \min\{f_n, g\}.$$

Then $0 \leq h_n \leq g$ and $|h_n| \leq \|g\|_\infty$ for all n. Moreover, $f_n \to f$ in measure implies $h_n \to g$ in measure, because the truncation at g cannot increase the set where f_n deviates from f by more than a given ε.

(Let $\wedge$ be taking the minimum. Really just $|g \wedge f_n - g \wedge f| \leq |f_n - f|$ for the cases when $g \leq f_n, f$ and $f_n \leq g \leq f$.)

Since the sequence $\{h_n\}$ is uniformly bounded and converges in measure to g, the bounded convergence theorem yields

$$\int g \, d\mathbb{P} = \lim_{n\to\infty} \int h_n \, d\mathbb{P} \leq \liminf_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Because g was an arbitrary bounded measurable function with $0 \leq g \leq f$, taking the supremum over all such g in the definition of $\int f \, d\mathbb{P}$ gives

$$\int f \, d\mathbb{P} \leq \liminf_{n\to\infty} \int f_n \, d\mathbb{P}.$$

□

An alternate and very common version of Fatou's lemma is the following.

Lemma 1.2.8 (Fatou, alternate formulation (Greedy inf)). *Let $\{f_n\}$ be a sequence of nonnegative measurable functions. Then*

$$\int \liminf_{n\to\infty} f_n \, d\mathbb{P} \le \liminf_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Proof. First assume

$$g(\omega) = \liminf_{n\to\infty} f_n(\omega) < \infty \quad \text{for almost every } \omega.$$

Define

$$g_n(\omega) = \inf_{m\ge n} f_m(\omega).$$

Then g_n is measurable, $0 \le g_n \le f_n$ for each n, and the sequence $\{g_n\}$ is increasing: $g_n \uparrow g = \liminf_{n\to\infty} f_n$ pointwise (and hence in measure). Applying the previous version of Fatou's lemma to $\{g_n\}$ gives

$$\int g \, d\mathbb{P} \le \liminf_{n\to\infty} \int g_n \, d\mathbb{P} \le \liminf_{n\to\infty} \int f_n \, d\mathbb{P},$$

since $g_n \le f_n$ implies $\int g_n \, d\mathbb{P} \le \int f_n \, d\mathbb{P}$.

If instead $g(\omega) = \infty$ on a set A with $\mathbb{P}(A) > 0$, then for any $M > 0$,

$$M\,\mathbb{P}(A) \le \int M\mathbf{1}_A \, d\mathbb{P} \le \int \liminf_{n\to\infty} (f_n \wedge M) \, d\mathbb{P}$$
$$\le \liminf_{n\to\infty} \int (f_n \wedge M) \, d\mathbb{P} \le \liminf_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Letting $M \to \infty$ shows that both sides of the desired inequality are $+\infty$, so the statement holds trivially. □

Corollary 1.2.9 (Monotone Convergence Theorem). *Let $\{f_n\}$ be a sequence of nonnegative measurable functions such that $f_n(\omega) \uparrow f(\omega)$ for all $\omega \in \Omega$ (or almost everywhere). Then*

$$\lim_{n\to\infty} \int f_n \, d\mathbb{P} = \int f \, d\mathbb{P}.$$

Proof. Since $0 \le f_n \le f$, monotonicity of the integral for nonnegative functions gives

$$\int f_n \, d\mathbb{P} \le \int f \, d\mathbb{P} \quad \text{for all } n,$$

so

$$\limsup_{n\to\infty} \int f_n \, d\mathbb{P} \le \int f \, d\mathbb{P}.$$

On the other hand, $f_n \uparrow f$ implies (Fatou's lemma)

$$\liminf_{n\to\infty} \int f_n \, d\mathbb{P} \ge \int \liminf_{n\to\infty} f_n \, d\mathbb{P} = \int f \, d\mathbb{P}$$

Combining the two inequalities yields

$$\lim_{n\to\infty} \int f_n \, d\mathbb{P} = \int f \, d\mathbb{P}.$$

□

1.2.8 Integrable functions

Definition 1.2.6 (Integrability: nonnegative functions). *A nonnegative measurable function $f : \Omega \to [0, \infty]$ is said to be* integrable *if*

$$\int f \, d\mathbb{P} < \infty.$$

Definition 1.2.7 (Integrability: general case). *A measurable function $f : \Omega \to \mathbb{R}$ is said to be integrable if*

$$\int |f| \, d\mathbb{P} < \infty.$$

For such an f, define the positive and negative parts

$$f_+ = \max\{f, 0\}, \qquad f_- = \max\{-f, 0\},$$

so that $f = f_+ - f_-$ and $|f| = f_+ + f_-$. Both f_+ and f_- are nonnegative measurable functions. The integral of f is then defined by

$$\int f \, d\mathbb{P} \triangleq \int f_+ \, d\mathbb{P} - \int f_- \, d\mathbb{P},$$

which is finite precisely when $\int |f| \, d\mathbb{P} < \infty$.

Exercise 1.2.4. Show that the integral satisfies the following properties:

1. If f, g are integrable and $a, b \in \mathbb{R}$, then $af + bg$ is integrable and
$$\int (af + bg) \, d\mathbb{P} = a \int f \, d\mathbb{P} + b \int g \, d\mathbb{P}.$$
2. If $f = 0$ almost everywhere, then f is integrable and $\int f \, d\mathbb{P} = 0$. Deduce that if f and g are integrable and $f = g$ almost everywhere, then $\int f \, d\mathbb{P} = \int g \, d\mathbb{P}$.

Proof. 1. **Linearity.** Let $a, b \in \mathbb{R}$. First, $af + bg$ is integrable since pointwisely

$$|af + bg| \le |a|\,|f| + |b|\,|g|$$

and hence

$$\int |af + bg| \, d\mathbb{P} \le |a| \int |f| \, d\mathbb{P} + |b| \int |g| \, d\mathbb{P} < \infty.$$

To prove the identity, write $f = f_+ - f_-$ and $g = g_+ - g_-$, where all four parts are nonnegative and integrable (because $\int f_+ \, d\mathbb{P} + \int f_- \, d\mathbb{P} = \int |f| \, d\mathbb{P} < \infty$, and similarly for g).

For nonnegative integrable functions, we already have linearity of the integral (from the construction via simple functions), so for $a \ge 0$ and $b \ge 0$,

$$\int (af + bg) \, d\mathbb{P} = \int \big(a(f_+ - f_-) + b(g_+ - g_-)\big) \, d\mathbb{P}$$
$$= a \int f_+ \, d\mathbb{P} - a \int f_- \, d\mathbb{P} + b \int g_+ \, d\mathbb{P} - b \int g_- \, d\mathbb{P}.$$

Thus

$$\int (af + bg) \, d\mathbb{P} = a \int f \, d\mathbb{P} + b \int g \, d\mathbb{P} \qquad (a, b \ge 0).$$

For general $a, b \in \mathbb{R}$, write $a = a_+ - a_-$ and $b = b_+ - b_-$ with $a_\pm, b_\pm \ge 0$, apply the case $a, b \ge 0$ to $a_+ f$, $a_- f$, $b_+ g$, $b_- g$, and use the definition of $\int f \, d\mathbb{P}$ as a difference of two finite numbers to conclude the same identity for all real a, b.

2. **Almost-everywhere equality.** If $f = 0$ a.e., then $|f| = 0$ a.e., so

$$\int |f| \, d\mathbb{P} = 0 < \infty,$$

hence f is integrable. Moreover, $f_+ = f_- = 0$ a.e., so

$$\int f \, d\mathbb{P} = \int f_+ \, d\mathbb{P} - \int f_- \, d\mathbb{P} = 0 - 0 = 0.$$

Now if $f = g$ a.e., then $f - g = 0$ a.e., so by the previous paragraph,

$$\int (f - g) \, d\mathbb{P} = 0.$$

Using linearity from (1),

$$0 = \int (f - g) \, d\mathbb{P} = \int f \, d\mathbb{P} - \int g \, d\mathbb{P},$$

which implies $\int f \, d\mathbb{P} = \int g \, d\mathbb{P}$.

□

1.2.9 Dominated convergence

Theorem 1.2.10 (Dominated Convergence Theorem)**.** *Let $\{f_n\}$ be a sequence of measurable functions and f a measurable function such that:*

- *$f_n \to f$ in measure, and*
- *there exists an non-negative integrable function g with $|f_n(\omega)| \le g(\omega)$ for all ω and all n.*

Then f is integrable and

$$\lim_{n\to\infty} \int f_n \, d\mathbb{P} = \int f \, d\mathbb{P}.$$

Proof. First note that $|f_n| \le g$ and g integrable imply by Fatou's lemma applied to $g - |f_n| \ge 0$ that f is integrable (details left as an exercise).

Consider the nonnegative sequences

$$g - f_n, \qquad g + f_n.$$

If $f_n \to f$ in measure, then $g - f_n \to g - f$ and $g + f_n \to g + f$ in measure as well, by basic properties of convergence in measure.

Observe that

$$\int (g - f_n) \, d\mathbb{P} = \int g \, d\mathbb{P} - \int f_n \, d\mathbb{P},$$

so

$$\liminf_{n\to\infty} \int (g - f_n) \, d\mathbb{P} = \int g \, d\mathbb{P} - \limsup_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Similarly,

$$\int (g + f_n) \, d\mathbb{P} = \int g \, d\mathbb{P} + \int f_n \, d\mathbb{P},$$

hence

$$\liminf_{n\to\infty} \int (g + f_n) \, d\mathbb{P} = \int g \, d\mathbb{P} + \liminf_{n\to\infty} \int f_n \, d\mathbb{P}.$$

Applying Fatou's lemma to the nonnegative functions $g - f_n$ and $g + f_n$ gives

$$\int (g - f) \, d\mathbb{P} \le \liminf_{n\to\infty} \int (g - f_n) \, d\mathbb{P} = \int g \, d\mathbb{P} - \limsup_{n\to\infty} \int f_n \, d\mathbb{P},$$

which rearranges to

$$\limsup_{n\to\infty} \int f_n \, d\mathbb{P} \le \int f \, d\mathbb{P}.$$

Similarly,

$$\int (g + f) \, d\mathbb{P} \le \liminf_{n\to\infty} \int (g + f_n) \, d\mathbb{P} = \int g \, d\mathbb{P} + \liminf_{n\to\infty} \int f_n \, d\mathbb{P},$$

which implies

$$\liminf_{n\to\infty} \int f_n \, d\mathbb{P} \ge \int f \, d\mathbb{P}.$$

Combining the two inequalities yields

$$\lim_{n\to\infty} \int f_n \, d\mathbb{P} = \int f \, d\mathbb{P}.$$

□

Remark 25. *Application of DCT: lets you pass the limit inside the integral.*

A useful trick is to rewrite an integral over a set as

$$\int_{A_n} X \, dP = \int \mathbf{1}_{A_n} X \, dP.$$

If $A_n \to A$ setwise (for example $A_n \uparrow A$ or $A_n \downarrow A$, often used along with the monotone class theorem), then

$$\mathbf{1}_{A_n} \to \mathbf{1}_A \quad \text{pointwise},$$

so

$$\mathbf{1}_{A_n} X \to \mathbf{1}_A X \quad \text{a.s.}$$

Moreover,

$$|\mathbf{1}_{A_n} X| \le |X|.$$

Thus, if $X \in L^1$, the Dominated Convergence Theorem applies and gives

$$\int_{A_n} X \, dP = \int \mathbf{1}_{A_n} X \, dP \longrightarrow \int \mathbf{1}_A X \, dP = \int_A X \, dP.$$

So the trick is: convert $\int_{A_n} X \, dP$ into an integral of $\mathbf{1}_{A_n} X$, then DCT lets you pass the limit inside the integral.

Exercise 1.2.5. Suppose f is integrable and

$$\int f(\omega) \, \mathbf{1}_A(\omega) \, d\mathbb{P} = 0 \quad \text{for all } A \in \mathcal{F}.$$

Show that $f = 0$ almost everywhere. (Hint: apply this condition to the sets $\{f > 0\}$ and $\{f < 0\}$ or suitable truncations.)

Remark 26 (DCT versus MCT). In many common applications of MCT, one can also use DCT, but not in general.

The reason is that MCT and DCT have different hypotheses:

- **MCT:** if $0 \le f_n \uparrow f$, then
$$\int f_n \, d\mu \to \int f \, d\mu,$$
possibly with value $+\infty$.
- **DCT:** if $f_n \to f$ pointwise and there exists $g \in L^1$ such that
$$|f_n| \le g \qquad \text{for all } n,$$
then
$$\int f_n \, d\mu \to \int f \, d\mu.$$

So DCT requires an *integrable dominating function*, while MCT does not.

For example, on $(\mathbb{R}, \mathcal{B}, \lambda)$, let

$$f_n = \mathbf{1}_{[0,n]}.$$

Then $0 \le f_n \uparrow 1$ pointwise, so by MCT,

$$\int f_n \, d\lambda \to \int 1 \, d\lambda = \infty.$$

But DCT cannot be used, because any dominating function g would have to satisfy

$$g \ge 1 \quad \text{a.e. on } \mathbb{R},$$

and such a g is not integrable.

Thus, MCT does *not* in general imply that DCT is applicable. DCT works only when an integrable dominator is available.

1.2.10 Transformations

Definition 1.2.8 (Measurable transformation)**.** *Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. A mapping*

$$T : (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$$

is called a measurable transformation if

$$T^{-1}(B) = \{\omega_1 \in \Omega_1 : T(\omega_1) \in B\} \in \mathcal{F}_1 \quad \textit{for all } B \in \mathcal{F}_2.$$

Remark 27. Generalizing the $(\mathbb{R}, \mathcal{F}_B)$ in the definition of random variables to $(\Omega_2, \mathcal{F}_2)$. Note that $\mathbb{R}$ is equipped with the Lebesgue measure.

If $(\Omega_1, \mathcal{F}_1)$ is equipped with a probability measure $\mathbb{P}$, a measurable transformation T induces a probability measure $\mathbb{Q}$ on $(\Omega_2, \mathcal{F}_2)$ by

$$\mathbb{Q}(B) = \underbrace{\mathbb{P}(T^{-1}(B))}_{\text{from } \mathcal{F}_1}, \quad B \in \mathcal{F}_2.$$

The measure $\mathbb{Q}$ is called the *pushforward* (or image) of $\mathbb{P}$ under T, and is denoted $\mathbb{Q} = \mathbb{P} \circ T^{-1}$.

Exercise 1.2.6. Assuming $\mathbb{P}$ is a countably additive probability measure on $(\Omega_1, \mathcal{F}_1)$, verify that $\mathbb{Q}(B) = \mathbb{P}(T^{-1}(B))$ defines a countably additive probability measure on $(\Omega_2, \mathcal{F}_2)$.

Proof.

- *Well-defined on $\mathcal{F}_2$.* If $B \in \mathcal{F}_2$, measurability of T gives $T^{-1}(B) \in \mathcal{F}_1$, hence $\mathbb{Q}(B) = \mathbb{P}(T^{-1}(B))$ is defined.

- *Nonnegativity.* For any $B \in \mathcal{F}_2$, $\mathbb{Q}(B) = \mathbb{P}(T^{-1}(B)) \geq 0$ since $\mathbb{P}$ is a measure.

- *Normalization.*
$$\mathbb{Q}(\Omega_2) = \mathbb{P}\big(T^{-1}(\Omega_2)\big) = \mathbb{P}(\Omega_1) = 1.$$
Also $\mathbb{Q}(\varnothing) = \mathbb{P}(T^{-1}(\varnothing)) = \mathbb{P}(\varnothing) = 0$.

- *Countable additivity.* Let $(B_n)_{n\geq 1} \subset \mathcal{F}_2$ be pairwise disjoint. Then $(T^{-1}(B_n))_{n\geq 1}$ are pairwise disjoint because
$$\begin{aligned} T^{-1}(B_m) \cap T^{-1}(B_n) &= T^{-1}(B_m \cap B_n) \\ &= T^{-1}(\varnothing) = \varnothing \quad (m \neq n). \end{aligned}$$
Moreover,
$$T^{-1}\Big(\bigcup_{n=1}^{\infty} B_n\Big) = \bigcup_{n=1}^{\infty} T^{-1}(B_n).$$
Therefore, by countable additivity of $\mathbb{P}$,
$$\begin{aligned} \mathbb{Q}\Big(\bigcup_{n=1}^{\infty} B_n\Big) &= \mathbb{P}\left(T^{-1}\Big(\bigcup_{n=1}^{\infty} B_n\Big)\right) \\ &= \mathbb{P}\left(\bigcup_{n=1}^{\infty} T^{-1}(B_n)\right) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(T^{-1}(B_n)) = \sum_{n=1}^{\infty} \mathbb{Q}(B_n). \end{aligned}$$

□

Theorem 1.2.11 (Change of variables / pushforward)**.** *Let $(\Omega_1, \mathcal{F}_1, \mathbb{P})$ be a probability space and $(\Omega_2, \mathcal{F}_2)$ a measurable space. Let*
$$T : (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$$
be a measurable transformation, and let
$$f : (\Omega_2, \mathcal{F}_2) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$$
be a measurable function (random variable). Define
$$g : \Omega_1 \to \mathbb{R}, \qquad g(\omega_1) = f(T(\omega_1)).$$
Then:

1. *g is measurable on $(\Omega_1, \mathcal{F}_1)$.*

   (Composition of measurability.)

2. *Let $\mathbb{Q} = \mathbb{P} \circ T^{-1}$ be the pushforward measure on $(\Omega_2, \mathcal{F}_2)$. Then g is integrable with respect to $\mathbb{P}$ if and only if f is integrable with respect to $\mathbb{Q}$, and in this case*
$$\int_{\Omega_2} f(\omega_2)\, d\mathbb{Q}(\omega_2) = \int_{\Omega_1} g(\omega_1)\, d\mathbb{P}(\omega_1).$$

Proof. For measurability of g, let $B \in \mathcal{B}_{\mathbb{R}}$. Then
$$\{\omega_1 : g(\omega_1) \in B\} = \{\omega_1 : f(T(\omega_1)) \in B\} = T^{-1}(f^{-1}(B)).$$
(The composition rule of preimage mapping.) Since f is measurable, $f^{-1}(B) \in \mathcal{F}_2$, and since T is measurable, $T^{-1}(f^{-1}(B)) \in \mathcal{F}_1$. Hence g is measurable.

For the integral identity, use the standard step-by-step extension (sometimes called the "standard machine"):

1. *Indicator functions.* Let $f = \mathbf{1}_B$ for some $B \in \mathcal{F}_2$. Then
$$g(\omega_1) = f(T(\omega_1)) = \mathbf{1}_B(T(\omega_1)) = \mathbf{1}_{T^{-1}(B)}(\omega_1).$$
Thus
$$\begin{aligned} \int_{\Omega_2} f\, d\mathbb{Q} &= \mathbb{Q}(B) = \mathbb{P}(T^{-1}(B)) \\ &= \int_{\Omega_1} \mathbf{1}_{T^{-1}(B)}\, d\mathbb{P} = \int_{\Omega_1} g\, d\mathbb{P}. \end{aligned}$$

2. *Simple functions.* If f is a simple function $f = \sum_{k=1}^{n} c_k \mathbf{1}_{B_k}$ with $B_k \in \mathcal{F}_2$, then
$$\begin{aligned} g(\omega_1) = f(T(\omega_1)) &= \sum_{k=1}^{n} c_k \mathbf{1}_{B_k}(T(\omega_1)) \\ &= \sum_{k=1}^{n} c_k \mathbf{1}_{T^{-1}(B_k)}(\omega_1). \end{aligned}$$
By linearity of the integral and the indicator case,
$$\int_{\Omega_2} f\, d\mathbb{Q} = \sum_{k=1}^{n} c_k \mathbb{Q}(B_k) = \sum_{k=1}^{n} c_k \mathbb{P}(T^{-1}(B_k)) = \int_{\Omega_1} g\, d\mathbb{P}.$$

3. *Bounded nonnegative functions.* Let $f \geq 0$ be bounded and measurable on Ω_2. Then there exist simple functions $f_n \uparrow f$ uniformly bounded, with $f_n \to f$ pointwise, and $g_n = f_n \circ T \to g$ pointwise with $0 \leq g_n \leq \sup f$. By the bounded convergence theorem and the simple-function case,
$$\int_{\Omega_2} f\, d\mathbb{Q} = \lim_{n\to\infty} \int_{\Omega_2} f_n\, d\mathbb{Q} = \lim_{n\to\infty} \int_{\Omega_1} g_n\, d\mathbb{P} = \int_{\Omega_1} g\, d\mathbb{P}.$$

4. *General nonnegative functions.* If $f \geq 0$ is measurable, define $f_n = \min\{f, n\}$ so that $f_n \uparrow f$. Then $g_n = f_n \circ T \uparrow g$. By the monotone convergence theorem applied on both spaces and the bounded case for each f_n,
$$\int_{\Omega_2} f\, d\mathbb{Q} = \lim_{n\to\infty} \int_{\Omega_2} f_n\, d\mathbb{Q} = \lim_{n\to\infty} \int_{\Omega_1} g_n\, d\mathbb{P} = \int_{\Omega_1} g\, d\mathbb{P}.$$

5. *Integrable functions.* For general (real-valued) integrable f, write $f = f_+ - f_-$ with $f_\pm \geq 0$ and integrable; then $g = f \circ T = f_+ \circ T - f_- \circ T$ with $(f_\pm \circ T) \geq 0$. Applying the nonnegative case to f_+ and f_- separately and using linearity,
$$\begin{aligned} \int_{\Omega_2} f\, d\mathbb{Q} &= \int_{\Omega_2} f_+\, d\mathbb{Q} - \int_{\Omega_2} f_-\, d\mathbb{Q} \\ &= \int_{\Omega_1} f_+ \circ T\, d\mathbb{P} - \int_{\Omega_1} f_- \circ T\, d\mathbb{P} = \int_{\Omega_1} g\, d\mathbb{P}. \end{aligned}$$

This completes the proof. □

Remark 28 (Distribution and expectation via pushforward).
$$\overbrace{\underbrace{(\Omega_1, \mathcal{F}_1, \mathbb{P}) \xrightarrow{T} (\Omega_2, \mathcal{F}_2)}_{\mathbb{Q}(\cdot)=\mathbb{P}(T^{-1}(\cdot))} \xrightarrow{f} (\mathbb{R}, \mathcal{B}_{\mathbb{R}})}^{g}$$

An important special case is obtained by taking $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ and f to be the identity map $f(\omega_2) = \omega_2$. Then a measurable map
$$T : (\Omega_1, \mathcal{F}_1, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$$
is simply a real-valued random variable, usually denoted $X(\omega_1) = T(\omega_1)$. The pushforward measure $\mathbb{Q} = \mathbb{P} \circ X^{-1}$ is exactly the *distribution* of X:
$$\mathbb{Q}((-\infty, x]) = \mathbb{P}(\{\omega_1 : X(\omega_1) \leq x\}) = F_X(x).$$

Writing $\omega_2 = x \in \mathbb{R}$ and $d\mathbb{Q}(x) = dF_X(x)$ as a notational shorthand, the theorem gives, for any integrable Borel function $f : \mathbb{R} \to \mathbb{R}$,
$$\int_{\mathbb{R}} f(x)\, dF_X(x) = \int_{\Omega_1} f(X(\omega_1))\, d\mathbb{P}(\omega_1).$$

In particular, taking $f(x) = x$ and $g(\omega_1) = f(X(\omega_1)) = X(\omega_1)$,

$$\int_{\mathbb{R}} x \, dF_X(x) = \int_{\Omega_1} X(\omega_1) \, d\mathbb{P}(\omega_1),$$

which is the usual identification of the expectation of X with the integral of x against its distribution.

1.2.11 Borel–Cantelli lemma and subsequences

Lemma 1.2.12 (Borel–Cantelli, first lemma)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\{A_n\}_{n=1}^{\infty} \subset \mathcal{F}$ satisfy*

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty.$$

Define

$$\overline{A} \triangleq \bigcap_{n=1}^{\infty} \bigcup_{m \geq n} A_m = \lim_{n\to\infty} \sup_{m \geq n} A_m = \limsup_{n\to\infty} A_n,$$

the event that infinitely many of the A_n occur. Then $\mathbb{P}(\overline{A}) = 0$.

Proof. Let

$$B_n = \bigcup_{m \geq n} A_m.$$

Then $\{B_n\}$ is a decreasing sequence: $B_1 \supseteq B_2 \supseteq \cdots$, and $\overline{A} = \bigcap_{n=1}^{\infty} B_n$. By the union bound,

$$\mathbb{P}(B_n) \leq \sum_{m \geq n} \mathbb{P}(A_m),$$

and the right-hand side tends to 0 as $n \to \infty$ because the tail of a convergent series converges to 0. Hence $\mathbb{P}(B_n) \downarrow 0$. Continuity from above of probability measures yields

$$\mathbb{P}(\overline{A}) = \mathbb{P}\Big(\bigcap_{n=1}^{\infty} B_n\Big) = \lim_{n\to\infty} \mathbb{P}(B_n) = 0.$$

□

Remark 29. The Borel–Cantelli lemma is frequently used to extract subsequences that converge almost surely from sequences that converge only in measure.

Lemma 1.2.13 (Subsequence convergence)**.** *Let $\{f_n\}$ be a sequence of measurable functions on $(\Omega, \mathcal{F}, \mathbb{P})$ that converges in measure to a measurable function f. Then there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \to f$ almost surely.*

Proof. Since $f_n \to f$ in measure, for each $k \in \mathbb{N}$,

$$\mathbb{P}\big(\{\omega : |f_n(\omega) - f(\omega)| > 1/k\}\big) \to 0 \quad \text{as } n \to \infty.$$

For each k, choose n_k large enough so that

$$\mathbb{P}(A_{n_k}) \leq \frac{1}{2^k},$$

where

$$A_n := \{\omega : |f_n(\omega) - f(\omega)| > 1/k\}$$

(with the understanding that A_n depends on k as well; write $A_{n,k}$ if desired). Then

$$\sum_{k=1}^{\infty} \mathbb{P}(A_{n_k}) \leq \sum_{k=1}^{\infty} \frac{1}{2^k} = 1 < \infty.$$

By the Borel–Cantelli lemma, the event

$$\overline{A} := \bigcap_{k=1}^{\infty} \bigcup_{m \geq k} A_{n_m}$$

has probability zero:

$$\mathbb{P}(\overline{A}) = 0.$$

Note that

$$\overline{A} \supseteq B := \{\omega : \limsup_{k\to\infty} |f_{n_k}(\omega) - f(\omega)| > 0\}.$$

Indeed, if $\omega \in B$, then for some $\varepsilon > 0$, the inequality $|f_{n_k}(\omega) - f(\omega)| > \varepsilon$ holds infinitely often. Therefore, it must be in infinitely many of A_{n_k}. Hence, $\omega \in \overline{A}$. Thus $B \subseteq \overline{A}$.

On B^c we must have

$$\lim_{k\to\infty} |f_{n_k}(\omega) - f(\omega)| = 0,$$

i.e. $f_{n_k}(\omega) \to f(\omega)$ for all $\omega \in B^c$. Since $\mathbb{P}(B^c) = 1$, this shows that $f_{n_k} \to f$ almost surely. □

Remark 30. This proof is essentially the same as the preceding argument (in the chapter). In practice, such subsequence results are typically proved using the Borel–Cantelli lemma, so it is useful to recognize the argument in that framework.

1.2.12 Product spaces

Let $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$ be probability spaces. The aim of this section is to construct a natural probability measure on the product space $\Omega_1 \times \Omega_2$.

A natural σ-algebra on $\Omega_1 \times \Omega_2$ is the one generated by *measurable rectangles*, i.e. sets of the form

$$A_1 \times A_2, \qquad A_1 \in \mathcal{F}_1,\ A_2 \in \mathcal{F}_2.$$

Denote by $\mathcal{F}$ the σ-algebra generated by all such rectangles; this is called the *product σ-algebra* on $\Omega_1 \times \Omega_2$.

We seek a countably additive probability measure $\mathbb{P}$ on $(\Omega_1 \times \Omega_2, \mathcal{F})$ that extends $\mathbb{P}_1$ and $\mathbb{P}_2$ in the expected way. On rectangles, a natural candidate is

$$\mathbb{P}(A_1 \times A_2) = \mathbb{P}_1(A_1)\,\mathbb{P}_2(A_2), \qquad A_1 \in \mathcal{F}_1,\ A_2 \in \mathcal{F}_2.$$

For finite disjoint unions of measurable rectangles, define

$$\mathbb{P}\Big(\bigsqcup_{i=1}^{n} A_{1i} \times A_{2i}\Big) \triangleq \sum_{i=1}^{n} \mathbb{P}_1(A_{1i})\,\mathbb{P}_2(A_{2i}).$$

Theorem 1.2.14 (Product Algebra)**.** *Let $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$ be probability spaces. Define:*

- $\mathcal{R} = \{A_1 \times A_2 : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\}$ *(measurable rectangles)*
- $\mathcal{A} = \{$*finite disjoint unions* $\bigsqcup_{i=1}^{n} R_i, R_i \in \mathcal{R}\}$

For $E = \bigsqcup_{i=1}^{n} R_i \in \mathcal{A}$ with $R_i = A_{1i} \times A_{2i}$,

$$\mathbb{P}(E) := \sum_{i=1}^{n} \mathbb{P}_1(A_{1i})\mathbb{P}_2(A_{2i}).$$

Then $\mathcal{A}$ is an algebra on $\Omega_1 \times \Omega_2$ and $\mathbb{P} : \mathcal{A} \to [0, 1]$ is well-defined.

Proof. Recall that a collection $\mathcal{C} \subset \mathcal{P}(\Omega_1 \times \Omega_2)$ is an *algebra* if:

1. $\emptyset \in \mathcal{C}$ and $\Omega_1 \times \Omega_2 \in \mathcal{C}$;
2. $\mathcal{C}$ is closed under finite unions;
3. $\mathcal{C}$ is closed under complements (equivalently, under relative complements).

Step 1: $\mathcal{A}$ is an algebra

(i) $\emptyset$ and $\Omega_1 \times \Omega_2$ are in $\mathcal{A}$. The empty set belongs to $\mathcal{A}$ as the empty disjoint union. Moreover,

$$\Omega_1 \times \Omega_2 = \Omega_1 \times \Omega_2$$

is a single measurable rectangle, hence an element of $\mathcal{R}$, and thus in $\mathcal{A}$.

(ii) Difference of two rectangles. Let $R = A_1 \times A_2$ and $S = B_1 \times B_2$ with $A_1, B_1 \in \mathcal{F}_1$ and $A_2, B_2 \in \mathcal{F}_2$. Then $R \setminus S = (A_1 \times A_2) \setminus (B_1 \times B_2) = (A_1 \times A_2) \cap \big((\Omega_1 \times \Omega_2) \setminus (B_1 \times B_2)\big)$. We have

$$(\Omega_1 \times \Omega_2) \setminus (B_1 \times B_2) = (\Omega_1 \setminus B_1) \times \Omega_2 \ \cup \ \Omega_1 \times (\Omega_2 \setminus B_2),$$

so

$$\begin{aligned} R \setminus S &= (A_1 \times A_2) \cap \big[(\Omega_1 \setminus B_1) \times \Omega_2\big] \\ &\quad \cup \ (A_1 \times A_2) \cap \big[\Omega_1 \times (\Omega_2 \setminus B_2)\big] \\ &= (A_1 \setminus B_1) \times A_2 \ \cup \ A_1 \times (A_2 \setminus B_2). \end{aligned}$$

Each term is a measurable rectangle. By further splitting A_1 and A_2 along B_1 and B_2 if necessary (e.g. using $A_1 \setminus B_1$, $A_1 \cap B_1$, and similarly for A_2, B_2), we can refine this representation so that $R \setminus S$ becomes a finite *disjoint* union of measurable rectangles. Hence $R \setminus S \in \mathcal{A}$.

(iii) Difference of finite disjoint unions. Let

$$E = \bigsqcup_{i=1}^{n} R_i, \quad R_i = A_{1i} \times A_{2i}, \quad F = \bigsqcup_{j=1}^{m} S_j, \quad S_j = B_{1j} \times B_{2j},$$

with both unions pairwise disjoint. Then

$$E \setminus F = \left(\bigsqcup_{i=1}^{n} R_i \right) \setminus F = \bigsqcup_{i=1}^{n} (R_i \setminus F),$$

where the union remains disjoint since the R_i are disjoint.

Fix i. Then

$$R_i \setminus F = R_i \setminus \left(\bigsqcup_{j=1}^{m} S_j \right) = \bigcap_{j=1}^{m} (R_i \setminus S_j).$$

By (ii), each $R_i \setminus S_j$ is in $\mathcal{A}$, hence a finite disjoint union of measurable rectangles. A finite intersection of finite unions of rectangles can be written as a finite union of intersections of rectangles, and

$$(A_1 \times A_2) \cap (B_1 \times B_2) = (A_1 \cap B_1) \times (A_2 \cap B_2),$$

so every such intersection is again a rectangle. Therefore each $R_i \setminus F$ is a finite (and after refinement, disjoint) union of rectangles, hence $R_i \setminus F \in \mathcal{A}$. Summing over i, we conclude $E \setminus F \in \mathcal{A}$.

(iv) Closure under finite unions. Let $E, F \in \mathcal{A}$. Then

$$E \cup F = E \sqcup (F \setminus E),$$

where the union on the right-hand side is disjoint. By assumption $E \in \mathcal{A}$, and by (iii) we have $F \setminus E \in \mathcal{A}$. Since $\mathcal{A}$ is defined as all finite disjoint unions of rectangles, the disjoint union $E \sqcup (F \setminus E)$ lies in $\mathcal{A}$. Hence $E \cup F \in \mathcal{A}$.

(v) Complements. For $E \in \mathcal{A}$, its complement in $\Omega_1 \times \Omega_2$ is

$$(\Omega_1 \times \Omega_2) \setminus E.$$

We already know $\Omega_1 \times \Omega_2 \in \mathcal{A}$, and by (iii) the difference of two elements of $\mathcal{A}$ is again in $\mathcal{A}$. Hence $(\Omega_1 \times \Omega_2) \setminus E \in \mathcal{A}$.

Combining (i)–(v), we see that $\mathcal{A}$ contains $\emptyset$ and $\Omega_1 \times \Omega_2$, and is closed under finite unions and complements. Thus $\mathcal{A}$ is an algebra on $\Omega_1 \times \Omega_2$.

Step 2: $\mathbb{P}$ is well-defined on $\mathcal{A}$

To show well-definedness, we must prove that if $E \in \mathcal{A}$ has two representations as a finite disjoint union of rectangles, then the value of $\mathbb{P}(E)$ does not depend on the chosen representation.

Suppose

$$E = \bigsqcup_{i=1}^{n} A_{1i} \times A_{2i} = \bigsqcup_{j=1}^{m} B_{1j} \times B_{2j},$$

with all rectangles pairwise disjoint in each representation.

Observe (pointwisely in the sample space)

$$1_E(\omega_1, \omega_2) = \sum_{i=1}^{n} 1_{A_{1i}}(\omega_1) 1_{A_{2i}}(\omega_2) = \sum_{j=1}^{m} 1_{B_{1j}}(\omega_1) 1_{B_{2j}}(\omega_2),$$

where the equality occurs because they are the indicator functions of the same set. For every fixed ω_2, we have a simple function of $(\Omega_1, \mathcal{F}_1)$. Therefore, for every $\omega_2 \in \Omega_2$,

$$\begin{aligned} \sum_{i=1}^{n} \mathbb{P}_1(A_{1i}) 1_{A_{2i}}(\omega_2) &= \int_{\omega_1} \sum_{i=1}^{n} 1_{A_{1i}}(\omega_1) 1_{A_{2i}}(\omega_2) d\mathbb{P}_1 \\ &= \int_{\omega_1} \sum_{j=1}^{m} 1_{B_{1j}}(\omega_1) 1_{B_{2j}}(\omega_2) d\mathbb{P}_1 = \sum_{j=1}^{m} \mathbb{P}_1(B_{1j}) 1_{B_{2j}}(\omega_2). \end{aligned}$$

These are again simple functions, therefore

$$\begin{aligned} \sum_{i=1}^{n} \mathbb{P}_1(A_{1i}) \mathbb{P}_2(A_{2i}) &= \int_{\omega_2} \sum_{i=1}^{n} \mathbb{P}_1(A_{1i}) 1_{A_{2i}}(\omega_2) d\mathbb{P}_2 \\ &= \int_{\omega_2} \sum_{j=1}^{m} \mathbb{P}_1(B_{1j}) 1_{B_{2j}}(\omega_2) d\mathbb{P}_2 = \sum_{j=1}^{m} \mathbb{P}_1(B_{1j}) \mathbb{P}_2(B_{2j}) \end{aligned}.$$

This completes the proof. □

Thus $\mathbb{P}$ is a finitely additive probability measure on the algebra $\mathcal{A}$.

Lemma 1.2.15. *The set function $\mathbb{P}$ is a countably additive probability measure on $\mathcal{A}$.*

Proof. Let $\{E_n\}_{n=1}^{\infty} \subset \mathcal{A}$ be a decreasing sequence,

$$E_1 \supseteq E_2 \supseteq \cdots, \qquad \bigcap_{n=1}^{\infty} E_n = \emptyset,$$

and we must show that $\mathbb{P}(E_n) \downarrow 0$.

For each n and each $\omega_2 \in \Omega_2$, define the *section*

$$E_{n,\omega_2} \triangleq \{\omega_1 \in \Omega_1 : (\omega_1, \omega_2) \in E_n\}.$$

Because $E_n \in \mathcal{A}$ is a finite disjoint union of rectangles, each E_{n,ω_2} belongs to $\mathcal{F}_1$, and for fixed n the set-function $\omega_2 \mapsto E_{n,\omega_2}$ takes only finitely many different values in $\mathcal{F}_1$. In particular, the function

$$f_n(\omega_2) \triangleq \mathbb{P}_1(E_{n,\omega_2})$$

is measurable and takes only finitely many distinct values, i.e. f_n is a simple function on $(\Omega_2, \mathcal{F}_2)$.

By construction,

$$\mathbb{P}(E_n) = \int_{\Omega_2} f_n(\omega_2) \, d\mathbb{P}_2(\omega_2).$$

Indeed, if explicitly write down, $E_n = \bigsqcup_{i=1}^{k} A_{1i} \times A_{2i}$, then $E_{n,\omega_2} = \bigcup_{i:\omega_2 \in A_{2i}} A_{1i}$ and hence

$$f_n(\omega_2) = \sum_{i:\omega_2 \in A_{2i}} \mathbb{P}_1(A_{1i}),$$

so integrating f_n with respect to $\mathbb{P}_2$ reproduces the finite sum that defines $\mathbb{P}(E_n)$.

Now $0 \le f_n(\omega_2) \le 1$ for all ω_2. Moreover, for each ω_2 the sections decrease:

$$E_{1,\omega_2} \supseteq E_{2,\omega_2} \supseteq \cdots, \qquad \bigcap_{n=1}^{\infty} E_{n,\omega_2} = \emptyset,$$

so by countable additivity (continuity from above) of $\mathbb{P}_1$,

$$\mathbb{P}_1(E_{n,\omega_2}) \downarrow 0 \quad \text{for all } \omega_2.$$

Thus $f_n(\omega_2) \downarrow 0$ pointwisely and $0 \le f_n \le 1$. By the bounded (or monotone) convergence theorem,

$$\mathbb{P}(E_n) = \int f_n \, d\mathbb{P}_2 \downarrow \int 0 \, d\mathbb{P}_2 = 0.$$

Therefore $\mathbb{P}$ is countably additive on $\mathcal{A}$. □

By Carathéodory's extension theorem, $\mathbb{P}$ extends uniquely to a countably additive probability measure on $\mathcal{F} = \sigma(\mathcal{A})$, the σ-algebra generated by the measurable rectangles. This extension is called the *product measure* of $\mathbb{P}_1$ and $\mathbb{P}_2$ and is commonly denoted by $\mathbb{P}_1 \otimes \mathbb{P}_2$.

Iterated integrals and Fubini's theorem

In this section, the objective is to establish the validity of exchanging the order of integration over product probability spaces. The proofs employ the standard stepwise approximation method, often referred to as the "standard machine" approach, progressing from indicator functions through simple functions to general integrable functions.

Let $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$ be probability spaces. Their product space is defined as $(\Omega, \mathcal{F}, \mathbb{P}) = (\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, \mathbb{P}_1 \otimes \mathbb{P}_2)$, where $\mathcal{F}_1 \otimes \mathcal{F}_2$ denotes the product σ-algebra generated by the measurable rectangles $\{A_1 \times A_2 : A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2\}$, and $\mathbb{P}_1 \otimes \mathbb{P}_2$ is the unique probability measure on $\mathcal{F}_1 \otimes \mathcal{F}_2$ satisfying $\mathbb{P}_1 \otimes \mathbb{P}_2(A_1 \times A_2) = \mathbb{P}_1(A_1)\mathbb{P}_2(A_2)$ for all measurable rectangles, as constructed in the previous subsection.

Lemma 1.2.16. *Let $A \in \mathcal{F}$. For each $\omega_1 \in \Omega_1$, define the ω_1-section of A by*

$$A_{\omega_1} = \{\omega_2 \in \Omega_2 : (\omega_1, \omega_2) \in A\},$$

and for each $\omega_2 \in \Omega_2$, define the ω_2-section by

$$A_{\omega_2} = \{\omega_1 \in \Omega_1 : (\omega_1, \omega_2) \in A\}.$$

Then the following hold:

1. *For every fixed $\omega_1 \in \Omega_1$, $A_{\omega_1} \in \mathcal{F}_2$, and for every fixed $\omega_2 \in \Omega_2$, $A_{\omega_2} \in \mathcal{F}_1$.*

2. *The functions $\omega_1 \mapsto \mathbb{P}_2(A_{\omega_1})$ (from $(\Omega_1, \mathcal{F}_1)$ to $[0,1]$) and $\omega_2 \mapsto \mathbb{P}_1(A_{\omega_2})$ (from $(\Omega_2, \mathcal{F}_2)$ to $[0,1]$) are measurable with respect to $\mathcal{F}_1$ and $\mathcal{F}_2$, respectively. Moreover,*

$$\mathbb{P}(A) = \int_{\Omega_1} \mathbb{P}_2(A_{\omega_1})\, \mathbb{P}_1(d\omega_1) = \int_{\Omega_2} \mathbb{P}_1(A_{\omega_2})\, \mathbb{P}_2(d\omega_2).$$

Proof. Part (a): Sections are measurable. First, verify the claim for measurable rectangles $A = A_1 \times A_2$ with $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$. In this case,

$$A_{\omega_1} = \begin{cases} A_2 & \text{if } \omega_1 \in A_1, \\ \emptyset & \text{if } \omega_1 \notin A_1. \end{cases}$$

Since both $A_2 \in \mathcal{F}_2$ and $\emptyset \in \mathcal{F}_2$, it follows that $A_{\omega_1} \in \mathcal{F}_2$ for all $\omega_1 \in \Omega_1$. Symmetrically,

$$A_{\omega_2} = \begin{cases} A_1 & \text{if } \omega_2 \in A_2, \\ \emptyset & \text{if } \omega_2 \notin A_2, \end{cases}$$

so $A_{\omega_2} \in \mathcal{F}_1$ for all $\omega_2 \in \Omega_2$.

Now, consider the collection $\mathcal{S} = \{A \subset \Omega_1 \times \Omega_2 : A_{\omega_1} \in \mathcal{F}_2 \ \forall \omega_1 \in \Omega_1 \text{ and } A_{\omega_2} \in \mathcal{F}_1 \ \forall \omega_2 \in \Omega_2\}$. This collection $\mathcal{S}$ forms a σ-algebra: it contains $\Omega_1 \times \Omega_2$ (with all sections equal to Ω_2 or Ω_1, respectively); it is closed under complements (since $(A^c)_{\omega_1} = (A_{\omega_1})^c \in \mathcal{F}_2$ and similarly for ω_2-sections); and it is closed under countable unions (by de Morgan's laws and σ-algebra properties of $\mathcal{F}_1, \mathcal{F}_2$). Since $\mathcal{S}$ contains all measurable rectangles and $\mathcal{F} = \sigma(\{A_1 \times A_2\})$, it follows that $\mathcal{S} \supseteq \mathcal{F}$. Thus, part (a) holds for all $A \in \mathcal{F}$.

Part (b): Section integrals equal the product measure. For a measurable rectangle $A = A_1 \times A_2$,

$$\mathbb{P}_2(A_{\omega_1}) = \begin{cases} \mathbb{P}_2(A_2) & \text{if } \omega_1 \in A_1, \\ 0 & \text{if } \omega_1 \notin A_1. \end{cases}$$

Hence, the map $\omega_1 \mapsto \mathbb{P}_2(A_{\omega_1})$ equals $\mathbb{P}_2(A_2)\mathbf{1}_{A_1}(\omega_1)$, which is $\mathcal{F}_1$-measurable as a product of a constant and a measurable indicator. Integrating yields

$$\int_{\Omega_1} \mathbb{P}_2(A_{\omega_1})\, \mathbb{P}_1(d\omega_1) = \mathbb{P}_2(A_2)\mathbb{P}_1(A_1) = \mathbb{P}(A).$$

The argument for $\omega_2 \mapsto \mathbb{P}_1(A_{\omega_2})$ is symmetric.

By linearity of the integral and measure, the identities extend to finite disjoint unions of measurable rectangles. Let $\mathcal{C} = \{A \in \mathcal{F} : \mathbb{P}(A) = \int_{\Omega_1} \mathbb{P}_2(A_{\omega_1})\, \mathbb{P}_1(d\omega_1) = \int_{\Omega_2} \mathbb{P}_1(A_{\omega_2})\, \mathbb{P}_2(d\omega_2)\}$. The collection $\mathcal{C}$ is a monotone class: it contains Ω and the empty set; is closed under increasing countable unions (by monotone convergence of measures); and closed under decreasing countable intersections (by continuity of measures from above). Since $\mathcal{C}$ contains the algebra of finite disjoint unions of rectangles (which generates $\mathcal{F}$), the monotone class theorem implies $\mathcal{C} = \mathcal{F}$. $\square$

Theorem 1.2.17 (Fubini's Theorem)**.** *Let $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ be an $\mathcal{F}$-measurable function. For each fixed $\omega_1 \in \Omega_1$, define the section function*

$$g_{\omega_1} : \Omega_2 \to \mathbb{R}, \quad g_{\omega_1}(\omega_2) = f(\omega_1, \omega_2),$$

and for each fixed $\omega_2 \in \Omega_2$, define

$$h_{\omega_2} : \Omega_1 \to \mathbb{R}, \quad h_{\omega_2}(\omega_1) = f(\omega_1, \omega_2).$$

Then:

1. *For every $\omega_1 \in \Omega_1$, g_{ω_1} is $\mathcal{F}_2$-measurable; similarly, for every $\omega_2 \in \Omega_2$, h_{ω_2} is $\mathcal{F}_1$-measurable.*

2. *If f is integrable, i.e., $\int_\Omega |f|\, d\mathbb{P} < \infty$, then $g_{\omega_1} \in L^1(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$ for $\mathbb{P}_1$-almost every $\omega_1 \in \Omega_1$, and $h_{\omega_2} \in L^1(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ for $\mathbb{P}_2$-almost every $\omega_2 \in \Omega_2$. (L^1 denotes ℓ_1 integrable.) Moreover, the functions*

$$G : \Omega_1 \to \mathbb{R}, \quad G(\omega_1) = \int_{\Omega_2} g_{\omega_1}(\omega_2)\, \mathbb{P}_2(d\omega_2),$$

$$H : \Omega_2 \to \mathbb{R}, \quad H(\omega_2) = \int_{\Omega_1} h_{\omega_2}(\omega_1)\, \mathbb{P}_1(d\omega_1)$$

are integrable on their respective spaces, and

$$\begin{aligned} &\int_\Omega f(\omega_1, \omega_2)\, \mathbb{P}(d\omega_1, d\omega_2) \\ &= \int_{\Omega_1} G(\omega_1)\, \mathbb{P}_1(d\omega_1) \\ &= \int_{\Omega_2} H(\omega_2)\, \mathbb{P}_2(d\omega_2). \end{aligned}$$

Proof. Part (a): Sections of measurable functions are measurable. Fix a Borel set $B \subset \mathbb{R}$. Since f is $\mathcal{F}$-measurable, the preimage $\{(\omega_1, \omega_2) : f(\omega_1, \omega_2) \in B\} \in \mathcal{F}$. By Lemma 1.2.16(a), for each fixed ω_1, the section $\{\omega_2 : f(\omega_1, \omega_2) \in B\} = g_{\omega_1}^{-1}(B) \in \mathcal{F}_2$, so g_{ω_1} is $\mathcal{F}_2$-measurable. The argument for h_{ω_2} is symmetric.

Part (b): Iterated integrals for integrable f. Proceed via the standard machine approximation.

1. *Indicators.* Suppose $f = \mathbf{1}_A$ for $A \in \mathcal{F}$. Then $g_{\omega_1} = \mathbf{1}_{A_{\omega_1}}$ and $h_{\omega_2} = \mathbf{1}_{A_{\omega_2}}$. By Lemma 1.2.16(b),

$$\begin{aligned} \int_\Omega f\, d\mathbb{P} = \mathbb{P}(A) &= \int_{\Omega_1} \mathbb{P}_2(A_{\omega_1})\, d\mathbb{P}_1(\omega_1) \\ &= \int_{\Omega_1} \left(\int_{\Omega_2} g_{\omega_1}(\omega_2)\, d\mathbb{P}_2(\omega_2) \right) d\mathbb{P}_1(\omega_1), \end{aligned}$$

and analogously for the reverse order. Note that $\mathbb{P}_2(A_{\omega_1}) < \infty$ everywhere since $\mathbb{P}_2$ is a probability measure.

2. *Simple functions.* Let $f = \sum_{k=1}^n c_k \mathbf{1}_{A_k}$ with $A_k \in \mathcal{F}$, $c_k \in \mathbb{R}$. Then $g_{\omega_1} = \sum_{k=1}^n c_k \mathbf{1}_{(A_k)_{\omega_1}}$ and similarly for h_{ω_2}. By

linearity of integration and the indicator case, the iterated integral equalities hold for f.

3. *Bounded measurable functions.* Let f be bounded and measurable, say $|f| \le M < \infty$. There exists a sequence of simple functions $f_n \to f$ pointwise with $|f_n| \le M$. For each ω_1, $g_{\omega_1,n} \to g_{\omega_1}$ pointwise and $|g_{\omega_1,n}| \le M$, so the bounded convergence theorem yields $\int_{\Omega_2} g_{\omega_1,n}\, d\mathbb{P}_2 \to \int_{\Omega_2} g_{\omega_1}\, d\mathbb{P}_2$ for each ω_1. Integrating over Ω_1 and using the simple-function case plus bounded convergence again passes the identities to the limit f.

4. *Nonnegative measurable functions.* Let $0 \le f < \infty$ be measurable. Choose bounded measurable functions $f_n \uparrow f$ pointwise (e.g., $f_n = \min\{f, n\}$). For each ω_1, $0 \le g_{\omega_1,n} \uparrow g_{\omega_1}$ pointwise, so the monotone convergence theorem gives $\int_{\Omega_2} g_{\omega_1,n}\, d\mathbb{P}_2 \uparrow \int_{\Omega_2} g_{\omega_1}\, d\mathbb{P}_2$ (possibly $+\infty$). The identities for f_n pass to the limit by monotone convergence over Ω_1.

5. *Integrable functions.* Let f be integrable, so $f = f_+ - f_-$ with $f_+, f_- \ge 0$ and $\int |f| = \int f_+ + \int f_- < \infty$. Both f_+ and f_- are nonnegative measurable, so their iterated integrals equal $\int f_\pm$ (finite) by the previous step. For almost every ω_1, both sections $g^+_{\omega_1}$ and $g^-_{\omega_1}$ are integrable (since $\int_{\Omega_1}\int_{\Omega_2} |g_{\omega_1}|\, d\mathbb{P}_2\, d\mathbb{P}_1 = \int |f| < \infty$ implies $\int_{\Omega_2} |g_{\omega_1}|\, d\mathbb{P}_2 < \infty$ a.e.), and $G(\omega_1) = G_+(\omega_1) - G_-(\omega_1)$ with $G_\pm$ integrable by the nonnegative case and linearity. Linearity of the integral then yields the result for f.

This completes the proof. □

Exercise 1.2.7. Let $\Omega = \mathbb{N} \times \mathbb{N}$ with

$$\mathbb{P}(i,j) = \frac{1}{2^{i+j}}, \quad i,j \ge 1.$$

For $i, j \ge 1$, define

$$f(i,j) = \begin{cases} 2^{i+j}, & j = i+1, \\ -2^{i+j}, & j = i-1,\ i \ge 2, \\ 0, & \text{otherwise.} \end{cases}$$

The marginals are $\mathbb{P}_1(i) = \mathbb{P}_2(i) = 1/2^i$ for $i \ge 1$. Compute

$$G(i) := \sum_{j=1}^{\infty} f(i,j)\, \mathbb{P}_2(j), \qquad H(j) := \sum_{i=1}^{\infty} f(i,j)\, \mathbb{P}_1(i),$$

and then the sums

$$\sum_{i \ge 1} G(i)\, \mathbb{P}_1(i), \qquad \sum_{j \ge 1} H(j)\, \mathbb{P}_2(j).$$

Observe that f is not integrable, and compare the two iterated sums. This illustrates the necessity of integrability in Fubini's theorem.

Lemma 1.2.18 (Jensen's inequality)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\Phi : \mathbb{R} \to \mathbb{R}$ be a convex function. If f is an integrable measurable function and $\Phi(f)$ is also integrable, then*

$$\int \Phi(f(\omega))\, d\mathbb{P}(\omega) \ge \Phi\left(\int f(\omega)\, d\mathbb{P}(\omega)\right).$$

Proof. A key fact about convex functions is that any convex Φ can be represented as the pointwise supremum of its supporting affine functions:

$$\Phi(x) = \sup_{(a,b)\in\mathcal{E}} (ax + b),$$

for some collection $\mathcal{E} \subset \mathbb{R}^2$ (Fenchel–Moreau theorem). Thus, for each fixed $(a,b) \in \mathcal{E}$ and all $x \in \mathbb{R}$,

$$\Phi(x) \ge ax + b.$$

Applying this with $x = f(\omega)$ gives (pointwisely)

$$\Phi(f(\omega)) \ge af(\omega) + b \quad \text{for all } \omega \in \Omega.$$

Integrating both sides,

$$\int \Phi(f(\omega))\, d\mathbb{P}(\omega) \ge a \int f(\omega)\, d\mathbb{P}(\omega) + b.$$

Since this inequality holds for all $(a,b) \in \mathcal{E}$, the left-hand side dominates the supremum over (a,b):

$$\begin{aligned} \int \Phi(f(\omega))\, d\mathbb{P}(\omega) &\ge \sup_{(a,b)\in\mathcal{E}} \left(a \int f(\omega)\, d\mathbb{P}(\omega) + b\right) \\ &= \Phi\left(\int f(\omega)\, d\mathbb{P}(\omega)\right), \end{aligned}$$

using again the representation of Φ as a supremum of affine functions. This proves the inequality. □

1.2.13 A few background facts

Preimages

The following proposition presents basic properties of preimages with respect to set operations for any function. Note that no measurability is involved here.

Proposition 1.2.19 (Basic properties of preimages)**.** *Let $f : X \to Y$ be a function. Define*

$$f^{-1}(A) = \{x \in X : f(x) \in A\}.$$

For any $A, B \subseteq Y$ and any family $\{A_i\}_{i\in I} \subseteq Y$, the following hold:

1. *Monotonicity: If $A \subseteq B$, then*
$$f^{-1}(A) \subseteq f^{-1}(B).$$

2. *Unions(countable):*
$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B),$$
$$f^{-1}\Big(\bigcup_{i\in I} A_i\Big) = \bigcup_{i\in I} f^{-1}(A_i).$$

3. *Intersections(countable):*
$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B),$$
$$f^{-1}\Big(\bigcap_{i\in I} A_i\Big) = \bigcap_{i\in I} f^{-1}(A_i).$$

4. *Complements (relative to Y):*
$$f^{-1}(A^c) = \big(f^{-1}(A)\big)^c.$$

5. *Differences and symmetric differences:*
$$f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B),$$
$$f^{-1}(A \triangle B) = f^{-1}(A) \triangle f^{-1}(B).$$

6. *Special cases:*
$$f^{-1}(\emptyset) = \emptyset, \qquad f^{-1}(Y) = X.$$

7. *Composition rule: If $g : Y \to Z$, then for any $C \subseteq Z$,*
$$(g \circ f)^{-1}(C) = f^{-1}\big(g^{-1}(C)\big).$$

8. *Disjointness: If $A \cap B = \emptyset$, then*
$$f^{-1}(A) \cap f^{-1}(B) = \emptyset.$$
More generally, if $\{A_i\}_{i\in I}$ are pairwise disjoint, then $\{f^{-1}(A_i)\}_{i\in I}$ are pairwise disjoint.

The preimage map automatically preserves the set operations that define a σ-algebra. So for a measurable function, the preimage mapping actually needs to make sure every Borel set is mapped to a set in the sigma algebra equipped in the sample space. (Note that it is common to have multiple Borel sets mapped to the same set in the sigma algebra.)

Also, note that continuous functions are Borel-measurable, by the definition of continuous functions.

Cardinality of Borel σ-algebra

Let $\mathcal{B}(\mathbb{R})$ be the Borel σ-algebra. Since the Borel σ-algebra can be generated by intervals in form of $(-\infty, a), a \in \mathbb{R}$, $|\mathcal{B}(\mathbb{R})| \geq |\mathbb{R}|$. Countable unions does not increase cardinality, thus $|\mathcal{B}(\mathbb{R})| = |\mathbb{R}|$. For the cardinality of $\mathbb{R}$, check Aleph number.

Lebesgue measure and Borel measure

Lebesgue measure is the completion of the Borel measure. Or equivalently, Borel measure is the restriction of Lebesgue measure over Borel sets.

The following definition presents the Lebesgue (outer) measure. Sets that satisfy Caratheodory criterion under Lebesgue outer measure formed the sigma algebra also known as Lebesgue set.

Definition 1.2.9 (Lebesgue measure)**.** *For any interval $I = (a, b)$ or $I = [a, b]$ in the set $\mathbb{R}$ of real numbers ($a, b \in \mathbb{R}$, $b \geq a$), let $\ell(I) = b - a$ denote its length. For any subset $E \subseteq \mathbb{R}$ the Lebesgue (outer) measure $\lambda^*(E)$ is defined as an infimum*

$$\lambda^*(E) = \inf\left\{\sum_{k=1}^{\infty} \ell(I_k) : E \subset \bigcup_{k=1}^{\infty} I_k\right\}.$$

A set E satisfies the Caratheodory criterion when for every $A \subseteq \mathbb{R}$, we have $\lambda^*(A) = \lambda^*(A \cap E) + \lambda^*(A \cap E^c)$. Sets satisfying the Caratheodory criterion are said to be Lebesgue-measurable.

The set of all Lebesgue-measurable sets forms a sigma-algebra, known as the Lebesgue σ-algebra, usually denoted as $\mathcal{L}(\mathbb{R})$.

Remark 31. The above definition can be generalized to higher dimensions using rectangular cuboids formed by Cartesian product of open intervals.

The above definition takes the usual notion of "length" on intervals and extends it to Lebesgue σ-algebra. This is a Caratheodory extension. When people say a set is "measurable" under certain setting, they mean the Caratheodory criterion is satisfied.

Remark 32. If assume the Axiom of Choice[7], then ZFC proves that non-Lebesgue-measurable sets do exist, such as the Vitali sets. Construction of Vitali sets directly invokes the Axiom of Choice.

Definition 1.2.10. *A measure space (Ω, Σ, μ) is said to be complete if and only if*

$$\forall N \in \Sigma \quad s.t.\ \mu(N) = 0, \implies \forall S \subseteq N, S \in \Sigma.$$

Proposition 1.2.20. *$(\mathbb{R}, \mathcal{L}(\mathbb{R}), \lambda^*)$ is a complete measure space.*

There exists sets that are in the Lebesgue σ-algebra but are not in the Borel σ-algebra. Restricting the Lebesgue measure to the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ yields the Borel measure, sometimes people also say "the Borel measure associated with Lebesgue measure".

Example 10 ($\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R})$). The Cantor set $C \subset [0, 1]$ is Borel and has Lebesgue measure 0. By completeness of Lebesgue measure space, any subset $A \subseteq C$ has Lebesgue measure 0. But there exists subset $A' \subseteq C$ that are not Borel. Since these A' does not even inside the Borel σ-algebra, they are not Borel measurable.

Refinement

Example 11 (refinement of partitions). Let

$$s = \sum_{i=1}^{n} c_i \mathbf{1}_{A_i}, \qquad t = \sum_{j=1}^{m} d_j \mathbf{1}_{B_j},$$

where $\{A_i\}_{i=1}^n$ and $\{B_j\}_{j=1}^m$ are (measurable) partitions of Ω. The *common refinement* is

$$C_{ij} := A_i \cap B_j \qquad (1 \leq i \leq n,\ 1 \leq j \leq m),$$

which is again a (measurable) partition of Ω (ignoring empty sets). On each C_{ij}, both s and t are constant, so

$$as + bt = \sum_{i=1}^{n}\sum_{j=1}^{m} (ac_i + bd_j)\, \mathbf{1}_{C_{ij}},$$

[7] "Informally put, the axiom of choice says that given any collection of non-empty sets, it is possible to construct a new set by choosing one element from each set, even if the collection is infinite."

i.e., $as+bt$ is a simple function written with respect to the refined partition.

Chapter 2

Characteristic Functions

2.1 Properties and Inversion Formula

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $f : \Omega \to \mathbb{C}$ is called a (complex-valued) random variable. Write $f = f_r + i f_i$, where $f_r, f_i : \Omega \to \mathbb{R}$ denote the real and imaginary parts of f, respectively. We say that f is (Lebesgue) measurable if both f_r and f_i are measurable real-valued functions. If, in addition, f_r and f_i are integrable, that is $\int |f_r| \, d\mathbb{P} < \infty$ and $\int |f_i| \, d\mathbb{P} < \infty$, we define the integral of f by

$$\int f \, d\mathbb{P} \;:=\; \int f_r \, d\mathbb{P} \;+\; i \int f_i \, d\mathbb{P}.$$

Exercise 2.1.1. Let $f : \Omega \to \mathbb{C}$ be a complex-valued measurable function such that $|f|$ is integrable. Show that

$$\left| \int f \, d\mathbb{P} \right| \;\le\; \int |f| \, d\mathbb{P}.$$

(First prove the inequality for simple functions, then extend by approximation, or use the fact that $z \mapsto |z|$ is convex and apply Jensen's inequality to the probability measure $\mathbb{P}$.)

For a real-valued random variable $X : \Omega \to \mathbb{R}$ we define the *characteristic function* of X by

$$\phi(t) \;=\; \mathbb{E}\big(e^{itX}\big) \;=\; \int_\Omega e^{itX(\omega)} \, \mathbb{P}(d\omega) \;=\; \int_{\mathbb{R}} e^{itx} \, F(dx),$$

where F is the distribution function of X, and $F(dx)$ denotes the corresponding probability measure on $\mathbb{R}$. By the exercise above, applied to $f(\omega) = e^{itX(\omega)}$, we obtain

$$|\phi(t)| \;=\; \left| \int e^{itX} \, d\mathbb{P} \right| \;\le\; \int |e^{itX}| \, d\mathbb{P} \;=\; \int 1 \, d\mathbb{P} \;=\; 1.$$

Theorem 2.1.1. *The characteristic function $\phi : \mathbb{R} \to \mathbb{C}$ of any probability distribution is uniformly continuous on $\mathbb{R}$ and positive definite. That is, for any real numbers $t_1, \dots, t_n$ and any complex numbers $\xi_1, \dots, \xi_n$, the matrix*

$$M \;\equiv\; \big[\phi(t_k - t_\ell)\big]_{k,\ell=1}^n$$

is nonnegative semidefinite, i.e.

$$\sum_{k,\ell=1}^n \xi_k \, \phi(t_k - t_\ell) \, \overline{\xi_\ell} \;\ge\; 0.$$

Proof. **Positive semi-definiteness.** Let $t_1, \dots, t_n \in \mathbb{R}$ and $\xi_1, \dots, \xi_n \in \mathbb{C}$. Then

$$\begin{aligned}
\vec{\xi} M \vec{\xi}^* &= \sum_{k,\ell=1}^n \xi_k \, \phi(t_k - t_\ell) \, \overline{\xi_\ell} \\
&= \sum_{k,\ell=1}^n \xi_k \, \mathbb{E}\big(e^{i(t_k - t_\ell)X}\big) \, \overline{\xi_\ell} \\
&= \mathbb{E}\left(\sum_{k,\ell=1}^n \xi_k \, e^{it_kX} \, \overline{\xi_\ell} \, e^{-it_\ell X} \right) \\
&= \mathbb{E}\left(\sum_{k=1}^n \xi_k e^{it_kX} \, \overline{\sum_{\ell=1}^n \xi_\ell e^{it_\ell X}} \right) \\
&= \mathbb{E}\Big(\Big| \sum_{k=1}^n \xi_k e^{it_kX} \Big|^2 \Big) \;\ge\; 0.
\end{aligned}$$

Thus M is nonnegative semidefinite. Moreover, equality holds if and only if

$$Y := \sum_{k=1}^n \xi_k e^{it_kX} = 0 \quad \text{almost surely.}$$

Uniform continuity. Let $s, t \in \mathbb{R}$. Then

$$\begin{aligned}
|\phi(t) - \phi(s)| &= \big|\mathbb{E}(e^{itX} - e^{isX})\big| \\
&\le \mathbb{E}\big(|e^{itX} - e^{isX}|\big) \\
&= \mathbb{E}\big(|e^{i(t-s)X} - 1|\big).
\end{aligned}$$

Thus it suffices to show that

$$\lim_{u \to 0} \mathbb{E}\big(|e^{iuX} - 1|\big) = 0.$$

Assume, by way of contradiction, that there exists $\varepsilon > 0$ and a sequence $u_n \to 0$ such that

$$\mathbb{E}\big(|e^{iu_nX} - 1|\big) \;\ge\; \varepsilon \quad \text{for all } n.$$

Define

$$Y_n := |e^{iu_nX} - 1|.$$

Then $0 \le Y_n \le 2$ for all n, and $Y_n(\omega) \to 0$ for every $\omega \in \Omega$ by continuity of $u \mapsto e^{iuX(\omega)}$. By the bounded convergence theorem, $\mathbb{E}(Y_n) \to 0$, contradicting $\mathbb{E}(Y_n) \ge \varepsilon$. Therefore $\mathbb{E}(|e^{iuX} - 1|) \to 0$ as $u \to 0$, which implies that ϕ is uniformly continuous on $\mathbb{R}$. $\square$

Remark 33. Characteristic functions satisfy three properties:

1. $|\phi(t)| \le 1, \phi(0) = 1$, (normalized)
2. positive semi-definite,
3. uniformly continuous.

Bochner's theorem shows that there exists a unique probability measure for any functions that satisfy the above three properties.

Lemma 2.1.2. *If $\int |X| \, d\mathbb{P} < \infty$, then ϕ is continuously differentiable on $\mathbb{R}$ and*

$$\phi'(0) = i \int X \, d\mathbb{P}.$$

Proof. Fix $t \in \mathbb{R}$ and $\delta \neq 0$. Consider the difference quotient

$$\begin{aligned}\frac{\phi(t)-\phi(t-\delta)}{\delta} &= \frac{1}{\delta}\,\mathbb{E}\big(e^{itX} - e^{i(t-\delta)X}\big)\\ &= \mathbb{E}\Big(e^{itX}\,\frac{1-e^{-i\delta X}}{\delta}\Big)\\ &= \mathbb{E}\Big(e^{itX}\,\frac{1-e^{-i\delta X}}{i\delta X}\,iX\Big).\end{aligned}$$

Define

$$Y_\delta := e^{itX}\,\frac{1-e^{-i\delta X}}{i\delta X}\,iX, \qquad Y := iXe^{itX}.$$

Then $Y_\delta(\omega) \to Y(\omega)$ for every ω as $\delta \to 0$, since

$$\frac{1-e^{-i\delta X(\omega)}}{i\delta X(\omega)} \to 1$$

whenever $X(\omega) \neq 0$, and the limit is trivially 1 when $X(\omega) = 0$. Moreover, using the estimate from the next lemma,

$$\left|\frac{1-e^{-i\delta X}}{i\delta X}\right| = \left|\frac{1-e^{i(-\delta X)}}{-\delta X}\right| \le 1,$$

we obtain

$$|Y_\delta| \;\le\; |X|.$$

Since $\int |X|\,d\mathbb{P} < \infty$, the dominated convergence theorem applies and yields

$$\lim_{\delta\to 0}\mathbb{E}(Y_\delta) = \mathbb{E}(Y).$$

Hence the derivative exists and is given by

$$\phi'(t) = \mathbb{E}(iXe^{itX}), \quad t \in \mathbb{R}.$$

In particular, at $t = 0$,

$$\phi'(0) = i\int X\,d\mathbb{P}.$$

The continuity of $\phi'(t)$ (and hence C^1-regularity of ϕ) follows by a similar dominated convergence argument applied to the mapping $t \mapsto iXe^{itX}$. □

Lemma 2.1.3. *For all $x \in \mathbb{R}$,*

$$\left|\frac{1-e^{ix}}{x}\right| \le 1.$$

Proof. Using the identity

$$1 - e^{ix} = e^{ix/2}\big(e^{-ix/2} - e^{ix/2}\big) = -2ie^{ix/2}\sin\left(\frac{x}{2}\right),$$

we have

$$\left|\frac{1-e^{ix}}{x}\right| = \left|2\,\frac{\sin\left(\frac{x}{2}\right)}{x}\right| = \left|\frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right| \le 1,$$

since $\big|\sin y\big| \le |y|$ for all $y \in \mathbb{R}$. The inequality also holds at $x = 0$ by continuity, interpreting the expression as its limit equal to 1. □

Exercise 2.1.2. Show that if $\mathbb{E}(|X|^r) < \infty$ for some integer $r \ge 1$, then ϕ is r times continuously differentiable on $\mathbb{R}$. (Hint: Differentiate under the integral sign and use dominated convergence with a suitable dominating function involving $|X|^r$.)
Extra credit: If r is even, show that the converse holds, i.e., if ϕ is r times continuously differentiable at 0, then $\mathbb{E}(|X|^r) < \infty$.

Let $\varphi(t) = \mathbb{E}(e^{itX})$.

1. Finite moments imply differentiability

   Assume $\mathbb{E}(|X|^r) < \infty$. For $h \neq 0$,

   $$\frac{e^{i(t+h)X} - e^{itX}}{h} = e^{itX}\,\frac{e^{ihX}-1}{h}.$$

   Using $|e^{iz} - 1| \le |z|$,

   $$\left|\frac{e^{ihX}-1}{h}\right| \le |X|.$$

   Thus the difference quotient is dominated by $|X|$.

   By induction, directly taking derivative over the integral gives

   $$\varphi^{(k)}(t) = \mathbb{E}\big[(iX)^k e^{itX}\big], \qquad k \le r,$$

   and

   $$|(iX)^k e^{itX}| \le |X|^k \in L^1.$$

   By dominated convergence, $\varphi^{(k)}$ is continuous, so φ is r-times differentiable.

2. Converse for even order

   Let $r = 2m$ and assume φ has a finite $2m$-th derivative at 0. Define

   $$S_m(t) := \sum_{k=0}^{2m}(-1)^k\binom{2m}{k}\varphi((m-k)t).$$

   Then

   $$S_m(t) = (-4)^m\,\mathbb{E}\big[\sin^{2m}(tX/2)\big].$$

   Hence

   $$\frac{(-1)^m S_m(t)}{t^{2m}} = \mathbb{E}\left[|X|^{2m}\left(\frac{\sin(tX/2)}{tX/2}\right)^{2m}\right].$$

   By Fatou's lemma,

   $$\mathbb{E}|X|^{2m} \le \liminf_{t\to 0}\frac{(-1)^m S_m(t)}{t^{2m}}.$$

   Since φ has a $2m$-th derivative at 0, $\lim_{t\to 0}\frac{S_m(t)}{t^{2m}} = \varphi^{(2m)}(0)$, so $\mathbb{E}|X|^{2m} < \infty$.

   Thus, for even r, $\mathbb{E}(|X|^r) < \infty$ is equivalent to $\varphi \in C^r$.

We now address the question: how can one recover the distribution function from the characteristic function?

Theorem 2.1.4 (Inversion formula)**.** *Let X be a real-valued random variable with distribution function $F(x) := \mathbb{P}(X \le x)$ and characteristic function ϕ. If $a, b \in \mathbb{R}$ are continuity points of F, then*

$$F(b) - F(a) = \lim_{T\to\infty}\frac{1}{2\pi}\int_{-T}^{T}\phi(t)\,\frac{e^{-itb}-e^{-ita}}{-it}\,dt.$$

Proof. For $T > 0$ consider

$$\begin{aligned}I_T &:= \frac{1}{2\pi}\int_{-T}^{T}\phi(t)\,\frac{e^{-itb}-e^{-ita}}{-it}\,dt\\ &= \frac{1}{2\pi}\int_{-T}^{T}\frac{e^{-itb}-e^{-ita}}{-it}\int e^{itX}\,d\mathbb{P}\,dt.\end{aligned}$$

By Fubini's theorem (justified by the integrability bounds $|\phi(t)| \le 1$ and the boundedness of the kernel), we may interchange the order of integration:

$$\begin{aligned}I_T &= \frac{1}{2\pi}\int\int_{-T}^{T}\frac{e^{it(X-b)}-e^{it(X-a)}}{-it}\,dt\,d\mathbb{P}\\ &= \frac{1}{2\pi}\int\int_{-T}^{T}\frac{\sin(t(X-a))-\sin(t(X-b))}{t}\,dt\,d\mathbb{P}.\end{aligned}$$

Using the symmetry of the integrand, this can be written as

$$I_T = \frac{1}{\pi}\int\int_{0}^{T}\frac{\sin(t(X-a))-\sin(t(X-b))}{t}\,dt\,d\mathbb{P}.$$

Define, for $T > 0$ and $x \in \mathbb{R}$,

$$u(T,x) := \int_0^T\frac{\sin(tx)}{\pi t}\,dt = \int_0^{T/\pi}\frac{\sin(\pi sx)}{\pi s}\,ds.$$

Then

$$I_T = \int\big(u(T,X-a) - u(T,X-b)\big)\,d\mathbb{P}.$$

It is known (Dirichlet integral) that:

(i) There exists a constant $C < \infty$ such that

$$\sup_{T>0,\,x\in\mathbb{R}}|u(T,x)| \le C.$$

(ii) For each fixed x,

$$\lim_{T\to\infty} u(T,x) = \begin{cases} \frac{1}{2}, & x>0, \\ -\frac{1}{2}, & x<0, \\ 0, & x=0. \end{cases}$$

Thus, for each ω,

$$\begin{aligned} &u(T, X(\omega)-a) - u(T, X(\omega)-b) \\ &\to \frac{1}{2}\mathbf{1}_{\{X>a\}}(\omega) - \frac{1}{2}\mathbf{1}_{\{X<a\}}(\omega) \\ &\quad - \frac{1}{2}\mathbf{1}_{\{X>b\}}(\omega) + \frac{1}{2}\mathbf{1}_{\{X<b\}}(\omega). \end{aligned}$$

Moreover, the uniform bound on u implies

$$\big|u(T, X-a) - u(T, X-b)\big| \le 2C,$$

so by the dominated convergence theorem,

$$\begin{aligned} &\lim_{T\to\infty} I_T \\ &= \int \lim_{T\to\infty} \big(u(T,X-a) - u(T,X-b)\big)\, d\mathbb{P} \\ &= \int \left(\frac{1}{2}\mathbf{1}_{\{X>a\}} - \frac{1}{2}\mathbf{1}_{\{X<a\}} - \frac{1}{2}\mathbf{1}_{\{X>b\}} + \frac{1}{2}\mathbf{1}_{\{X<b\}}\right) d\mathbb{P} \\ &= \frac{1}{2}\Big(\mathbb{P}(X>b) - \mathbb{P}(X<b) - \mathbb{P}(X>a) + \mathbb{P}(X<a)\Big). \end{aligned}$$

Using $\mathbb{P}(X<b) = F(b-)$ and $\mathbb{P}(X\le b) = F(b)$, we can rewrite this as

$$\lim_{T\to\infty} I_T = F(b) - F(a) - \frac{1}{2}\big(\mathbb{P}(X=b) - \mathbb{P}(X=a)\big).$$

If a and b are continuity points of F, then $\mathbb{P}(X=a) = \mathbb{P}(X=b) = 0$, and hence

$$\lim_{T\to\infty} I_T = F(b) - F(a),$$

as claimed. □

Exercise 2.1.3. Show that a distribution function F can have at most countable discontinuity points.

Solution. Let $F:\mathbb{R}\to\mathbb{R}$ be a distribution function. Then F is nondecreasing and bounded. For $x\in\mathbb{R}$ define the jump size

$$\Delta F(x) := F(x) - F(x-), \qquad F(x-) := \lim_{t\uparrow x} F(t).$$

Since F is monotone, the left limit $F(x-)$ exists for every x. Moreover, x is a point of discontinuity of F if and only if $\Delta F(x) > 0$.

For each $n\in\mathbb{N}$ set

$$D_n := \{x\in\mathbb{R} : \Delta F(x) \ge 1/n\}.$$

We claim that D_n is finite. Indeed, let $x_1 < \cdots < x_k$ be distinct points of D_n. Then each jump is at least $1/n$, so

$$\sum_{i=1}^{k} \Delta F(x_i) \ge \frac{k}{n}.$$

On the other hand, the total sum of jumps cannot exceed the total variation of F on $\mathbb{R}$, and since F is a distribution function we have $0\le F\le 1$, hence

$$\sum_{i=1}^{k} \Delta F(x_i) \le \sup_x F(x) - \inf_x F(x) \le 1.$$

Therefore $k/n \le 1$, i.e. $k\le n$. This shows D_n is finite.

Now let D be the set of all discontinuity points of F. If $x\in D$, then $\Delta F(x) > 0$, so choose n such that $1/n \le \Delta F(x)$. Then $x\in D_n$. Hence

$$D = \{x : \Delta F(x) > 0\} = \bigcup_{n=1}^{\infty} D_n,$$

a countable union of finite sets. Thus D is at most countable.

Exercise 2.1.4. Prove that if two distribution functions F and G agree at all their points of continuity, then $F = G$ everywhere.

Hint: Show that the set of discontinuity points of a distribution function is at most countable (use monotonicity). Then combine this with right-continuity of distribution functions.

Solution. Let F and G be distribution functions and assume that $F(x) = G(x)$ at every point x at which both functions are continuous.

By the previous exercise, the sets of discontinuity points D_F and D_G are at most countable, hence so is $D_F\cup D_G$. Therefore the set

$$C := \mathbb{R}\setminus(D_F\cup D_G)$$

(the set of points where both F and G are continuous) is dense in $\mathbb{R}$: indeed, a countable set cannot contain an interval, so every open interval meets C.

Fix arbitrary $x\in\mathbb{R}$. Since C is dense, for each $m\in\mathbb{N}$ we can choose a subsequence

$$x_m \in C\cap(x,\, x+1/m),$$

so that $x_m > x$ and $x_m \downarrow x$. By right-continuity of distribution functions,

$$F(x) = \lim_{m\to\infty} F(x_m), \qquad G(x) = \lim_{m\to\infty} G(x_m).$$

But $x_m\in C$ implies $F(x_m) = G(x_m)$ for every m. Taking limits yields

$$F(x) = \lim_{m\to\infty} F(x_m) = \lim_{m\to\infty} G(x_m) = G(x).$$

Since x was arbitrary, $F = G$ on $\mathbb{R}$.

2.1.1 Supplement: Dirichlet Integral

Lemma 2.1.5. *Consider the Dirichlet integral*

$$u(T,x) \;=\; \int_0^T \frac{\sin(tx)}{\pi t}\, dt, \qquad T>0,\ x\in\mathbb{R}.$$

Then the following hold:

1. *There exists a constant $C<\infty$ such that*
$$\sup_{T>0,\ x\in\mathbb{R}} |u(T,x)| \le C,$$
and in fact one may take $C=2$.

2. *For each fixed $x\in\mathbb{R}$,*
$$\lim_{T\to\infty} u(T,x) = \begin{cases} \frac{1}{2}, & x>0, \\ -\frac{1}{2}, & x<0, \\ 0, & x=0. \end{cases}$$

Proof. **Proof of (i).** By symmetry in x, it suffices to consider $x>0$. Further, let k be such that $T\in(2k\frac{\pi}{x}, 2(k+1)\frac{\pi}{x}]$. If $k=0$ then

$$\left|\int_0^T \frac{\sin tx}{\pi t} dt\right| \le \int_0^T \frac{x}{\pi} dt = \frac{Tx}{\pi} \le 2.$$

For $k\ge 1$, we express

$$\begin{aligned} u(T,x) &= \sum_{j=0}^{k-1} \int_{2j\frac{\pi}{x}}^{(2j+1)\frac{\pi}{x}} \left(\frac{\sin tx}{\pi t} + \frac{\sin x(t+\frac{\pi}{x})}{\pi(t+\frac{\pi}{x})}\right) dt \\ &\quad + \int_{2k\frac{\pi}{x}}^{T} \frac{\sin tx}{\pi t} dt \\ &= \sum_{j=0}^{k-1} \int_{2j\frac{\pi}{x}}^{(2j+1)\frac{\pi}{x}} \frac{\sin tx}{t(tx+\pi)} dt + \int_{2k\frac{\pi}{x}}^{T} \frac{\sin tx}{\pi t} dt. \end{aligned}$$

Thus, we have (the second integral below only appears when

$k \geq 2$)

$$
\begin{aligned}
&|u(T,x)| \\
&\leq \int_0^{\frac{\pi}{x}} \frac{\sin tx}{t(tx+\pi)} dt + \int_{2\frac{\pi}{x}}^{(2k-1)\frac{\pi}{x}} \frac{1}{t^2 x} dt + \int_{2k\frac{\pi}{x}}^{T} \frac{1}{\pi t} dt \\
&\leq \int_0^{\frac{\pi}{x}} \frac{x}{(tx+\pi)} dt + \frac{1}{2\pi} - \frac{1}{(2k-1)\pi} + \frac{1}{\pi} \ln\left(\frac{Tx}{2k\pi}\right) \\
&\leq \ln 2 + \frac{1}{2\pi} + \frac{1}{\pi} \ln\left(\frac{k+1}{k}\right) \leq \left(1 + \frac{1}{\pi}\right) \ln 2 + \frac{1}{2\pi}.
\end{aligned}
$$

This establishes part (i). We used $\sin(tx) \leq |tx|$ in the first inequality.

Proof of (ii). Again, by symmetry it suffices to consider $x > 0$. Write $u(T,x)$ as a complex contour integral:

$$u(T,x) = \frac{1}{2} \int_{L_T} \frac{e^{izx}}{i\pi z} dz,$$

where L_T denotes the line segment from $-T$ to T along the real axis.

For each $T > \varepsilon > 0$, consider the closed contour $\Gamma_{T,\varepsilon}$ consisting of:

1. the line segment from $-T$ to $-\varepsilon$ on the real axis;
2. the semicircle of radius ε centered at 0 in the upper half-plane, traversed clockwise from $-\varepsilon$ to ε;
3. the line segment from ε to T on the real axis;
4. the semicircle of radius T centered at 0 in the upper half-plane, traversed counterclockwise from T to $-T$.

The integrand $\frac{e^{izx}}{i\pi z}$ is analytic inside and on $\Gamma_{T,\varepsilon}$ except at $z = 0$, which is excluded by the small semicircle of radius ε. Hence, by Cauchy's theorem, the integral of $\frac{e^{izx}}{i\pi z}$ around $\Gamma_{T,\varepsilon}$ is zero:

$$\int_{\Gamma_{T,\varepsilon}} \frac{e^{izx}}{i\pi z} dz = 0.$$

Decomposing this contour integral into its four pieces and regrouping, we obtain

$$u(T,x) - \frac{1}{2} \int_{L_\varepsilon} \frac{e^{izx}}{i\pi z} dz + \int_\pi^0 \frac{1}{2\pi} e^{i\varepsilon e^{i\theta} x} d\theta + \int_0^\pi \frac{1}{2\pi} e^{iTe^{i\theta} x} d\theta = 0,$$

where L_ε is the line segment from $-\varepsilon$ to ε along the real axis.

We now estimate each term as $\varepsilon \to 0$ and $T \to \infty$.

The small semicircle and the truncated line segment. First,

$$
\begin{aligned}
\left| \frac{1}{2} \int_{L_\varepsilon} \frac{e^{izx}}{i\pi z} dz \right| &= \left| \frac{1}{2} \int_{L_\varepsilon} \frac{\sin(zx)}{\pi z} dz \right| \\
&\leq \frac{1}{2} \int_{L_\varepsilon} \frac{|x|}{\pi} |dz| = \frac{|x|}{2\pi} \cdot 2\varepsilon = \frac{|x|\varepsilon}{\pi},
\end{aligned}
$$

which tends to 0 as $\varepsilon \to 0$.

Next, for the small semicircle of radius ε,

$$\int_\pi^0 \frac{1}{2\pi} e^{i\varepsilon e^{i\theta} x} d\theta = -\frac{1}{2} + \int_\pi^0 \frac{1}{2\pi} \left(e^{i\varepsilon e^{i\theta} x} - 1\right) d\theta.$$

We estimate the error term:

$$
\begin{aligned}
&\left| \int_\pi^0 \frac{1}{2\pi} \left(e^{i\varepsilon e^{i\theta} x} - 1\right) d\theta \right| \\
&= \left| \int_\pi^0 \frac{1}{2\pi} \left(e^{-\varepsilon x \sin\theta} e^{i\varepsilon x \cos\theta} - 1\right) d\theta \right| \\
&= \left| \int_\pi^0 \frac{1}{2\pi} \left(e^{-\varepsilon x \sin\theta} \left(e^{i\varepsilon x \cos\theta} - 1\right) + e^{-\varepsilon x \sin\theta} - 1\right) d\theta \right| \\
&\leq \int_0^\pi \frac{1}{2\pi} e^{-\varepsilon x \sin\theta} \left|e^{i\varepsilon x \cos\theta} - 1\right| d\theta + \int_0^\pi \frac{1}{2\pi} \left|1 - e^{-\varepsilon x \sin\theta}\right| d\theta.
\end{aligned}
$$

Using $|e^{iy} - 1| \leq |y|$ and $|1 - e^{-a}| \leq a$ for $a \geq 0$, we get

$$
\begin{aligned}
&\leq \int_0^\pi \frac{1}{2\pi} e^{-\varepsilon x \sin\theta} \varepsilon x |\cos\theta| \, d\theta + \int_0^\pi \frac{1}{2\pi} \varepsilon x \sin\theta \, d\theta \\
&\leq \frac{\varepsilon x}{2\pi} \int_0^\pi |\cos\theta| \, d\theta + \frac{\varepsilon x}{2\pi} \int_0^\pi \sin\theta \, d\theta = \frac{\varepsilon x}{2\pi} \cdot 2 + \frac{\varepsilon x}{2\pi} \cdot 2 = \frac{2\varepsilon x}{\pi}.
\end{aligned}
$$

Thus this error also tends to 0 as $\varepsilon \to 0$.

Combining the last two estimates and letting $\varepsilon \to 0$ in the contour identity gives

$$u(T,x) - \frac{1}{2} + \int_0^\pi \frac{1}{2\pi} e^{iTe^{i\theta} x} d\theta = 0 \quad \text{for all } T > 0.$$

The large semicircle. We now show that the remaining integral tends to 0 as $T \to \infty$. Write

$$\int_0^\pi \frac{1}{2\pi} e^{iTe^{i\theta} x} d\theta = \frac{1}{2\pi} \int_0^\pi e^{-Tx\sin\theta} e^{iTx\cos\theta} d\theta.$$

Split the integral at $\theta_0 := \sin^{-1}(1/\sqrt{T})$ (for $T > 1$):

$$
\begin{aligned}
\int_0^\pi \frac{1}{2\pi} e^{iTe^{i\theta} x} d\theta &= \frac{1}{2\pi} \int_0^{\theta_0} e^{-Tx\sin\theta} e^{iTx\cos\theta} d\theta \\
&\quad + \frac{1}{2\pi} \int_{\theta_0}^\pi e^{-Tx\sin\theta} e^{iTx\cos\theta} d\theta.
\end{aligned}
$$

Bounding the integrands in modulus, using $|e^{iTx\cos\theta}| \leq 1$ and $e^{-Tx\sin\theta} \leq 1$:

$$
\begin{aligned}
&\left| \frac{1}{2\pi} \int_0^{\theta_0} e^{-Tx\sin\theta} e^{iTx\cos\theta} d\theta \right| \\
&\leq \frac{1}{2\pi} \int_0^{\theta_0} 1 \, d\theta = \frac{1}{2\pi} \theta_0 = \frac{1}{2\pi} \sin^{-1}\left(\frac{1}{\sqrt{T}}\right), \\
&\left| \frac{1}{2\pi} \int_{\theta_0}^\pi e^{-Tx\sin\theta} e^{iTx\cos\theta} d\theta \right| \\
&\leq \frac{1}{2\pi} \int_{\theta_0}^\pi e^{-Tx\sin\theta} d\theta \leq \frac{1}{2\pi} \int_{\theta_0}^\pi e^{-\sqrt{T}x} d\theta \leq \frac{1}{2} e^{-\sqrt{T}x},
\end{aligned}
$$

because on $[\theta_0, \pi]$ one has $\sin\theta \geq \sin\theta_0 = 1/\sqrt{T}$. Hence

$$\left| \int_0^\pi \frac{1}{2\pi} e^{iTe^{i\theta} x} d\theta \right| \leq \frac{1}{2\pi} \sin^{-1}\left(\frac{1}{\sqrt{T}}\right) + \frac{1}{2} e^{-\sqrt{T}x},$$

which clearly tends to 0 as $T \to \infty$.

Conclusion. Letting first $\varepsilon \to 0$ and then $T \to \infty$ in the contour identity yields

$$\lim_{T\to\infty} u(T,x) = \frac{1}{2} \quad \text{for } x > 0.$$

By oddness in x and continuity at $x = 0$, this extends to

$$\lim_{T\to\infty} u(T,x) = \begin{cases} \frac{1}{2}, & x > 0, \\ -\frac{1}{2}, & x < 0, \\ 0, & x = 0, \end{cases}$$

which completes the proof. □

2.2 Weak Convergence

Definition 2.2.1. *A sequence $(P_n)_{n\in\mathbb{N}}$ of probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is said to converge weakly to a probability measure P, written $P_n \Rightarrow P$, if*

$$\lim_{n\to\infty} P_n([a,b]) = P([a,b])$$

for every closed interval $[a,b]$ such that a and b are continuity points of the distribution function $F_P(x) := P((-\infty, x])$.

Remark 34. Relation to convergence in distribution:

Let $X_n : (\Omega, \mathcal{F}, \mathbb{P}) \to \mathbb{R}$ and $X : (\Omega, \mathcal{F}, \mathbb{P}) \to \mathbb{R}$ be random variables. The *distribution* (or *law*) of a random variable X is the

pushforward measure of $\mathbb{P}$ under X, defined by

$$\mu_X(A) := \mathbb{P}(X \in A) = \mathbb{P}\big(X^{-1}(A)\big), \qquad A \in \mathcal{B}(\mathbb{R}).$$

Equivalently, $\mu_X = \mathbb{P} \circ X^{-1}$.

Likewise, for each n, the distribution of X_n is

$$\mu_{X_n}(A) := \mathbb{P}(X_n \in A) = \mathbb{P}\big(X_n^{-1}(A)\big), \qquad A \in \mathcal{B}(\mathbb{R}).$$

We say that X_n *converges in distribution* to X, written

$$X_n \xrightarrow{d} X,$$

if and only if the corresponding probability measures converge weakly:

$$\mu_{X_n} \Rightarrow \mu_X.$$

Thus, *convergence in distribution of random variables is exactly weak convergence of their distributions.*

Exercise 2.2.1. Show that weak convergence admits the following equivalent characterization in terms of distribution functions. Let $F_n(x) := P_n((-\infty, x])$ and $F(x) := P((-\infty, x])$. Then $P_n \Rightarrow P$ if and only if

$$\lim_{n\to\infty} F_n(x) = F(x)$$

at every continuity point x of F. (Hint: Use right-continuity and monotonicity of distribution functions.)

Proof. This is the Portmanteau theorem specialized to half-lines.

($\Rightarrow$) Assume $P_n \Rightarrow P$. By the Portmanteau theorem, for every closed set C and open set G,

$$\limsup_{n\to\infty} P_n(C) \le P(C), \qquad \liminf_{n\to\infty} P_n(G) \ge P(G).$$

Fix $x \in \mathbb{R}$. Since $(-\infty, x]$ is closed,

$$\limsup_{n\to\infty} F_n(x) = \limsup_{n\to\infty} P_n((-\infty, x]) \le P((-\infty, x]) = F(x).$$

Also $(-\infty, x)$ is open, and $P_n((-\infty, x)) = \lim_{y\uparrow x} F_n(y) =: F_n(x^-)$, hence

$$\liminf_{n\to\infty} F_n(x^-) = \liminf_{n\to\infty} P_n((-\infty, x)) \ge P((-\infty, x)) =: F(x^-).$$

By monotonicity of F_n, we have $F_n(x^-) \le F_n(x)$ for all n, so

$$F(x^-) \le \liminf_{n\to\infty} F_n(x) \le \limsup_{n\to\infty} F_n(x) \le F(x).$$

If x is a continuity point of F, then $F(x^-) = F(x)$, and the squeeze implies

$$\lim_{n\to\infty} F_n(x) = F(x).$$

($\Leftarrow$) Assume that $F_n(x) \to F(x)$ for every continuity point x of F. Let $a < b$ be continuity points of F. Then

$$P_n((a, b]) = F_n(b) - F_n(a) \;\longrightarrow\; F(b) - F(a) = P((a, b]).$$

Let $\mathcal{I}$ be the collection of all finite disjoint unions of such half-open intervals:

$$\mathcal{I} := \left\{ \bigcup_{k=1}^{m} (a_k, b_k] : m \in \mathbb{N}, a_k < b_k, a_k, b_k \text{continuity points of } F \right\}.$$

Then $P_n(A) \to P(A)$ for all $A \in \mathcal{I}$.

Now let $G \subset \mathbb{R}$ be open. Write $G = \bigcup_{j\ge1} (\alpha_j, \beta_j)$ as a countable disjoint union of open intervals. For each $m \in \mathbb{N}$ choose numbers $\alpha_j^{(m)}, \beta_j^{(m)}$ such that

$$\alpha_j < \alpha_j^{(m)} < \beta_j^{(m)} < \beta_j;$$
$$\alpha_j^{(m)}, \beta_j^{(m)} \text{ are continuity points of } F,$$

and set

$$G_m := \bigcup_{j=1}^{m} (\alpha_j^{(m)}, \beta_j^{(m)}] \in \mathcal{I}.$$

Then $G_m \uparrow G$, hence $P(G_m) \uparrow P(G)$ and $P_n(G_m) \uparrow P_n(G)$ for each n. Therefore,

$$\liminf_{n\to\infty} P_n(G) \;\ge\; \liminf_{n\to\infty} P_n(G_m) \;=\; P(G_m)$$

(using $P_n(G) \ge P_n(G_m)$ and $P_n(G_m) \to P(G_m)$). Letting $m \to \infty$ yields

$$\liminf_{n\to\infty} P_n(G) \ge P(G).$$

For a closed set C, apply this to $G = C^c$:

$$\limsup_{n\to\infty} P_n(C) = 1 - \liminf_{n\to\infty} P_n(C^c) \le 1 - P(C^c) = P(C).$$

By Portmanteau theorem, these two inequalities imply $P_n \Rightarrow P$. □

Theorem 2.2.1 (Lévy–Cramér Continuity Theorem)**.** *Let $(P_n)_{n\in\mathbb{N}}$ be a sequence of probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with distribution functions F_n and characteristic functions ϕ_n, and let P be a probability measure with distribution function F and characteristic function ϕ. The following are equivalent:*

1. *$P_n \Rightarrow P$ (or equivalently, $F_n \Rightarrow F$);*
2. *For every bounded continuous function $f : \mathbb{R} \to \mathbb{R}$,*
$$\lim_{n\to\infty} \int_{\mathbb{R}} f(x)\, P_n(dx) = \int_{\mathbb{R}} f(x)\, P(dx);$$
or equivalently
$$\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$$
3. *$\phi_n(t) \to \phi(t)$ pointwise for all $t \in \mathbb{R}$.*

Proof. The equivalence $(i) \Leftrightarrow (ii) \Leftrightarrow (iii)$ will be established by proving the cycle $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$.

$(i) \Rightarrow (ii)$**.** Let $f : \mathbb{R} \to \mathbb{R}$ be bounded and continuous, say $|f| \le M$. Fix $\varepsilon > 0$. Choose continuity points $a < b$ of F such that $F(a) \le \varepsilon$ and $F(b) \ge 1 - \varepsilon$. By weak convergence, for sufficiently large n we have $F_n(a) \le 2\varepsilon$ and $F_n(b) \ge 1 - 2\varepsilon$.

Since f is uniformly continuous on the compact interval $[a, b]$, for any $\delta > 0$ there exists a partition

$$a =: a_1 < a_2 < \cdots < a_{N+1} =: b$$

with $N = N_\delta \in \mathbb{N}$ such that all partition points a_j are continuity points of F and

$$|f(x) - f(y)| < \delta \qquad \text{for all } x, y \in [a, b] \text{ with } |x - y| < \delta.$$

Define the step function

$$\hat{f}(x) := \sum_{j=1}^{N} f(a_j) \mathbf{1}_{[a_j, a_{j+1})}(x).$$

Then $|\hat{f}| \le M$ and $\hat{f} = 0$ outside $[a, b]$. By weak convergence at the continuity points a_j,

$$\lim_{n\to\infty} \int \hat{f}\, dP_n = \lim_{n\to\infty} \sum_{j=1}^{N} f(a_j)\big(P_n([a_{j+1}, a_j))\big)$$
$$= \sum_{j=1}^{N} f(a_j)\big(P([a_{j+1}, a_j))\big) = \int \hat{f}\, dP.$$

Now estimate

$$\left| \int f\, dP_n - \int f\, dP \right| \le \left| \int f\, dP_n - \int \hat{f}\, dP_n \right| + \left| \int \hat{f}\, dP_n - \int \hat{f}\, dP \right| + \left| \int \hat{f}\, dP - \int f\, dP \right|.$$

For the first and third terms,

$$\left| \int f\, dP_n - \int \hat{f}\, dP_n \right| \le \int_{[a,b]} |f - \hat{f}|\, dP_n + M\big(P_n((-\infty, a]) + P_n((b, \infty))\big) \le \delta \cdot P_n([a, b]) + 4M\varepsilon \le \delta + 4M\varepsilon,$$

and similarly

$$\left|\int \hat{f}\,dP - \int f\,dP\right| \le \delta + 2M\varepsilon.$$

Taking $n \to \infty$ and using the convergence of the middle term yields

$$\limsup_{n\to\infty}\left|\int f\,dP_n - \int f\,dP\right| \le 2\delta + 6M\varepsilon.$$

Since $\delta, \varepsilon > 0$ are arbitrary, the result follows.

$(ii) \Rightarrow (iii)$**.** The function $f(x) := e^{itx}$ is bounded ($|f| \le 1$) and continuous for any fixed $t \in \mathbb{R}$. Thus

$$\lim_{n\to\infty}\int e^{itx}\,P_n(dx) = \int e^{itx}\,P(dx),$$

i.e., $\phi_n(t) \to \phi(t)$.

$(iii) \Rightarrow (i)$**.** This direction is the deepest. The argument proceeds in several steps, first establishing a stronger version under a technical assumption, then removing it.

Remark 35. In Lévy–Cramér Continuity Theorem, $\phi_n(t) \to \phi(t)$ and $\phi_n(t)$, $\phi(t)$ are characteristic functions. In the following proof we only use that $\phi(t)$ has $\phi(0) = 1$ and $\phi(t)$ is continuous at $t = 0$. Note that $\phi_n(t) \to \phi(t)$ and $\phi_n(t)$ are characteristic functions does not imply that $\phi(t)$ is a characteristic function. A counter example could be $\frac{\sin^2(\pi x)}{(\pi x)^2}$ whose Fourier pair is the triangular (tent) function. One could construct a sequence of tent function that pointwise converge to $\mathbf{1}(\{t = 0\})$.

Step 1: Subsequence convergence via diagonalization. Let $(r_j)_{j\in\mathbb{N}}$ enumerate the rationals $\mathbb{Q}$. Since $(F_n(r_1))_{n\in\mathbb{N}}$ is bounded in $[0,1]$, it has a convergent subsequence $(F_{n_k^{(1)}}(r_1))_{k\in\mathbb{N}}$. From this, extract a further subsequence $(F_{n_k^{(2)}}(r_1), F_{n_k^{(2)}}(r_2))_{k\in\mathbb{N}}$[1] such that both converge (as $k \to \infty$). Continuing inductively, we obtain a chain of subsequences such that the "diagonal" sequence

$$G_k(x) := F_{n_k^{(k)}}(x)$$

converges at every rational r_j, $j \in \mathbb{N}$. Denote this pointwise limit on $\mathbb{Q}$ by $G_\infty(r_j) := \lim_{k\to\infty} G_k(r_j)$.

Step 2: Extension to a distribution function. Define $G : \mathbb{R} \to [0,1]$ by

$$G(x) := \inf\{G_\infty(r) : r \in \mathbb{Q},\ r > x\}.$$

Then G is nondecreasing (as infima of nondecreasing limits). Right-continuity follows from: for $x_n \downarrow x$,

$$G(x) = \inf_{r>x} G_\infty(r) \le \inf_n G(x_n) \le G_\infty(r)$$

for any $r > x$, and letting $r \downarrow x$ yields equality.

Step 3: Convergence at continuity points. Let x be a continuity point of G. Then

$$G(x) = \inf_{r>x} G_\infty(r) \le \limsup_{k\to\infty} G_k(x),$$

since $G_k(r) \ge G_k(x)$ for $r > x$ implies $G_\infty(r) \ge \limsup G_k(x)$. Conversely, for $y < x$ choose rational $r \in (y, x)$; then

$$\liminf_{k\to\infty} G_k(x) \ge G_\infty(r) \ge G(y).$$

Letting $y \uparrow x$ (using continuity of G) gives $G(x) \le \liminf G_k(x)$. Thus $G_k(x) \to G(x)$ at every continuity point x of G, and $0 \le G(-\infty) \le G(\infty) \le 1$.

Remark 36. Any sequence of distribution functions (F_n) admits a subsequence (F_{n_k}) that converges pointwise to some right-continuous nondecreasing G with $0 \le G(-\infty) \le G(\infty) \le 1$ at all continuity points of G. This is widely known as *Helly selection theorem.*

Step 4: Normalization. We show $G(-\infty) = 0$ and $G(\infty) = 1$, so G is a distribution function. By pointwise convergence $\phi_n(0) = 1 \to \phi(0) = 1$, and continuity of ϕ at 0 is given. By Fubini's theorem (justified below),

$$\begin{aligned}&\frac{1}{2T}\int_{-T}^{T}\phi_n(t)\,dt\\ &= \frac{1}{2T}\int_{-T}^{T}\int_{\mathbb{R}} e^{itx}\,P_n(dx)\,dt = \int_{\mathbb{R}}\frac{\sin(Tx)}{Tx}\,P_n(dx).\end{aligned}$$

The integrand is bounded by 1, so

$$\begin{aligned}&\left|\int_{\mathbb{R}}\frac{\sin(Tx)}{Tx}\,P_n(dx)\right|\\ &\le \int_{[-l,l]} 1\,P_n(dx) + \int_{(-\infty,-l)\cup(l,\infty)}\frac{1}{Tl}\,P_n(dx)\\ &= F_n(l) - F_n(-l) + \frac{1 - F_n(l) + F_n(-l)}{Tl}.\end{aligned}$$

Along the diagonal subsequence,

$$\begin{aligned}&\left|\frac{1}{2T}\int_{-T}^{T}\phi(t)\,dt\right|\\ &\le \limsup_{k\to\infty}\Big(F_{n_k^{(k)}}(l) - F_{n_k^{(k)}}(-l)\Big)\Big(1 - \frac{1}{Tl}\Big) + \frac{1}{Tl}\\ &= \Big(G(l) - G(-l)\Big)\Big(1 - \frac{1}{Tl}\Big) + \frac{1}{Tl},\end{aligned}$$

where the limit and interchange follow by bounded convergence (as $l \in \mathbb{N} \subset \mathbb{Q}$). Choose $T = 1/\sqrt{l}$; letting $l \to \infty$ and using $\phi(0) = 1$ and continuity at 0 ($\frac{1}{2T}\int_{-T}^{T}\phi_n(t)\,dt \to 0$ when $T \to 0$) yields

$$1 = G(\infty) - G(-\infty).$$

Thus G is a distribution function, $F_{n_k^{(k)}} \Rightarrow G$ by definition, and by $(i) \Rightarrow (ii)$ we have $\phi(t) = \int e^{itx}\,G(dx)$. By uniqueness of characteristic functions (to be proved later), G is uniquely determined by ϕ.

Remark 37. Set $T = \frac{1}{\sqrt{\ell}}$, $x \in [-\sqrt[4]{\ell}, \sqrt[4]{\ell}]$, then $Tx \in [-\frac{1}{\sqrt[4]{\ell}}, \frac{1}{\sqrt[4]{\ell}}]$. When $\ell \to \infty$, Tx concentrates around 0.

Step 5: Full sequence convergence. It remains to show the original sequence $F_n \Rightarrow G$. Suppose not; then there exist $\varepsilon > 0$, a continuity point x_0 of G, and a subsequence F_{n_m} such that

$$\lim_{m\to\infty}|F_{n_m}(x_0) - G(x_0)| > \varepsilon.$$

By Remark 36, this subsequence has a further subsequence $F_{n_{m_\ell}} \Rightarrow H$ for some distribution function H. But $\phi_{n_{m_\ell}}(t) \to \phi(t)$ pointwise, so by Step 4 applied to this subsequence, $H = G$, contradicting $|H(x_0) - G(x_0)| > \varepsilon$. Thus $F_n \Rightarrow G$, completing the proof under the assumption that ϕ is continuous at 0.

Remark 38. The key for Step 5 is $\phi_n(t) \to \phi(t)$ therefore for characteristic functions we have pointwise convergence to the same function for all subsequences. Then subsequences of the distribution functions must converge to the same distribution.

To remove the continuity assumption at 0: note that $\phi_n(t) \to \phi(t)$ for all t and $|\phi_n(t)| \le 1$ imply uniform continuity of each ϕ_n (by preliminaries). Thus ϕ is uniformly continuous on compact sets $[-R, R]$. By (iii), for each fixed R the convergence $\phi_n|_{[-R,R]} \to \phi|_{[-R,R]}$ is uniform, so

$$\lim_{R\to\infty}\limsup_{n\to\infty}\sup_{|t|\le R}|\phi_n(t) - \phi(t)| = 0.$$

The characteristic function ϕ of any probability measure is continuous at 0, so the assumption holds. $\square$

[1] Bolzano-Weierstrass theorem states that each infinite bounded sequence in $\mathbb{R}^n$ has a convergent subsequence. Helly's selection theorem (also called the Helly selection principle) states that a uniformly bounded sequence of monotone real functions admits a convergent subsequence. Here we iteratively apply this.

2.3 Tightness and the Portmanteau Theorem

Definition 2.3.1. *A sequence of distribution functions $(F_n)_{n\in\mathbb{N}}$ on $\mathbb{R}$ is called uniformly tight (or simply tight) if for every $\varepsilon > 0$ there exists $M_\varepsilon > 0$ such that*

$$F_n(M_\varepsilon) \geq 1-\varepsilon \quad \text{and} \quad F_n(-M_\varepsilon) \leq \varepsilon \qquad \text{for all } n \in \mathbb{N}.$$

Equivalently, for each $\varepsilon > 0$ there exists a compact interval $[-M_\varepsilon, M_\varepsilon]$ that carries at least $1-\varepsilon$ of the mass of every F_n.

Theorem 2.3.1 (Prokhorov)**.** *If $(F_n)_{n\in\mathbb{N}}$ is a uniformly tight sequence of distribution functions on $\mathbb{R}$, then there exists a subsequence $(F_{n_k})_{k\in\mathbb{N}}$ and a distribution function G such that $F_{n_k} \Rightarrow G$ weakly.*

Proof. By Remark 36, there exists a subsequence (F_{n_k}) and a right-continuous nondecreasing function $G : \mathbb{R} \to [0,1]$ such that $F_{n_k}(x) \to G(x)$ at all continuity points x of G, and

$$0 \leq G(-\infty) \leq G(\infty) \leq 1.$$

Let $\varepsilon > 0$ be arbitrary and choose $M_\varepsilon > 0$ such that, by tightness,

$$F_n(M_\varepsilon) \geq 1-\varepsilon, \quad F_n(-M_\varepsilon) \leq \varepsilon \qquad \text{for all } n.$$

Pick continuity points $b > a$ of G with $a > M_\varepsilon$ and $-b < -M_\varepsilon$ (e.g. $a = M_\varepsilon + 1$, $b = M_\varepsilon + 2$, then adjust slightly to continuity points). Then for these a, b,

$$G(a) = \lim_{k\to\infty} F_{n_k}(a) \geq 1-\varepsilon, \qquad G(b) = \lim_{k\to\infty} F_{n_k}(b) \leq \varepsilon,$$

by monotonicity of F_{n_k} and the tightness bounds at $\pm M_\varepsilon$. Since G is nondecreasing and $\varepsilon > 0$ is arbitrary, this implies

$$\lim_{x\to\infty} G(x) = 1, \qquad \lim_{x\to-\infty} G(x) = 0.$$

Thus G is a distribution function on $\mathbb{R}$, and by construction $F_{n_k} \Rightarrow G$. □

Remark 39. We can actually show necessity of tightness of existing a subsequence converging to a distribution function:

Necessity. If $\{\mu_n\}$ is not tight, there exists a positive ϵ such that for each finite interval $(a,b]$, $\mu_n((a,b]) \leq 1-\epsilon$ for some n. Choose n_k so that $\mu_{n_k}((-k,k]) \leq 1-\epsilon$. Suppose that some subsequence $(\mu_{n_{k(j)}})$ of (μ_{n_k}) were to converge weakly to some probability measure μ. Choose $(a,b]$ so that $\mu\{a\} = \mu\{b\} = 0$ and $\mu((a,b]) > 1-\epsilon$. For large enough j, $(a,b] \subset (-k(j), k(j)]$, and so

$$1-\epsilon \geq \mu_{n_{k(j)}}((-k(j),k(j)]) \geq \mu_{n_{k(j)}}((a,b]) \to \mu((a,b]).$$

Thus $\mu((a,b]) \leq 1-\epsilon$, a contradiction.

Theorem. Let (Ω, d) be a complete separable metric space with Borel σ-algebra $\mathcal{B}(\Omega)$, and let $C \subset \Omega$ be Borel. Then $X(\omega) = \inf_{y\in C} d(\omega, y)$ is Borel measurable.

Proof. Let $\{z_n\} \subset \overline{C}$ be countable dense in $\overline{C}$. Each $\omega \mapsto d(\omega, z_n)$ is continuous, hence Borel measurable. For $t \in \mathbb{R}$,

$$\{\omega : X(\omega) < t\} = \bigcup_n \{\omega : d(\omega, z_n) < t\} \in \mathcal{B}(\Omega),$$

a countable union of open balls. Thus X is Borel measurable. □

Theorem 2.3.2 (Portmanteau)**.** *Let (Ω, d) be a complete metric space with Borel σ-algebra $\mathcal{B}$, and let $(P_n)_{n\in\mathbb{N}}$ be a sequence of probability measures on $(\Omega, \mathcal{B})$ such that $P_n \Rightarrow P$ weakly[2] Then:*

1. *For every closed set $C \subset \Omega$,*
$$\limsup_{n\to\infty} P_n(C) \leq P(C).$$
2. *For every open set $O \subset \Omega$,*
$$\liminf_{n\to\infty} P_n(O) \geq P(O).$$
3. *For every Borel set $A \subset \Omega$ such that $P(\partial A) = 0$ (a* continuity set *of P),*
$$\lim_{n\to\infty} P_n(A) = P(A).$$

Proof. 1. Let $C \subset \Omega$ be closed and define the distance function

$$X(\omega) := d(\omega, C) = \inf_{y\in C} d(\omega, y), \qquad \omega \in \Omega.$$

In complete metric space, this distance function is measurable using the afore mentioned theorem.

Then $X(\omega) = 0$ if $\omega \in C$ and $X(\omega) > 0$ otherwise (by the closed set assumption). For $k \in \mathbb{N}$, define

$$f_k(\omega) := \Big(\frac{1}{1+X(\omega)}\Big)^k.$$

Each f_k is bounded, continuous, and satisfies $1_C \leq f_k \leq 1$ and $f_k(\omega) \downarrow 1_C(\omega)$ pointwise as $k \to \infty$.

Since $P_n \Rightarrow P$, by the extremal characterization via bounded continuous functions,

$$\lim_{n\to\infty} \int f_k \, dP_n = \int f_k \, dP \qquad \text{for each fixed } k.$$

Moreover,

$$P_n(C) = \int 1_C \, dP_n \leq \int f_k \, dP_n,$$

so taking lim sup over n,

$$\limsup_{n\to\infty} P_n(C) \leq \int f_k \, dP.$$

Since $f_k \downarrow 1_C$ and $0 \leq f_k \leq 1$, the monotone convergence theorem yields

$$\int f_k \, dP \downarrow \int 1_C \, dP = P(C) \quad \text{as } k \to \infty.$$

Thus

$$\limsup_{n\to\infty} P_n(C) \leq \lim_{k\to\infty} \int f_k \, dP = P(C).$$

2. For an open set $O \subset \Omega$, consider the closed complement $C := O^c$. Then

$$P_n(O) = 1 - P_n(C), \quad P(O) = 1 - P(C),$$

and by part (1),

$$\limsup_{n\to\infty} P_n(C) \leq P(C) \quad \Longrightarrow \quad \liminf_{n\to\infty} P_n(O) \geq P(O).$$

3. Let $A \subset \Omega$ be a Borel set with $P(\partial A) = 0$, where $\partial A := \overline{A} \backslash A^\circ$ is the boundary of A. Then $\overline{A}$ is closed and A° is open, so by (1) and (2),

$$\limsup_{n\to\infty} P_n(A) \leq \limsup_{n\to\infty} P_n(\overline{A}) \leq P(\overline{A}),$$

and

$$\liminf_{n\to\infty} P_n(A) \geq \liminf_{n\to\infty} P_n(A^\circ) \geq P(A^\circ).$$

Using $A^\circ \subset A \subset \overline{A}$ and $P(\partial A) = 0$,

$$P(A^\circ) \leq P(A) \leq P(\overline{A}), \quad P(\overline{A}) - P(A^\circ) = P(\partial A) = 0,$$

so $P(A^\circ) = P(A) = P(\overline{A})$. Combining the inequalities yields

$$P(A) \leq \liminf_{n\to\infty} P_n(A) \leq \limsup_{n\to\infty} P_n(A) \leq P(A),$$

hence $\lim_{n\to\infty} P_n(A) = P(A)$.

□

Remark 40. *1.* and *2.* are actually equivalent definition of weakly convergence. We present the proof of getting $P_n \Rightarrow P$ from *2.*

Proof that (2) ⇒ (1). Assume that for every open set $G \subset \mathbb{R}$,

$$\liminf_{n\to\infty} P_n(G) \;\geq\; P(G). \tag{2.1}$$

[2] some other version of the Portmanteau theorem.

We show that $P_n \Rightarrow P$, i.e. $\int f\,dP_n \to \int f\,dP$ for every bounded continuous f.

Step 1: reduction to $0 \le f \le 1$. Let f be bounded and continuous and set $m := \inf f$, $M := \sup f$. If $M = m$ there is nothing to prove. Otherwise define

$$g(x) := \frac{f(x) - m}{M - m}, \qquad x \in \mathbb{R},$$

so that g is continuous and $0 \le g \le 1$. Since

$$\int f\,dP_n = m + (M - m) \int g\,dP_n,$$

$$\int f\,dP = m + (M - m) \int g\,dP,$$

it suffices to prove $\int g\,dP_n \to \int g\,dP$. Hence assume from now on that $f : \mathbb{R} \to [0, 1]$ is continuous.

Step 2: layer-cake representation. For any probability measure μ on $\mathbb{R}$,

$$\int f\,d\mu \;=\; \int_0^1 \mu(\{x : f(x) > t\})\,dt. \tag{2.2}$$

Indeed, for each x we have $f(x) = \int_0^1 \mathbf{1}_{\{f(x)>t\}}\,dt$, and then Tonelli's theorem yields

$$\int f\,d\mu = \int \left(\int_0^1 \mathbf{1}_{\{f>t\}}\,dt \right) d\mu = \int_0^1 \mu(\{f > t\})\,dt.$$

Step 3: the $\liminf$ *inequality.* For $t \in (0, 1)$ set $G_t := \{x : f(x) > t\}$. Since f is continuous, each G_t is open. Applying (2.1) to G_t gives

$$\liminf_{n\to\infty} P_n(G_t) \;\ge\; P(G_t) \qquad \text{for all } t \in (0, 1).$$

Using (2.2) and Fatou's lemma,

$$\liminf_{n\to\infty} \int f\,dP_n = \liminf_{n\to\infty} \int_0^1 P_n(G_t)\,dt$$

$$\ge \int_0^1 \liminf_{n\to\infty} P_n(G_t)\,dt \ge \int_0^1 P(G_t)\,dt = \int f\,dP.$$

Step 4: the $\limsup$ *inequality via* $1 - f$. The function $1 - f$ is continuous and takes values in $[0, 1]$, so by the same argument,

$$\liminf_{n\to\infty} \int (1 - f)\,dP_n \;\ge\; \int (1 - f)\,dP.$$

Since $\int (1 - f)\,dP_n = 1 - \int f\,dP_n$ and likewise for P, this becomes

$$1 - \limsup_{n\to\infty} \int f\,dP_n \;\ge\; 1 - \int f\,dP,$$

i.e.

$$\limsup_{n\to\infty} \int f\,dP_n \;\le\; \int f\,dP.$$

Step 5: conclusion. Combining the previous two inequalities yields

$$\int f\,dP \le \liminf_{n\to\infty} \int f\,dP_n \le \limsup_{n\to\infty} \int f\,dP_n \le \int f\,dP,$$

hence $\int f\,dP_n \to \int f\,dP$. Therefore $P_n \Rightarrow P$. $\square$

Chapter 3

Independence, Law of Large Numbers, and Limit Theorems

3.1 Independence

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which all random variables are defined.

Definition 3.1.1 (Independence of events)**.** *Events $A, B \in \mathcal{F}$ are* independent *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Definition 3.1.2 (Independence of families)**.** *Let $\mathcal{A}, \mathcal{B} \subseteq \mathcal{F}$. The families $\mathcal{A}$ and $\mathcal{B}$ are* independent *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \qquad \forall\, A \in \mathcal{A},\ B \in \mathcal{B}.$$

Lemma 3.1.1 (Extension to generated σ-algebra)**.** *Let $\mathcal{A}, \mathcal{B} \subseteq \mathcal{F}$ be independent families, and suppose $\mathcal{B}$ is a π-system. Then $\mathcal{A}$ and $\sigma(\mathcal{B})$ are independent.*

Proof. Define

$$\mathcal{I}(\mathcal{A}) = \big\{F \in \mathcal{F} : \mathbb{P}(F \cap A) = \mathbb{P}(F)\mathbb{P}(A) \quad \forall\, A \in \mathcal{A}\big\}.$$

We show $\mathcal{I}(\mathcal{A})$ is a λ-system containing $\mathcal{B}$, hence $\mathcal{I}(\mathcal{A}) \supseteq \mathcal{D}(\mathcal{B}) = \sigma(\mathcal{B})$ by the π-λ theorem.

(i) $\Omega \in \mathcal{I}(\mathcal{A})$ since $\mathbb{P}(\Omega \cap A) = \mathbb{P}(A) = \mathbb{P}(\Omega)\mathbb{P}(A)$.

(ii) Let $F_2 \supseteq F_1 \in \mathcal{I}(\mathcal{A})$ and $A \in \mathcal{A}$. Then

$$\begin{aligned}\mathbb{P}((F_2 \setminus F_1) \cap A) &= \mathbb{P}(F_2 \cap A) - \mathbb{P}(F_1 \cap A)\\ &= \big[\mathbb{P}(F_2) - \mathbb{P}(F_1)\big]\mathbb{P}(A)\\ &= \mathbb{P}(F_2 \setminus F_1)\mathbb{P}(A).\end{aligned}$$

Thus $F_2 \setminus F_1 \in \mathcal{I}(\mathcal{A})$.

(iii) Let $\{F_n\}_{n\in\mathbb{N}} \subset \mathcal{I}(\mathcal{A})$ be increasing with $F_\infty = \bigcup_n F_n$ and $A \in \mathcal{A}$. By continuity of probability measures from below,

$$\begin{aligned}\mathbb{P}(F_\infty \cap A) &= \lim_{n\to\infty} \mathbb{P}(F_n \cap A)\\ &= \lim_{n\to\infty} \mathbb{P}(F_n)\mathbb{P}(A)\\ &= \mathbb{P}(F_\infty)\mathbb{P}(A).\end{aligned}$$

Thus $F_\infty \in \mathcal{I}(\mathcal{A})$.

Since $\mathcal{I}(\mathcal{A})$ is a λ-system containing the π-system $\mathcal{B}$, we have $\sigma(\mathcal{B}) \subseteq \mathcal{I}(\mathcal{A})$, as required. □

Definition 3.1.3 (Mutual independence)**.** *Finite families $\{\mathcal{A}_i\}_{i=1}^k \subseteq \mathcal{F}$ are* mutually independent *if for all $1 \le l \le k$ and choices $i_1, \dots, i_l \in \{1, \dots, k\}$ with i_j distinct,*

$$\mathbb{P}\Big(\bigcap_{j=1}^{l} A_{i_j}\Big) = \prod_{j=1}^{l} \mathbb{P}(A_{i_j}) \qquad \forall\, A_{i_j} \in \mathcal{A}_{i_j}.$$

3.2 Independence of random variables

Definition 3.2.1 (Independence of random variables)**.** *Random variables $X, Y : \Omega \to \mathbb{R}$ are* independent *if*

$$\mathbb{P}(X \in A,\, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B) \qquad \forall\, A, B \in \mathcal{B}(\mathbb{R}).$$

Remark 41. This is equivalent to $\sigma(X)$ and $\sigma(Y)$ being independent σ-algebras.

For $n \ge 2$, random variables $X_1, \dots, X_n$ are *independent* if

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in A_i) \qquad \forall\, A_i \in \mathcal{B}(\mathbb{R}).$$

Note that pairwise independence does not imply mutual independence. A counter example could be: take X_1, X_2 be independent Bernoulli R.V. and X_3 be the XOR of them.

Remark 42. For general events, mutual independence needs to be true for all subsets. For random variables, due to the specification of using Borel sigma-algebra, mutual independence only need to be stated for the n-letter product (no need going through all subset indices) as one can use cylinder sets, fixing some $A_i \in \mathcal{B}(\mathbb{R})$ with $\mathbf{P}(X_i \in A_i) = 1$, then mutual independence is automatically implied by the n-letter form definition.

The difference of independence between random variables and events is merely a matter of definition from "$\forall\, A_i \in \mathcal{B}(\mathbb{R})$". And $\mathbb{P}(X_1 = x_1, X_2 = x_2) = \mathbb{P}(X_1^{-1}(x_1) \cap X_2^{-1}(x_2))$, the distribution of random variable is its push-forward measure.

Remark 43 (Product construction of i.i.d. uniforms). On $\Omega = (0, 1]$ with Lebesgue measure λ, $X(\omega) = \omega$ is uniform$(0, 1]$. To construct two i.i.d. uniforms, take the product space $(0, 1]^2$ with product Lebesgue measure λ^2, and define $X(\omega_1, \omega_2) = \omega_1$, $Y(\omega_1, \omega_2) = \omega_2$. Independence follows from Fubini-Tonelli.

For Bernoulli(1/2), partition $(0, 1]$ into four equal-length intervals and assign $(X, Y) = (0, 0), (0, 1), (1, 0), (1, 1)$ respectively on each.

Independence and product measures. Let $X_1 : (\Omega, \mathcal{F}, \mathbb{P}) \to (S_1, \mathcal{F}_1)$ and $X_2 : (\Omega, \mathcal{F}, \mathbb{P}) \to (S_2, \mathcal{F}_2)$ be random variables. They are said to be *independent* if for every measurable sets $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$,

$$\mathbb{P}(X_1 \in A,\ X_2 \in B) = \mathbb{P}(X_1 \in A)\,\mathbb{P}(X_2 \in B).$$

If we denote by (push-forward measures)

$$\mu_1 = \mathbb{P} \circ X_1^{-1}, \qquad \mu_2 = \mathbb{P} \circ X_2^{-1}$$

the marginal laws of X_1 and X_2, and by

$$\mu = \mathbb{P} \circ (X_1, X_2)^{-1}$$

the joint law of (X_1, X_2), then independence is equivalent to

$$\mu = \mu_1 \otimes \mu_2.$$

Thus, independence means that the *joint distribution* is exactly the *product measure of the marginals*. This does *not* mean that the original probability space $(\Omega, \mathcal{F}, \mathbb{P})$ itself must be a product

space; rather, it is the pushforward measure on the target space $S_1 \times S_2$ that factors as a product.

Example: two independent $U[0,1]$ random variables. Suppose X_1, X_2 are independent and each uniformly distributed on $[0,1]$. Then each marginal law is Lebesgue measure λ on $[0,1]$, and the joint law is

$$\lambda \otimes \lambda$$

on $[0,1]^2$. In other words, (X_1, X_2) is uniformly distributed over the unit square.

The event

$$\{X_1 = X_2\}$$

corresponds to the diagonal set

$$\{(x_1, x_2) \in [0,1]^2 : x_1 = x_2\}.$$

This is a line in the plane, so it has two-dimensional Lebesgue measure 0. Hence

$$\mathbb{P}(X_1 = X_2) = 0.$$

Conclusion. The key idea is:

$$\text{independence} \iff \text{joint law} = \text{product of marginal laws}.$$

In the uniform example, this product law is area measure on the unit square, and the diagonal has area 0, which explains why

$$\mathbb{P}(X_1 = X_2) = 0.$$

Exercise 3.2.1. Random variables X, Y are independent if and only if their joint pushforward measure satisfies $\mathbb{P}_{X,Y} = \mathbb{P}_X \otimes \mathbb{P}_Y$. (Hint: starts with measurable rectangles.)

Exercise 3.2.2. Let X, Y be independent random variables and $f, g : \mathbb{R} \to \mathbb{R}$ Borel measurable with $\mathbb{E}[|f(X)|], \mathbb{E}[|g(Y)|] < \infty$. Prove

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)].$$

(Hint: standard-machine.)

Solution: First prove the claim for indicator functions. For Borel A, B,

$$\begin{aligned}\mathbb{E}[\mathbf{1}_A(X)\mathbf{1}_B(Y)] &= \mathbb{P}(X \in A, Y \in B)\\ &= \mathbb{P}(X \in A)\mathbb{P}(Y \in B)\\ &= \mathbb{E}[\mathbf{1}_A(X)]\,\mathbb{E}[\mathbf{1}_B(Y)].\end{aligned}$$

Extend to simple functions. Let $f = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$ and $g = \sum_{j=1}^n b_j \mathbf{1}_{B_j}$ with Borel A_i, B_j. Then by linearity,

$$\begin{aligned}\mathbb{E}[f(X)g(Y)] &= \sum_{i,j} a_i b_j\, \mathbb{E}[\mathbf{1}_{A_i}(X)\mathbf{1}_{B_j}(Y)]\\ &= \sum_{i,j} a_i b_j\, \mathbb{E}[\mathbf{1}_{A_i}(X)]\,\mathbb{E}[\mathbf{1}_{B_j}(Y)]\\ &= \mathbb{E}[f(X)]\,\mathbb{E}[g(Y)].\end{aligned}$$

Finally, for bounded Borel f, g, choose sequences of simple functions f_k, g_k such that $f_k \to f$ and $g_k \to g$ pointwise with $|f_k| \le \|f\|_\infty$, $|g_k| \le \|g\|_\infty$. Then $f_k(X)g_k(Y) \to f(X)g(Y)$ pointwise and

$$|f_k(X)g_k(Y)| \le \|f\|_\infty \|g\|_\infty,$$

so by dominated convergence,

$$\begin{aligned}\mathbb{E}[f_k(X)g_k(Y)] &\to \mathbb{E}[f(X)g(Y)],\\ \mathbb{E}[f_k(X)] &\to \mathbb{E}[f(X)],\\ \mathbb{E}[g_k(Y)] &\to \mathbb{E}[g(Y)].\end{aligned}$$

Since $\mathbb{E}[f_k(X)g_k(Y)] = \mathbb{E}[f_k(X)]\mathbb{E}[g_k(Y)]$ for each k, passing to the limit gives

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)].$$

Remark 44. This implies that (for the same t) $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$ for characteristic functions ϕ, but the converse fails. However,

$$\mathbb{E}[e^{i(t_1X+t_2Y)}] = \mathbb{E}[e^{it_1X}]\mathbb{E}[e^{it_2Y}] \qquad \forall\, t_1, t_2 \in \mathbb{R}$$

if and only if X, Y are independent.

3.3 Weak Law of Large Numbers

Remark 45. The weak law is often easier to prove and needs less machinery. (For example, bounded variance with Chebyshev inequality.) The strong law is stronger since convergence in measure is implied by convergence almost surely. But the string law is more tecnical to prove as we usually needs to control infinitely many n at once such as Borel-Cantelli, tail σ-slgebras, and truncation.

Remark 46. One could say that WLLN is essentially Chebyshev's inequality while SLLN is essentially Borel-Cantelli lemma.

Theorem 3.3.1 (Pairwise independence, bounded variance, convergence in prob/measure)**.** *Let $X_1, \dots, X_n$ be pairwise independent random variables with $\mathbb{E}[X_i] = 0$ and $0 \le \mathrm{Var}(X_i) = \sigma_i^2 \le B < \infty$ for all i. Then*

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} 0, \qquad \textit{where } S_n = X_1 + \cdots + X_n.$$

Proof. By pairwise independence (covariances are zero) and zero means,

$$\mathrm{Var}\Big(\frac{S_n}{n}\Big) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) \le \frac{B}{n}.$$

By Chebyshev's inequality, for any $\varepsilon > 0$,

$$\mathbb{P}\Big(\Big|\frac{S_n}{n}\Big| > \varepsilon\Big) \le \frac{\mathrm{Var}(S_n/n)}{\varepsilon^2} \le \frac{B}{n\varepsilon^2} \to 0.$$

Thus $\frac{S_n}{n} \to 0$ in probability. □

Definition 3.3.1 (Uniform integrability)**.** *A collection $\{X_\alpha\}_\alpha$ of random variables is uniformly integrable if*

$$\lim_{M\to\infty} \sup_\alpha \mathbb{E}\big[|X_\alpha|\mathbf{1}_{\{|X_\alpha|>M\}}\big] = 0.$$

Remark 47. The definition of Uniform integrability is saying "All the random variables in the family have tails that are small in a uniform way."

- $\mathbf{1}_{\{|X|>M\}}$ picks out the "tail event" $|X| > M$,
- $|X|\mathbf{1}_{\{|X|>M\}}$ is the contribution of the large values of X,
- $\mathbb{E}(|X|\mathbf{1}_{\{|X|>M\}})$ is the expected mass in that tail.

Identically distributed assumption would allow us to remove the $\sup_\alpha$ in the definition of uniform integrability.

Remark 48. Uniform integrability is stronger than mere boundedness in $L^1 : \sup_k E(|X_k|) < \infty$.

Indeed, if $\{X_k\}_k$ is uniformly integrable, then for any $M > 0$,

$$\begin{aligned}\mathbb{E}|X_k| &= \mathbb{E}\big[|X_k|\mathbf{1}_{\{|X_k|\le M\}}\big] + \mathbb{E}\big[|X_k|\mathbf{1}_{\{|X_k|>M\}}\big]\\ &\le M + \mathbb{E}\big[|X_k|\mathbf{1}_{\{|X_k|>M\}}\big].\end{aligned}$$

Taking supremum over k gives

$$\sup_k \mathbb{E}|X_k| \le M + \sup_k \mathbb{E}\big[|X_k|\mathbf{1}_{\{|X_k|>M\}}\big].$$

For M large enough, the second term is small, hence $\sup_k \mathbb{E}|X_k| < \infty$.

However, the converse is false. Consider

$$X_k = \begin{cases} k, & \text{with probability } \frac{1}{k},\\ 0, & \text{with probability } 1 - \frac{1}{k}.\end{cases}$$

Then

$$\mathbb{E}|X_k| = 1 \quad \text{for all } k,$$

so $\sup_k \mathbb{E}|X_k| < \infty$.

But for any $M > 0$, choosing $k > M$ gives

$$\mathbb{E}\big[|X_k|\mathbf{1}_{\{|X_k|>M\}}\big] = 1.$$

Hence

$$\sup_k \mathbb{E}\big[|X_k|\mathbf{1}_{\{|X_k|>M\}}\big] \geq 1 \quad \text{for every } M,$$

so the family is not uniformly integrable.

Therefore, (trivially)

$$\text{uniform integrability} \implies \sup_k \mathbb{E}|X_k| < \infty,$$

but the converse does not hold.

Theorem 3.3.2 (Pairwise independence, uniform integrable, convergence in prob/measure)**.** *Let $X_1, X_2, \ldots$ be pairwise independent, mean-zero, uniformly integrable random variables. Then*

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} 0, \qquad \text{where } S_n = X_1 + \cdots + X_n.$$

Proof. Fix $\varepsilon, \delta > 0$. By uniform integrability, choose $M = M(\delta)$ such that

$$\mathbb{E}\big[|X_i|\mathbf{1}_{\{|X_i|>M\}}\big] < \delta \qquad \forall\, i.$$

Decompose $X_i = X_i^T + Y_i^T$, where

$$X_i^T = X_i\mathbf{1}_{\{|X_i|\leq M\}}, \qquad Y_i^T = X_i\mathbf{1}_{\{|X_i|>M\}}.$$

Let $a_i = \mathbb{E}[X_i^T]$, so $\mathbb{E}[Y_i^T] = -a_i$ and $|a_i| \leq \mathbb{E}[|Y_i^T|] \leq \delta$. Define

$$G_n^T = \sum_{i=1}^n (X_i^T - a_i), \qquad B_n^T = \sum_{i=1}^n (Y_i^T + a_i),$$

so $S_n = G_n^T + B_n^T$ and

$$\mathbb{E}\Big[\Big|\frac{S_n}{n}\Big|\Big] \leq \mathbb{E}\Big[\Big|\frac{G_n^T}{n}\Big|\Big] + \mathbb{E}\Big[\Big|\frac{B_n^T}{n}\Big|\Big].$$

Estimate B_n^T:

$$\mathbb{E}\Big[\Big|\frac{B_n^T}{n}\Big|\Big] \leq \frac{1}{n}\sum_{i=1}^n \big(\mathbb{E}[|Y_i^T|] + |a_i|\big) \leq 2\delta.$$

Estimate G_n^T: Each $X_i^T - a_i$ is mean-zero, bounded by $M + |a_i| \leq M + \delta$, and pairwise independent. Thus

$$\mathrm{Var}(X_i^T - a_i) \leq \mathbb{E}[(X_i^T - a_i)^2] \leq (M+\delta)^2.$$

Remark 49. (The above inequality is a direct consequence of truncation, which avoids the issue that second order moment could be unbounded.)

By pairwise independence,

$$\mathrm{Var}\Big(\frac{G_n^T}{n}\Big) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i^T - a_i) \leq \frac{(M+\delta)^2}{n}.$$

By Chebyshev,

$$\mathbb{E}\Big[\Big|\frac{G_n^T}{n}\Big|\Big] \leq \sqrt{\mathrm{Var}\Big(\frac{G_n^T}{n}\Big)} \leq \frac{M+\delta}{\sqrt{n}}.$$

Combine:

$$\mathbb{E}\Big[\Big|\frac{S_n}{n}\Big|\Big] \leq \frac{M+\delta}{\sqrt{n}} + 2\delta.$$

By Markov's inequality,

$$\mathbb{P}\Big(\Big|\frac{S_n}{n}\Big| > \varepsilon\Big) \leq \frac{1}{\varepsilon}\mathbb{E}\Big[\Big|\frac{S_n}{n}\Big|\Big] \leq \frac{M+\delta}{\varepsilon\sqrt{n}} + \frac{2\delta}{\varepsilon}.$$

Taking $n \to \infty$,

$$\limsup_{n\to\infty} \mathbb{P}\Big(\Big|\frac{S_n}{n}\Big| > \varepsilon\Big) \leq \frac{2\delta}{\varepsilon}.$$

Since $\varepsilon > 0$ is fixed and $\delta > 0$ is arbitrary, the limsup is 0. Thus $\frac{S_n}{n} \xrightarrow{\mathbb{P}} 0$. □

3.4 Strong Law of Large Numbers

So far, we have considered convergence in measure for sequences of random variables. Until now, we have not needed to explicitly show that one can define an infinite sequence of independent random variables. In fact, such a construction is possible, and to formalize this idea, we begin with a preparatory lemma.

Lemma 3.4.1. *Let $A \subset \mathbb{R}^n$ be a Borel set and P a probability measure. For every $\epsilon > 0$, there exists a closed and bounded set $K_\epsilon \subset A$ such that*

$$P(A \setminus K_\epsilon) < \epsilon.$$

Proof. **Step 0: Reduction to a bounded set.** Choose $R > 0$ so large that

$$P\big(\{x \in \mathbb{R}^n : \|x\| > R\}\big) < \epsilon/3.$$

Let $B_R = \{x : \|x\| \leq R\}$. Then

$$P\big(A \setminus (A \cap B_R)\big) \leq P(B_R^c) < \epsilon/3.$$

Thus it suffices to find a closed and bounded $K_\epsilon \subset A \cap B_R$ with

$$P\big((A \cap B_R) \setminus K_\epsilon\big) < 2\epsilon/3,$$

because then

$$P(A \setminus K_\epsilon) \leq P\big(A \setminus (A \cap B_R)\big) + P\big((A \cap B_R) \setminus K_\epsilon\big) < \epsilon.$$

From now on we may therefore assume that A is bounded; it remains to approximate A from inside by a closed set.

Step 1: The case $n = 1$ for basic intervals. First, consider $A = (a, b]$. For each $m \in \mathbb{N}$, let

$$K_m = \big[a + \tfrac{1}{m}, b\big],$$

which is closed and satisfies $K_m \subset (a, b]$. By the right-continuity of measures,

$$P(K_m) = P\big([a + \tfrac{1}{m}, b]\big) \longrightarrow P\big((a, b]\big) \quad \text{as } m \to \infty.$$

Hence $P\big((a, b] \setminus K_m\big) \to 0$. Given $\epsilon > 0$, choose m large enough that

$$P\big((a, b] \setminus K_m\big) < \epsilon.$$

Thus the lemma holds for sets of the form $(a, b]$. By taking finite disjoint unions of such intervals (and intersecting with a large compact interval as in Step 0 if needed), we obtain the result for all bounded sets that are finite disjoint unions of half-open intervals.

Step 2: The class $\mathcal{F}$ on $\mathbb{R}$. Let $\mathcal{F}$ be the collection of all Borel sets $A \subset \mathbb{R}$ with the property:

For every $\epsilon > 0$ there exists a closed and bounded $K \subset A$ such that $P(A \setminus K) < \epsilon$.

We have just shown that all finite disjoint unions of intervals of the form $(a, b]$ lie in $\mathcal{F}$. These sets form an algebra that generates the Borel σ-algebra on $\mathbb{R}$.

Step 3: $\mathcal{F}$ is a monotone class. We show that $\mathcal{F}$ is closed under increasing unions and decreasing intersections.

(i) Increasing unions. Let $A_1 \subset A_2 \subset \cdots$ be in $\mathcal{F}$, and set $A = \bigcup_{j=1}^\infty A_j$. Fix $\epsilon > 0$. By continuity from below,

$$P(A) = \lim_{j\to\infty} P(A_j).$$

Choose N such that

$$P(A) - P(A_N) < \epsilon/3.$$

Since $A_N \in \mathcal{F}$, there exists a closed and bounded $K_N \subset A_N$ such that

$$P(A_N \setminus K_N) < \epsilon/3.$$

Then $K_N \subset A$ and

$$P(A \setminus K_N) \leq P(A \setminus A_N) + P(A_N \setminus K_N) < \epsilon/3 + \epsilon/3 < \epsilon.$$

Thus $A \in \mathcal{F}$.

(ii) Decreasing intersections. Let $A_1 \supset A_2 \supset \cdots$ be in $\mathcal{F}$, and set $A = \bigcap_{j=1}^{\infty} A_j$. Fix $\epsilon > 0$. For each j, choose closed and bounded $K_j \subset A_j$ such that

$$P(A_j \setminus K_j) < \epsilon/2^{j+1}.$$

Define

$$K = \bigcap_{j=1}^{\infty} K_j.$$

Then K is closed and $K \subset A$. Moreover,

$$A \subset A_j \subset K_j \cup (A_j \setminus K_j) \quad \Rightarrow \quad A \setminus K \subset \bigcup_{j=1}^{\infty} (A_j \setminus K_j).$$

Hence

$$P(A \setminus K) \le \sum_{j=1}^{\infty} P(A_j \setminus K_j) \le \sum_{j=1}^{\infty} \frac{\epsilon}{2^{j+1}} < \epsilon.$$

Thus $A \in \mathcal{F}$.

Therefore, $\mathcal{F}$ is a monotone class. Since it contains the algebra of finite disjoint unions of half-open intervals that generates the Borel σ-algebra, the monotone class theorem implies that $\mathcal{F}$ contains all Borel subsets of $\mathbb{R}$. This completes the proof for $n = 1$.

Step 4: Higher dimensions. For $n > 1$, the argument is similar but we start from rectangles instead of intervals. Consider sets of the form

$$R = (a_1, b_1] \times \cdots \times (a_n, b_n],$$

and for each $m \in \mathbb{N}$ define

$$K_m = [a_1 + \tfrac{1}{m}, b_1] \times \cdots \times [a_n + \tfrac{1}{m}, b_n].$$

Then K_m is a closed rectangle with $K_m \subset R$, and by the product structure and right-continuity in each coordinate, we have $P(K_m) \to P(R)$ as $m \to \infty$. Thus the lemma holds for such rectangles, and hence for finite disjoint unions of such rectangles.

Finite disjoint unions of these rectangles form an algebra that generates the Borel σ-algebra on $\mathbb{R}^n$. Repeating the construction of the class $\mathcal{F}$ and the monotone-class argument in $\mathbb{R}^n$, we conclude that every Borel set in $\mathbb{R}^n$ can be approximated from within by closed and bounded sets in measure.

Combining this with the initial truncation to a large ball (Step 0) yields, for each $\epsilon > 0$, a closed and bounded set $K_\epsilon \subset A$ such that $P(A \setminus K_\epsilon) < \epsilon$. □

A family of probability measures $\{P_n\}$ on $\mathbb{R}^n$ is said to be *consistent* if for all Borel sets $A \subset \mathbb{R}^n$,

$$P_{n+1}(A \times \mathbb{R}) = P_n(A).$$

Let $\Omega = \mathbb{R}^\infty = \{x_1, x_2, \ldots\}$ denote the space of all real sequences. Consider the natural σ-algebra Σ generated by the finite-dimensional cylinder sets of the form

$$A \times \mathbb{R} \times \mathbb{R} \times \cdots, \quad A \in \mathcal{B}(\mathbb{R}^n) \text{ for some } n.$$

Theorem 3.4.2 (Kolmogorov's consistency theorem)**.** *Let $\Omega := \mathbb{R}^{\mathbb{N}}$ and let Σ be the σ-field generated by the finite-dimensional cylinder sets*

$$\Pi_n^{-1}(A) := \{\omega \in \Omega : (\omega_1, \ldots, \omega_n) \in A\}, \quad A \in \mathcal{B}(\mathbb{R}^n),\ n \in \mathbb{N}.$$

Given a consistent family of Borel probability measures $\{P_n\}$ on $\mathbb{R}^n$, i.e.

$$P_n(A) = P_{n+1}(A \times \mathbb{R}), \qquad A \in \mathcal{B}(\mathbb{R}^n),$$

there exists a unique probability measure P on (Ω, Σ) such that

$$P(\Pi_n^{-1}(A)) = P_n(A), \qquad \forall n \in \mathbb{N},\ A \in \mathcal{B}(\mathbb{R}^n).$$

(In words, any consistent Borel probability measure family induces a unique probability measure on (Ω, Σ).)

Proof. Let $\mathcal{C}$ be the algebra of all finite-dimensional cylinder sets:

$$\mathcal{C} := \{\Pi_n^{-1}(A) : n \in \mathbb{N},\ A \in \mathcal{B}(\mathbb{R}^n)\}.$$

For $C \in \mathcal{C}$ of the form $C = \Pi_n^{-1}(A)$, define

$$P(C) := P_n(A).$$

Consistency of $\{P_n\}$ implies that this is well-defined and that P is finitely additive on $\mathcal{C}$.

By Carathéodory's extension theorem, P extends to a (unique) probability measure on Σ provided P is countably additive on $\mathcal{C}$. Since $\mathcal{C}$ is an algebra, it suffices to prove continuity from above at $\emptyset$:

If (A_n) is a decreasing sequence in $\mathcal{C}$ with $A_n \downarrow \emptyset$, then $P(A_n) \downarrow 0$.

Assume, for contradiction, that this fails. Then there exists a decreasing sequence (A_n) in $\mathcal{C}$ with

$$A_n \downarrow \emptyset \quad \text{and} \quad P(A_n) \ge \delta > 0 \text{ for all } n.$$

Step 1: Represent A_n and extract compact cores. Each A_n is a finite-dimensional cylinder, so we can write

$$A_n = \Pi_{k_n}^{-1}(B_{k_n})$$

for some $k_n \in \mathbb{N}$ and $B_{k_n} \in \mathcal{B}(\mathbb{R}^{k_n})$. By enlarging coordinates if necessary, assume (k_n) is strictly increasing.

By inner regularity of P_{k_n} on $\mathbb{R}^{k_n}$, for each n choose a compact set $K_{k_n} \subset B_{k_n}$ such that

$$P_{k_n}(B_{k_n} \setminus K_{k_n}) \ \le\ \frac{\delta}{2^{n+2}}.$$

Set

$$C_n := \Pi_{k_n}^{-1}(K_{k_n}) \subset A_n, \qquad D_n := \bigcap_{m \le n} C_m.$$

Then (D_n) is decreasing, $D_n \subset A_n$, and

$$\begin{aligned} P(A_n \setminus D_n) &= P\Big(A_n \setminus \bigcap_{m \le n} C_m\Big) = P\Big(\bigcup_{m \le n} (A_n \setminus C_m)\Big) \\ &\le \sum_{m=1}^{n} P(A_n \setminus C_m) \le \sum_{m=1}^{n} P(A_m \setminus C_m) \\ &= \sum_{m=1}^{n} P_{k_m}(B_{k_m} \setminus K_{k_m}) \\ &\le \sum_{m=1}^{n} \frac{\delta}{2^{m+2}} \le \frac{\delta}{4}. \end{aligned}$$

Hence

$$P(D_n) \ \ge\ P(A_n) - P(A_n \setminus D_n) \ \ge\ \delta - \frac{\delta}{4} = \frac{3\delta}{4} > 0,$$

so every D_n is nonempty.

Step 2: Projections and compact containment. For $k \in \mathbb{N}$ and $n \in \mathbb{N}$, set

$$E_{n,k} := \Pi_k(D_n) \subset \mathbb{R}^k.$$

Then $(E_{n,k})_{n \ge 1}$ is decreasing in n.

For each k, define

$$n(k) := \min\{n \in \mathbb{N} : k_n \ge k\},$$

which is well-defined and nondecreasing in k since (k_n) is strictly increasing.

Fix k and let $n_0 := n(k)$. For all $n \ge n_0$ we have $D_n \subset C_{n_0}$, so

$$E_{n,k} = \Pi_k(D_n) \subset \Pi_k(C_{n_0}) = \Pi_k\big(\Pi_{k_{n_0}}^{-1}(K_{k_{n_0}})\big).$$

Let

$$\pi_k^{(n_0)} : \mathbb{R}^{k_{n_0}} \to \mathbb{R}^k, \quad \pi_k^{(n_0)}(x_1, \ldots, x_{k_{n_0}}) := (x_1, \ldots, x_k),$$

and define

$$K_k^* := \pi_k^{(n_0)}(K_{k_{n_0}}).$$

Then K_k^* is compact in $\mathbb{R}^k$ and

$$E_{n,k} \subset K_k^* \quad \text{for all } n \ge n_0.$$

Thus, for each fixed k, the tail $(E_{n,k})_{n \ge n(k)}$ is a nested family

of closed sets inside the compact set K_k^*, so

$$F_k := \bigcap_{n \geq n(k)} E_{n,k} \neq \emptyset.$$

Step 3: Projection relation between F_k and F_{k-1}. Fix $k \geq 2$. For all n sufficiently large so that $n \geq n(k)$ and $n \geq n(k-1)$, we have $k_n \geq k$ and $k_n \geq k-1$, hence

$$E_{n,k} = \Pi_k(D_n), \qquad E_{n,k-1} = \Pi_{k-1}(D_n),$$

and thus

$$\Pi_{k-1}(E_{n,k}) = \Pi_{k-1}(\Pi_k(D_n)) = \Pi_{k-1}(D_n) = E_{n,k-1}.$$

Let $m_k := \max\{n(k), n(k-1)\}$. Then

$$F_k = \bigcap_{n \geq n(k)} E_{n,k} = \bigcap_{n \geq m_k} E_{n,k},$$

and similarly

$$F_{k-1} = \bigcap_{n \geq n(k-1)} E_{n,k-1} = \bigcap_{n \geq m_k} E_{n,k-1}.$$

Therefore

$$\begin{aligned}
\Pi_{k-1}(F_k) &= \Pi_{k-1}\Big(\bigcap_{n \geq m_k} E_{n,k}\Big) \\
&= \bigcap_{n \geq m_k} \Pi_{k-1}(E_{n,k}) \\
&= \bigcap_{n \geq m_k} E_{n,k-1} \\
&= F_{k-1}.
\end{aligned}$$

In particular, every point in F_{k-1} has an extension in F_k.

Step 4: Construct a point and derive a contradiction. For each $k \in \mathbb{N}$ we defined

$$F_k := \bigcap_{n \geq n(k)} E_{n,k}, \qquad E_{n,k} := \Pi_k(D_n),$$

and we showed $F_k \neq \emptyset$ and $\Pi_{k-1}(F_k) = F_{k-1}$ for all $k \geq 2$.

We now construct $\mathbf{x} = (x_1, x_2, \dots) \in \Omega$ inductively. Choose $x_1 \in F_1$. Suppose $(x_1, \dots, x_{k-1}) \in F_{k-1}$ has been chosen. Since $\Pi_{k-1}(F_k) = F_{k-1}$, there exists $x_k \in \mathbb{R}$ such that

$$(x_1, \dots, x_{k-1}, x_k) \in F_k.$$

This defines $\mathbf{x} \in \Omega$ with

$$\Pi_k(\mathbf{x}) \in F_k \subset E_{n,k} = \Pi_k(D_n) \quad \text{for all } n \geq n(k)$$

and all k.

Assume, towards a contradiction, that

$$\mathbf{x} \notin \bigcap_{n \geq 1} D_n.$$

Then there exists some n_0 with $\mathbf{x} \notin D_{n_0}$. Let n_0 be the *smallest* such index. Since (D_n) is decreasing, we have

$$\mathbf{x} \in D_n \quad \text{for all } n < n_0, \qquad \mathbf{x} \notin D_{n_0}.$$

By definition, $D_{n_0} = \bigcap_{m \leq n_0} C_m$, and each C_m is a cylinder depending only on finitely many coordinates (those up to k_m). Since (D_n) is decreasing and k_m is increasing, we have

$$\Pi_{k_{n_0}}(\mathbf{x}) \notin \Pi_{k_{n_0}}(D_n) \quad \text{for all } n \geq n_0.$$

On the other hand, by construction of $\mathbf{x}$ we have

$$\Pi_{k_{n_0}}(\mathbf{x}) \in F_{k_{n_0}},$$

where

$$F_{k_{n_0}} := \bigcap_{n \geq n(k_{n_0})} E_{n,k_{n_0}}, \qquad E_{n,k} := \Pi_k(D_n).$$

Choose $n \geq \max\{n_0, n(k_{n_0})\}$. Then $n \geq n(k_{n_0})$, so by the definition of $F_{k_{n_0}}$,

$$\Pi_{k_{n_0}}(\mathbf{x}) \in F_{k_{n_0}} \subset E_{n,k_{n_0}} = \Pi_{k_{n_0}}(D_n),$$

which contradicts the fact that $\Pi_{k_{n_0}}(\mathbf{x}) \notin \Pi_{k_{n_0}}(D_n)$ for all $n \geq n_0$.

Thus our assumption that $\mathbf{x} \notin \bigcap_{n \geq 1} D_n$ was false, so

$$\mathbf{x} \in \bigcap_{n \geq 1} D_n.$$

Since $D_n \subset A_n$ for all n, we also have

$$\mathbf{x} \in \bigcap_{n \geq 1} A_n,$$

contradicting $A_n \downarrow \emptyset$.

Hence P is continuous from above at $\emptyset$ on $\mathcal{C}$, and therefore countably additive on $\mathcal{C}$. By Carathéodory's extension theorem, P extends uniquely to a probability measure on (Ω, Σ) with the given finite-dimensional marginals $\{P_n\}$.

□

We first prove a basic version of the strong law under a fourth-moment constraint.

Theorem 3.4.3 (fourwise independent fourth order finite moment, almost surely convergence)**.** *Let $X_1, \dots, X_n$ be fourwise independent random variables with*

$$\mathbb{E}[X_i] = 0 \quad \text{and} \quad \mathbb{E}[X_i^4] = c < \infty$$

for all i. Then $\frac{S_n}{n} \to 0$ almost surely, where $S_n := \sum_{i=1}^n X_i$.

Proof. We first compute the fourth moment:

$$\mathbb{E}\left[\left(\frac{S_n}{n}\right)^4\right] = \frac{1}{n^4}\,\mathbb{E}\left[\left(\sum_{i=1}^n X_i\right)^4\right] \tag{3.1}$$

$$= \frac{1}{n^4}\left(\sum_{i=1}^n \mathbb{E}[X_i^4] + 6 \sum_{1 \leq i < j \leq n} \mathbb{E}[X_i^2 X_j^2]\right). \tag{3.2}$$

All other cross terms vanish by fourwise independence: any term with at least three distinct indices has expectation zero (using $\mathbb{E}[X_k] = 0$), and terms of the form $\mathbb{E}[X_i^3 X_j]$ also vanish.

By Cauchy–Schwarz,

$$\mathbb{E}[X_i^2] = \mathbb{E}[X_i^2 \cdot 1] \leq \sqrt{\mathbb{E}[X_i^4]\mathbb{E}[1^2]} \leq \sqrt{c},$$

so $\mathbb{E}[X_i^2 X_j^2] \leq c$ for all $i \neq j$. Hence

$$\mathbb{E}\left[\left(\frac{S_n}{n}\right)^4\right] \leq \frac{1}{n^4}\left(nc + 6\binom{n}{2}c\right) = \frac{c}{n^4}\big(n + 3n(n-1)\big) \leq \frac{3c}{n^2}.$$

By Markov's inequality, for any $\epsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \epsilon\right) \leq \frac{1}{\epsilon^4}\mathbb{E}\left[\left(\frac{S_n}{n}\right)^4\right] \leq \frac{3c}{\epsilon^4 n^2}.$$

Let $A_n^\epsilon := \{\omega : \left|\frac{S_n(\omega)}{n}\right| > \epsilon\}$. Then $\sum_{n=1}^\infty \mathbb{P}(A_n^\epsilon) < \infty$, so by the Borel–Cantelli lemma,

$$\mathbb{P}(A_n^\epsilon \text{ i.o.}) = 0.$$

The event $\{A_n^\epsilon \text{ i.o.}\}$ is exactly $\{\limsup_{n\to\infty} |\frac{S_n}{n}| > \epsilon\}$. Thus

$$\mathbb{P}\left(\limsup_{n\to\infty}\left|\frac{S_n}{n}\right| > \epsilon\right) = 0 \quad \text{for every } \epsilon > 0.$$

Taking the countable union over $\epsilon = 1/m$, $m \in \mathbb{N}$, we obtain

$$\mathbb{P}\left(\limsup_{n\to\infty}\left|\frac{S_n}{n}\right| > 0\right) = 0,$$

which means $\frac{S_n}{n} \to 0$ almost surely. □

Remark 50. The idea of finite-fourth-moment assumption is to invoke Borel-Cantelli lemma.

Theorem 3.4.4 (pairwise independent, identical, nonnegative, bounded mean, almost surely convergence)**.** *Let $X_1, X_2, \dots$ be pairwise independent and identically distributed nonnegative random variables with common distribution X and $\mathbb{E}[X] = \mu < \infty$.*

Let $S_n := \sum_{i=1}^n X_i$. Then

$$\frac{S_n}{n} \to \mu \quad \textit{almost surely.}$$

Proof. This is Etamadi's proof.

Step 1 (Truncation). Define

$$Y_k := X_k \mathbf{1}_{\{X_k \le k\}}, \quad T_n := \sum_{k=1}^n Y_k, \quad Z_k := X_k - Y_k = X_k \mathbf{1}_{\{X_k > k\}}.$$

Then

$$\begin{aligned}
&\sum_{k=1}^\infty \mathbb{P}(Z_k > 0) \\
&= \sum_{k=1}^\infty \mathbb{P}(X > k) \le \int_0^\infty \mathbb{P}(X > t)\, dt \\
&= \int_0^\infty \int \mathbf{1}_{\{X>t\}}\, d\mathbb{P}\, dt = \int \int_0^X 1\, dt\, d\mathbb{P} = \int X\, d\mathbb{P} = \mu < \infty.
\end{aligned}$$

By the Borel–Cantelli lemma,

$$\mathbb{P}(Z_k > 0 \text{ i.o.}) = 0.$$

Let

$$A := \{\omega : Z_k(\omega) = 0 \text{ eventually}\}, \quad A_m := \{\omega : Z_k(\omega) = 0\ \forall k \ge m\}.$$

Then $A_m \uparrow A$ and $\mathbb{P}(A) = 1$.

For $\omega \in A_m$ we have $Z_k(\omega) = 0$ for all $k \ge m$, so for $n \ge m$,

$$\sum_{k=1}^n Z_k(\omega) = \sum_{k=1}^{m-1} Z_k(\omega),$$

and hence

$$\begin{aligned}
0 &\le \liminf_{n\to\infty} \frac{1}{n} \sum_{k=1}^n Z_k(\omega) \\
&\le \limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^n Z_k(\omega) \\
&\le \limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^{m-1} Z_k(\omega) = 0.
\end{aligned}$$

Thus, for every $\omega \in A$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^n Z_k(\omega) = \lim_{n\to\infty} \frac{S_n(\omega) - T_n(\omega)}{n} = 0.$$

Next, note that

$$\mathbb{E}[Z_n] = \mathbb{E}\big[X\, \mathbf{1}_{\{X>n\}}\big] \to 0 \quad \text{as } n \to \infty.$$

Hence, by Cesàro summability,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^n \mathbb{E}[Z_k] = \lim_{n\to\infty} \frac{1}{n}\, \mathbb{E}\left[\sum_{k=1}^n Z_k\right] = \lim_{n\to\infty} \frac{1}{n}\, \mathbb{E}[S_n - T_n] = 0,$$

which implies

$$\lim_{n\to\infty} \frac{\mathbb{E}[T_n]}{n} = \mu.$$

Since $T_n \le S_n$ and $\frac{1}{n}(S_n - T_n) \to 0$ a.s., it will be enough to show that

$$\frac{T_n - \mathbb{E}[T_n]}{n} \to 0 \quad \text{almost surely.} \tag{3.3}$$

Step 2 (Subsequence convergence). Fix $\alpha > 1$ and let $k_n := \lfloor \alpha^n \rfloor$. By Chebyshev's inequality,

$$\mathbb{P}\big(|T_{k_n} - \mathbb{E}[T_{k_n}]| \ge \varepsilon k_n\big) \le \frac{\mathrm{Var}(T_{k_n})}{\varepsilon^2 k_n^2}.$$

Since the Y_m are pairwise independent and nonnegative,

$$\mathrm{Var}(T_{k_n}) = \sum_{m=1}^{k_n} \mathrm{Var}(Y_m),$$

and thus

$$\begin{aligned}
\sum_{n=1}^\infty \mathbb{P}\big(|T_{k_n} - \mathbb{E}[T_{k_n}]| \ge \varepsilon k_n\big) &\le \sum_{n=1}^\infty \frac{1}{\varepsilon^2 k_n^2} \sum_{m=1}^{k_n} \mathrm{Var}(Y_m) \\
&= \sum_{m=1}^\infty \mathrm{Var}(Y_m) \sum_{n: k_n \ge m} \frac{1}{\varepsilon^2 k_n^2}.
\end{aligned}$$

For each fixed m, let n_0 be the smallest n with $k_n \ge m$. Then for all $n \ge n_0$,

$$k_n = \lfloor \alpha^n \rfloor \ge \lfloor \alpha^{n_0} \rfloor \lfloor \alpha^{n-n_0} \rfloor \ge m\, \frac{\alpha^{n-n_0}}{2},$$

so

$$\sum_{n: k_n \ge m} \frac{1}{k_n^2} \le \sum_{n=n_0}^\infty \frac{4}{m^2 \alpha^{2(n-n_0)}} \le \frac{4}{m^2} \sum_{j=0}^\infty \frac{1}{\alpha^{2j}} = \frac{4\alpha^2}{m^2(\alpha^2 - 1)}.$$

Hence

$$\sum_{n=1}^\infty \mathbb{P}\big(|T_{k_n} - \mathbb{E}[T_{k_n}]| \ge \varepsilon k_n\big) \le \frac{4\alpha^2}{\varepsilon^2(\alpha^2 - 1)} \sum_{m=1}^\infty \frac{\mathrm{Var}(Y_m)}{m^2}.$$

A simple bound (see Lemma 3.4.5) gives

$$\sum_{m=1}^\infty \frac{\mathrm{Var}(Y_m)}{m^2} \le 4\, \mathbb{E}[X],$$

so

$$\sum_{n=1}^\infty \mathbb{P}\big(|T_{k_n} - \mathbb{E}[T_{k_n}]| \ge \varepsilon k_n\big) \le 16\, \frac{\alpha^2}{\varepsilon^2(\alpha^2 - 1)}\, \mathbb{E}[X] < \infty.$$

By the Borel–Cantelli lemma,

$$\frac{T_{k_n} - \mathbb{E}[T_{k_n}]}{k_n} \to 0 \quad \text{almost surely.}$$

Using $\frac{\mathbb{E}[T_{k_n}]}{k_n} \to \mu$, we obtain

$$\frac{T_{k_n}}{k_n} \to \mu \quad \text{almost surely.}$$

Step 3 (Sandwich). Observe that

$$\frac{\alpha^{n+1} - 1}{\alpha^n} \le \frac{k_{n+1}}{k_n} \le \frac{\alpha^{n+1}}{\alpha^n - 1},$$

so $\lim_{n\to\infty} \frac{k_{n+1}}{k_n} = \alpha$.

For each m choose n_m such that $k_{n_m} \le m < k_{n_m+1}$. Then, by monotonicity of (T_n) in n,

$$\frac{k_{n_m}}{k_{n_m+1}} \cdot \frac{T_{k_{n_m}}}{k_{n_m}} \le \frac{T_m}{m} \le \frac{k_{n_m+1}}{k_{n_m}} \cdot \frac{T_{k_{n_m+1}}}{k_{n_m+1}}.$$

On the event where $\frac{T_{k_n}}{k_n} \to \mu$, we deduce

$$\frac{\mu}{\alpha} \le \liminf_{m\to\infty} \frac{T_m}{m} \le \limsup_{m\to\infty} \frac{T_m}{m} \le \alpha\mu.$$

Since $\alpha > 1$ was arbitrary, letting $\alpha \downarrow 1$ yields

$$\frac{T_m}{m} \to \mu \quad \text{almost surely.}$$

Combining this with $\frac{1}{n}(S_n - T_n) \to 0$ a.s., we conclude that

$$\frac{S_n}{n} \to \mu \quad \text{almost surely.}$$

□

Lemma 3.4.5. *Let $X_1, X_2, \ldots$ be pairwise independent and identically distributed nonnegative random variables with common distribution X and $\mathbb{E}[X] = \mu < \infty$. Define $Y_k := X_k \mathbf{1}_{\{X_k \le k\}}$. Then*

$$\sum_{k=1}^\infty \frac{\mathrm{var}(Y_k)}{k^2} \le 4\, \mathbb{E}[X].$$

Proof. Since $Y_k \ge 0$, we have

$$\mathrm{var}(Y_k) \le \mathbb{E}[Y_k^2],$$

so

$$\begin{aligned}\sum_{k=1}^{\infty}\frac{\operatorname{var}(Y_k)}{k^2} &\le \sum_{k=1}^{\infty}\frac{\mathbb{E}[Y_k^2]}{k^2}\\ &= \sum_{k=1}^{\infty}\frac{1}{k^2}\int\left(\int_0^{Y_k} 2t\,dt\right)d\mathbb{P}\\ &= \sum_{k=1}^{\infty}\frac{1}{k^2}\int\int_0^{\infty} 2t\,\mathbf{1}_{\{Y_k>t\}}\,dt\,d\mathbb{P}\\ &\overset{(a)}{=} \sum_{k=1}^{\infty}\frac{1}{k^2}\int_0^{\infty} 2t\,\mathbb{P}(Y_k>t)\,dt.\end{aligned}$$

Since $Y_k = X_k\mathbf{1}_{\{X_k\le k\}}$, we have

$$\mathbb{P}(Y_k>t) = \mathbb{P}(X_k>t,\ X_k\le k) \le \mathbb{P}(X>t) \quad \text{for } t\le k,$$

and $\mathbb{P}(Y_k>t)=0$ for $t>k$. Thus

$$\mathbb{P}(Y_k>t) \le \mathbf{1}_{\{t\le k\}}\mathbb{P}(X>t),$$

and hence

$$\begin{aligned}\sum_{k=1}^{\infty}\frac{\operatorname{var}(Y_k)}{k^2} &\le \sum_{k=1}^{\infty}\frac{1}{k^2}\int_0^{k} 2t\,\mathbb{P}(X>t)\,dt\\ &\overset{(b)}{=} \int_0^{\infty} 2t\,\mathbb{P}(X>t)\left(\sum_{k:\,k\ge\max\{t,1\}}\frac{1}{k^2}\right)dt.\end{aligned}$$

We now bound the inner sum by splitting into three ranges of t:

$$\sum_{k:\,k\ge\max\{t,1\}}\frac{1}{k^2} \le \begin{cases}\displaystyle\sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}, & 0<t\le 1,\\ \displaystyle\sum_{k=1}^{\infty}\frac{1}{k^2} - 1 \le \frac{\pi^2}{6}-1, & 1<t\le 2,\\ \displaystyle\sum_{k=\lceil t\rceil}^{\infty}\frac{1}{k^2} \le \int_{t-1}^{\infty}\frac{dx}{x^2} = \frac{1}{t-1}, & t>2.\end{cases}$$

Therefore

$$\begin{aligned}&\sum_{k=1}^{\infty}\frac{\operatorname{var}(Y_k)}{k^2}\\ &\le \int_0^{\infty}\mathbb{P}(X>t)\Big[2t\,\frac{\pi^2}{6}\,\mathbf{1}_{\{t\le 1\}}\\ &\quad + 2t\Big(\frac{\pi^2}{6}-1\Big)\mathbf{1}_{\{1<t\le 2\}} + \frac{2t}{t-1}\mathbf{1}_{\{t>2\}}\Big]dt\\ &\le \int_0^{\infty} 4\,\mathbb{P}(X>t)\,dt = 4\,\mathbb{E}[X].\end{aligned}$$

Here, (a) and (b) follow from Fubini's theorem, the second inequality uses $\mathbb{P}(Y_k>t)\le\mathbb{P}(X>t)$, and the last inequality is a straightforward bound on the piecewise coefficient in parentheses by 4. □

Theorem 3.4.6 (Kolmogorov's zero–one law)**.** *Suppose $X_1, X_2, \dots$ are independent random variables. Let $\mathcal{F}_n := \sigma(X_n, X_{n+1}, \dots)$ and define*

$$\mathcal{F}_\infty := \bigcap_{n\ge 1}\mathcal{F}_n.$$

Then for every $A\in\mathcal{F}_\infty$,

$$\mathbb{P}(A)\in\{0,1\}.$$

In my words, if something is known(described by the sigma algebra $\mathcal{F}_\infty$) in the infinity tail, then it is trivial.

Remark 51. Note that the sigma-algebra of a random variable X is defined to be $\sigma(X) = \{X^{-1}(B) : B\in\mathcal{B}(\mathbb{R})\}$. It is the collection of all events that can be determined by just knowing the value of X.

Note that early we define random variable from a reverse direction of the above notion of sigma-algebra of a random variable. When we define random variable, we want it to use some real values to represent outcome of events in the sample space Ω. Here we want to find what are these represented events in the sample space Ω from the random variable X they linked to.

Proof. We show that any $A\in\mathcal{F}_\infty$ is independent of itself. Assume without loss of generality that $\mathbb{P}(A)>0$.

For each $n\ge 1$, we have $A\in\mathcal{F}_{n+1}$ by definition of $\mathcal{F}_\infty$. Since $\mathcal{F}_{n+1}$ is generated by $\{X_{n+1}, X_{n+2}, \dots\}$ and these are independent of $\{X_1,\dots,X_n\}$, it follows that A is independent of

$$\mathcal{F}'_n := \sigma(X_1,\dots,X_n).$$

Consequently, A is independent of every set in

$$\mathcal{F}' := \bigcup_{n\ge 1}\mathcal{F}'_n.$$

Note that for notations here, without $'$ means tails and with $'$ means head. The keypoint of this proof is first understand the definition $\mathcal{F}_n := \sigma(X_n, X_{n+1}, \dots)$.

Let $\mathcal{A}$ be the collection of sets that are independent of A. Suppose $\mathbb{P}(A)\ne 0$. Define two probability measures on $\sigma(\mathcal{F}')$ by

$$\mathbb{P}_1(B) := \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)}, \qquad \mathbb{P}_2(B) := \mathbb{P}(B).$$

For any $B\in\mathcal{F}'$, we have $B\in\mathcal{A}$, so $\mathbb{P}(A\cap B) = \mathbb{P}(A)\mathbb{P}(B)$, and hence $\mathbb{P}_1(B) = \mathbb{P}_2(B)$. Thus $\mathbb{P}_1$ and $\mathbb{P}_2$ agree on the field $\mathcal{F}'$, and by uniqueness of measure extension, they agree on $\sigma(\mathcal{F}')$.

Each $\mathcal{F}_n$ is generated by finite-dimensional cylinder sets, and every such cylinder set belongs to some $\mathcal{F}'_{n'}$. Therefore

$$\mathcal{F}_n\subset\sigma(\mathcal{F}') \quad \text{for all } n,$$

and hence

$$\mathcal{F}_\infty = \bigcap_{n\ge 1}\mathcal{F}_n \subset \sigma(\mathcal{F}').$$

Since $A\in\mathcal{F}_\infty\subset\sigma(\mathcal{F}')$, we have in particular

$$\mathbb{P}_1(A) = \mathbb{P}_2(A),$$

i.e.

$$\frac{\mathbb{P}(A\cap A)}{\mathbb{P}(A)} = \mathbb{P}(A),$$

or

$$\mathbb{P}(A) = \mathbb{P}(A)^2.$$

Thus $\mathbb{P}(A)\in\{0,1\}$. □

Remark 52. *Rough idea of Kolmogorov's zero–one law.*

Let

$$\mathcal{T} = \bigcap_{n=1}^{\infty}\sigma(X_n, X_{n+1}, X_{n+2}, \dots)$$

be the tail σ-algebra of an independent sequence $(X_n)_{n\ge 1}$, and let

$$A\in\mathcal{T}.$$

Since $A\in\mathcal{T}$, for every n we have

$$A\in\sigma(X_{n+1}, X_{n+2}, \dots).$$

On the other hand,

$$\sigma(X_1,\dots,X_n)$$

is independent of

$$\sigma(X_{n+1}, X_{n+2}, \dots)$$

because the random variables are independent. Hence, for every n, the event A is independent of

$$\sigma(X_1,\dots,X_n).$$

Now the information from all finite initial blocks generates

$$\sigma(X_1, X_2, \dots),$$

so one concludes that A is independent of the whole σ-algebra generated by the sequence. In particular, since

$$A \in \sigma(X_1, X_2, \dots),$$

the event A is independent of itself.

Therefore,

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)\mathbb{P}(A) = \mathbb{P}(A)^2.$$

So

$$\mathbb{P}(A) \in \{0, 1\}.$$

Intuition.

A tail event depends only on what happens arbitrarily far out in the sequence, so changing finitely many coordinates cannot affect it. Because of independence, such an event is independent of every finite initial segment, and hence ends up being independent of itself. That forces its probability to be either 0 or 1.

3.5 Central limit theorem

3.5.1 Some useful lemmas regarding real and complex numbers

Lemma 3.5.1. *For $z \in \mathbb{C}$ with $0 \le |z| \le 1$,*

$$\left|e^{-z} - 1 + z\right| \;\le\; |z|^2.$$

Proof. From the Taylor expansion

$$e^{-z} = 1 - z + \sum_{n=2}^{\infty} \frac{(-z)^n}{n!},$$

it suffices to show

$$\sum_{n=2}^{\infty} \frac{|z|^n}{n!} \;\le\; |z|^2.$$

For $|z| \le 1$ and $n \ge 2$ we have

$$\frac{|z|^n}{n!} \;\le\; \frac{|z|^n}{2^{n-1}} \;\le\; \frac{|z|^2}{2^{n-1}},$$

so

$$\sum_{n=2}^{\infty} \frac{|z|^n}{n!} \;\le\; |z|^2 \sum_{n=2}^{\infty} \frac{1}{2^{n-1}} = |z|^2.$$

□

Lemma 3.5.2. *For $x \in \mathbb{R}$ and $n \ge 0$,*

$$\left|e^{ix} - \sum_{m=0}^{n} \frac{(ix)^m}{m!}\right| \;\le\; \min\left\{\frac{|x|^{n+1}}{(n+1)!}, \frac{2|x|^n}{n!}\right\}.$$

Proof. Integrating by parts (or using Taylor's theorem with integral remainder) one obtains

$$e^{ix} - \sum_{m=0}^{n} \frac{(ix)^m}{m!} = \frac{i^{n+1}}{n!} \int_0^x (x-s)^n e^{is}\, ds.$$

Hence

$$\left|e^{ix} - \sum_{m=0}^{n} \frac{(ix)^m}{m!}\right| \le \frac{1}{n!} \int_0^{|x|} s^n\, ds = \frac{|x|^{n+1}}{(n+1)!},$$

giving the first bound.

For the second bound, apply the same formula with n replaced by $n-1$:

$$\left|e^{ix} - \sum_{m=0}^{n-1} \frac{(ix)^m}{m!}\right| \le \frac{|x|^n}{n!}.$$

Subtracting the nth term and using the triangle inequality,

$$\left|e^{ix} - \sum_{m=0}^{n} \frac{(ix)^m}{m!}\right| \le \left|e^{ix} - \sum_{m=0}^{n-1} \frac{(ix)^m}{m!}\right| + \left|\frac{(ix)^n}{n!}\right|$$
$$\le \frac{|x|^n}{n!} + \frac{|x|^n}{n!} = \frac{2|x|^n}{n!}.$$

□

Remark 53. Later in the proof we apply lemma 3.5.2 by setting $n = 2$.

Lemma 3.5.3. *Let $z_1, \dots, z_n$ and $\omega_1, \dots, \omega_n$ be complex numbers with $|z_m| \le \theta$ and $|\omega_m| \le \theta$ for all m. Then*

$$\left|\prod_{m=1}^{n} z_m - \prod_{m=1}^{n} \omega_m\right| \;\le\; \theta^{n-1} \sum_{m=1}^{n} |z_m - \omega_m|.$$

Proof. We first write

$$\prod_{m=1}^{n} z_m - \prod_{m=1}^{n} \omega_m$$
$$= z_1 \prod_{m=2}^{n} z_m - \omega_1 \prod_{m=2}^{n} \omega_m$$
$$= z_1 \left(\prod_{m=2}^{n} z_m - \prod_{m=2}^{n} \omega_m\right) + (z_1 - \omega_1) \prod_{m=2}^{n} \omega_m,$$

so

$$\left|\prod_{m=1}^{n} z_m - \prod_{m=1}^{n} \omega_m\right| \le \theta \left|\prod_{m=2}^{n} z_m - \prod_{m=2}^{n} \omega_m\right| + \theta^{n-1}|z_1 - \omega_1|.$$

Iterating this estimate (induction on n) yields

$$\left|\prod_{m=1}^{n} z_m - \prod_{m=1}^{n} \omega_m\right| \le \theta^{n-1} \sum_{m=1}^{n} |z_m - \omega_m|.$$

□

Lemma 3.5.4. *Let X be a real random variable with $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = \sigma^2 < \infty$. Let $\phi_X(t) := \mathbb{E}[e^{itX}]$ be its characteristic function. Then for each fixed $t \in \mathbb{R}$,*

$$\lim_{n\to\infty} n\left(\phi_X\left(\frac{t}{\sqrt{n}}\right) - 1 + \frac{\mathbb{E}[X^2]}{2}\frac{t^2}{n}\right) = 0.$$

Proof. Write

$$n\left(\phi_X\left(\tfrac{t}{\sqrt{n}}\right) - 1 + \tfrac{\sigma^2 t^2}{2n}\right) = n\,\mathbb{E}\left(e^{iXt/\sqrt{n}} - 1 - iXt/\sqrt{n} + \frac{X^2 t^2}{2n}\right).$$

Fix $\varepsilon > 0$ and decompose according to

$$A_n := \left\{|X| > \frac{\varepsilon\sqrt{n}}{2|t|}\right\}, \qquad A_n^c := \Omega \setminus A_n.$$

Large-X part. On A_n,

$$\left|n\left(e^{iXt/\sqrt{n}} - 1\right) - iXt\sqrt{n} + \frac{X^2 t^2}{2}\right|$$
$$\le 2n + |t|\sqrt{n}\,|X| + \frac{t^2}{2}X^2.$$

Hence

$$\mathbb{E}\Big[\Big|n\left(e^{iXt/\sqrt{n}} - 1\right) - iXt\sqrt{n} + \tfrac{X^2t^2}{2}\Big|\,\mathbf{1}_{A_n}\Big]$$
$$\le 2n\,\mathbb{P}(A_n) + |t|\sqrt{n}\,\mathbb{E}[|X|\mathbf{1}_{A_n}] + \frac{t^2}{2}\,\mathbb{E}[X^2\mathbf{1}_{A_n}].$$

On A_n we have $n\mathbf{1}_{A_n} \le \frac{4t^2}{\varepsilon^2}X^2\mathbf{1}_{A_n}$, and similarly

$$\sqrt{n}\,|X|\mathbf{1}_{A_n} \le \frac{2|t|}{\varepsilon}X^2\mathbf{1}_{A_n}.$$

Thus each of the first two terms is bounded by a constant multiple of $\mathbb{E}[X^2\mathbf{1}_{A_n}]$, and by dominated convergence (push limit inside) $\mathbb{E}[X^2\mathbf{1}_{A_n}] \to 0$ as $n \to \infty$. Hence the contribution from A_n tends to 0.

Small-X part. On A_n^c we have

$$|X| \le \frac{\varepsilon\sqrt{n}}{2|t|}, \qquad \left|\frac{Xt}{\sqrt{n}}\right| \le \frac{\varepsilon}{2}.$$

Using Lemma 3.5.2 with $n = 2$ and $|z| \le 1$ gives a bound of the form

$$\left|e^{iz} - 1 - iz + \tfrac{z^2}{2}\right| \;\le\; C|z|^3$$

for some universal constant C and all $|z| \le 1$. Taking $z := Xt/\sqrt{n}$ and n large so that $|z| \le 1$, we obtain

$$\left|n\Big(e^{iXt/\sqrt{n}} - 1 - iXt/\sqrt{n} + \frac{X^2t^2}{2n}\Big)\right| \;\le\; C\,|X|^3\frac{|t|^3}{\sqrt{n}},$$

and on A_n^c this is further bounded by

$$C\,|X|^2|t|^2\,\varepsilon,$$

for all large n. Since $|X|^2$ is integrable, the expectation over A_n^c is therefore bounded by

$$\mathbb{E}\Big(\Big|n\big(e^{iXt/\sqrt{n}} - 1\big) - iXt\sqrt{n} + \tfrac{X^2t^2}{2}\Big|\mathbf{1}_{A_n^c}\Big) \le C\sigma^2|t|^2\varepsilon.$$

(ε is arbitrary) Letting $\varepsilon \downarrow 0$ finishes the proof.

Combining the two parts, the full expectation converges to 0, i.e.

$$\lim_{n\to\infty} n\left(\phi_X\left(\frac{t}{\sqrt{n}}\right) - 1 + \frac{\sigma^2t^2}{2n}\right) = 0.$$

□

Theorem 3.5.5 (Central limit theorem, i.i.d. case)**.** *Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[X_i^2] = \sigma^2 < \infty$. Then*

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \;\overset{w}{\Longrightarrow}\; \mathcal{N}(0, \sigma^2).$$

Proof. Let $\phi_X(t) := \mathbb{E}[e^{itX}]$ be the characteristic function of X_1. Then

$$\mathbb{E}\Big[e^{it\frac{X_1+\cdots+X_n}{\sqrt{n}}}\Big] = \prod_{i=1}^n \mathbb{E}\Big[e^{itX_i/\sqrt{n}}\Big] = \big(\phi_X(t/\sqrt{n})\big)^n.$$

By Lemma 3.5.4,

$$n\Big(\phi_X\Big(\tfrac{t}{\sqrt{n}}\Big) - 1 + \tfrac{\sigma^2t^2}{2n}\Big) \to 0,$$

so

$$\phi_X\Big(\tfrac{t}{\sqrt{n}}\Big) = 1 - \tfrac{\sigma^2t^2}{2n} + o\big(\tfrac{1}{n}\big),$$

and hence

$$\big(\phi_X(t/\sqrt{n})\big)^n = \left(1 - \frac{\sigma^2t^2}{2n} + o\big(\tfrac{1}{n}\big)\right)^n \longrightarrow e^{-\sigma^2t^2/2}.$$

Thus the characteristic functions converge pointwise to that of $\mathcal{N}(0, \sigma^2)$, and Lévy's continuity theorem yields the claimed weak convergence. □

Remark 54. The following proof of Lindeberg-Feller theorem relaxes the assumption of the vanilla central limit theorem. The proof mechanics can be viewed as standard-machine steps to show distribution/weak convergence, as shown in the later subsection "convergence to a Poisson random variable" in which the random variable is integer valued and thus non-negativity is required.

Note that the second requirement in Lindeberg-Feller theorem says the total contribution to the second moments from large deviations goes to zero, the second moments "concentrate" around zero. This is naturally aroused from the proof structure.

Theorem 3.5.6 (Lindeberg–Feller theorem)**.** *For each n, let $X_{n,1}, \dots, X_{n,n}$ be independent (triangular array) random variables with $\mathbb{E}[X_{n,m}] = 0$. Assume:*

1. $\displaystyle\sum_{m=1}^n \mathbb{E}[X_{n,m}^2] \;\longrightarrow\; \sigma^2 > 0;$
2. *For every $\varepsilon > 0$,*
$$\sum_{m=1}^n \mathbb{E}\big[X_{n,m}^2\,\mathbf{1}_{\{|X_{n,m}|>\varepsilon\}}\big] \;\longrightarrow\; 0.$$

Then $S_n := X_{n,1} + \cdots + X_{n,n} \overset{w}{\Longrightarrow} \mathcal{N}(0, \sigma^2)$.

Proof. Let $\phi_{n,m}(t) := \mathbb{E}[e^{itX_{n,m}}]$ and $\sigma_{n,m}^2 := \mathbb{E}[X_{n,m}^2]$. By Lévy's continuity theorem it suffices to show

$$\lim_{n\to\infty}\prod_{m=1}^n \phi_{n,m}(t) = e^{-t^2\sigma^2/2} \quad \text{for each fixed } t \in \mathbb{R}.$$

Set $a_n := \sup_{1\le m\le n}\sigma_{n,m}^2$. For any $\varepsilon > 0$,

$$a_n = \sup_{1\le m\le n}\mathbb{E}[X_{n,m}^2] \le \varepsilon^2 + \sum_{m=1}^n \mathbb{E}\big[X_{n,m}^2\mathbf{1}_{\{|X_{n,m}|>\varepsilon\}}\big],$$

so $\limsup_n a_n \le \varepsilon^2$, and since $\varepsilon > 0$ is arbitrary we have $a_n \to 0$.

Fix $t \in \mathbb{R}$. For n large enough we have $|1 - (t^2\sigma_{n,m}^2/2)| \le 1$ for all $1 \le m \le n$. Applying Lemma 3.5.3 with $\theta = 1$ gives

$$\begin{aligned}
&\left|\prod_{m=1}^n \phi_{n,m}(t) - \prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right)\right| \\
&\le \sum_{m=1}^n \left|\phi_{n,m}(t) - \Big(1 - \frac{t^2\sigma_{n,m}^2}{2}\Big)\right| \\
&= \sum_{m=1}^n \left|\mathbb{E}\Big(e^{itX_{n,m}} - 1 - itX_{n,m} + \tfrac{t^2X_{n,m}^2}{2}\Big)\right|.
\end{aligned}$$

By Lemma 3.5.2,

$$\left|e^{iz} - 1 - iz + \tfrac{z^2}{2}\right| \le \min\left\{\frac{|z|^3}{6},\, |z|^2\right\},$$

so

$$\begin{aligned}
&\left|\prod_{m=1}^n \phi_{n,m}(t) - \prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right)\right| \\
&\le \sum_{m=1}^n \mathbb{E}\left[\min\left\{\frac{|tX_{n,m}|^3}{6},\, |tX_{n,m}|^2\right\}\right].
\end{aligned}$$

For any fixed $\varepsilon > 0$,

$$\begin{aligned}
&\min\left\{\frac{|tX_{n,m}|^3}{6},\, |tX_{n,m}|^2\right\} \\
&\le \frac{|t|\,\varepsilon}{6}|tX_{n,m}|^2\,\mathbf{1}_{\{|X_{n,m}|\le\varepsilon\}} + |tX_{n,m}|^2\,\mathbf{1}_{\{|X_{n,m}|>\varepsilon\}}.
\end{aligned}$$

Thus

$$\begin{aligned}
&\left|\prod_{m=1}^n \phi_{n,m}(t) - \prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right)\right| \\
&\le \frac{|t|^3\varepsilon}{6}\sum_{m=1}^n \mathbb{E}\big[X_{n,m}^2\big] + |t|^2\sum_{m=1}^n \mathbb{E}\big[X_{n,m}^2\mathbf{1}_{\{|X_{n,m}|>\varepsilon\}}\big].
\end{aligned}$$

By assumptions (1)–(2), as $n \to \infty$ the second sum tends to 0, while the first is bounded by $(|t|^3\varepsilon/6)\,\sigma^2$ for all large n. Hence

$$\limsup_{n\to\infty}\left|\prod_{m=1}^n \phi_{n,m}(t) - \prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right)\right| \le \frac{|t|^3\varepsilon\sigma^2}{6}.$$

Since $\varepsilon > 0$ is arbitrary, the left-hand side is 0; that is,

$$\prod_{m=1}^n \phi_{n,m}(t) - \prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right) \longrightarrow 0.$$

It remains to show

$$\prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right) - \prod_{m=1}^n e^{-t^2\sigma_{n,m}^2/2} \longrightarrow 0.$$

Again apply Lemma 3.5.3 with $\theta = 1$:

$$\begin{aligned}
&\left|\prod_{m=1}^n\left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right) - \prod_{m=1}^n e^{-t^2\sigma_{n,m}^2/2}\right| \\
&\le \sum_{m=1}^n \left|e^{-t^2\sigma_{n,m}^2/2} - \left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right)\right|.
\end{aligned}$$

By Lemma 3.5.1 with $z := t^2\sigma_{n,m}^2/2$,

$$\left| e^{-t^2\sigma_{n,m}^2/2} - \left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right)\right| \le \left(\frac{t^2\sigma_{n,m}^2}{2}\right)^2 = \frac{t^4\sigma_{n,m}^4}{4}.$$

Hence

$$\left|\prod_{m=1}^n \left(1 - \frac{t^2\sigma_{n,m}^2}{2}\right) - \prod_{m=1}^n e^{-t^2\sigma_{n,m}^2/2}\right| \le \frac{t^4}{4}\sum_{m=1}^n \sigma_{n,m}^4 \le \frac{t^4 a_n}{4}\sum_{m=1}^n \sigma_{n,m}^2,$$

where $a_n := \sup_m \sigma_{n,m}^2$. By assumption (1), $\sum_m \sigma_{n,m}^2 \to \sigma^2$, and we already showed $a_n \to 0$. Therefore the right-hand side tends to 0.

Combining the two approximations,

$$\prod_{m=1}^n \phi_{n,m}(t) \to \exp\left(-\tfrac{t^2}{2}\sum_{m=1}^n \sigma_{n,m}^2\right) \longrightarrow e^{-t^2\sigma^2/2},$$

and Lévy's continuity theorem gives $S_n \xrightarrow{w} \mathcal{N}(0,\sigma^2)$. □

Lemma 3.5.7 (Borel–Cantelli lemma, version II)**.** *If $(A_i)_{i\ge1}$ are mutually independent events with $\sum_i \mathbb{P}(A_i) = \infty$, then*

$$\mathbb{P}(A_n \text{ i.o.}) = 1.$$

Proof. For $k \ge 1$ define

$$B_k := \bigcup_{m\ge k} A_m.$$

Then $\{B_k\}$ is a decreasing sequence of events, and

$$\{A_n \text{ i.o.}\} = \bigcap_{k\ge1} B_k.$$

By independence,

$$\mathbb{P}(B_k) = 1 - \prod_{m\ge k}(1 - \mathbb{P}(A_m)).$$

Using $1 - x \le e^{-x}$ for $0 \le x \le 1$ we obtain

$$\prod_{m\ge k}(1 - \mathbb{P}(A_m)) \le \exp\Big(-\sum_{m\ge k}\mathbb{P}(A_m)\Big).$$

Since $\sum_{m\ge1}\mathbb{P}(A_m) = \infty$, the tail sums $\sum_{m\ge k}\mathbb{P}(A_m) \to \infty$, hence

$$\prod_{m\ge k}(1 - \mathbb{P}(A_m)) \to 0, \quad \text{so} \quad \mathbb{P}(B_k) \to 1.$$

By continuity from above,

$$\mathbb{P}(A_n \text{ i.o.}) = \mathbb{P}\Big(\bigcap_{k\ge1} B_k\Big) = \lim_{k\to\infty}\mathbb{P}(B_k) = 1.$$

□

Remark 55. Note that this version of Borel-Cantelli lemma requires mutual independent events. But all the rest steps in the proof is taking complement of the proof of the original version of Borel-Cantelli lemma.

3.5.2 Convergence to a Poisson random variable

A Poisson(λ) random variable Z takes values in $\mathbb{N} = \{0, 1, 2, \dots\}$ and satisfies

$$\mathbb{P}(Z = n) = \frac{\lambda^n e^{-\lambda}}{n!}, \qquad n \in \mathbb{N}.$$

Theorem 3.5.8. *Let $X_{n,1}, \dots, X_{n,n}$ be independent nonnegative integer-valued random variables such that*

$$\mathbb{P}(X_{n,m} = 1) = p_{n,m}, \qquad \mathbb{P}(X_{n,m} \ge 2) = \varepsilon_{n,m}.$$

Assume:

1. $\sum_{m=1}^n p_{n,m} \to \lambda \in (0,\infty)$ *(the total rate is λ);*
2. $\max_{1\le m\le n} p_{n,m} \to 0$ *(most queues are empty);*
3. $\sum_{m=1}^n \varepsilon_{n,m} \to 0$ *(very rarely there is a coincidence).*

Then $S_n := \sum_{m=1}^n X_{n,m} \xrightarrow{w} Z$, where $Z \sim$ Poisson(λ).

Proof. We work with characteristic functions. Note that

$$\mathbb{E}\big[e^{itS_n}\big] = \prod_{m=1}^n \mathbb{E}\big[e^{itX_{n,m}}\big].$$

Step 1: Replace $X_{n,m}$ by Bernoulli($p_{n,m}$). By Lemma 3.5.3 with $\theta = 1$,

$$\left|\prod_{m=1}^n \mathbb{E}\big[e^{itX_{n,m}}\big] - \prod_{m=1}^n \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right| \le \sum_{m=1}^n \left|\mathbb{E}\big[e^{itX_{n,m}}\big] - \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right|.$$

Now

$$\mathbb{E}\big[e^{itX_{n,m}}\big] = (1 - p_{n,m} - \varepsilon_{n,m})\cdot 1 + p_{n,m}e^{it} + \mathbb{E}\big[e^{itX_{n,m}}\mathbf{1}_{\{X_{n,m}\ge2\}}\big],$$

so

$$\mathbb{E}\big[e^{itX_{n,m}}\big] - \big(1 - p_{n,m} + p_{n,m}e^{it}\big) = -\varepsilon_{n,m} + \mathbb{E}\big[e^{itX_{n,m}}\mathbf{1}_{\{X_{n,m}\ge2\}}\big].$$

Hence

$$\left|\mathbb{E}\big[e^{itX_{n,m}}\big] - \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right| \le 2\varepsilon_{n,m},$$

and therefore

$$\left|\prod_{m=1}^n \mathbb{E}\big[e^{itX_{n,m}}\big] - \prod_{m=1}^n \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right| \le 2\sum_{m=1}^n \varepsilon_{n,m} \to 0.$$

Step 2: Approximate Bernoulli factors by exponentials. For $z := p_{n,m}(1 - e^{it})$ we have $|z| \le 2p_{n,m}$ and $p_{n,m} \to 0$, so for n large enough $|z| \le 1$. Lemma 3.5.1 then gives

$$\left|e^{-p_{n,m}(1-e^{it})} - \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right| \le 4p_{n,m}^2,$$

where we used that $|1 - e^{it}| \le 2$ to adjust the constant.

Applying Lemma 3.5.3 again with $\theta = 1$,

$$\left|\prod_{m=1}^n e^{-p_{n,m}(1-e^{it})} - \prod_{m=1}^n \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right| \le \sum_{m=1}^n \left|e^{-p_{n,m}(1-e^{it})} - \big(1 - p_{n,m} + p_{n,m}e^{it}\big)\right| \le 4\sum_{m=1}^n p_{n,m}^2 \le 4\Big(\max_{1\le m\le n} p_{n,m}\Big)\sum_{m=1}^n p_{n,m} \to 0,$$

by assumptions (1) and (2).

Step 3: Take the limit. We have

$$\prod_{m=1}^n e^{-p_{n,m}(1-e^{it})} = \exp\Big(-(1 - e^{it})\sum_{m=1}^n p_{n,m}\Big),$$

and by (1),

$$\sum_{m=1}^n p_{n,m} \to \lambda,$$

so

$$\prod_{m=1}^n e^{-p_{n,m}(1-e^{it})} \longrightarrow \exp\big(-\lambda(1 - e^{it})\big),$$

which is the characteristic function of Poisson(λ).

Combining Steps 1–3,

$$\mathbb{E}\big[e^{itS_n}\big] = \prod_{m=1}^n \mathbb{E}\big[e^{itX_{n,m}}\big] \longrightarrow \exp\big(-\lambda(1 - e^{it})\big),$$

and Lévy's continuity theorem yields $S_n \overset{w}{\Longrightarrow} Z$ with $Z \sim$ Poisson(λ). $\square$

Remark 56. The idea of Lindeberg-Fellers theorem is by considering characteristic functions, we can decompose summation into n-letter products (given independence) which allows us to approximate the products of characteristic functions with lower order terms.

In some other cases, for example, minimum of n-random variables. The cdfs can be break into n-letter product forms.

Chapter 4

Signed measures and Conditional expectation

4.1 Signed measures

Definition 4.1.1. *Let $(\Omega, \mathcal{F})$ be a measurable space. A* finite signed measure *is a mapping $\lambda : \mathcal{F} \to \mathbb{R}$ such that whenever $\{A_i\}_{i\geq 1}$ are pairwise disjoint sets in $\mathcal{F}$,*

$$\lambda\Big(\bigcup_{i\geq 1} A_i\Big) = \sum_{i\geq 1} \lambda(A_i).$$

(Finite as it is not using extended real system. Lebesgue measure is not finite as the measure on $\mathbb{R}$ is ∞.)

Clearly $\lambda(\emptyset) = 0$ $(\lambda(\Omega \cup \emptyset) = \lambda(\Omega))$.

Lemma 4.1.1. *If λ is a finite signed measure, then*

$$\sup_{A\in\mathcal{F}} |\lambda(A)| < \infty.$$

Proof. For $A \in \mathcal{F}$, define

$$\lambda_+(A) := \sup_{B\subseteq A} |\lambda(B)|.$$

If there exists a set A such that both $\lambda_+(A)$ and $\lambda_+(A^c)$ are finite, then for any $B \in \mathcal{F}$,

$$|\lambda(B)| = |\lambda(B \cap A) + \lambda(B \cap A^c)| \leq \lambda_+(A) + \lambda_+(A^c),$$

and we are done.

Assume, on the contrary, that for every A at least one of $\lambda_+(A)$ or $\lambda_+(A^c)$ is infinite. Without loss of generality there is some A with $\lambda_+(A) = \infty$. We will construct a decreasing sequence $A_n \downarrow A_*$ with $|\lambda(A_n)| \to \infty$, which contradicts finiteness of λ.

Since $\lambda_+(A) = \infty$, for any integer k there exists $B \subseteq A$ with $|\lambda(B)| \geq k(1 + |\lambda(A)|)$. For such a B,

$$|\lambda(A\backslash B)| \geq |\lambda(B)| - |\lambda(A)| \geq k + (k-1)|\lambda(A)| \geq (k-1)(1+|\lambda(A)|).$$

Moreover,

$$\lambda_+(A) \leq \lambda_+(B) + \lambda_+(A \setminus B),$$

(Slicing makes no decrease to measure.) so at least one of $\lambda_+(B)$ or $\lambda_+(A \setminus B)$ is infinite.

Choose $A_1 \subset A$ such that

$$|\lambda(A_1)| > 1 + |\lambda(A)|, \qquad \lambda_+(A_1) = \infty.$$

(Keep slicing into two parts and one of them has infinite measure. Slicing gives a decreasing sequence of sets. Countably additive gives continuity. But slicing eventually leads to infinity.) Repeat: having chosen A_{n-1} with $\lambda_+(A_{n-1}) = \infty$, pick $A_n \subset A_{n-1}$ such that

$$|\lambda(A_n)| > n\big(1 + |\lambda(A_{n-1})|\big) > n^2(1 + |\lambda(A)|), \qquad \lambda_+(A_n) = \infty.$$

Thus (A_n) is decreasing, and $A_n \downarrow A_*$. By countable additivity, $\lambda(A_n) \to \lambda(A_*)$, but by construction $|\lambda(A_n)| \to \infty$, which is impossible since $\lambda(A_*)$ is finite. This contradiction proves the lemma. □

Definition 4.1.2. *A set $A \in \mathcal{F}$ is called* totally positive *(with respect to λ) if $\lambda(B) \geq 0$ for every $B \subseteq A$, $B \in \mathcal{F}$.*

Lemma 4.1.2. *Let $A \in \mathcal{F}$ with $\lambda(A) = \ell \geq 0$. Then there exists a totally positive set $A_+ \subseteq A$ such that $\lambda(A_+) \geq \ell$.*

Proof. (The idea of this proof is simply keep dropping the sets with negative measures. The $\frac{m}{2}$ is from countably additivity.)

Let

$$m := \inf_{B\subseteq A} \lambda(B).$$

If $m \geq 0$, then A itself is totally positive and we may take $A_+ = A$. Otherwise $m < 0$.

Choose $B_1 \subseteq A$ such that $\lambda(B_1) < m/2$, and set $A_1 := A \setminus B_1$. Then

$$\lambda(A_1) = \lambda(A) - \lambda(B_1) \geq \ell,$$

and

$$m_1 := \inf_{B\subseteq A_1} \lambda(B) > \frac{m}{2}.$$

Repeat: choose $B_2 \subseteq A_1$ such that $\lambda(B_2) < m_1/2$, and set $A_2 := A_1 \setminus B_2$. Then $\lambda(A_2) \geq \ell$ and

$$m_2 := \inf_{B\subseteq A_2} \lambda(B) > \frac{m_1}{2} > \frac{m}{4}.$$

Proceeding inductively, we obtain a decreasing sequence $A_n \downarrow A_*$ such that

$$\lambda(A_n) \geq \ell, \qquad m_n := \inf_{B\subseteq A_n} \lambda(B) > \frac{m}{2^n} \quad \forall n.$$

By continuity from above,

$$\lambda(A_*) = \lim_{n\to\infty} \lambda(A_n) \geq \ell,$$

and for any $B \subseteq A_*$ we have $B \subseteq A_n$ for all large n, so

$$\lambda(B) \geq m_n \to 0.$$

Thus $\lambda(B) \geq 0$ for all $B \subseteq A_*$, i.e. A_* is totally positive. Setting $A_+ := A_*$ gives the claim. □

Note that:

- the class of totally positive sets is closed under countable unions;
- any subset of a totally positive set is also totally positive.

Lemma 4.1.3 (Hahn decomposition)**.** *(For any signed measure λ.) There exists a partition of Ω into a totally positive set Ω_+ and a totally negative set Ω_- (i.e. $-\lambda$ is totally positive on Ω_-).*

Proof. Define

$$\lambda_p(A) := \sup_{B\subseteq A} \lambda(B), \qquad m := \lambda_p(\Omega) = \sup_{A\subseteq\Omega} \lambda(A).$$

If $m \leq 0$, set $\Omega_+ := \emptyset$ and $\Omega_- := \Omega$. Otherwise $m > 0$.

By the previous lemma there exists a totally positive set $A_1 \subseteq \Omega$ with $\lambda(A_1) > m/2$. (we want increase to m, but the supremum may not be directly achieved, hence use $m/2$ and iterates upwards.) Set $B_1 := \Omega \setminus A_1$. Clearly $\lambda_p(B_1) \leq m/2$. Now choose a totally positive set $A_2 \subseteq B_1$ such that

$$\lambda(A_2) > \frac{\lambda_p(B_1)}{2},$$

and set $B_2 := B_1 \setminus A_2 = \Omega \setminus (A_1 \cup A_2)$. Then $\lambda_p(B_2) \le \lambda_p(B_1)/2 \le m/4$.

Proceeding inductively, we obtain disjoint totally positive sets $A_1, A_2, \dots$ and decreasing sets B_k defined by

$$B_k := \Omega \setminus \bigcup_{i=1}^{k} A_i,$$

such that

$$\lambda_p(B_k) \le \frac{m}{2^k}.$$

Define

$$\Omega_+ := \bigcup_{k \ge 1} A_k, \qquad \Omega_- := \bigcap_{k \ge 1} B_k.$$

Then Ω_+ is totally positive (countable union of totally positive sets), and Ω_+ and Ω_- form a partition of Ω.

To see that Ω_- is totally negative, note that for any $C \subseteq \Omega_-$ and any k we have $C \subseteq B_k$, so

$$\lambda(C) \le \lambda_p(B_k) \le \frac{m}{2^k}.$$

Letting $k \to \infty$ yields $\lambda(C) \le 0$. Thus $-\lambda$ is totally positive on Ω_-. □

Remark. The Hahn decomposition (Ω_+, Ω_-) is not unique; however any two such decompositions differ only by a set that is both totally positive and totally negative, and hence has measure 0 along with all its subsets. We call such sets *totally zero* sets.

Given a finite signed measure λ and a Hahn decomposition (Ω_+, Ω_-), define two finite nonnegative

$$\mu_+(A) := \lambda(A \cap \Omega_+), \qquad \mu_-(A) := -\lambda(A \cap \Omega_-).$$

Then $\lambda = \mu_+ - \mu_-$.

Now let μ be a (nonnegative) finite measure on $(\Omega, \mathcal{F})$ and let f be an integrable real-valued function. Define a signed measure

$$\lambda(A) := \int f \, \mathbf{1}_A \, d\mu, \qquad A \in \mathcal{F}.$$

Then λ is a finite signed measure (countable additivity follows from the dominated convergence theorem). Moreover, if $\mu(A) = 0$ then $\lambda(A) = 0$. Indeed, if $\mu(A) = 0$, then for each n,

$$0 = n\,\mu(A) \;\ge\; \int (|f| \wedge n)\, \mathbf{1}_A \, d\mu \;\uparrow\; \int |f|\, \mathbf{1}_A \, d\mu \;\ge\; |\lambda(A)|.$$

Thus $\lambda(A) = 0$ whenever $\mu(A) = 0$.

Definition 4.1.3. *Let $(\Omega, \mathcal{F})$ be a measurable space. Let λ be a finite signed measure on $(\Omega, \mathcal{F})$ and μ a nonnegative measure on $(\Omega, \mathcal{F})$. We say that λ is* absolutely continuous *with respect to μ, written $\lambda \ll \mu$, if*

$$\mu(A) = 0 \;\Rightarrow\; \lambda(A) = 0 \quad \textit{for all } A \in \mathcal{F}.$$

Theorem 4.1.4 (Radon–Nikodym)**.** *Let λ be a finite signed measure on $(\Omega, \mathcal{F})$ and μ a finite nonnegative measure on $(\Omega, \mathcal{F})$ (i.e. $\mu(\Omega) < \infty$). Assume $\lambda \ll \mu$. Then there exists an integrable function f, $\mathcal{F}$-measurable, such that*

$$\lambda(A) = \int_A f \, d\mu, \quad \forall A \in \mathcal{F}.$$

Such f is also known as the Radon-Nikodym derivative $\frac{d\lambda}{d\mu}$.

Proof. Note that $(\Omega, \mathcal{F})$ is fixed at the beginning. For each $q \in \mathbb{Q}$ define a new finite signed measure

$$\lambda_q(A) := \lambda(A) - q\,\mu(A), \qquad A \in \mathcal{F}.$$

Let Ω_+^q be the (Hahn) totally positive part of Ω associated to λ_q, i.e. Ω_+^q is totally positive for λ_q and $\Omega_-^q := \Omega \setminus \Omega_+^q$ is totally negative.

If necessary, modify each Ω_+^q on a totally zero set (one that is both totally positive and totally negative for λ_q); this does not change λ_q and allows us to arrange that for rationals $q_1 < q_2$,

$$\Omega_+^{q_2} \subseteq \Omega_+^{q_1}.$$

(Indeed, if $q_1 < q_2$ and $\omega \in \Omega_+^{q_2} \setminus \Omega_+^{q_1}$, one can move such points into the negative part for λ_{q_1} without affecting the Hahn decomposition, and repeat this consistently across the countable set of rationals.)

Define

$$f(\omega) := \sup\{q \in \mathbb{Q} : \omega \in \Omega_+^q\}.$$

(f is the transportation ratio from μ to λ. Note that the input of f is $\omega \in \Omega$ but the measure μ and finite signed measure λ is meaningful for subsets in the sigma algebra.) Then, for $x \in \mathbb{R}$,

$$\{\omega : f(\omega) > x\} = \bigcup_{q > x} \Omega_+^q,$$

so f is $\mathcal{F}$-measurable.

Finiteness of f μ-a.e. Let

$$A := \bigcap_{q \in \mathbb{Q}} \Omega_+^q.$$

For any q, total positivity of Ω_+^q for λ_q implies $\lambda_q(A) \ge 0$, i.e.

$$\lambda(A) - q\,\mu(A) \ge 0, \quad \forall q \in \mathbb{Q}.$$

If $\mu(A) > 0$, then letting $q \to +\infty$ gives a contradiction. Hence $\mu(A) = 0$, and by $\lambda \ll \mu$, also $\lambda(A) = 0$. On A we clearly have $f(\omega) = +\infty$, so

$$A_\infty := \{\omega : f(\omega) = +\infty\} \subseteq A,$$

and thus $\mu(A_\infty) = \lambda(A_\infty) = 0$.

Similarly, suppose there is a set C such that $C \cap \Omega_+^q = \emptyset$ for all $q \in \mathbb{Q}$. Then $C \subseteq (\bigcup_q \Omega_+^q)^c$, which lies in the negative part for every λ_q. Hence

$$\lambda_q(C) = \lambda(C) - q\,\mu(C) \le 0, \quad \forall q \in \mathbb{Q},$$

which again forces $\mu(C) = 0$ and $\lambda(C) = 0$. On such C we must have $f(\omega) = -\infty$, so the set

$$A_{-\infty} := \{\omega : f(\omega) = -\infty\}$$

also has μ- and λ-measure zero.

Thus f is finite on a set of full μ-measure (and hence full λ-measure).

Basic inequalities on level sets. Let $a < b$ be real numbers and set

$$I_{(a,b]} := \{\omega : f(\omega) \in (a, b]\}.$$

If $q \le a$, then $f(\omega) > q$ for $\omega \in I_{(a,b]}$, so $I_{(a,b]} \subseteq \Omega_+^q$ and thus, by total positivity of Ω_+^q for λ_q,

$$\lambda_q(I_{(a,b]}) = \lambda(I_{(a,b]}) - q\,\mu(I_{(a,b]}) \ge 0.$$

If $q > b$, then $f(\omega) \le b < q$ on $I_{(a,b]}$, so $I_{(a,b]} \subseteq (\Omega_+^q)^c$, which lies in the negative part for λ_q, and hence

$$\lambda_q(I_{(a,b]}) = \lambda(I_{(a,b]}) - q\,\mu(I_{(a,b]}) \le 0.$$

Combining the two bounds and letting $q \uparrow a$ and $q \downarrow b$, we obtain

$$a\,\mu(I_{(a,b]}) \le \lambda(I_{(a,b]}) \le b\,\mu(I_{(a,b]}).$$

If $\mu(I_{(a,b]}) \ne 0$, then the following form

$$a \le \frac{\lambda(I_{(a,b]})}{\mu(I_{(a,b]})} \le b.$$

is stating the same thing as the mean value theorem.

Step function approximation. Fix $h > 0$ and consider the grid intervals $((n-1)h, nh]$, $n \in \mathbb{Z}$, and the corresponding sets

$$I_n := \{\omega : (n-1)h < f(\omega) \le nh\}.$$

For each n the above inequality with $a = (n-1)h$, $b = nh$ gives

$$(n-1)h\,\mu(I_n) \le \lambda(I_n) \le nh\,\mu(I_n).$$

Equivalently,

$$\lambda(I_n) - h\,\mu(I_n) \le (n-1)h\,\mu(I_n) \le \int_{I_n} f\,d\mu,$$

$$\int_{I_n} f\,d\mu \le nh\,\mu(I_n) \le \lambda(I_n) + h\,\mu(I_n). \tag{4.1}$$

Let $A \in \mathcal{F}$ be arbitrary. Multiplying (4.1) by $\mathbf{1}_A$ and summing over n (using disjointness of the I_n and countable additivity) yields

$$\lambda(A) - h\,\mu(A) \le \int_A f\,d\mu \le \lambda(A) + h\,\mu(A).$$

In the details,

$$\sum_{n=1}^{m}[\lambda(I_n \cap A) - h \cdot \mu(I_n \cap A)] \le \sum_{n=1}^{m} \int_{I_n \cap A} f\ d\mu$$
$$\le \sum_{n=1}^{m} \lambda(I_n \cap A) + h \cdot \mu(I_n \cap A).$$

Letting $h \downarrow 0$ gives $\lambda(A) = \int_A f\,d\mu$ for all $A \in \mathcal{F}$.

To see that f is integrable, apply the previous inequality with $A_+ := \{f > 0\}$ and $A_- := \{f \le 0\}$ to deduce that $\int f_+\,d\mu < \infty$ and $\int f_-\,d\mu < \infty$, hence $f \in L^1(\mu)$. □

Remark 57 (Uniqueness). The Radon–Nikodym derivative is unique μ-almost surely. Indeed, suppose f and g both satisfy

$$\lambda(A) = \int_A f\,d\mu = \int_A g\,d\mu, \quad \forall A \in \mathcal{F}.$$

For $\varepsilon > 0$ set

$$A_\varepsilon := \{\omega : f(\omega) - g(\omega) \ge \varepsilon\}.$$

Then

$$0 = \lambda(A_\varepsilon) = \int_{A_\varepsilon} f\,d\mu = \int_{A_\varepsilon} g\,d\mu,$$

so

$$\int_{A_\varepsilon} (f-g)\,d\mu = 0,$$

but the integrand $\int_{A_\varepsilon}(f-g)\,d\mu \ge \varepsilon \cdot \mu(A_\varepsilon)$ on A_ε, hence $\varepsilon\,\mu(A_\varepsilon) \le 0$, which forces $\mu(A_\varepsilon) = 0$. Similarly, $\mu(\{g - f \ge \varepsilon\}) = 0$, so $f = g$ μ-almost surely.

Lemma 4.1.5 (Change of measure formula)**.** *Let $(\Omega, \mathcal{F})$ be a measurable space, and let λ, μ be σ-finite measures such that $\lambda \ll \mu$. Let*

$$g = \frac{d\lambda}{d\mu}.$$

Then for every nonnegative measurable function h,

$$\int h\,d\lambda = \int h\,g\,d\mu.$$

More generally, if $h \in L^1(\lambda)$, then

$$\int h\,d\lambda = \int h\,g\,d\mu.$$

Proof. By the Radon–Nikodym theorem,

$$\lambda(A) = \int_A g\,d\mu \qquad \text{for all } A \in \mathcal{F}.$$

First take $h = \mathbf{1}_A$. Then

$$\int h\,d\lambda = \lambda(A) = \int_A g\,d\mu = \int h\,g\,d\mu.$$

By linearity, the formula also holds for all nonnegative simple functions

$$h = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}.$$

Now let $h \ge 0$ be measurable. Choose simple functions $h_n \uparrow h$. Then by the Monotone Convergence Theorem,

$$\int h\,d\lambda = \lim_{n\to\infty} \int h_n\,d\lambda = \lim_{n\to\infty} \int h_n g\,d\mu = \int hg\,d\mu.$$

Finally, if $h \in L^1(\lambda)$, write

$$h = h^+ - h^-.$$

Apply the nonnegative case to h^+ and h^-, and subtract:

$$\int h\,d\lambda = \int h\,g\,d\mu.$$

□

Standard machine: prove the identity first for indicator functions, extend to simple functions by linearity, then to nonnegative measurable functions by MCT, and finally to integrable signed functions by decomposing into positive and negative parts.

4.2 Conditional expectation

Proposition 4.2.1. *Let f be integrable on $(\Omega, \mathcal{F}, \mu)$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-field. Then there exists a $\mathcal{G}$-measurable, integrable function g such that*

$$\int_A f\,d\mu = \int_A g\,d\mu \quad \forall A \in \mathcal{G}.$$

Moreover, if g_1 and g_2 both satisfy this property, then $g_1 = g_2$ almost surely.

Proof. Define a finite signed measure on $(\Omega, \mathcal{G})$ by

$$\lambda(A) := \int_A f\,d\mu, \qquad A \in \mathcal{G}.$$

Clearly $\lambda \ll \mu$. (We can choose the sub-sigma-algebra $\mathcal{G}$ arbitrarily.) By the Radon–Nikodym theorem, there exists an integrable $\mathcal{G}$-measurable function $\hat{g}$ such that

$$\lambda(A) = \int_A \hat{g}\,d\mu, \quad \forall A \in \mathcal{G}.$$

Setting $g := \hat{g}$ proves existence.

For uniqueness, let g_1 and g_2 both satisfy the defining property, and for $\varepsilon > 0$ define

$$A_\varepsilon := \{\omega : g_1(\omega) - g_2(\omega) > \varepsilon\} \in \mathcal{G}.$$

Then

$$\int_{A_\varepsilon} g_1\,d\mu = \int_{A_\varepsilon} f\,d\mu = \int_{A_\varepsilon} g_2\,d\mu,$$

so

$$\int_{A_\varepsilon} (g_1 - g_2)\,d\mu = 0,$$

but the integrand is $\ge \varepsilon$ on A_ε, hence $\varepsilon\mu(A_\varepsilon) \le 0$ and $\mu(A_\varepsilon) = 0$. Letting $\varepsilon \downarrow 0$ and applying the same argument to $g_2 - g_1$ shows $g_1 = g_2$ almost surely. □

Any such g is called the *conditional expectation* of f given $\mathcal{G}$ and is denoted by $\mathbb{E}(f \mid \mathcal{G})$.

Remark 58. $\mathbb{E}(f|\mathcal{G})$ is a random variable constructed over random variable f, the actual form is $\mathbb{E}(f|\mathcal{G})(\omega)$. This function based on f is a "pooled" version of f and it is only $\mathcal{G}$-measurable.

For elementary conditional expectation, $E(X|Z = z_j) = \sum_{x_i \in \mathcal{X}} x_i P(X = x_i | Z = z_j)$. The random variable $Y = E(X|Z)$ is defined as if $Z(\omega) = z_j$, then $Y(\omega) = E(X|Z = z_j) =: y_j$. It is partitioning Ω into atoms $\mathcal{G}$ defined by z_j. And within each z_j-atom, $Y = E(X|Z)$ is a constant, because Y needs to be $\mathcal{G}$-measurable.

Conditioning on a random variable: $E(X|Y)$ is short-hand for $E(X|\sigma(Y))$. Again it is a function of Y, once the outcome of Y is known, then $E(X|Y)$ is known.

Theorem 4.2.2 (Properties of conditional expectation)**.** *Let f be integrable and $\mathcal{F}$-measurable, and let $\mathcal{G} \subset \mathcal{F}$. Then:*

1. $\mathbb{E}(f) = \mathbb{E}\big(\mathbb{E}(f \mid \mathcal{G})\big)$. *(The conditional expectation's integral agrees with the original function over sample space Ω.)*

2. *If $f \geq 0$ almost surely, then $\mathbb{E}(f \mid \mathcal{G}) \geq 0$ almost surely.*

3. *If f_1, f_2 are integrable and $a, b \in \mathbb{R}$, then*
$$\mathbb{E}(af_1 + bf_2 \mid \mathcal{G}) = a\,\mathbb{E}(f_1 \mid \mathcal{G}) + b\,\mathbb{E}(f_2 \mid \mathcal{G}) \quad a.s.$$
($\forall A \in \mathcal{G}$, apply linearity of expecation.)

4. $\mathbb{E}(|f|) \geq \mathbb{E}\big(|\mathbb{E}(f \mid \mathcal{G})|\big)$. *(Jensen's inequality.)*

5. *If h is $\mathcal{G}$-measurable, and h is bounded (can be further relaxed, see later proof), then*
$$\mathbb{E}(hf \mid \mathcal{G}) = h\,\mathbb{E}(f \mid \mathcal{G}) \quad a.s.$$
(h is $\mathcal{G}$-measureable thus is a constant over $\mathcal{G}$, also, $\mathbb{E}(f \mid \mathcal{G})$ is constant over $\mathcal{G}$.)

6. *If $\mathcal{G}_1 \subseteq \mathcal{G}$, then*
$$\mathbb{E}(f \mid \mathcal{G}_1) = \mathbb{E}\big(\mathbb{E}(f \mid \mathcal{G}) \mid \mathcal{G}_1\big) \quad a.s.$$
(Constrained to a smaller sigma-algebra)

7. *(Jensen's inequality) If $\Phi : \mathbb{R} \to \mathbb{R}$ is convex and $\Phi(f)$ is integrable, then*
$$\mathbb{E}\big(\Phi(f) \mid \mathcal{G}\big) \;\geq\; \Phi\big(\mathbb{E}(f \mid \mathcal{G})\big) \quad a.s.$$

8. *(Independence) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X \in L^1(\mathbb{P})$. Let $\mathcal{G}, \mathcal{H}$ be sub-σ-algebras of $\mathcal{F}$.*

   (a) *Assume $\mathcal{G}$ and $\mathcal{H}$ are independent (i.e. $\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H)$ for all $G \in \mathcal{G}$, $H \in \mathcal{H}$), and that X is $\mathcal{H}$-measurable. Show that*
$$\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X] \qquad \text{almost surely.}$$

   (b) *Suppose that $X \in L^1(\mathbb{P})$ and that $\mathcal{H}$ is independent of $\sigma(\sigma(X), \mathcal{G})$. Show that then*
$$\mathbb{E}[X \mid \mathcal{G} \vee \mathcal{H}] = \mathbb{E}[X \mid \mathcal{G}] \qquad \text{almost surely,}$$
where $\mathcal{G} \vee \mathcal{H}$ is the σ-algebra generated by $\mathcal{G} \cup \mathcal{H}$. (Hint: $\mathcal{G} \vee \mathcal{H}$ is generated by sets of the form $G \cap H$, for $G \in \mathcal{G}$ and $H \in \mathcal{H}$.)

Proof. We just sketch the standard arguments.

1. Taking $A = \Omega$ in the defining identity gives
$$\int_\Omega f\,d\mu = \int_\Omega \mathbb{E}(f \mid \mathcal{G})\,d\mu.$$

2. For $\varepsilon > 0$ let
$$A_\varepsilon^- := \{\omega : \mathbb{E}(f \mid \mathcal{G})(\omega) \leq -\varepsilon\} \in \mathcal{G}.$$
Then
$$\int_{A_\varepsilon^-} f\,d\mu = \int_{A_\varepsilon^-} \mathbb{E}(f \mid \mathcal{G})\,d\mu \leq -\varepsilon\,\mu(A_\varepsilon^-),$$
but $f \geq 0$ implies the left-hand side is ≥ 0, so $\mu(A_\varepsilon^-) = 0$. Letting $\varepsilon \downarrow 0$ shows $\mathbb{E}(f \mid \mathcal{G}) \geq 0$ a.s.

3. For any $A \in \mathcal{G}$,
$$\int_A (af_1 + bf_2)\,d\mu = a\int_A f_1\,d\mu + b\int_A f_2\,d\mu$$
$$= a\int_A \mathbb{E}(f_1 \mid \mathcal{G})\,d\mu + b\int_A \mathbb{E}(f_2 \mid \mathcal{G})\,d\mu.$$
Thus $a\mathbb{E}(f_1 \mid \mathcal{G}) + b\mathbb{E}(f_2 \mid \mathcal{G})$ satisfies the defining property of $\mathbb{E}(af_1 + bf_2 \mid \mathcal{G})$, and uniqueness gives the claim.

4. Let $g := \mathbb{E}(f \mid \mathcal{G})$. Define
$$A_+ := \{\omega : g(\omega) \geq 0\}, \qquad A_- := \{\omega : g(\omega) < 0\},$$
so $A_+, A_- \in \mathcal{G}$ and form a partition. Then
$$\int_{A_+} g\,d\mu = \int_{A_+} f\,d\mu \leq \int_{A_+} |f|\,d\mu,$$
and
$$-\int_{A_-} g\,d\mu = -\int_{A_-} f\,d\mu \leq \int_{A_-} |f|\,d\mu.$$
Adding the two inequalities gives
$$\int |g|\,d\mu = \int_{A_+} g\,d\mu - \int_{A_-} g\,d\mu$$
$$\leq \int_{A_+} |f|\,d\mu + \int_{A_-} |f|\,d\mu = \int |f|\,d\mu,$$
i.e. $\mathbb{E}|g| \leq \mathbb{E}|f|$.

5. First take $h = \mathbf{1}_A$ with $A \in \mathcal{G}$. For any $B \in \mathcal{G}$, we have
$$\int_B \mathbb{E}(hf \mid \mathcal{G})\,d\mu = \int_B hf\,d\mu = \int_{A\cap B} f\,d\mu$$
$$= \int_{A\cap B} \mathbb{E}(f \mid \mathcal{G})\,d\mu = \int_B \mathbf{1}_A \mathbb{E}(f \mid \mathcal{G})\,d\mu.$$
Thus $\mathbb{E}(hf \mid \mathcal{G}) = h\mathbb{E}(f \mid \mathcal{G})$ a.s. for indicators h. Linearity extends this to simple $\mathcal{G}$-measurable h, and a uniform approximation by simple functions plus dominated convergence gives the result for bounded $\mathcal{G}$-measurable h.

6. Let $g := \mathbb{E}(f \mid \mathcal{G})$. Then $\mathbb{E}(g \mid \mathcal{G}_1)$ is $\mathcal{G}_1$-measurable, and for $A \in \mathcal{G}_1$,
$$\int_A f\,d\mu = \int_A g\,d\mu = \int_A \mathbb{E}(g \mid \mathcal{G}_1)\,d\mu.$$
Thus $\mathbb{E}(g \mid \mathcal{G}_1)$ satisfies the defining property of $\mathbb{E}(f \mid \mathcal{G}_1)$; uniqueness implies $\mathbb{E}(f \mid \mathcal{G}_1) = \mathbb{E}(g \mid \mathcal{G}_1)$ a.s.

7. Let Φ be convex. It is a standard fact that
$$\Phi(x) = \sup_\alpha (a_\alpha x + b_\alpha)$$
for some collection of affine functions $x \mapsto a_\alpha x + b_\alpha$. For each α,
$$\Phi(f) \geq a_\alpha f + b_\alpha,$$
so
$$\mathbb{E}(\Phi(f) \mid \mathcal{G}) \geq a_\alpha\,\mathbb{E}(f \mid \mathcal{G}) + b_\alpha.$$
Taking the supremum over all affine functions in the representation of Φ gives
$$\mathbb{E}(\Phi(f) \mid \mathcal{G}) \geq \Phi\big(\mathbb{E}(f \mid \mathcal{G})\big) \quad \text{a.s.}$$

□

Lemma 4.2.3. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G} \subset \mathcal{F}$ a sub-σ-field. Let $f \in L^1(\mathbb{P})$ and let h be $\mathcal{G}$-measurable. Assume*
$$hf \in L^1(\mathbb{P}) \quad \text{and} \quad h\,\mathbb{E}(f \mid \mathcal{G}) \in L^1(\mathbb{P}).$$
Then
$$\mathbb{E}(hf \mid \mathcal{G}) = h\,\mathbb{E}(f \mid \mathcal{G}) \quad \text{almost surely.}$$

Proof. Set $g := \mathbb{E}(f \mid \mathcal{G})$. For each $n \geq 1$ define the truncations
$$h_n := h\,\mathbf{1}_{\{|h| \leq n\}}.$$
Then h_n is bounded and $\mathcal{G}$-measurable, so by the bounded case of the property we already proved,
$$\mathbb{E}(h_n f \mid \mathcal{G}) = h_n g \quad \text{a.s.}$$
Equivalently, for every $A \in \mathcal{G}$,
$$\int_A h_n f\,d\mathbb{P} = \int_A h_n g\,d\mathbb{P}. \tag{4.2}$$
We now let $n \to \infty$ in (4.2). Since $h_n f \to hf$ pointwise and $|h_n f| \leq |hf| \in L^1$, the dominated convergence theorem yields
$$\int_A h_n f\,d\mathbb{P} \;\longrightarrow\; \int_A hf\,d\mathbb{P}.$$

Similarly, $h_n g \to hg$ pointwise and $|h_n g| \le |hg| \in L^1$ by assumption, so again by dominated convergence,

$$\int_A h_n g \, d\mathbb{P} \longrightarrow \int_A hg \, d\mathbb{P}.$$

Passing to the limit in (4.2) gives

$$\int_A hf \, d\mathbb{P} = \int_A hg \, d\mathbb{P}, \qquad \forall A \in \mathcal{G}.$$

By the defining property and uniqueness of conditional expectation,

$$\mathbb{E}(hf \mid \mathcal{G}) = hg = h\,\mathbb{E}(f \mid \mathcal{G}) \quad \text{a.s.}$$

□

Remark. A convenient sufficient condition for the assumptions above is $h, f \in L^2(\mathbb{P})$, since then $hf \in L^1$ and $h\,\mathbb{E}(f \mid \mathcal{G}) \in L^1$ by Cauchy–Schwarz:

$$\int |h\,\mathbb{E}(f \mid \mathcal{G})| \, d\mathbb{P} \le \|h\|_2 \, \|\mathbb{E}(f \mid \mathcal{G})\|_2 \le \|h\|_2 \, \|f\|_2 < \infty.$$

4.2.1 Conditional probability and regular conditional probability

The goal of this section is to define a *conditional probability* kernel $\omega \mapsto \mu(\omega, \cdot)$ that represents conditional probabilities given a sub-σ-field.

Definition 4.2.1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G} \subseteq \mathcal{F}$. A mapping*

$$\mu : \Omega \times \mathcal{F} \to [0,1], \qquad (\omega, A) \mapsto \mu(\omega, A),$$

is called a regular conditional probability *of $\mathbb{P}$ given $\mathcal{G}$ if:*

1. *For every $A \in \mathcal{F}$, the function $\omega \mapsto \mu(\omega, A)$ is $\mathcal{G}$-measurable and*

   $$\mu(\omega, A) = \mathbb{E}(1_A \mid \mathcal{G})(\omega) \quad \text{for almost every } \omega.$$

   (Inherited this "almost sure" from conditional expectation. The pooled conditional expectation does not control over the zero-measure sets.)

2. *For almost every ω, the set function $\mathcal{F} \ni A \mapsto \mu(\omega, A)$ is a probability measure on $(\Omega, \mathcal{F})$. (This is the extra structure that the above ordinary expectation does not automatically gives us. This definition may not yield countably additive or normalized probability measure.*

   Conditioned on a singleton $\{\omega\}$ could be problematic. There are uncountably many and zero measure. The following proof shows actually the above definition is water-tight under Polish sample space, Borel sigma-algebra, and countably generated sigma-algebra.)

Remark 59. Conditional probability is defined based on conditional expectation taking the indicator function.

Conditional probability itself is not controlled by the definition of conditional expectation over the null zero-measure set!

μ here is a probability kernel (random measure):

- For each fixed A, the map $\omega \mapsto \mu(\omega, A)$ is a random variable and is a version of $\mathbb{P}(A|\mathcal{G})$ (it is a random variable!) (there are uncontrolled null sets where multiple version exists).
- For almost everywhere fixed ω, the map $A \mapsto \mu(\omega, A)$ is a probability measure!

Remark 60. There is often a notation abuse here, keep in mind that:

- **Ordinary conditional probability:** for a fixed event $A \in \mathcal{F}$,

  $$\mathbb{P}(A \mid \mathcal{G}) = \mathbb{E}(1_A \mid \mathcal{G})$$

  is a $\mathcal{G}$-measurable random variable, defined up to almost sure equality.

- **Regular conditional probability:** a map

  $$\mu : \Omega \times \mathcal{F} \to [0,1]$$

  such that for each fixed $A \in \mathcal{F}$, the function

  $$\omega \mapsto \mu(\omega, A)$$

  is a version of $\mathbb{P}(A \mid \mathcal{G})$, while for almost every fixed ω, the map

  $$A \mapsto \mu(\omega, A)$$

  is a probability measure on $(\Omega, \mathcal{F})$.

Example 12 (A trivial example for ordinary conditional probability and regular conditional probability). Assume $\mathbb{P}$ is the uniform probability measure on $[0,1]$, and let

$$A_1 := [0, 1/2], A_2 = (1/2, 3/4], A_3 = (3/4, 7/8], A_4 := (7/8, 1].$$

Then

$$\mathcal{F} = \sigma(A_1, A_2, A_3) = \sigma(A_1, A_2, A_3, A_4),$$

and

$$\mathcal{G} = \sigma(A_1, A_2)$$

has atoms

$$A_1, \qquad A_2, \qquad A_3 \cup A_4.$$

Hence every $\mathcal{G}$-measurable random variable is constant on A_1, A_2, and $A_3 \cup A_4$.

- Since $A_1 \in \mathcal{G}$,

  $$\mathbb{P}(A_1 \mid \mathcal{G}) = 1_{A_1}.$$

- For $A_3 \notin \mathcal{G}$, the conditional probability must be constant on each $\mathcal{G}$-atom. Thus

  $$\mathbb{P}(A_3 \mid \mathcal{G}) = 0 \quad \text{on } A_1 \cup A_2,$$

  and on $A_3 \cup A_4$,

  $$\mathbb{P}(A_3 \mid \mathcal{G}) = \frac{\mathbb{P}(A_3)}{\mathbb{P}(A_3 \cup A_4)} = \frac{1/8}{1/4} = \frac{1}{2}.$$

  (It has to be "flatten" as the conditional expectation must be constants over sets in $\mathcal{G}$!) Therefore

  $$\mathbb{P}(A_3 \mid \mathcal{G}) = \frac{1}{2} 1_{A_3 \cup A_4}.$$

A regular conditional probability $\mu(\omega, \cdot)$ given $\mathcal{G}$ is obtained by conditioning on the $\mathcal{G}$-atom containing ω:

$$\mu(\omega, E) = \begin{cases} \dfrac{\mathbb{P}(E \cap A_1)}{\mathbb{P}(A_1)}, & \omega \in A_1, \\ \dfrac{\mathbb{P}(E \cap A_2)}{\mathbb{P}(A_2)}, & \omega \in A_2, \\ \dfrac{\mathbb{P}(E \cap (A_3 \cup A_4))}{\mathbb{P}(A_3 \cup A_4)}, & \omega \in A_3 \cup A_4. \end{cases}$$

Equivalently, for $\omega \in A_1$ it is the probability measure concentrated on A_1, for $\omega \in A_2$ it is concentrated on A_2, and for $\omega \in A_3 \cup A_4$ it assigns mass $1/2$ to each of A_3 and A_4.

Example 13 (To understand the above condition 2, on existence of a regular conditional probability).

Condition (1) alone does not imply Condition (2). Even for Polish sample space, and countably generated sub-sigma-algebra.

Let

$$\Omega = [0,1], \qquad \mathcal{F} = \mathcal{G} = \mathcal{B}([0,1]), \qquad \mathbb{P} = \lambda,$$

where λ is Lebesgue measure. Since $\mathcal{G} = \mathcal{F}$, for every $A \in \mathcal{F}$,

$$\mathbb{P}(A \mid \mathcal{F}) = \mathbb{E}(1_A \mid \mathcal{F}) = 1_A \quad \text{a.s.}$$

Now define

$$\mu(\omega, A) := \begin{cases} 0, & \text{if } A \text{ is countable,} \\ 1_A(\omega), & \text{if } A \text{ is uncountable.} \end{cases}$$

For each fixed $A \in \mathcal{F}$, the map $\omega \mapsto \mu(\omega, A)$ is $\mathcal{F}$-measurable, and it is a version of $\mathbb{E}(1_A \mid \mathcal{F})$ (Note that conditional expectation has no control over zero measure sets. This gives use the freedom to choose!):

- if A is uncountable, then $\mu(\omega, A) = 1_A(\omega)$;
- if A is countable, then $\lambda(A) = 0$, so $1_A = 0$ a.s., and hence $\mu(\omega, A) = 0$ is also a valid version.

Thus Condition (1) holds.

However, Condition (2) fails. Indeed, for any fixed $\omega \in [0, 1]$,

$$\mu(\omega, \{\omega\}) = 0, \qquad \mu(\omega, [0, 1] \setminus \{\omega\}) = 0,$$

because $\{\omega\}$ is countable and $\omega \notin [0, 1] \setminus \{\omega\}$. But

$$\mu(\omega, [0, 1]) = 1.$$

Hence

$$\mu(\omega, [0, 1]) \neq \mu(\omega, \{\omega\}) + \mu(\omega, [0, 1] \setminus \{\omega\}),$$

so $A \mapsto \mu(\omega, A)$ is not even finitely additive, and therefore is not a probability measure.

This shows that choosing a version of $\mathbb{P}(A \mid \mathcal{G})$ separately for each A need not produce a regular conditional probability. The issue is not existence: in this example a regular version does exist, namely

$$\mu_{\text{reg}}(\omega, A) = 1_A(\omega).$$

The point is that Condition (2) requires a *coherent* simultaneous choice of versions in the variable A.

In general such regular conditional probabilities need not exist. However, they do exist under standard "niceness" assumptions. One convenient sufficient condition is:

> If Ω is a Polish[1] space, $\mathcal{F}$ is its Borel σ-algebra, and $\mathcal{G} \subseteq \mathcal{F}$ is countably generated, then a regular conditional probability exists.

Below we sketch the construction in a simple but representative case: $\Omega = [0, 1]$ with $\mathcal{F}$ the Borel σ-algebra.

Step 1: Conditional probabilities on a countable generating class.

Let $\mathbb{Q}$ be the rationals and consider the countable family

$$A_q := (-\infty, q], \qquad q \in \mathbb{Q}.$$

For each $q \in \mathbb{Q}$, define

$$\mu(\omega, A_q) := \mathbb{E}(1_{A_q} \mid \mathcal{G})(\omega).$$

Each $\mu(\cdot, A_q)$ is $\mathcal{G}$-measurable and defined $\mathbb{P}$-a.s. Since the family $\{A_q : q \in \mathbb{Q}\}$ is countable, we can choose a set $S \subset \Omega$ with $\mathbb{P}(S) = 1$ such that for every $\omega \in S$:

- the map $q \mapsto \mu(\omega, A_q)$ is nondecreasing in q;
- $\mu(\omega, A_q) = 0$ for all $q < 0$;
- $\mu(\omega, A_q) = 1$ for all $q \geq 1$.

(Throw away violations between rationals. There could be only contably many such violations.)

Step 2: Extend to all half-lines and define a distribution function.

For $y \in \mathbb{R}$ set

$$A_y := (-\infty, y], \qquad \mu(\omega, A_y) := \inf_{q > y,\, q \in \mathbb{Q}} \mu(\omega, A_q), \quad \omega \in S.$$

Then for each fixed $\omega \in S$ the function

$$F_\omega(y) := \mu(\omega, A_y)$$

is nondecreasing in y, right-continuous, satisfies $\lim_{y \to -\infty} F_\omega(y) = 0$ and $\lim_{y \to +\infty} F_\omega(y) = 1$. Thus F_ω is a distribution function on $\mathbb{R}$.

By the standard one-to-one correspondence (Lebesgue's theorem), for each $\omega \in S$ there exists a unique probability measure $\mu(\omega, \cdot)$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that

$$\mu(\omega, (-\infty, y]) = F_\omega(y), \quad y \in \mathbb{R}.$$

For $\omega \notin S$, define $\mu(\omega, \cdot)$ arbitrarily as some fixed probability measure (say, a Dirac mass at 0).

Thus, for every ω, $A \mapsto \mu(\omega, A)$ is a probability measure.

Step 3: Check the conditional expectation property.

For $q \in \mathbb{Q}$ we have, by construction,

$$\mu(\omega, A_q) = \mathbb{E}(1_{A_q} \mid \mathcal{G})(\omega) \quad \text{for a.e. } \omega,$$

so for every $G \in \mathcal{G}$,

$$\int_G 1_{A_q} \, d\mathbb{P} = \int_G \mu(\omega, A_q) \, d\mathbb{P}(\omega).$$

Now fix $y \in \mathbb{R}$ and let $q \downarrow y$, $q \in \mathbb{Q}$. Then $1_{A_q} \downarrow 1_{A_y}$ pointwise and $\mu(\omega, A_q) \downarrow \mu(\omega, A_y)$ pointwise for all $\omega \in S$. By the monotone convergence theorem on both sides,

$$\int_G 1_{A_y} \, d\mathbb{P} = \int_G \mu(\omega, A_y) \, d\mathbb{P}(\omega), \quad \forall G \in \mathcal{G}.$$

By linearity and finite additivity, the same equality holds for all sets in the algebra of finite disjoint unions of intervals of the form $(a, b]$.

Step 4: Extension via the monotone class theorem.

Let $\mathcal{S}$ be the collection of all Borel sets $A \subset \mathbb{R}$ such that

$$\int_G 1_A \, d\mathbb{P} = \int_G \mu(\omega, A) \, d\mathbb{P}(\omega) \quad \forall G \in \mathcal{G}.$$

We know that $\mathcal{S}$ contains the algebra of finite disjoint unions of intervals $(a, b]$. It is straightforward to check that:

- If $A_n \uparrow A$ or $A_n \downarrow A$ with all $A_n \in \mathcal{S}$, then $A \in \mathcal{S}$ (monotone convergence on both sides).

Thus $\mathcal{S}$ is a monotone class. By the monotone class theorem, $\mathcal{S}$ contains the σ-field generated by the algebra of finite disjoint unions of intervals, which is exactly the Borel σ-field $\mathcal{B}(\mathbb{R})$. Therefore

$$\int_G 1_A \, d\mathbb{P} = \int_G \mu(\omega, A) \, d\mathbb{P}(\omega) \quad \forall A \in \mathcal{B}(\mathbb{R}),\ G \in \mathcal{G}.$$

Since each map $\omega \mapsto \mu(\omega, A)$ is $\mathcal{G}$-measurable (by construction as a limit of $\mathcal{G}$-measurable functions), we have shown that μ is a regular conditional probability.

To summarize: on a "nice" space such as $\Omega = [0, 1]$ with its Borel σ-algebra and a countably generated sub-σ-algebra $\mathcal{G}$, one can construct a regular conditional probability kernel $\omega \mapsto \mu(\omega, \cdot)$ by:

- first defining $\mu(\omega, A_q) = \mathbb{E}(1_{A_q} \mid \mathcal{G})(\omega)$ on a countable generating class $\{A_q\}_{q \in \mathbb{Q}}$, - then extending to all Borel sets via distribution functions and the monotone class theorem.

4.3 A few more explanations

Conditional expectation via indicator functions. A simple way to understand conditional expectation is that

$$\mathbb{E}[X \mid \mathcal{G}]$$

[1] In the mathematical discipline of general topology, a Polish space is a separable completely metrizable topological space; that is, a space homeomorphic to a complete metric space that has a countable dense subset.

is the $\mathcal{G}$-measurable random variable that represents X using only the information in $\mathcal{G}$.

The key role of indicator functions is the following definition: a random variable Y is a version of $\mathbb{E}[X \mid \mathcal{G}]$ if

1. Y is $\mathcal{G}$-measurable;
2. for every event $A \in \mathcal{G}$,
$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A].$$

So indicator functions $\mathbf{1}_A$ are used to "test" whether Y and X have the same average on every event that is visible under $\mathcal{G}$.

Why this is natural.

Since $\mathcal{G}$ represents the available information, the events $A \in \mathcal{G}$ are exactly the events we are allowed to distinguish. Thus $\mathbb{E}[X \mid \mathcal{G}]$ should be a $\mathcal{G}$-measurable random variable Y such that on every such event,
$$\int_A X \, d\mathbb{P} = \int_A Y \, d\mathbb{P}.$$
Using indicators, this is exactly
$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[Y\mathbf{1}_A].$$

Chapter 5

Martingales

5.1 Martingales

In this section we study an important class of sequences of random variables that arise naturally in many settings, and also as a technical tool in proving various results.

Definition 5.1.1. *Given a measurable space* $(\Omega, \mathcal{F})$*, a* filtration *is an increasing sequence of sub–σ-fields*

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}$$

such that $\mathcal{F}_i \subseteq \mathcal{F}_j$ *whenever* $i \leq j$*.*

Definition 5.1.2. *A sequence of random variables* $\{X_i\}_{i\geq 1}$ *is said to be* adapted *to a filtration* $\{\mathcal{F}_i\}_{i\geq 1}$ *if* X_i *is* $\mathcal{F}_i$*-measurable for every* i*.*

Definition 5.1.3. *Let* $\{X_i\}_{i\geq 1}$ *be a sequence of integrable random variables adapted to a filtration* $\{\mathcal{F}_i\}_{i\geq 1}$*.*

- $\{X_i\}$ *is a* martingale *if, for all* $i \geq 1$*,*
$$\mathbb{E}(X_{i+1} \mid \mathcal{F}_i) = X_i \quad a.s.$$
- $\{X_i\}$ *is a* submartingale *if, for all* $i \geq 1$*,*
$$\mathbb{E}(X_{i+1} \mid \mathcal{F}_i) \geq X_i \quad a.s.$$
- $\{X_i\}$ *is a* supermartingale *if, for all* $i \geq 1$*,*
$$\mathbb{E}(X_{i+1} \mid \mathcal{F}_i) \leq X_i \quad a.s.$$

Remark 61. Given a sequence of random variables $\{X_n\}$, a natural filtration is $\mathcal{F}_n := \sigma(X_1, \ldots, X_n)$. Unless stated otherwise, this will be the underlying filtration.

Remark 62. Suppose we take the natural filtration $\mathcal{F}_n := \sigma(X_1, \ldots, X_n)$. Recall that by definition of conditional expectation, $\mathbb{E}(X_{i+1} \mid \mathcal{F}_i)$ is a function of random variables $(X_j), j \leq i$. For martingales, such a function is exactly X_i.

Example 14 (Three coin tosses and filtration). Consider three tosses of a fair coin. The sample space is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Let X_1, X_2, X_3 denote the outcomes of the first, second, and third tosses, respectively.

Time 0: before any toss. At this time, no information is available, so we take

$$\mathcal{F}_0 = \{\emptyset, \Omega\}.$$

Time 1: after the first toss. Now we know whether the first toss is H or T. Thus

$$\mathcal{F}_1 = \sigma(X_1) = \big\{\emptyset, \Omega, \{HHH, HHT, HTH, HTT\}, \{THH, THT, TTH, TTT\}\big\}.$$

Time 2: after the first two tosses. Now we know the pair (X_1, X_2), so

$$\mathcal{F}_2 = \sigma(X_1, X_2),$$

which is generated by the partition

$$\{HHH, HHT\}, \quad \{HTH, HTT\},$$
$$\{THH, THT\}, \quad \{TTH, TTT\}.$$

Hence $\mathcal{F}_2$ consists of all unions of these four sets.

Time 3: after all three tosses. At this stage, the entire outcome is known, so

$$\mathcal{F}_3 = \sigma(X_1, X_2, X_3) = 2^{\Omega}.$$

Therefore,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3,$$

which is a filtration.

This filtration describes the gradual arrival of information: at time n, the σ-algebra $\mathcal{F}_n$ contains exactly those events that can be determined from the first n tosses.

The following statements are standard and easy to verify.

1. Let $\{X_i\}$ be a sequence of independent integrable random variables and set $S_n := X_1 + \cdots + X_n$. Then, for any constants α, β, the process
$$M_n := \alpha\big(S_n - \mathbb{E}(S_n)\big) + \beta$$
is a martingale (with respect to the natural filtration).

2. If $\{X_i\}$ is a martingale and Φ is concave with $\Phi(X_i)$ integrable for all i, then $\{\Phi(X_i)\}$ is a supermartingale. Similarly, if Φ is convex with $\Phi(X_i)$ integrable, then $\{\Phi(X_i)\}$ is a submartingale.

   Proof. By Jensen's inequality,
$$\mathbb{E}(\Phi(X_{i+1}) \mid \mathcal{F}_i) \leq \Phi\big(\mathbb{E}(X_{i+1} \mid \mathcal{F}_i)\big) = \Phi(X_i) \quad \text{a.s.}$$
□

3. If $\{X_i\}$ is a submartingale and Φ is nondecreasing and convex with $\Phi(X_i)$ integrable, then $\{\Phi(X_i)\}$ is a submartingale. Similarly, if $\{X_i\}$ is a supermartingale and Φ is nonincreasing and concave with $\Phi(X_i)$ integrable, then $\{\Phi(X_i)\}$ is a supermartingale.

4. Given a filtration $\{\mathcal{F}_i\}$ and an integrable random variable X, the process $X_i := \mathbb{E}(X \mid \mathcal{F}_i)$ is a martingale.

5. If $\{X_i\}$ is a martingale, then for $i \leq j$,
$$\mathbb{E}(X_j \mid \mathcal{F}_i) = X_i \quad \text{a.s.}$$
For submartingales (respectively supermartingales) the equality becomes the corresponding inequality.

Definition 5.1.4. *Given a filtration* $\{\mathcal{F}_i\}_{i\geq 1}$*, a sequence* $\{H_i\}_{i\geq 1}$ *of integrable random variables is called* predictable *(with respect to* $\{\mathcal{F}_i\}$*) if* H_{i+1} *is* $\mathcal{F}_i$*-measurable for every* $i \geq 1$*.*

Theorem 5.1.1 (Doob's maximal inequality, L^1 version)**.** *Suppose $\{X_j\}_{j=1}^n$ is a martingale (or a nonnegative submartingale). For $l > 0$ define*

$$A_l := \left\{\omega : \sup_{1\le j\le n} |X_j(\omega)| \ge l\right\}.$$

Then

$$\mathbb{P}(A_l) \;\le\; \frac{1}{l}\int_{A_l} |X_n|\, d\mathbb{P} \;\le\; \frac{1}{l}\,\mathbb{E}(|X_n|).$$

Proof. First note that $\{|X_j|\}_{j=1}^n$ is a submartingale. Define

$$B_j := \{\omega : |X_1(\omega)| < l, \dots, |X_{j-1}(\omega)| < l,\ |X_j(\omega)| \ge l\}, \quad 1 \le j \le n.$$

Then each $B_j \in \mathcal{F}_j$, the B_j are disjoint, and

$$A_l = \bigsqcup_{j=1}^n B_j.$$

Since $\{|X_j|\}$ is a submartingale, we have

$$\mathbb{E}(|X_n| \mid \mathcal{F}_j) \ge |X_j| \quad \text{a.s.}$$

Hence

$$\int_{B_j} |X_n|\, d\mathbb{P} = \int_{B_j} \mathbb{E}(|X_n| \mid \mathcal{F}_j)\, d\mathbb{P} \;\ge\; \int_{B_j} |X_j|\, d\mathbb{P} \;\ge\; l\,\mathbb{P}(B_j).$$

Summing over j,

$$\int_{A_l} |X_n|\, d\mathbb{P} = \sum_{j=1}^n \int_{B_j} |X_n|\, d\mathbb{P} \;\ge\; l\sum_{j=1}^n \mathbb{P}(B_j) = l\,\mathbb{P}(A_l),$$

which gives the first inequality. The second inequality is immediate since $\int_{A_l} |X_n|\, d\mathbb{P} \le \mathbb{E}(|X_n|)$. □

Lemma 5.1.2 (Tail integral formula)**.** *Let $Y \ge 0$ be a random variable and $p > 0$. Then*

$$\int Y^p\, d\mathbb{P} = p\int_0^\infty y^{p-1}\,\mathbb{P}(Y > y)\, dy.$$

Proof. Observe that for $y \ge 0$,

$$Y^p = \int_0^Y p\, y^{p-1}\, dy = \int_0^\infty p\, y^{p-1}\, \mathbf{1}_{\{Y>y\}}\, dy.$$

Integrating both sides and applying Fubini's theorem (the integrand is nonnegative) gives

$$\begin{aligned}\int Y^p\, d\mathbb{P} &= \int_\Omega \int_0^\infty p\, y^{p-1}\, \mathbf{1}_{\{Y(\omega)>y\}}\, dy\, d\mathbb{P}(\omega)\\ &= \int_0^\infty p\, y^{p-1}\left(\int_\Omega \mathbf{1}_{\{Y(\omega)>y\}}\, d\mathbb{P}(\omega)\right) dy\\ &= p\int_0^\infty y^{p-1}\,\mathbb{P}(Y > y)\, dy.\end{aligned}$$

□

Lemma 5.1.3. *Let X, Y be nonnegative random variables on the same probability space such that for all $l > 0$,*

$$\mathbb{P}(Y \ge l) \;\le\; \frac{1}{l}\int_{\{Y\ge l\}} X\, d\mathbb{P}.$$

Then for every $p > 1$,

$$\int Y^p\, d\mathbb{P} \;\le\; \left(\frac{p}{p-1}\right)^p \int X^p\, d\mathbb{P}.$$

Proof. Let $F(y) = \mathbb{P}(Y \le y)$ and $T(y) = 1 - F(y) = \mathbb{P}(Y > y)$. Then

$$\begin{aligned}\int Y^p\, d\mathbb{P} &= \int_0^\infty y^p\, dF(y) = -\int_0^\infty y^p\, dT(y)\\ &= p\int_0^\infty y^{p-1}T(y)\, dy \qquad \text{(Tail integral formula).}\end{aligned}$$

By the assumption with $l = y$,

$$T(y) = \mathbb{P}(Y > y) \le \frac{1}{y}\int_{\{Y>y\}} X\, d\mathbb{P} \quad \text{for } y > 0,$$

hence

$$\begin{aligned}\int Y^p\, d\mathbb{P} &\le p\int_0^\infty y^{p-2}\left(\int_{\{Y>y\}} X\, d\mathbb{P}\right) dy\\ &= p\int X\left(\int_0^Y y^{p-2}\, dy\right) d\mathbb{P}\\ &= \frac{p}{p-1}\int XY^{p-1}\, d\mathbb{P}.\end{aligned}$$

Applying Hölder's inequality with conjugate exponents p and $p/(p-1)$ gives

$$\int XY^{p-1}\, d\mathbb{P} \le \left(\int X^p\, d\mathbb{P}\right)^{1/p}\left(\int Y^p\, d\mathbb{P}\right)^{(p-1)/p}.$$

Combining the last two displays,

$$\int Y^p\, d\mathbb{P} \le \frac{p}{p-1}\left(\int X^p\, d\mathbb{P}\right)^{1/p}\left(\int Y^p\, d\mathbb{P}\right)^{(p-1)/p}.$$

If $\mathbb{E}(Y^p) < \infty$, divide both sides by $\left(\int Y^p\, d\mathbb{P}\right)^{(p-1)/p}$ to obtain

$$\left(\int Y^p\, d\mathbb{P}\right)^{1/p} \le \frac{p}{p-1}\left(\int X^p\, d\mathbb{P}\right)^{1/p},$$

which implies the desired inequality.

If $\mathbb{E}(Y^p) = \infty$, apply the argument to $Y_m := \min\{Y, m\}$, use monotone convergence as $m \to \infty$, and obtain the same bound in the limit. □

Corollary 5.1.4 (Doob's maximal inequality in L^p)**.** *Let $\{X_j\}_{j=1}^n$ be a martingale or a nonnegative submartingale, and set*

$$S := \sup_{1\le j\le n} |X_j|.$$

Then for every $p > 1$,

$$\int S^p\, d\mathbb{P} \;\le\; \left(\frac{p}{p-1}\right)^p \int |X_n|^p\, d\mathbb{P}.$$

Proof. Apply Doob's L^1 inequality with $Y := S$ and $X := |X_n|$: for each $l > 0$,

$$\mathbb{P}(S \ge l) \le \frac{1}{l}\int_{\{S\ge l\}} |X_n|\, d\mathbb{P}.$$

Then Lemma 5.1.3 with this choice of X, Y gives

$$\int S^p\, d\mathbb{P} \le \left(\frac{p}{p-1}\right)^p \int |X_n|^p\, d\mathbb{P}.$$

□

5.1.1 Martingale transforms

Definition 5.1.5 (Martingale transform)**.** *Let $(X_n)_{n\ge0}$ be a martingale with respect to a filtration $(\mathcal{F}_n)_{n\ge0}$, and let $(C_n)_{n\ge1}$ be a* predictable *process, i.e. C_n is $\mathcal{F}_{n-1}$-measurable for each $n \ge 1$. The* martingale transform *of X by C is the process*

$$(C \circ X)_n := \sum_{m=1}^n C_m\,(X_m - X_{m-1}), \quad n \ge 0,$$

with the convention $(C \circ X)_0 := 0$.

Remark 63. This is the discrete analogue of a stochastic integral. Note that if (C_n) is bounded and predictable and (X_n) is a martingale, then $(C \circ X)_n$ is again a martingale.

Proposition 5.1.5 (Martingale transform of a martingale)**.** *Let $(X_n)_{n\ge0}$ be a martingale with respect to $(\mathcal{F}_n)_{n\ge0}$, and let $(C_n)_{n\ge1}$ be a bounded predictable process, i.e. C_n is $\mathcal{F}_{n-1}$-measurable and $|C_n| \le K < \infty$ a.s. for all $n \ge 1$. Then $(C \circ X)_n$ is a martingale with respect to $(\mathcal{F}_n)$.*

Proof. Clearly $(C \circ X)_n$ is $\mathcal{F}_n$-measurable. Moreover

$$|(C \circ X)_n| \le \sum_{m=1}^{n} |C_m||X_m - X_{m-1}| \le K \sum_{m=1}^{n} |X_m - X_{m-1}|,$$

so $(C \circ X)_n$ is integrable.

For the martingale property, for $n \ge 0$,

$$\begin{aligned}\mathbb{E}\big((C \circ X)_{n+1} \mid \mathcal{F}_n\big) &= \mathbb{E}\Big((C \circ X)_n + C_{n+1}(X_{n+1} - X_n) \;\Big|\; \mathcal{F}_n\Big) \\ &= (C \circ X)_n + \mathbb{E}\big(C_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n\big).\end{aligned}$$

Since C_{n+1} is $\mathcal{F}_n$-measurable and (X_n) is a martingale,

$$\mathbb{E}\big(C_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n\big) = C_{n+1}\,\mathbb{E}(X_{n+1} - X_n \mid \mathcal{F}_n) = C_{n+1}\,(X_n - X_n) = 0,$$

so

$$\mathbb{E}\big((C \circ X)_{n+1} \mid \mathcal{F}_n\big) = (C \circ X)_n.$$

Thus $(C \circ X)_n$ is a martingale. □

Proposition 5.1.6 (Transforms of sub/supermartingales)**.** *Let $(X_n)_{n\ge0}$ be a submartingale (respectively supermartingale) with respect to $(\mathcal{F}_n)_{n\ge0}$, and let $(C_n)_{n\ge1}$ be a bounded predictable process such that $C_n \ge 0$ a.s. for all n (respectively $C_n \le 0$ a.s. for all n). Then $(C \circ X)_n$ is a submartingale (respectively supermartingale) with respect to $(\mathcal{F}_n)$.*

Proof. As before, $(C \circ X)_n$ is $\mathcal{F}_n$-measurable and integrable. For $n \ge 0$,

$$\mathbb{E}\big((C \circ X)_{n+1} \mid \mathcal{F}_n\big) = (C \circ X)_n + \mathbb{E}\big(C_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n\big).$$

Using predictability and the submartingale (or supermartingale) property,

$$\mathbb{E}\big(C_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n\big) = C_{n+1}\,\mathbb{E}(X_{n+1} - X_n \mid \mathcal{F}_n).$$

If (X_n) is a submartingale and $C_{n+1} \ge 0$, then $\mathbb{E}(X_{n+1} - X_n \mid \mathcal{F}_n) \ge 0$ a.s., so

$$C_{n+1}\,\mathbb{E}(X_{n+1} - X_n \mid \mathcal{F}_n) \ge 0,$$

and hence

$$\mathbb{E}\big((C \circ X)_{n+1} \mid \mathcal{F}_n\big) \ge (C \circ X)_n.$$

Thus $(C \circ X)_n$ is a submartingale.

If (X_n) is a supermartingale and $C_{n+1} \le 0$, then $\mathbb{E}(X_{n+1} - X_n \mid \mathcal{F}_n) \le 0$ a.s., so the product is again nonnegative and the same inequality holds, showing that $(C \circ X)_n$ is a supermartingale. □

5.2 Stopping times

Definition 5.2.1 (Stopping time)**.** *Let $(\mathcal{F}_n)_{n\ge0}$ be a filtration. A mapping*

$$T : \Omega \to \{0, 1, 2, \dots, \infty\}$$

is called a stopping time *(with respect to $(\mathcal{F}_n)$) if*

$$\{\omega : T(\omega) \le n\} \in \mathcal{F}_n \quad \text{for all } n \ge 0.$$

Theorem 5.2.1 (Stopped process)**.** *Let $(X_n)_{n\ge0}$ be a supermartingale with respect to $(\mathcal{F}_n)_{n\ge0}$, and let T be a stopping time. Define the stopped process*

$$X_n^T(\omega) := X_{T(\omega)\wedge n}(\omega), \quad n \ge 0.$$

Then $(X_n^T)_{n\ge0}$ is a supermartingale. In particular,

$$\mathbb{E}(X_{T\wedge n}) \le \mathbb{E}(X_0), \quad n \ge 0.$$

If (X_n) is a martingale, then (X_n^T) is a martingale and, for all n,

$$\mathbb{E}(X_{T\wedge n}) = \mathbb{E}(X_0).$$

Proof. First note that X_n^T is $\mathcal{F}_n$-measurable, since

$$X_n^T = \sum_{k=0}^{n} X_k\,\mathbf{1}_{\{T=k\}} + X_n\,\mathbf{1}_{\{T>n\}},$$

and $\{T \le k\} \in \mathcal{F}_k \subseteq \mathcal{F}_n$ for $k \le n$.

For the supermartingale property, we compute for $n \ge 0$:

$$\begin{aligned}\mathbb{E}(X_{T\wedge(n+1)} \mid \mathcal{F}_n) &= \mathbb{E}\big(X_{T\wedge(n+1)}\mathbf{1}_{\{T\le n\}} + X_{T\wedge(n+1)}\mathbf{1}_{\{T>n\}} \;\big|\; \mathcal{F}_n\big) \\ &= \mathbb{E}\big(X_{T\wedge n}\mathbf{1}_{\{T\le n\}} + X_{n+1}\mathbf{1}_{\{T>n\}} \;\big|\; \mathcal{F}_n\big) \\ &= X_{T\wedge n}\mathbf{1}_{\{T\le n\}} + \mathbf{1}_{\{T>n\}}\,\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \quad \text{a.s.} \\ &\le X_{T\wedge n}\mathbf{1}_{\{T\le n\}} + \mathbf{1}_{\{T>n\}}X_n \quad \text{a.s.} \\ &= X_{T\wedge n}.\end{aligned}$$

Thus (X_n^T) is a supermartingale.

If (X_n) is a martingale, the same computation with equality $\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = X_n$ shows that

$$\mathbb{E}(X_{T\wedge(n+1)} \mid \mathcal{F}_n) = X_{T\wedge n},$$

so (X_n^T) is a martingale. Taking expectations gives $\mathbb{E}(X_{T\wedge n}) = \mathbb{E}(X_0)$ for all n. □

5.2.1 Doob's optional stopping theorem

Theorem 5.2.2 (Doob's optional stopping theorem)**.** *The following results hold.*

1. *Let $(X_n)_{n\ge0}$ be a supermartingale and T a stopping time. If any one of the following conditions is satisfied:*
   - *(a) T is bounded;*
   - *(b) (X_n) is uniformly bounded and $T < \infty$ a.s.;*
   - *(c) $\mathbb{E}(T) < \infty$ and the increments $|X_{T\wedge n} - X_{T\wedge(n-1)}|$ are uniformly bounded (say by K),*

   then $\mathbb{E}(X_T) \le \mathbb{E}(X_0)$.

2. *If (X_n) is a martingale and any of the above three conditions holds, then*
   $$\mathbb{E}(X_T) = \mathbb{E}(X_0).$$

3. *Let $(C_n)_{n\ge1}$ be a uniformly bounded predictable sequence and $(X_n)_{n\ge0}$ a martingale such that $|X_n - X_{n-1}|$ is uniformly bounded, and let T be a stopping time with $\mathbb{E}(T) < \infty$. Then*
   $$\mathbb{E}\big((C \circ X)_T\big) = 0,$$
   where $(C \circ X)_n := \sum_{m=1}^{n} C_m(X_m - X_{m-1})$.

4. *If $(X_n)_{n\ge0}$ is a nonnegative supermartingale and T is finite a.s., then*
   $$\mathbb{E}(X_T) \le \mathbb{E}(X_0).$$

Proof. We prove each part in turn.

(1) Assume (X_n) is a supermartingale.

(a) Suppose $T \le N$ a.s. for some integer N. Then $X_{T\wedge N} = X_T$. By the stopped-process theorem, $(X_{T\wedge n})_{n\ge0}$ is a supermartingale, so

$$\mathbb{E}(X_T) = \mathbb{E}(X_{T\wedge N}) \le \mathbb{E}(X_0).$$

(b) Suppose (X_n) is uniformly bounded, say $|X_n| \le M$ for all n, and $T < \infty$ a.s. Then $X_{T\wedge n} \to X_T$ a.s. as $n \to \infty$, and all $X_{T\wedge n}, X_T$ are bounded by M. By the bounded convergence theorem,

$$\mathbb{E}(X_{T\wedge n}) \to \mathbb{E}(X_T).$$

But by the stopped process result,

$$\mathbb{E}(X_{T\wedge n}) \le \mathbb{E}(X_0) \quad \text{for all } n,$$

so taking the limit gives $\mathbb{E}(X_T) \le \mathbb{E}(X_0)$.

(c) Suppose $\mathbb{E}(T) < \infty$ and $|X_{T\wedge n} - X_{T\wedge(n-1)}| \le K$ for all n. Then

$$|X_{T\wedge n} - X_0| = \left|\sum_{k=1}^{T\wedge n}\big(X_{T\wedge k} - X_{T\wedge(k-1)}\big)\right|$$
$$\le \sum_{k=1}^{T\wedge n}\big|X_{T\wedge k} - X_{T\wedge(k-1)}\big| \le K\,(T\wedge n) \le KT.$$

Since KT is integrable (because $\mathbb{E}(T) < \infty$), dominated convergence applies to $X_{T\wedge n} - X_0$:

$$\mathbb{E}(X_T - X_0) = \mathbb{E}\big(\lim_{n\to\infty}(X_{T\wedge n} - X_0)\big) = \lim_{n\to\infty}\mathbb{E}(X_{T\wedge n} - X_0) \le 0,$$

since each $\mathbb{E}(X_{T\wedge n}) \le \mathbb{E}(X_0)$. Hence $\mathbb{E}(X_T) \le \mathbb{E}(X_0)$.

(2) If (X_n) is a martingale, then (X_n) and $(-X_n)$ are both supermartingales. Applying part (1) to X_n gives $\mathbb{E}(X_T) \le \mathbb{E}(X_0)$, and applying it to $-X_n$ gives

$$\mathbb{E}(-X_T) \le \mathbb{E}(-X_0) \quad\Longrightarrow\quad \mathbb{E}(X_T) \ge \mathbb{E}(X_0).$$

Thus $\mathbb{E}(X_T) = \mathbb{E}(X_0)$.

(3) Under the stated assumptions, $(C\circ X)_n$ is a martingale with mean zero: $\mathbb{E}((C\circ X)_n) = 0$ for all n. Moreover,

$$|(C\circ X)_n - (C\circ X)_{n-1}| = |C_n|\cdot|X_n - X_{n-1}| \le K_1K_2$$

for suitable constants K_1, K_2 coming from the uniform bounds on (C_n) and on the increments $(X_n - X_{n-1})$. Hence the increments of $(C\circ X)_n$ are uniformly bounded, and we can apply part (2) with X_n replaced by $(C\circ X)_n$ to obtain

$$\mathbb{E}\big((C\circ X)_T\big) = \mathbb{E}\big((C\circ X)_0\big) = 0.$$

(4) Suppose (X_n) is a nonnegative supermartingale and $T < \infty$ a.s. Then $X_{T\wedge n} \to X_T$ a.s. and $X_{T\wedge n} \ge 0$. By Fatou's lemma,

$$\mathbb{E}(X_T) = \mathbb{E}\big(\liminf_{n\to\infty} X_{T\wedge n}\big) \le \liminf_{n\to\infty}\mathbb{E}(X_{T\wedge n}) \le \mathbb{E}(X_0),$$

where we again used that $(X_{T\wedge n})$ is a supermartingale in the last step. □

Lemma 5.2.3. *Let T be a stopping time and suppose there exist $N \in \mathbb{N}$ and $\varepsilon > 0$ such that*

$$\mathbb{P}(T \le n + N \mid \mathcal{F}_n) \ge \varepsilon \quad \textit{a.s. for all } n.$$

Then $\mathbb{E}(T) < \infty$.

Proof. Fix $k \ge 1$. Then

$$\begin{aligned}\mathbb{P}(kN < T) &= \mathbb{E}\big(\mathbf{1}_{\{T>(k-1)N\}}\mathbf{1}_{\{T>kN\}}\big)\\ &= \mathbb{E}\Big(\mathbb{E}\big(\mathbf{1}_{\{T>(k-1)N\}}\mathbf{1}_{\{T>kN\}} \mid \mathcal{F}_{(k-1)N}\big)\Big)\\ &= \mathbb{E}\Big(\mathbf{1}_{\{T>(k-1)N\}}\,\mathbb{E}\big(\mathbf{1}_{\{T>kN\}} \mid \mathcal{F}_{(k-1)N}\big)\Big)\\ &= \mathbb{E}\Big(\mathbf{1}_{\{T>(k-1)N\}}\big(1 - \mathbb{P}(T \le kN \mid \mathcal{F}_{(k-1)N})\big)\Big)\\ &\le (1-\varepsilon)\,\mathbb{E}\big(\mathbf{1}_{\{T>(k-1)N\}}\big) = (1-\varepsilon)\,\mathbb{P}(T > (k-1)N).\end{aligned}$$

By induction on k, this yields

$$\mathbb{P}(T > kN) \le (1-\varepsilon)^k \quad \text{for all } k \ge 0.$$

Now write

$$\begin{aligned}\mathbb{E}(T) &= \sum_{n\ge0}\mathbb{P}(T > n) = \sum_{k\ge0}\sum_{m=0}^{N-1}\mathbb{P}(T > kN + m)\\ &\le \sum_{k\ge0}\sum_{m=0}^{N-1}\mathbb{P}(T > kN) \le N\sum_{k\ge0}(1-\varepsilon)^k = \frac{N}{\varepsilon} < \infty.\end{aligned}$$

Thus $\mathbb{E}(T) < \infty$. □

5.2.2 A counterexample to naive optional stopping

Here is a simple example showing that even if (X_n) is a martingale with bounded increments and T is an a.s. finite stopping time, it may still happen that $\mathbb{E}(X_T) \ne \mathbb{E}(X_0)$.

Consider a simple symmetric random walk $(X_n)_{n\ge0}$ on $\mathbb{Z}$ with $X_0 = 0$ and increments ±1 with probability $1/2$ each. Clearly (X_n) is a martingale and $|X_n - X_{n-1}| = 1$ for all n. Define the stopping time

$$T := \inf\{n \ge 0 : X_n = -1\}.$$

If $T < \infty$ a.s., then $X_T = -1$ a.s., so

$$\mathbb{E}(X_T) = -1 \ne 0 = \mathbb{E}(X_0).$$

Thus it suffices to show that T is almost surely finite.

To this end, fix $n \ge 1$ and let

$$\mathcal{T}_n := \{\omega : T(\omega) > n\}$$

be the event that the walk has not yet hit -1 by time n. For $0 \le k \le n$ define

$\mathcal{E}_k^n := \{\omega : X_0(\omega) = 0, \dots, X_n(\omega) = k\}$,

$\mathcal{H}_k^n := \{\omega : X_0(\omega) = 0, \dots, X_n(\omega) = k,\ \exists m \in \{1,\dots,n\} \text{ with } X_m(\omega) =$

Then

$$\mathcal{T}_n = \bigcup_{k=0}^{n}\big(\mathcal{E}_k^n \setminus \mathcal{H}_k^n\big),$$

a disjoint union. Specializing to even times and endpoints, we obtain

$$\mathbb{P}(T > 2n) = \sum_{k=0}^{n}\mathbb{P}\big(\mathcal{E}_{2k}^{2n} \setminus \mathcal{H}_{2k}^{2n}\big).$$

A standard reflection-principle argument (as in Lemma 5.2.4) gives

$$\mathbb{P}\big(\mathcal{E}_{2k}^{2n} \setminus \mathcal{H}_{2k}^{2n}\big) = \frac{1}{2^{2n}}\left(\binom{2n}{n+k} - \binom{2n}{n+k+1}\right),$$

so

$$\begin{aligned}\mathbb{P}(T > 2n) &= \sum_{k=0}^{n}\frac{1}{2^{2n}}\left(\binom{2n}{n+k} - \binom{2n}{n+k+1}\right)\\ &= \frac{1}{2^{2n}}\binom{2n}{n}.\end{aligned}$$

Using the standard bound

$$\binom{2n}{n} \le \frac{2^{2n}}{\sqrt{\pi n}},$$

we obtain

$$\mathbb{P}(T > 2n) \le \frac{1}{\sqrt{\pi n}} \xrightarrow[n\to\infty]{} 0.$$

Hence $\mathbb{P}(T < \infty) = 1$, so T is finite almost surely, and yet $\mathbb{E}(X_T) \ne \mathbb{E}(X_0)$.

Reflection Principle

The following elegant combinatorial lemma is often regarded as an example of the reflection principle.

Lemma 5.2.4. *Let $\mathcal{S}$ be the set of integer-valued sequences $(s_n)_{n\ge0}$ such that $|s_{n+1} - s_n| = 1$ for all $n \ge 0$. Fix integers k, l and m with $m \le \min\{k-1, l\}$. For $n \ge 1$ let $\mathcal{S}_{k,l,m}^n \subset \mathcal{S}$ be the set of sequences such that $s_0 = k$, $s_n = l$, and there exists some $q \in \{1,\dots,n\}$ with $s_q = m$. Then*

$$|\mathcal{S}_{k,l,m}^n| = \begin{cases} 0, & \textit{if } n - k + l \equiv 1 \pmod 2,\\ 0, & \textit{if } |l - k| > n,\\ \binom{n}{\frac{n+k+l-2m}{2}}, & \textit{otherwise.}\end{cases}$$

Proof. Let $\widehat{\mathcal{S}}^n_{k,l,m} \subset \mathcal{S}$ be the set of sequences with $s_0 = 2m - k$ and $s_n = l$. Since $2m - k < m \le l$, every such sequence must hit the level m at some time $q \in \{1, \dots, n\}$.

We construct a bijection between $\widehat{\mathcal{S}}^n_{k,l,m}$ and $\mathcal{S}^n_{k,l,m}$ as follows: given a sequence $(s_0, \dots, s_n) \in \widehat{\mathcal{S}}^n_{k,l,m}$, let q be the first time with $s_q = m$, and reflect the path about the level m up to time q, i.e. replace s_j by $2m - s_j$ for $0 \le j \le q$, leaving s_j unchanged for $j > q$. This produces a sequence in $\mathcal{S}^n_{k,l,m}$ with $s_0 = k$, $s_n = l$, and a first visit to m at time q. The map is clearly invertible by the same reflection, so it is a bijection.

Thus $|\mathcal{S}^n_{k,l,m}| = |\widehat{\mathcal{S}}^n_{k,l,m}|$. The latter is just the number of simple symmetric random walk paths from $2m - k$ to l in n steps, which is well known to be

$$\binom{n}{\frac{n+k+l-2m}{2}}$$

when the parity and range conditions are satisfied, and 0 otherwise. This gives the stated formula. □

5.2.3 Applications of martingales

Example 15 (Monkey typing). Consider a monkey typing characters on a 26-key keyboard. Assume that the characters typed are i.i.d. and each character is chosen uniformly at random. What is the expected time before the monkey types the word "WOWOWOW"?

Proof. The expected time is

$$\mathbb{E}(T) = 26^7 + 26^5 + 26^3 + 26.$$

A beautiful argument using martingales for such problems was developed by Li (1980).

Let $(X_n)_{n\ge1}$ denote the characters typed by the monkey. Define the stopping time

$$T := \inf\{n \ge 7 : (X_{n-6}, \dots, X_n) = (W, O, W, O, W, O, W)\}.$$

Clearly this stopping time satisfies the hypothesis of the lemma on $\mathbb{E}(T) < \infty$ with $N = 7$ and $\varepsilon = 26^{-7}$, so $\mathbb{E}(T) < \infty$.

For each $m \ge 1$ define a process $(Y^{(m)}_n)_{n\ge m-1}$ as follows:

$$Y^{(m)}_{m-1} := 1,$$

and for $l \ge 1$ set

$$Y^{(m)}_{m+l-1} := \begin{cases} 26^{\,l\wedge 7}, & \text{if } (X_m, \dots, X_{m+(l\wedge7)-1}) = \\ & (W, O, W, O, W, O, W)_{1:(l\wedge7)}, \\ 0, & \text{otherwise.} \end{cases}$$

In particular, once a mismatch occurs, $Y^{(m)}_n$ stays at 0. It is clear that $Y^{(m)}_n$ is $\mathcal{F}_n$-measurable, and a direct check using the uniform distribution of the letters shows that

$$\mathbb{E}\big(Y^{(m)}_{n+1} \mid \mathcal{F}_n\big) = Y^{(m)}_n, \qquad n \ge m - 1.$$

Moreover, for $n \ge m + 6$ we have $Y^{(m)}_{n+1} = Y^{(m)}_n$, and $Y^{(m)}_n \le 26^7$ for all n.

Now define

$$M_n := \sum_{m=1}^{n+1} Y^{(m)}_n - (n + 1), \qquad n \ge 0.$$

Then (M_n) is a martingale with $\mathbb{E}(M_n) = 0$ for all n (a zero-mean martingale). Furthermore,

$$\begin{aligned} M_n - M_{n-1} &= \sum_{m=1}^{n+1} Y^{(m)}_n - \sum_{m=1}^{n} Y^{(m)}_{n-1} \\ &= \sum_{m=1}^{n} \big(Y^{(m)}_n - Y^{(m)}_{n-1}\big) + Y^{(n+1)}_n. \end{aligned}$$

Only those m with $m \in \{n-6, \dots, n+1\}$ can contribute nonzero terms, and each $Y^{(m)}_n$ is bounded by 26^7. Hence

$$|M_n - M_{n-1}| \le C := 2 \cdot 7 \cdot 26^7$$

for all n (this bound can be improved, but is sufficient for our purposes).

Since T is a stopping time with $\mathbb{E}(T) < \infty$ and (M_n) has uniformly bounded increments, we may apply part (c) of Doob's optional stopping theorem to conclude that

$$\mathbb{E}(M_T) = \mathbb{E}(M_0) = 0,$$

that is,

$$\mathbb{E}\Big(\sum_{m=1}^{T+1} Y^{(m)}_T - (T + 1)\Big) = 0.$$

From the definition of T and of the processes $Y^{(m)}$, one checks that on the event $\{T < \infty\}$ we have

$$\begin{aligned} &Y^{(T-6)}_T = 26^7, \quad Y^{(T-4)}_T = 26^5, \\ &Y^{(T-2)}_T = 26^3, \quad Y^{(T)}_T = 26, \quad Y^{(T+1)}_T = 1, \end{aligned}$$

and all other $Y^{(m)}_T$ are zero. Hence

$$\sum_{m=1}^{T+1} Y^{(m)}_T = 26^7 + 26^5 + 26^3 + 26 + 1.$$

Taking expectations in the identity for M_T gives

$$\mathbb{E}\big(26^7 + 26^5 + 26^3 + 26 + 1 - (T + 1)\big) = 0,$$

so

$$\mathbb{E}(T) = 26^7 + 26^5 + 26^3 + 26.$$

□

Example 16 (Li's martingale for the pattern HH, simpler to understand). Let $(X_n)_{n\ge1}$ be i.i.d. fair coin tosses, with

$$\mathbb{P}(X_n = H) = \mathbb{P}(X_n = T) = \frac{1}{2}.$$

Define the stopping time

$$\tau := \inf\{n \ge 2 : (X_{n-1}, X_n) = (H, H)\},$$

the first time the pattern HH appears. We show that

$$\mathbb{E}[\tau] = 6.$$

Consider the following gambling scheme. At each time $t \ge 1$, a new gambler enters and pays \$1 to start betting that a copy of the word HH begins at time t.

- First, the gambler bets the entire \$1 on the event $\{X_t = H\}$.
- If this bet wins, then the \$1 becomes \$2, and the gambler immediately bets the entire \$2 on the event $\{X_{t+1} = H\}$.
- If the second bet also wins, then the gambler finishes with \$4.

Since each individual bet is fair, the casino's net gain process is a martingale. Equivalently, if Y_n denotes the total amount owed by the casino to all gamblers still alive after time n, then

$$M_n := Y_n - n$$

is a martingale, because by time n the casino has collected exactly \$1 from each of the n gamblers who have entered.

Now stop at time τ, the first occurrence of HH. At time τ:

- the gambler who started at time $\tau - 1$ has correctly guessed both letters of HH, hence is owed \$4;
- the gambler who started at time τ has correctly guessed the last H, hence is currently owed \$2.

No earlier gambler can still be alive, since that would imply that HH had already occurred before time τ. Therefore

$$Y_\tau = 4 + 2 = 6,$$

and so

$$M_\tau = 6 - \tau.$$

By optional stopping,

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0] = 0.$$

Hence

$$0 = \mathbb{E}[6 - \tau] = 6 - \mathbb{E}[\tau],$$

which gives

$$\mathbb{E}[\tau] = 6.$$

Remark 64. This example illustrates the basic idea of Li's martingale method: one turns the pattern-matching problem into a fair gambling game, and then reads off $\mathbb{E}[\tau]$ from the deterministic payoff at the stopping time.

The terminology "martingale" comes from the historical connection with gambling systems. In the present construction, after a gambler wins the first bet, they stake the entire current fortune on the next letter, which reflects that gambling origin. Historically, however, the classical betting system called a martingale refers more specifically to a doubling strategy after losses, rather than exactly this "bet everything after a win" rule.

5.3 Convergence theorems

Definition 5.3.1 (Upcrossing)**.** *Let $(X_n)_{n\geq 0}$ be a sequence of random variables and let $a < b$. The (up to time n) number of upcrossings of the interval $[a, b]$ is the nonnegative, nondecreasing integer-valued random variable*

$$U_n[a,b] := \max\Big\{k : \exists\, 0 \leq s_1 < t_1 < \cdots < s_k < t_k \leq n \text{ with } X_{s_i} \leq a,\ X_{t_i} > b\ \forall i\Big\}.$$

If (X_n) is adapted to a filtration $(\mathcal{F}_n)$, then $(U_n[a, b])$ is also adapted to $(\mathcal{F}_n)$.

Lemma 5.3.1 (Doob's upcrossing inequality)**.** *The following hold.*

1. *Let (X_n) be a supermartingale. Then for any $n \geq 1$ and $a < b$,*
$$(b-a)\,\mathbb{E}(U_n[a,b]) \leq \mathbb{E}\big((X_n - a)_-\big) \leq \mathbb{E}(|X_n|) + |a|.$$
2. *Let (X_n) be a submartingale. Then for any $n \geq 1$ and $a < b$,*
$$(b-a)\,\mathbb{E}(U_n[a,b]) \leq \mathbb{E}\big((X_n-a)_+\big) - \mathbb{E}\big((X_0-a)_+\big) \leq \mathbb{E}(|X_n|) + |a|.$$

Proof. 1. Define a non-negative bounded predictable process $(C_n)_{n\geq 1}$ inductively by

$$C_1 := \mathbf{1}_{\{X_0 \leq a\}},$$
$$C_n := \mathbf{1}_{\{C_{n-1}=1\}}\mathbf{1}_{\{X_{n-1}\leq b\}} + \mathbf{1}_{\{C_{n-1}=0\}}\mathbf{1}_{\{X_{n-1}\leq a\}}, \quad n \geq 2.$$

In words, (C_n) takes the value 1 starting from the time the process goes at or below a, and stays 1 until the time it first goes above b; then it becomes 0 again, and turns back to 1 only when the process returns to at or below a.

Define $Y_0 := 0$ and $Y_n := (C \circ X)_n$ for $n \geq 1$. Since (X_n) is a supermartingale and (C_n) is predictable, non-negative, and bounded, (Y_n) is also a supermartingale. Moreover, by construction,

$$Y_n \ \geq\ (b-a)\,U_n[a,b] - (X_n - a)_-.$$

Taking expectations and using $\mathbb{E}(Y_n) \leq \mathbb{E}(Y_0) = 0$ gives

$$(b-a)\,\mathbb{E}(U_n[a,b]) \leq \mathbb{E}\big((X_n - a)_-\big),$$

and the final bound follows from $(X_n - a)_- \leq |X_n| + |a|$.

2. Let $Z_n := (X_n - a)_+ + a$. Then (Z_n) is a submartingale and it has the same upcrossings of $[a, b]$ as (X_n). Define $Y_n := (C \circ Z)_n$ for $n \geq 1$ as above. As before,

$$Y_n \ \geq\ (b-a)\,U_n[a,b].$$

Similarly define $\widetilde{Y}_n := ((1 - C) \circ Z)_n$ for $n \geq 1$. One checks that $(\widetilde{Y}_n)$ is also a submartingale and $\mathbb{E}(\widetilde{Y}_n) \geq 0$ for all n. Note that

$$Y_n + \widetilde{Y}_n = Z_n - Z_0,$$

so

$$\begin{aligned}\mathbb{E}(Y_n) &= \mathbb{E}(Z_n - Z_0) - \mathbb{E}(\widetilde{Y}_n)\\ &\leq \mathbb{E}(Z_n - Z_0) = \mathbb{E}\big((X_n - a)_+\big) - \mathbb{E}\big((X_0 - a)_+\big).\end{aligned}$$

Combining this with $Y_n \geq (b - a)U_n[a, b]$ yields the desired inequality. The last bound follows from $(X_n - a)_+ \leq |X_n| + |a|$.

□

Theorem 5.3.2 (Martingale convergence theorem)**.** *Let (X_n) be a supermartingale or a submartingale such that*

$$\sup_{n\geq 0} \mathbb{E}(|X_n|) < \infty.$$

Then X_n converges almost surely to a finite limit.

Proof. First note that

$$\begin{aligned}&\Big\{\omega : \liminf_n X_n(\omega) < \limsup_n X_n(\omega)\Big\}\\ &= \bigcup_{\substack{a,b\in\mathbb{Q}\\ a<b}} \Big\{\omega : \liminf_n X_n(\omega) < a < b < \limsup_n X_n(\omega)\Big\}\\ &= \bigcup_{\substack{a,b\in\mathbb{Q}\\ a<b}} \Big\{\omega : U_\infty[a,b](\omega) = \infty\Big\},\end{aligned}$$

where $U_\infty[a, b] := \lim_{n\to\infty} U_n[a, b]$ exists in $[0, \infty]$ by monotonicity. By Doob's upcrossing inequality and the monotone convergence theorem,

$$(b-a)\,\mathbb{E}\big(U_\infty[a,b]\big) = (b-a)\lim_{n\to\infty}\mathbb{E}(U_n[a,b]) \leq \sup_n \mathbb{E}(|X_n|) + |a| < \infty.$$

Hence for each rational $a < b$,

$$\mathbb{P}\big(U_\infty[a,b] = \infty\big) = 0.$$

Therefore, with probability one, for every rational $a < b$ the total number of upcrossings of $[a, b]$ is finite. This implies that for almost every ω either $(X_n(\omega))$ converges to a finite limit or $|X_n(\omega)| \to \infty$.

Define

$$A := \{\omega : |X_n(\omega)| \to \infty\}.$$

On A we have $\liminf_{n\to\infty} |X_n|(\omega) = \infty$, so

$$\int \liminf_{n\to\infty} |X_n|\, d\mathbb{P} \ \geq\ \int_A \liminf_{n\to\infty} |X_n|\, d\mathbb{P} = \infty \cdot \mathbb{P}(A),$$

with the convention $\infty \cdot 0 = 0$.

On the other hand, by Fatou's lemma,

$$\int \liminf_{n\to\infty} |X_n|\, d\mathbb{P} \ \leq\ \liminf_{n\to\infty} \int |X_n|\, d\mathbb{P} \ \leq\ \sup_n \mathbb{E}(|X_n|) < \infty.$$

Hence $\infty \cdot \mathbb{P}(A) < \infty$, which forces $\mathbb{P}(A) = 0$. Thus

$$\mathbb{P}\big(|X_n| \to \infty\big) = 0,$$

and so X_n converges almost surely to a finite limit. □

Corollary 5.3.3. *If (X_n) is a nonnegative supermartingale, then*

$$\mathbb{E}(|X_n|) = \mathbb{E}(X_n) \leq \mathbb{E}(X_0) < \infty \quad \text{for all } n,$$

and therefore (X_n) converges almost surely to a finite limit.

Theorem 5.3.4. *Let $(X_n)_{n\geq 0}$ be a martingale with bounded increments, i.e.*

$$|X_n - X_{n-1}| \leq K \quad \text{for all } n \geq 1 \text{ and all } \omega,$$

for some constant $K > 0$. Define

$$C := \big\{\omega : \lim_n X_n(\omega) \text{ exists and is finite}\big\},$$
$$D := \big\{\omega : \liminf_n X_n(\omega) = -\infty \text{ and } \limsup_n X_n(\omega) = +\infty\big\}.$$

Then $\mathbb{P}(C \cup D) = 1$.

Proof. Since $(X_n - X_0)$ is also a martingale with bounded increments, we may assume without loss of generality that $X_0 = 0$.

Fix $M > 0$ and define the stopping time

$$T_M := \inf\{n \geq 0 : X_n < -M\}.$$

Then $(X_{n\wedge T_M} + M + K)_{n\geq 0}$ is a nonnegative martingale (adding the constant $M + K$ simply shifts the process up). By Corollary 5.3.3, this process converges almost surely to a finite limit. In particular, on the event $\{T_M = \infty\}$ the sequence (X_n) itself converges almost surely to a finite limit, since $X_{n\wedge T_M} = X_n$ there.

Letting $M \to \infty$ along the integers, we see that (X_n) has a finite limit almost surely on the event

$$\Big\{\omega : \inf_n X_n(\omega) > -\infty\Big\}.$$

Applying the same argument to $(-X_n)$ shows that (X_n) has a finite limit almost surely on

$$\Big\{\omega : \sup_n X_n(\omega) < +\infty\Big\}.$$

Thus, with probability one, either (X_n) converges to a finite limit (this is the event C), or

$$\inf_n X_n = -\infty \quad \text{and} \quad \sup_n X_n = +\infty,$$

which is precisely the event D. Hence $\mathbb{P}(C \cup D) = 1$. □

Theorem 5.3.5. *Let $(X_n)_{n\geq 0}$ be a martingale adapted to $(\mathcal{F}_n)_{n\geq 0}$. If*

$$\sup_n \mathbb{E}(X_n^2) < \infty,$$

then there exists an integrable random variable X_∞ such that

$$X_n \to X_\infty \quad \text{a.s. and in } L^2,$$

i.e. $\mathbb{E}\big((X_n - X_\infty)^2\big) \to 0$. Moreover $\mathbb{E}(X_\infty^2) < \infty$ and

$$X_n = \mathbb{E}(X_\infty \mid \mathcal{F}_n) \quad \text{a.s. for all } n.$$

Proof. Since $\sup_n \mathbb{E}(X_n^2) < \infty$, we also have $\sup_n \mathbb{E}(|X_n|) < \infty$, so Doob's martingale convergence theorem implies that $X_n \to X_\infty$ a.s. for some finite X_∞. By Fatou's lemma,

$$\mathbb{E}(X_\infty^2) \leq \liminf_{n\to\infty} \mathbb{E}(X_n^2) \leq \sup_n \mathbb{E}(X_n^2) < \infty.$$

Without loss of generality we may assume $X_0 = 0$ (otherwise replace X_n by $X_n - X_0$). Define the martingale differences

$$Y_n := X_n - X_{n-1}, \quad n \geq 1.$$

Then

$$\begin{aligned}\mathbb{E}(Y_n X_{n-1}) &= \mathbb{E}\big(X_{n-1}(X_n - X_{n-1})\big)\\ &= \mathbb{E}\big(\mathbb{E}(X_{n-1}(X_n - X_{n-1}) \mid \mathcal{F}_{n-1})\big).\end{aligned}$$

Since X_{n-1} is $\mathcal{F}_{n-1}$-measurable and X_n is integrable,

$$\mathbb{E}(X_n X_{n-1} \mid \mathcal{F}_{n-1}) = X_{n-1}\,\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) = X_{n-1}^2 \quad \text{a.s.},$$

so

$$\mathbb{E}(Y_n X_{n-1}) = \mathbb{E}(X_{n-1}X_n) - \mathbb{E}(X_{n-1}^2) = 0.$$

Thus (Y_n) is orthogonal in L^2, and by induction,

$$\mathbb{E}(X_n^2) = \sum_{m=1}^{n} \mathbb{E}(Y_m^2).$$

Since $\sup_n \mathbb{E}(X_n^2) < \infty$, the series $\sum_{m\geq 1} \mathbb{E}(Y_m^2)$ converges and

$$\sum_{m=n+1}^{\infty} \mathbb{E}(Y_m^2) \xrightarrow[n\to\infty]{} 0.$$

For $m > n$ we have

$$\mathbb{E}\big((X_m - X_n)^2\big) = \mathbb{E}\Big(\Big(\sum_{k=n+1}^{m} Y_k\Big)^2\Big) = \sum_{k=n+1}^{m} \mathbb{E}(Y_k^2),$$

again by orthogonality. Letting $m \to \infty$ and using Fatou's lemma gives

$$\mathbb{E}\big((X_\infty - X_n)^2\big) \leq \lim_{m\to\infty} \mathbb{E}\big((X_m - X_n)^2\big) = \sum_{k=n+1}^{\infty} \mathbb{E}(Y_k^2) \xrightarrow[n\to\infty]{} 0.$$

Thus $X_n \to X_\infty$ in L^2.

Finally, for any $m > n$,

$$\mathbb{E}(X_\infty - X_m \mid \mathcal{F}_n) = \mathbb{E}(X_\infty \mid \mathcal{F}_n) - X_n \quad \text{a.s.},$$

since (X_n) is a martingale. As $\mathbb{E}\big((X_\infty - X_m)^2\big) \to 0$, we get $X_\infty - X_m \to 0$ in L^2 and hence in L^1, so

$$\begin{aligned}&\mathbb{E}\big(|\mathbb{E}(X_\infty \mid \mathcal{F}_n) - X_n|\big)\\ &\leq \mathbb{E}\big(\mathbb{E}(|X_\infty - X_m| \mid \mathcal{F}_n)\big)\\ &= \mathbb{E}(|X_\infty - X_m|) \xrightarrow[m\to\infty]{} 0.\end{aligned}$$

Therefore $\mathbb{E}(X_\infty \mid \mathcal{F}_n) = X_n$ a.s. □

Lemma 5.3.6. *Let (X_n) and X be integrable random variables. If*

1. *$X_n \to X$ in probability, and*
2. *$\{X_n\}$ is uniformly integrable,*

then $\mathbb{E}(|X_n - X|) \to 0$.

Proof. For $K > 0$ define the truncations

$$Y_n^K := X_n \mathbf{1}_{\{|X_n|\leq K\}} + \operatorname{sgn}(X_n)\, K\, \mathbf{1}_{\{|X_n|>K\}},$$
$$Y^K := X \mathbf{1}_{\{|X|\leq K\}} + \operatorname{sgn}(X)\, K\, \mathbf{1}_{\{|X|>K\}}.$$

Then $|Y_n^K - Y^K| \leq |X_n - X|$, and $Y_n^K \to Y^K$ in probability (as truncation is continuous and bounded). Since Y_n^K, Y^K are uniformly bounded by K, the bounded convergence theorem implies

$$\mathbb{E}\big(|Y_n^K - Y^K|\big) \xrightarrow[n\to\infty]{} 0.$$

Uniform integrability of (X_n) gives

$$\sup_n \mathbb{E}(|X_n - Y_n^K|) \leq \sup_n \mathbb{E}(|X_n|\mathbf{1}_{\{|X_n|>K\}}) \xrightarrow[K\to\infty]{} 0.$$

Also $\mathbb{E}(|X - Y^K|) = \mathbb{E}(|X|\mathbf{1}_{\{|X|>K\}}) \to 0$ as $K \to \infty$ by integrability of X. Therefore

$$\begin{aligned}&\limsup_{n\to\infty} \mathbb{E}(|X_n - X|)\\ &\leq \limsup_{n\to\infty} \mathbb{E}(|X_n - Y_n^K|) + \limsup_{n\to\infty} \mathbb{E}(|Y_n^K - Y^K|) + \mathbb{E}(|X - Y^K|)\\ &\leq \sup_n \mathbb{E}(|X_n|\mathbf{1}_{\{|X_n|>K\}}) + \mathbb{E}(|X|\mathbf{1}_{\{|X|>K\}}).\end{aligned}$$

Letting $K \to \infty$ shows that $\limsup_n \mathbb{E}(|X_n - X|) = 0$, hence $\mathbb{E}(|X_n - X|) \to 0$. □

Theorem 5.3.7. *Let $(X_n)_{n\geq 0}$ be a uniformly integrable martingale adapted to $(\mathcal{F}_n)_{n\geq 0}$. Then there exists an integrable X_∞ such that*

$$X_n \to X_\infty \quad \text{a.s. and in } L^1,$$

i.e. $\mathbb{E}(|X_n - X_\infty|) \to 0$. Moreover $\mathbb{E}(|X_\infty|) < \infty$ and

$$X_n = \mathbb{E}(X_\infty \mid \mathcal{F}_n) \quad \text{a.s. for all } n.$$

Proof. Uniform integrability implies $\sup_n \mathbb{E}(|X_n|) < \infty$. By the martingale convergence theorem, $X_n \to X_\infty$ a.s. for some finite X_∞. Fatou's lemma yields $\mathbb{E}(|X_\infty|) < \infty$. Applying Lemma 5.3.6

with $X_n \to X_\infty$ in probability and uniform integrability gives $\mathbb{E}(|X_n - X_\infty|) \to 0$.

For $m > n$,

$$\mathbb{E}(X_\infty - X_m \mid \mathcal{F}_n) = \mathbb{E}(X_\infty \mid \mathcal{F}_n) - X_n \quad \text{a.s.}$$

Taking expectations of absolute values and using the tower property,

$$\begin{aligned}
&\mathbb{E}\big(|\mathbb{E}(X_\infty \mid \mathcal{F}_n) - X_n|\big) \\
&\le \mathbb{E}\big(\mathbb{E}(|X_\infty - X_m| \mid \mathcal{F}_n)\big) \\
&= \mathbb{E}(|X_\infty - X_m|) \xrightarrow[m\to\infty]{} 0.
\end{aligned}$$

Hence $\mathbb{E}(X_\infty \mid \mathcal{F}_n) = X_n$ a.s. for each n. □

Theorem 5.3.8. *Fix $p > 1$. Let $(X_n)_{n\ge0}$ be a martingale adapted to $(\mathcal{F}_n)_{n\ge0}$. If*

$$\sup_n \mathbb{E}(|X_n|^p) < \infty,$$

then there exists a random variable X_∞ such that

$$X_n \to X_\infty \quad \textit{a.s. and in } L^p,$$

i.e. $\|X_n - X_\infty\|_p \to 0$. Moreover $\mathbb{E}(|X_\infty|^p) < \infty$ and

$$X_n = \mathbb{E}(X_\infty \mid \mathcal{F}_n) \quad \textit{a.s. for all } n.$$

Proof. The bound $\sup_n \mathbb{E}(|X_n|^p) < \infty$ implies uniform integrability, so by the martingale convergence theorem there exists X_∞ such that $X_n \to X_\infty$ a.s. Fatou's lemma yields

$$\mathbb{E}(|X_\infty|^p) \le \liminf_{n\to\infty} \mathbb{E}(|X_n|^p) \le \sup_n \mathbb{E}(|X_n|^p) < \infty.$$

By the L^1-version of the martingale convergence theorem, we also have $\mathbb{E}(|X_n - X_\infty|) \to 0$ and $X_n = \mathbb{E}(X_\infty \mid \mathcal{F}_n)$ a.s.

To upgrade to L^p convergence, fix $K > 0$ and define the truncation

$$Y_\infty^K := X_\infty \mathbf{1}_{\{|X_\infty|\le K\}} + \operatorname{sgn}(X_\infty)\, K\, \mathbf{1}_{\{|X_\infty|>K\}}.$$

Set

$$\widehat{Y}_n^K := \mathbb{E}(Y_\infty^K \mid \mathcal{F}_n), \quad n \ge 0.$$

By Jensen's inequality for conditional expectation,

$$\|X_n - \widehat{Y}_n^K\|_p = \|\mathbb{E}(X_\infty - Y_\infty^K \mid \mathcal{F}_n)\|_p \le \|X_\infty - Y_\infty^K\|_p.$$

Hence

$$\|X_n - X_\infty\|_p \le \|X_n - \widehat{Y}_n^K\|_p + \|X_\infty - Y_\infty^K\|_p + \|Y_\infty^K - \widehat{Y}_n^K\|_p \le 2\|X_\infty - Y_\infty^K\|_p + \|Y_\infty^K - \widehat{Y}_n^K\|_p.$$

For fixed K, the random variable Y_∞^K is bounded, so $(\widehat{Y}_n^K)$ is a bounded martingale. By the L^2 martingale convergence theorem, there exists $\widehat{Y}_\infty^K$ such that

$$\widehat{Y}_n^K \to \widehat{Y}_\infty^K \quad \text{a.s. and in } L^2,$$

and in particular in L^p since $p \le 2$ or using boundedness to bounf a $p-2$ power if $p > 2$ and we have a uniform L^p bound. Moreover,

$$\mathbb{E}(\widehat{Y}_\infty^K \mid \mathcal{F}_n) = \lim_{m\to\infty} \mathbb{E}(\widehat{Y}_m^K \mid \mathcal{F}_n) = \lim_{m\to\infty} \widehat{Y}_n^K = \widehat{Y}_n^K \quad \text{a.s.},$$

so $(\widehat{Y}_n^K)$ is the martingale with terminal value $\widehat{Y}_\infty^K$.

We now show that $\widehat{Y}_\infty^K = Y_\infty^K$ a.s. For each n and each $A \in \mathcal{F}_n$,

$$\int_A Y_\infty^K \, d\mathbb{P} = \int_A \mathbb{E}(Y_\infty^K \mid \mathcal{F}_n)\, d\mathbb{P} = \int_A \widehat{Y}_n^K \, d\mathbb{P}.$$

Letting $n \to \infty$ and using L^1 convergence $\widehat{Y}_n^K \to \widehat{Y}_\infty^K$, we obtain

$$\int_A Y_\infty^K \, d\mathbb{P} = \int_A \widehat{Y}_\infty^K \, d\mathbb{P} \quad \text{for all } A \in \bigcup_n \mathcal{F}_n.$$

The collection of all A for which this equality holds is a monotone class, and it contains the algebra $\bigcup_n \mathcal{F}_n$. Since $Y_\infty^K, \widehat{Y}_\infty^K$ are integrable and $\sigma(\bigcup_n \mathcal{F}_n)$-measurable, the monotone class theorem implies that

$$\int_A Y_\infty^K \, d\mathbb{P} = \int_A \widehat{Y}_\infty^K \, d\mathbb{P} \quad \text{for all } A \in \sigma\Big(\bigcup_n \mathcal{F}_n\Big),$$

hence $Y_\infty^K = \widehat{Y}_\infty^K$ a.s. In particular,

$$\|Y_\infty^K - \widehat{Y}_n^K\|_p \xrightarrow[n\to\infty]{} 0.$$

Putting everything together, for each fixed K,

$$\limsup_{n\to\infty} \|X_n - X_\infty\|_p \le 2\|X_\infty - Y_\infty^K\|_p.$$

Since $|Y_\infty^K| \le |X_\infty|$ and $Y_\infty^K \to X_\infty$ pointwise as $K \to \infty$, dominated convergence in L^p gives $\|X_\infty - Y_\infty^K\|_p \to 0$ as $K \to \infty$. Hence

$$\lim_{n\to\infty} \|X_n - X_\infty\|_p = 0,$$

as claimed. □

Theorem 5.3.9 (Doob's decomposition)**.** *Let $(X_n)_{n\ge0}$ be a process adapted to $(\mathcal{F}_n)_{n\ge0}$. Then there exist processes $(M_n)_{n\ge0}$ and $(A_n)_{n\ge0}$ such that*

$$X_n = X_0 + M_n + A_n, \quad n \ge 0,$$

where (M_n) is a martingale with $M_0 = 0$ and (A_n) is predictable, i.e. $A_0 = 0$ and A_n is $\mathcal{F}_{n-1}$-measurable for $n \ge 1$. This decomposition is unique up to almost sure equality. Moreover, (X_n) is a submartingale if and only if (A_n) is almost surely nondecreasing.

Proof. Suppose such a decomposition exists. Then

$$X_n - X_{n-1} = (M_n - M_{n-1}) + (A_n - A_{n-1}),$$

and taking conditional expectations with respect to $\mathcal{F}_{n-1}$ gives $\mathbb{E}(X_n - X_{n-1} \mid \mathcal{F}_{n-1}) = \mathbb{E}(M_n - M_{n-1} \mid \mathcal{F}_{n-1}) + (A_n - A_{n-1}) = A_n - A_n$ since (M_n) is a martingale. Thus necessarily

$$A_n = \sum_{k=1}^n \mathbb{E}(X_k - X_{k-1} \mid \mathcal{F}_{k-1}), \quad n \ge 1,$$

with $A_0 = 0$, and A_n is predictable by construction.

Define

$$M_n := X_n - X_0 - A_n, \quad n \ge 0.$$

Then $M_0 = 0$, M_n is $\mathcal{F}_n$-measurable, and

$$\begin{aligned}
\mathbb{E}(M_n \mid \mathcal{F}_{n-1}) &= \mathbb{E}(X_n - X_0 - A_n \mid \mathcal{F}_{n-1}) \\
&= X_{n-1} - X_0 - A_{n-1} = M_{n-1},
\end{aligned}$$

so (M_n) is a martingale. This proves existence.

For uniqueness, suppose

$$X_n = X_0 + M_n + A_n = X_0 + \widetilde{M}_n + \widetilde{A}_n,$$

with $(M_n), (\widetilde{M}_n)$ martingales (null at zero) and $(A_n), (\widetilde{A}_n)$ predictable. Then

$$M_n - \widetilde{M}_n = \widetilde{A}_n - A_n.$$

The left-hand side is a martingale, the right-hand side is predictable. Taking conditional expectations with respect to $\mathcal{F}_{n-1}$,

$$M_{n-1} - \widetilde{M}_{n-1} = \mathbb{E}(M_n - \widetilde{M}_n \mid \mathcal{F}_{n-1}) = \widetilde{A}_n - A_n,$$

so $\widetilde{A}_n - A_n$ is both $\mathcal{F}_{n-1}$- and $\mathcal{F}_n$-measurable; iterating this argument and using $M_0 = \widetilde{M}_0 = 0$, we conclude $M_n - \widetilde{M}_n$ is constant a.s., hence zero a.s., and thus $A_n = \widetilde{A}_n$ a.s. as well. This gives uniqueness up to almost sure equality.

Finally, (X_n) is a submartingale if and only if

$$\mathbb{E}(X_n \mid \mathcal{F}_{n-1}) \ge X_{n-1} \quad \text{a.s.},$$

i.e.

$$\mathbb{E}(M_n + A_n \mid \mathcal{F}_{n-1}) \ge M_{n-1} + A_{n-1}.$$

Using the martingale property of (M_n) and predictability of (A_n),

$$M_{n-1} + A_n \ge M_{n-1} + A_{n-1} \quad \text{a.s.},$$

which is equivalent to $A_n \ge A_{n-1}$ a.s. Thus (X_n) is a submartingale if and only if (A_n) is almost surely nondecreasing. □

Definition 5.3.2 (Angle-bracket (predictable quadratic variation))**.** *Let $(X_n)_{n\ge0}$ be a square-integrable martingale with $X_0 = 0$. Then (X_n^2) is a submartingale, and its Doob decomposition yields a predictable process $(A_n)_{n\ge0}$ such that*

$$X_n^2 - A_n, \quad n \ge 0,$$

is a martingale. We write

$$A_n := \langle X\rangle_n, \quad n \ge 0,$$

and call $(\langle X\rangle_n)$ *the* angle-bracket process *of* (X_n).

Note that if $\mathbb{E}(\langle X\rangle_\infty) < \infty$, then $\sup_n \mathbb{E}(X_n^2) \le \mathbb{E}(\langle X\rangle_\infty) < \infty$ and (X_n) converges almost surely and in L^2.

Theorem 5.3.10. *Let* $(X_n)_{n\ge 0}$ *be a square-integrable martingale with* $X_0 = 0$ *and* $\langle X\rangle_n = A_n$ *its angle-bracket process. Then:*

(i) If $A_\infty(\omega) < \infty$*, then* $\lim_n X_n(\omega)$ *exists (except possibly on a null set).*

(ii) If (X_n) *has uniformly bounded increments, then the converse holds: if* $\lim_n X_n(\omega)$ *exists and is finite, then* $A_\infty(\omega) < \infty$ *(except possibly on a null set).*

Proof. (i) For $k \in \mathbb{N}$ define the stopping time

$$S(k) := \inf\{n \ge 0 : A_{n+1} > k\}.$$

Note that

$$A_{n\wedge S(k)} = A_{S(k)}\,\mathbf{1}_{\{S(k)\le n-1\}} + A_n\,\mathbf{1}_{\{S(k)>n-1\}},$$

so $(A_{n\wedge S(k)})$ is predictable. Moreover,

$$X_{n\wedge S(k)}^2 - A_{n\wedge S(k)}, \quad n \ge 0,$$

is a martingale, and $A_{n\wedge S(k)} \le k$ for all n. Hence

$$\sup_n \mathbb{E}\big(X_{n\wedge S(k)}^2\big) \le \mathbb{E}(A_{S(k)}) \le k < \infty,$$

so by the L^2 martingale convergence theorem, $(X_{n\wedge S(k)})$ converges almost surely.

On the event $\{S(k) = \infty\}$ we have $A_{n+1} \le k$ for all n, so $A_\infty \le k$ there, and $X_{n\wedge S(k)} = X_n$. Thus X_n converges almost surely on $\{S(k) = \infty\}$. If $A_\infty(\omega) < \infty$, then $S(k)(\omega) = \infty$ for all sufficiently large k, so $\lim_n X_n(\omega)$ exists whenever $A_\infty(\omega) < \infty$.

(ii) Suppose now that (X_n) has uniformly bounded increments: $|X_n - X_{n-1}| \le K$ for all n. For $k \in \mathbb{N}$ define the stopping time

$$T(k) := \inf\{n \ge 0 : |X_n| > k\}.$$

Then on $\{T(k) = \infty\}$ we have $|X_n| \le k$ for all n. Moreover, for each n,

$$X_{n\wedge T(k)}^2 - A_{n\wedge T(k)}$$

is a martingale starting at 0, so

$$\mathbb{E}\big(X_{n\wedge T(k)}^2\big) = \mathbb{E}\big(A_{n\wedge T(k)}\big).$$

By bounded increments and the definition of $T(k)$,

$$|X_{n\wedge T(k)}| \le |X_{T(k)}| \le k + K \quad \text{on } \{T(k) \le n\},$$

and $|X_{n\wedge T(k)}| \le k$ on $\{T(k) > n\}$, so

$$\mathbb{E}\big(X_{n\wedge T(k)}^2\big) \le (k+K)^2.$$

Hence

$$\mathbb{E}\big(A_{n\wedge T(k)}\big) \le (k+K)^2 \quad \text{for all } n,$$

and by monotone convergence

$$\mathbb{E}\big(A_\infty \mathbf{1}_{\{T(k)=\infty\}}\big) = \lim_{n\to\infty} \mathbb{E}\big(A_{n\wedge T(k)}\big) \le (k+K)^2 < \infty.$$

Thus $A_\infty < \infty$ almost surely on $\{T(k) = \infty\}$.

If $\lim_n X_n(\omega)$ exists and is finite, then in particular the path is bounded, so $|X_n(\omega)| \le k$ for some k and all sufficiently large n. Hence $\omega \in \{T(k) = \infty\}$ for some k, and so $A_\infty(\omega) < \infty$. □

Theorem 5.3.11 (Lévy's extension of the Borel–Cantelli lemmas)**.** *Let* $(\mathcal{F}_n)_{n\ge 0}$ *be a filtration and let* $G_n \in \mathcal{F}_n$*. Define*

$$Z_n := \sum_{k=1}^n \mathbf{1}_{G_k}, \qquad B_n := \mathbb{E}(\mathbf{1}_{G_n} \mid \mathcal{F}_{n-1}), \qquad Y_n := \sum_{k=1}^n B_k.$$

Then, almost surely,

- *if* $Y_\infty(\omega) < \infty$*, then* $Z_\infty(\omega) < \infty$ *(except possibly on a set of measure zero);*
- *if* $Y_\infty(\omega) = \infty$*, then* $Z_n(\omega)/Y_n(\omega) \to 1$ *(except possibly on a set of measure zero).*

Proof. Observe that

$$\mathbb{E}(Z_n - Y_n \mid \mathcal{F}_{n-1}) = Z_{n-1} - Y_{n-1},$$

so $(M_n)_{n\ge 0}$ defined by $M_n := Z_n - Y_n$ with $M_0 = 0$ is a martingale. Let $A_n := \langle M\rangle_n$ be its angle-bracket process. Then

$$\begin{aligned} A_n &= \sum_{k=1}^n \mathbb{E}(M_k^2 - M_{k-1}^2 \mid \mathcal{F}_{k-1}) = \sum_{k=1}^n \mathbb{E}\big((M_k - M_{k-1})^2 \mid \mathcal{F}_{k-1}\big) \\ &= \sum_{k=1}^n \mathbb{E}\big((\mathbf{1}_{G_k} - B_k)^2 \mid \mathcal{F}_{k-1}\big) = \sum_{k=1}^n B_k(1-B_k) \le \sum_{k=1}^n B_k = Y_n. \end{aligned}$$

If $Y_\infty(\omega) < \infty$, then (almost surely) $A_\infty(\omega) \le Y_\infty(\omega) < \infty$, and by the previous theorem, $M_n(\omega)$ converges almost surely. Hence $Z_n(\omega) = M_n(\omega) + Y_n(\omega)$ converges almost surely to a finite limit, so $Z_\infty(\omega) < \infty$.

For the second part, if $A_\infty(\omega) < \infty$ then $M_n(\omega)$ converges, and so if $Y_\infty = \infty$ we trivially have $M_n(\omega)/Y_n(\omega) \to 0$ and hence $Z_n(\omega)/Y_n(\omega) \to 1$. Thus the interesting case is when $A_\infty(\omega) = \infty$.

Define the martingale transform

$$U_n := \sum_{k=1}^n \frac{M_k - M_{k-1}}{1 + A_k}, \quad U_0 := 0.$$

Since (M_n) is a martingale and $(1 + A_k)^{-1}$ is predictable and bounded, (U_n) is a martingale. We claim that (U_n) is square-integrable and that $\langle U\rangle_n$ is uniformly bounded.

The angle-bracket of (U_n) is

$$\begin{aligned} \langle U\rangle_n &= \sum_{k=1}^n \mathbb{E}\big((U_k - U_{k-1})^2 \mid \mathcal{F}_{k-1}\big) \\ &= \sum_{k=1}^n \mathbb{E}\left(\frac{(M_k - M_{k-1})^2}{(1+A_k)^2} \,\middle|\, \mathcal{F}_{k-1}\right) \\ &= \sum_{k=1}^n \frac{A_k - A_{k-1}}{(1+A_k)^2} \\ &\le \sum_{k=1}^n \left(\frac{A_k}{1+A_k} - \frac{A_{k-1}}{1+A_{k-1}}\right) = \frac{A_n}{1+A_n} \le 1. \end{aligned}$$

Further, $\mathbb{E}(U_n^2) = \mathbb{E}(\langle U\rangle_n) \le 1$. By the L^2 martingale convergence theorem, (U_n) converges almost surely to a finite limit.

Now write

$$\frac{M_n}{1+A_n} = \frac{1}{1+A_n}\sum_{k=1}^n (M_k - M_{k-1}) = \sum_{k=1}^n \frac{1+A_k}{1+A_n}\cdot\frac{M_k - M_{k-1}}{1+A_k}.$$

Set

$$x_k := \frac{M_k - M_{k-1}}{1+A_k}, \qquad b_k := 1 + A_k.$$

Then $U_n = \sum_{k=1}^n x_k$ converges almost surely, and b_k is increasing with $b_k \to \infty$ on $\{A_\infty = \infty\}$. By Kronecker's lemma,

$$\frac{1}{b_n}\sum_{k=1}^n b_k x_k \xrightarrow[n\to\infty]{} 0 \quad \text{a.s. on } \{A_\infty = \infty\},$$

i.e. $M_n/(1 + A_n) \to 0$ almost surely on $\{A_\infty = \infty\}$. Since $A_n \le Y_n$, and we are restricted to the set $\{Y_\infty = \infty\}$,

$$\frac{M_n}{1+Y_n} \le \frac{M_n}{1+A_n} \to 0.$$

Observe that

$$\frac{M_n}{1+Y_n} = \frac{Y_n}{1+Y_n}\left(\frac{Z_n}{Y_n} - 1\right).$$

On the set $\{Y_\infty = \infty\}$, $\frac{Y_n}{1+Y_n} \to 1$ as $n \to \infty$, implying that $\frac{Z_n(\omega)}{Y_n(\omega)} \to 1$ as $n \to \infty$, completing the proof. $\square$

Lemma 5.3.12 (Kronecker)**.** *Let $\sum_{m=1}^\infty x_m$ be a convergent series and let $(b_n)_{n\geq 1}$ be a sequence of positive numbers with $b_n \uparrow \infty$. Then*

$$\frac{\sum_{k=1}^n b_k x_k}{b_n} \xrightarrow[n\to\infty]{} 0.$$

Proof. Let $s_n := \sum_{m=1}^n x_m$ with $s_n \to s$ and $s_0 := 0$. Then

$$x_k = s_k - s_{k-1},$$

so

$$\begin{aligned}\sum_{k=1}^n \frac{b_k x_k}{b_n} &= \sum_{k=1}^n \frac{b_k(s_k - s_{k-1})}{b_n} \\ &= \frac{b_n s_n}{b_n} - \sum_{k=1}^{n-1} \frac{(b_{k+1} - b_k)s_k}{b_n} \\ &= s_n - s - \sum_{k=1}^{n-1} \frac{(b_{k+1} - b_k)(s_k - s)}{b_n}.\end{aligned}$$

Given $\varepsilon > 0$ there exists n_0 such that $|s_k - s| \leq \varepsilon$ for all $k \geq n_0$. For $n > n_0$,

$$\begin{aligned}\left|\sum_{k=1}^n \frac{b_k x_k}{b_n}\right| &\leq |s_n - s| + \sum_{k=1}^{n-1} \frac{(b_{k+1} - b_k)|s_k - s|}{b_n} + \left|\frac{b_1 s}{b_n}\right| \\ &\leq \varepsilon + \frac{1}{b_n}\sum_{k=1}^{n_0}(b_{k+1} - b_k)|s_k - s| \\ &\quad + \frac{1}{b_n}\sum_{k=n_0+1}^{n-1}(b_{k+1} - b_k)\varepsilon + \left|\frac{b_1 s}{b_n}\right| \\ &\leq \varepsilon + \frac{1}{b_n}\sum_{k=1}^{n_0}(b_{k+1} - b_k)|s_k - s| + \varepsilon\,\frac{b_n - b_{n_0}}{b_n} + \left|\frac{b_1 s}{b_n}\right|.\end{aligned}$$

Taking $\limsup_{n\to\infty}$ and using $b_n \to \infty$ yields

$$\limsup_{n\to\infty}\left|\sum_{k=1}^n \frac{b_k x_k}{b_n}\right| \leq 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the result follows. $\square$

Theorem 5.3.13 (Kakutani's theorem for product martingales)**.** *Let $X_1, X_2, \ldots$ be independent nonnegative random variables with $\mathbb{E}(X_n) = 1$ for all n. Define $M_0 := 1$ and*

$$M_n := X_1 X_2 \cdots X_n, \quad n \geq 1.$$

Then (M_n) is a nonnegative martingale, so

$$M_\infty := \lim_{n\to\infty} M_n$$

exists almost surely.

Define $a_n := \mathbb{E}(\sqrt{X_n})$. The following five statements are equivalent:

(a) $\displaystyle\sum_{n=1}^\infty (1 - a_n) < \infty.$

(b) $\displaystyle\prod_{n=1}^\infty a_n > 0.$

(c) (M_n) is uniformly integrable.

(d) $M_n \to M_\infty$ in L^1.

(e) $\mathbb{E}(M_\infty) = 1.$

If one of the statements fails, then $\mathbb{P}(M_\infty = 0) = 1$.

Proof. Note that $0 < a_n \leq 1$, the upper bound by Jensen's inequality applied to the concave function $x \mapsto \sqrt{x}$.

We first show $(a) \iff (b)$, then $(b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (e) \Rightarrow (b)$, and finally that failure of any of these implies $\mathbb{P}(M_\infty = 0) = 1$.

$(a) \iff (b)$: Since $e^{-x} \geq 1 - x$ for all $x \geq 0$,

$$\prod_{n=1}^\infty a_n \leq \exp\Big(-\sum_{n=1}^\infty (1 - a_n)\Big).$$

Thus if $\prod_n a_n > 0$ then necessarily $\sum_n (1 - a_n) < \infty$, so $(b) \Rightarrow (a)$.

Conversely, suppose $\sum_n (1 - a_n) = B < \infty$. Set

$$\mathcal{I} := \{n : a_n \leq 1/2\}.$$

Then

$$\sum_{n\in\mathcal{I}} (1 - a_n) \geq \sum_{n\in\mathcal{I}} \frac{1}{2} = \frac{|\mathcal{I}|}{2} \leq B,$$

so $|\mathcal{I}| \leq 2B < \infty$. Let $a_* := \min_{n\in\mathcal{I}} a_n$ (if $\mathcal{I}$ is empty, interpret the product over $\mathcal{I}$ as 1). Then

$$\prod_{n\in\mathcal{I}} a_n \geq a_*^{|\mathcal{I}|} > 0.$$

For $n \notin \mathcal{I}$ we have $a_n > 1/2$, so $0 \leq 1 - a_n \leq 1/2$. On $[0, 1/2]$ we have $e^{-2x} \leq 1 - x$, hence

$$a_n \geq e^{-2(1-a_n)} \quad \text{for } n \notin \mathcal{I}.$$

Therefore

$$\prod_{n\notin\mathcal{I}} a_n \geq \exp\Big(-2\sum_{n\notin\mathcal{I}} (1 - a_n)\Big) > 0.$$

Combining both parts,

$$\prod_{n=1}^\infty a_n = \Big(\prod_{n\in\mathcal{I}} a_n\Big)\Big(\prod_{n\notin\mathcal{I}} a_n\Big) > 0,$$

so $(a) \Rightarrow (b)$.

$(b) \Rightarrow (c)$: Define

$$b_n := \prod_{k=1}^n a_k, \qquad Y_n := \frac{\sqrt{X_1 \cdots X_n}}{b_n} = \frac{\sqrt{M_n}}{b_n}, \quad n \geq 1.$$

Then (Y_n) is a nonnegative martingale (adapted to the natural filtration), so Y_n converges a.s. to some $Y_\infty \geq 0$.

If $b_\infty := \prod_{n=1}^\infty a_n > 0$, then

$$\mathbb{E}(Y_n^2) = \mathbb{E}\left(\frac{M_n}{b_n^2}\right) = \frac{\mathbb{E}(M_n)}{b_n^2} = \frac{1}{b_n^2} \leq \frac{1}{b_\infty^2} < \infty,$$

so $\sup_n \mathbb{E}(Y_n^2) < \infty$.

By Doob's L^2 maximal inequality applied to (Y_n^2),

$$\mathbb{E}\Big(\sup_{1\leq k\leq n} Y_k^2\Big) \leq 4\,\mathbb{E}(Y_n^2) \leq 4\sup_m \mathbb{E}(Y_m^2) < \infty.$$

Since $M_k = b_k^2 Y_k^2$ and $b_k \geq b_\infty > 0$, we have

$$M_k \leq Y_k^2 / b_\infty^2 \quad \text{for all } k,$$

so

$$\sup_{1\leq k\leq n} M_k \leq \frac{1}{b_\infty^2} \sup_{1\leq k\leq n} Y_k^2.$$

Thus, by monotone convergence,

$$\mathbb{E}\Big(\sup_{k\geq 1} M_k\Big) \leq \frac{1}{b_\infty^2}\mathbb{E}\Big(\sup_{k\geq 1} Y_k^2\Big) < \infty.$$

Let $M_* := \sup_{k\geq 1} M_k$. Then M_* is integrable, and for every n,

$$\mathbb{E}(|M_n|\mathbf{1}_{\{|M_n|>K\}}) \leq \mathbb{E}(|M_*|\mathbf{1}_{\{|M_*|>K\}}).$$

Therefore

$$\lim_{K\to\infty} \sup_n \mathbb{E}(|M_n|\mathbf{1}_{\{|M_n|>K\}}) \leq \lim_{K\to\infty} \mathbb{E}(|M_*|\mathbf{1}_{\{|M_*|>K\}}) = 0,$$

so (M_n) is uniformly integrable. Thus $(b) \Rightarrow (c)$.

$(c) \Rightarrow (d)$: Since (M_n) is a nonnegative martingale with $\mathbb{E}(M_n) = 1$ and is uniformly integrable, Lemma 5.3.6 (with $p = 1$) implies that $M_n \to M_\infty$ in L^1.

$(d) \Rightarrow (e)$: We have

$$|1 - \mathbb{E}(M_\infty)| = |\mathbb{E}(M_n - M_\infty)| \le \mathbb{E}(|M_n - M_\infty|).$$

Letting $n \to \infty$ and using L^1 convergence gives $\mathbb{E}(M_\infty) = 1$.

$(e) \Rightarrow (b)$: Consider again $Y_n := \sqrt{M_n}/b_n$ as above. As (Y_n) is a nonnegative martingale, $Y_n \to Y_\infty$ a.s. for some finite $Y_\infty \ge 0$. If $b_n \downarrow 0$, then

$$M_n = b_n^2 Y_n^2 \xrightarrow[n\to\infty]{} 0$$

almost surely, so $M_\infty = 0$ a.s., contradicting $\mathbb{E}(M_\infty) = 1$. Thus b_n cannot converge to 0, and in fact

$$b_\infty = \prod_{n=1}^{\infty} a_n > 0.$$

So $(e) \Rightarrow (b)$.

Failure of the conditions: If one of the conditions (a)–(e) fails, then in particular (b) fails, so

$$\prod_{n=1}^{\infty} a_n = 0.$$

Then $b_n \downarrow 0$ and from $Y_n \to Y_\infty$ a.s. we get

$$M_n = b_n^2 Y_n^2 \xrightarrow[n\to\infty]{} 0 \quad \text{a.s.},$$

so $M_\infty = 0$ almost surely. □

5.3.1 Backwards martingales

Definition 5.3.3 (Backwards martingale)**.** *A* backwards martingale *is a sequence of random variables $(X_n)_{n\le 0}$ adapted to a filtration $(\mathcal{F}_n)_{n\le 0}$ such that*

$$X_n := \mathbb{E}(X_0 \mid \mathcal{F}_n), \quad n \le 0,$$

for some integrable random variable X_0.

Lemma 5.3.14. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X an integrable random variable. Consider the collection*

$$\mathcal{C} := \{Y : Y = \mathbb{E}(X \mid \mathcal{G}) \text{ for some } \mathcal{G} \subset \mathcal{F}\}.$$

Then $\mathcal{C}$ is uniformly integrable.

Remark. Formally, $\mathcal{C}$ consists of all versions of such conditional expectations.

Proof. Let $\varepsilon > 0$ be given. For $\delta > 0$ define

$$c_\delta := \sup\big\{\mathbb{E}(|X|\mathbf{1}_A) : A \in \mathcal{F},\ \mathbb{P}(A) \le \delta\big\}.$$

By dominated convergence, $c_\delta \downarrow 0$ as $\delta \downarrow 0$. Choose $\delta_0 > 0$ such that $c_{\delta_0} \le \varepsilon$. Also choose $M > 0$ such that

$$\frac{1}{M}\,\mathbb{E}(|X|) \le \delta_0.$$

For $Y \in \mathcal{C}$, Jensen's inequality gives

$$|Y| \le \mathbb{E}(|X| \mid \mathcal{G}) \quad \text{a.s.},$$

so in particular $\mathbb{E}(|Y|) \le \mathbb{E}(|X|)$. Hence

$$\mathbb{P}(|Y| > M) \le \frac{1}{M}\,\mathbb{E}(|Y|) \le \frac{1}{M}\,\mathbb{E}(|X|) \le \delta_0.$$

By the definition of c_{δ_0}, for the set $A := \{|Y| > M\}$ we have $\mathbb{P}(A) \le \delta_0$ and

$$\mathbb{E}(|Y|\mathbf{1}_{\{|Y|>M\}}) \le \mathbb{E}(|X|\mathbf{1}_{\{|Y|>M\}}) \le c_{\delta_0} \le \varepsilon.$$

Since this bound is uniform over all $Y \in \mathcal{C}$, it follows that for this choice of M,

$$\sup_{Y\in\mathcal{C}} \mathbb{E}(|Y|\mathbf{1}_{\{|Y|>M\}}) \le \varepsilon.$$

Thus $\mathcal{C}$ is uniformly integrable. □

Theorem 5.3.15. *Let $(X_n)_{n\le 0}$ be a backwards martingale with respect to $(\mathcal{F}_n)_{n\le 0}$, i.e. $X_n = \mathbb{E}(X_0 \mid \mathcal{F}_n)$ for an integrable X_0. Then the limit*

$$X_{-\infty} := \lim_{n\to-\infty} X_n$$

exists almost surely and in L^1. Moreover, if

$$\mathcal{F}_{-\infty} := \bigcap_{n\le 0} \mathcal{F}_n,$$

then

$$X_{-\infty} = \mathbb{E}(X_0 \mid \mathcal{F}_{-\infty}) \quad a.s.$$

Proof. Apply Doob's upcrossing inequality to the finite sequence $X_{-n}, \dots, X_0$. For any $a < b$,

$$(b-a)\,\mathbb{E}(U_n[a,b]) \le \mathbb{E}\big((X_0 - a)_+\big) < \infty,$$

where $U_n[a,b]$ is the number of upcrossings of $[a,b]$ by $X_{-n}, \dots, X_0$. Letting $n \to \infty$ shows that the total number of upcrossings is finite for every rational $a < b$, hence $(X_n)_{n\le 0}$ has an almost sure limit $X_{-\infty}$ as $n \to -\infty$.

By the previous lemma on uniform integrability of conditional expectations, the family $\{X_n : n \le 0\}$ is uniformly integrable, so convergence in probability plus uniform integrability yields L^1 convergence:

$$\mathbb{E}(|X_n - X_{-\infty}|) \xrightarrow[n\to-\infty]{} 0.$$

Let $A \in \mathcal{F}_{-\infty}$. Then $A \in \mathcal{F}_n$ for all $n \le 0$, and $X_n = \mathbb{E}(X_0 \mid \mathcal{F}_n)$ implies

$$\mathbb{E}(X_n\mathbf{1}_A) = \mathbb{E}(X_0\mathbf{1}_A) \quad \text{for all } n \le 0.$$

On the other hand, $X_n\mathbf{1}_A \to X_{-\infty}\mathbf{1}_A$ in L^1, so

$$\mathbb{E}(X_n\mathbf{1}_A) \xrightarrow[n\to-\infty]{} \mathbb{E}(X_{-\infty}\mathbf{1}_A).$$

Thus $\mathbb{E}(X_0\mathbf{1}_A) = \mathbb{E}(X_{-\infty}\mathbf{1}_A)$ for all $A \in \mathcal{F}_{-\infty}$, i.e. $X_{-\infty}$ is a version of $\mathbb{E}(X_0 \mid \mathcal{F}_{-\infty})$ provided we show that $X_{-\infty}$ is $\mathcal{F}_{-\infty}$-measurable.

To see this, fix $a < b$ and define

$$G_n((a,b)) := \bigcap_{m\ge n} \{\omega : X_m(\omega) \in (a,b)\}, \quad n \le 0.$$

Since X_m is $\mathcal{F}_m$-measurable and $\mathcal{F}_m \subseteq \mathcal{F}_n$ for $m \ge n$ (reverse filtration), each $G_n((a,b)) \in \mathcal{F}_n$. The sets $G_n((a,b))$ form an increasing family in n and

$$\bigcup_{n\le 0} G_n((a,b)) = \{\omega : X_{-\infty}(\omega) \in (a,b)\}.$$

Hence $\{X_{-\infty} \in (a,b)\} \in \mathcal{F}_n$ for every $n \le 0$, so

$$\{X_{-\infty} \in (a,b)\} \in \bigcap_{n\le 0} \mathcal{F}_n = \mathcal{F}_{-\infty}.$$

Since intervals (a,b) generate the Borel σ-algebra, $X_{-\infty}$ is $\mathcal{F}_{-\infty}$-measurable. This completes the proof. □

5.3.2 Exchangeable processes and de Finetti's theorem

Let $X_1, X_2, \dots$ be random variables. For any measurable function $f(X_1, X_2, \dots)$ and $n \ge 1$, define the symmetrization

$$A_n(f) := \frac{1}{n!} \sum_{\pi\in S_n} f\big(X_{\pi(1)}, X_{\pi(2)}, \dots, X_{\pi(n)}, X_{n+1}, X_{n+2}, \dots\big),$$

where S_n is the permutation group on $\{1, \dots, n\}$.

Let $\mathfrak{F}_n$ be the class of all measurable functions $f(X_1, X_2, \dots)$ that are symmetric in $X_1, \dots, X_n$. Let $\mathcal{E}_n$ be the σ-algebra generated by $\mathfrak{F}_n$; equivalently, $\mathcal{E}_n$ is generated by all random variables of the form $A_n(f)$. Clearly $\mathfrak{F}_{n+1} \subseteq \mathfrak{F}_n$, hence $\mathcal{E}_{n+1} \subseteq \mathcal{E}_n$. Define the *exchangeable* σ-algebra

$$\mathcal{E} := \bigcap_{n\ge 1} \mathcal{E}_n.$$

Definition 5.3.4. *A sequence $(X_n)_{n\ge 1}$ is called* exchangeable *if for every n and every permutation $\pi \in S_n$, the random vectors*

$$(X_1, \dots, X_n) \quad \text{and} \quad (X_{\pi(1)}, \dots, X_{\pi(n)})$$

have the same distribution.

Theorem 5.3.16 (de Finetti)**.** *If $(X_n)_{n\geq 1}$ is exchangeable, then conditionally on $\mathcal{E}$ the random variables $X_1, X_2, \dots$ are independent and identically distributed.*

Proof. **Step 1: Symmetric U-statistics and backwards martingale.** Let $f : \mathbb{R}^k \to \mathbb{R}$ be bounded and measurable. For $n \geq k$ let $\mathcal{P}_{n,k}$ be the set of ordered k–tuples of distinct indices from $\{1, \dots, n\}$, i.e.

$$\mathcal{P}_{n,k} := \{(i_1, \dots, i_k) : i_r \in [n],\ i_r \neq i_s \text{ for } r \neq s\},$$

so $|\mathcal{P}_{n,k}| = P(n,k) := n!/(n-k)!$. Define the (symmetrized) k–fold average

$$\begin{aligned} B_n(f) &:= \frac{(n-k)!}{n!} \sum_{(i_1,\dots,i_k)\in\mathcal{P}_{n,k}} f(X_{i_1}, \dots, X_{i_k}) \\ &= \frac{1}{P(n,k)} \sum_{(i_1,\dots,i_k)\in\mathcal{P}_{n,k}} f(X_{i_1}, \dots, X_{i_k}). \end{aligned}$$

By construction, permuting the indices $1, \dots, n$ just permutes the summands in the average, so $B_n(f)$ is invariant under any permutation of $(X_1, \dots, X_n)$. Let $\mathcal{E}_n$ be the σ-algebra generated by all events invariant under S_n. Then $B_n(f)$ is $\mathcal{E}_n$-measurable.

By exchangeability, for any $(i_1, \dots, i_k) \in \mathcal{P}_{n,k}$,

$$\mathbb{E}\big(f(X_{i_1}, \dots, X_{i_k}) \mid \mathcal{E}_n\big) = \mathbb{E}\big(f(X_1, \dots, X_k) \mid \mathcal{E}_n\big).$$

Averaging this identity over all $(i_1, \dots, i_k) \in \mathcal{P}_{n,k}$ gives

$$B_n(f) = \mathbb{E}\big(f(X_1, \dots, X_k) \mid \mathcal{E}_n\big).$$

The σ-algebras $(\mathcal{E}_n)_{n\geq 1}$ form a decreasing sequence, and

$$\mathcal{E} := \bigcap_{n\geq 1} \mathcal{E}_n$$

is the exchangeable σ-algebra. Thus $(B_n(f), \mathcal{E}_n)_{n\geq k}$ is a backwards martingale. By the backwards martingale convergence theorem,

$$B_n(f) \xrightarrow[n\to\infty]{\text{a.s. and in } L^1} \mathbb{E}\big(f(X_1, \dots, X_k) \mid \mathcal{E}\big).$$

Step 2: Factorization identity. To prove conditional independence, let $f : \mathbb{R}^{k-1} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be bounded measurable. Write

$$\varphi(x_1, \dots, x_k) := f(x_1, \dots, x_{k-1})\, g(x_k).$$

We analyze $B_n(f)B_n(g)$.

By definition,

$$B_n(f) = \frac{1}{P(n,k-1)} \sum_{I\in\mathcal{P}_{n,k-1}} f(X_I), \qquad B_n(g) = \frac{1}{n}\sum_{j=1}^{n} g(X_j),$$

so

$$B_n(f)B_n(g) = \frac{1}{P(n,k-1)\,n} \left(\sum_{I\in\mathcal{P}_{n,k-1}} f(X_I)\right)\left(\sum_{j=1}^{n} g(X_j)\right).$$

Expand the product and split according to whether $j \notin I$ or $j \in I$:

$$\sum_{I\in\mathcal{P}_{n,k-1}} \sum_{j\notin I} f(X_I)g(X_j) \ + \sum_{I\in\mathcal{P}_{n,k-1}} \sum_{j\in I} f(X_I)g(X_j).$$

For the first sum, each pair (I, j) with $I \in \mathcal{P}_{n,k-1}$ and $j \notin I$ determines a unique ordered k–tuple $(i_1, \dots, i_k) \in \mathcal{P}_{n,k}$ by appending j to I (in order), and conversely each $(i_1, \dots, i_k) \in \mathcal{P}_{n,k}$ can be obtained in this way in exactly k different ways (choosing which coordinate plays the role of j). Moreover, for such pairs

$$f(X_I)g(X_j) = \varphi(X_{i_1}, \dots, X_{i_k}).$$

Thus the first sum equals

$$\begin{aligned} &\sum_{I\in\mathcal{P}_{n,k-1}} \sum_{j\notin I} f(X_I)g(X_j) \\ &= k \sum_{(i_1,\dots,i_k)\in\mathcal{P}_{n,k}} \varphi(X_{i_1}, \dots, X_{i_k}) \\ &= k\, P(n,k)\, B_n(\varphi). \end{aligned}$$

For the second sum, use the crude bound

$$\left|\sum_{I\in\mathcal{P}_{n,k-1}} \sum_{j\in I} f(X_I)g(X_j)\right| \leq P(n,k-1)\,(k-1)\,\|f\|_\infty \|g\|_\infty,$$

because for each I there are at most $k-1$ values of j with $j \in I$. Hence

$$\begin{aligned} &\left|\frac{1}{P(n,k-1)\,n} \sum_{I\in\mathcal{P}_{n,k-1}} \sum_{j\in I} f(X_I)g(X_j)\right| \\ &\leq \frac{k-1}{n}\,\|f\|_\infty \|g\|_\infty \xrightarrow[n\to\infty]{} 0. \end{aligned}$$

Putting this together,

$$\begin{aligned} &B_n(f)B_n(g) \\ &= \frac{1}{P(n,k-1)\,n}\{k\,P(n,k)\,B_n(\varphi) \\ &\qquad + O\big(P(n,k-1)\,(k-1)\|f\|_\infty\|g\|_\infty\big)\} \\ &= \frac{k\,P(n,k)}{P(n,k-1)\,n} B_n(\varphi) + O(1/n). \end{aligned}$$

Since

$$\frac{k\,P(n,k)}{P(n,k-1)\,n} = \frac{k\,\frac{n!}{(n-k)!}}{\frac{n!}{(n-k+1)!}\,n} = \frac{n-k+1}{n},$$

we obtain the identity

$$B_n(f)B_n(g) = \frac{n-k+1}{n}\, B_n(\varphi) + O\left(\frac{1}{n}\right),$$

where the error term is uniform in n (for fixed f, g, k).

Step 3: Passing to the limit and conditional independence. As $n \to \infty$, we know from Step 1 that

$$\begin{aligned} B_n(f) &\to \mathbb{E}\big(f(X_1, \dots, X_{k-1}) \mid \mathcal{E}\big), \\ B_n(g) &\to \mathbb{E}\big(g(X_k) \mid \mathcal{E}\big), \\ B_n(\varphi) &\to \mathbb{E}\big(\varphi(X_1, \dots, X_k) \mid \mathcal{E}\big) \end{aligned}$$

almost surely and in L^1, and the $O(1/n)$ term tends to 0. Therefore, taking limits in

$$B_n(f)B_n(g) = \frac{n-k+1}{n}\, B_n(\varphi) + O\left(\frac{1}{n}\right)$$

gives

$$\begin{aligned} &\mathbb{E}\big(f(X_1, \dots, X_{k-1}) \mid \mathcal{E}\big)\, \mathbb{E}\big(g(X_k) \mid \mathcal{E}\big) \\ &= \mathbb{E}\big(f(X_1, \dots, X_{k-1})g(X_k) \mid \mathcal{E}\big). \end{aligned}$$

Since this holds for all bounded measurable f and g, it is exactly the statement that X_k is conditionally independent of $(X_1, \dots, X_{k-1})$ given $\mathcal{E}$.

Because k was arbitrary, it follows that $(X_n)_{n\geq 1}$ is conditionally independent given $\mathcal{E}$.

Step 4: Conditional identical distribution. Finally, by exchangeability, for any bounded measurable $h : \mathbb{R} \to \mathbb{R}$ and any i, j,

$$\mathbb{E}\big(h(X_i) \mid \mathcal{E}\big) = \mathbb{E}\big(h(X_j) \mid \mathcal{E}\big) \quad \text{a.s.},$$

so the regular conditional laws of X_i and X_j given $\mathcal{E}$ are identical. Therefore, conditionally on $\mathcal{E}$, the sequence $(X_n)_{n\geq 1}$ is i.i.d., which is de Finetti's theorem. □

Theorem 5.3.17 (Law of the iterated logarithm for Gaussians)**.** *Let $(X_n)_{n\geq 1}$ be i.i.d. $\mathcal{N}(0,1)$ random variables and $S_n := \sum_{i=1}^n X_i$. Then*

$$\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n\log\log n}} = 1 \quad \textit{a.s.}$$

Proof. For $n \geq 3$ set $h(n) := \sqrt{2n\log\log n}$. For any $\theta \in \mathbb{R}$,

$$\mathbb{E}(e^{\theta S_n}) = e^{\frac{1}{2}\theta^2 n}.$$

Thus $(e^{\theta S_n})$ is a submartingale, and Doob's inequality yields

$$\mathbb{P}\Big(\sup_{1\le k\le n} S_k \ge \ell\Big) \le e^{-\theta\ell}\, e^{\frac{1}{2}\theta^2 n}.$$

Optimizing in θ gives

$$\mathbb{P}\Big(\sup_{1\le k\le n} S_k \ge \ell\Big) \le e^{-\ell^2/(2n)}.$$

Fix $K > 1$ and define $\ell_n := K\, h(K^{n-1})$. Then

$$\begin{aligned}\mathbb{P}\Big(\sup_{1\le k\le K^n} S_k \ge \ell_n\Big) &\le \exp\Big(-\frac{\ell_n^2}{2K^n}\Big)\\ &= \exp\big(-K\log\log K^{n-1}\big) = \frac{1}{\big((n-1)\log K\big)^K}.\end{aligned}$$

Since $\sum_n \big((n-1)\log K\big)^{-K} < \infty$, the first Borel–Cantelli lemma implies that almost surely, for all large n,

$$\sup_{1\le k\le K^n} S_k < \ell_n.$$

Thus, for all large n and all $K^{n-1} \le k \le K^n$,

$$S_k \le \sup_{1\le j\le K^n} S_j \le \ell_n = K\, h(K^{n-1}) \le K\, h(k),$$

so

$$\limsup_{k\to\infty} \frac{S_k}{h(k)} \le K \quad \text{a.s.}$$

Since $K > 1$ is arbitrary, we obtain

$$\limsup_{k\to\infty} \frac{S_k}{h(k)} \le 1 \quad \text{a.s.}$$

By symmetry, the same argument applied to $-S_k$ yields

$$\liminf_{k\to\infty} \frac{S_k}{h(k)} \ge -1 \quad \text{a.s.}$$

For the lower bound on the limsup, fix an integer $N > 4$. Note that

$$\frac{S(N^{n+1}) - S(N^n)}{\sqrt{N^{n+1} - N^n}} \sim \mathcal{N}(0,1)$$

by independence and variance additivity. For $\varepsilon \in (0,1)$, set

$$F_n := \left\{ \frac{S(N^{n+1}) - S(N^n)}{\sqrt{N^{n+1} - N^n}} \ge (1-\varepsilon)\, \frac{h(N^{n+1} - N^n)}{\sqrt{N^{n+1} - N^n}} \right\}.$$

By the Gaussian tail lemma below,

$$\begin{aligned}\mathbb{P}(F_n) &\ge g\big((1-\varepsilon)\sqrt{2\log\log(N^{n+1} - N^n)}\big)\\ &= g\big((1-\varepsilon)\sqrt{2\log(n\log N + \log(N-1))}\big),\end{aligned}$$

where

$$g(x) := \frac{1}{\sqrt{2\pi}(x + x^{-1})} e^{-x^2/2}.$$

One checks (exercise) that

$$\sum_{n=1}^{\infty} g\big((1-\varepsilon)\sqrt{2\log(n\log N + \log(N-1))}\big) = \infty.$$

Hence $\sum_n \mathbb{P}(F_n) = \infty$, and by the second Borel–Cantelli lemma, $\mathbb{P}(F_n \text{ i.o.}) = 1$. Thus, infinitely often,

$$S(N^{n+1}) - S(N^n) \ge (1-\varepsilon)\, h(N^{n+1} - N^n).$$

From the upper bound already proved, for any $K > 1$ we have eventually $S(N^n) \ge -K\, h(N^n)$, so infinitely often,

$$S(N^{n+1}) \ge (1-\varepsilon)\, h(N^{n+1} - N^n) - K\, h(N^n).$$

Dividing by $h(N^{n+1})$ and taking lim sup,

$$\limsup_{n\to\infty} \frac{S(N^{n+1})}{h(N^{n+1})} \ge (1-\varepsilon) + \theta\Big(\frac{1}{\sqrt{N}}\Big),$$

where the error term tends to 0 as $N \to \infty$. Letting $N \to \infty$ and then $\varepsilon \downarrow 0$ gives

$$\limsup_{k\to\infty} \frac{S_k}{h(k)} \ge 1 \quad \text{a.s.}$$

Combining with the upper bound completes the proof. □

Lemma 5.3.18. *Let X be standard normal, $X \sim \mathcal{N}(0,1)$. Then for all $x > 0$,*

$$\mathbb{P}(X > x) \ge \frac{1}{\sqrt{2\pi}(x + x^{-1})}\, e^{-x^2/2}.$$

Proof. Let $\varphi(x)$ be the standard normal density. A direct computation gives

$$\big(x^{-1}\varphi(x)\big)' = -\big(1 + x^{-2}\big)\varphi(x).$$

Thus

$$\begin{aligned}x^{-1}\varphi(x) &= -\int_x^\infty \big(y^{-1}\varphi(y)\big)' dy\\ &= \int_x^\infty (1 + y^{-2})\varphi(y)\, dy \le (1 + x^{-2})\, \mathbb{P}(X > x).\end{aligned}$$

Rearranging yields

$$\mathbb{P}(X > x) \ge \frac{x^{-1}\varphi(x)}{1 + x^{-2}} = \frac{1}{\sqrt{2\pi}(x + x^{-1})} e^{-x^2/2}.$$

□